



HPE ProLiant DL360 Gen10 Server Maintenance and Service Guide

Part Number: 30-BC3F38A8-409

Published: March 2026

Edition: 19

HPE ProLiant DL360 Gen10 Server Maintenance and Service Guide

Abstract

This guide describes identification and maintenance procedures, diagnostic tools, specifications, and requirements for hardware components and software. This guide is for an experienced service technician. Hewlett Packard Enterprise assumes you are qualified in the servicing of computer equipment, trained in recognizing hazards in products, and are familiar with weight and stability precautions.

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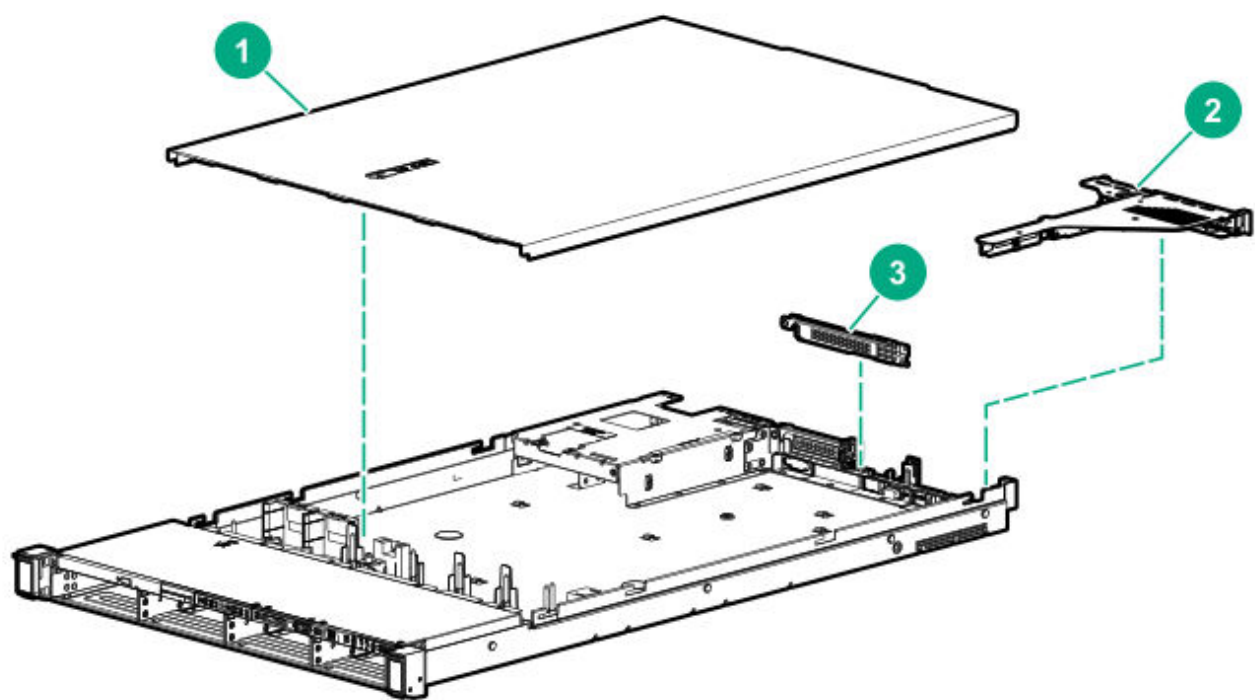
Illustrated parts catalog

Subtopics

- [Mechanical components](#)
- [System components](#)
- [Server options](#)

Mechanical components

Hewlett Packard Enterprise continually improves and changes product parts. For complete and current supported parts information, see the [Hewlett Packard Enterprise PartSurfer website](#).



Item	Description
1	Access panel spare parts
2	PCIe riser cage spare part
3	PCIe riser blank spare parts
4	Miscellaneous blank spare parts*

Item	Description
5	Server ear and front bezel spare parts*
6	Rack mounting hardware spare parts*
7	Cable management arm spare parts*

*Not shown

Subtopics

[Access panel spare parts](#)

[PCIe riser cage spare part](#)

[PCIe riser blank spare parts](#)

[Miscellaneous blank spare parts](#)

[Server ear and front bezel spare parts](#)

[Rack mounting hardware spare parts](#)

[Cable management arm spare parts](#)

Access panel spare parts

Customer self repair: **mandatory**

Description	Spare part number
Access panel	875536-001

PCIe riser cage spare part

Customer self repair: **optional**

Description	Spare part number
Assembly, secondary riser x16 LP	875539-001
Assembly, secondary riser x16 FH	875540-001

For primary riser board spares, see [PCIe riser board spare parts](#)

PCIe riser blank spare parts

Customer self repair: **mandatory**

Description	Spare part number
PCIe riser blank, slotted	875537-001

Miscellaneous blank spare parts

Customer self repair: **mandatory**

Description	Spare part number
Blank cover kit - Contains power supply blank, Flexible LOM blank, fan blank, and heatsink blank	775423-001
LATCH, MEGA CBLANK, FAN	P39788-001

Server ear and front bezel spare parts

Customer self repair: **mandatory**

Description	Spare part number
Left ear	875541-001
Right ear	875542-001
1U Bezel	875561-001

Rack mounting hardware spare parts

Customer self repair: **mandatory**

Description	Spare part number
SFF Easy Install Rail kit	875544-001
LFF Easy Install Rail kit	744112-001
SFF Ball Bearing Rail kit	875543-001
LFF Ball Bearing Rail kit	675041-001

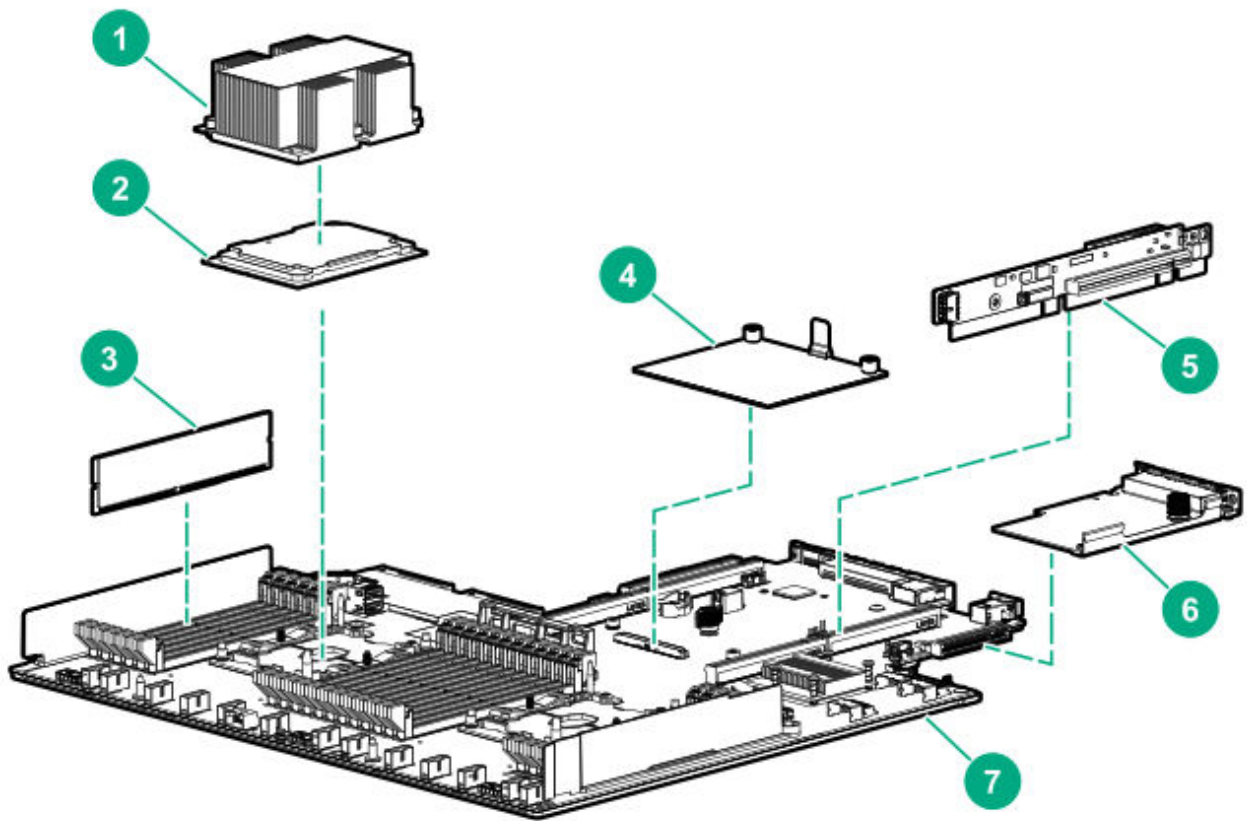
Cable management arm spare parts

Customer self repair: **mandatory**

Description	Spare part number
1U Cable Management Arm	744113-001

System components

Hewlett Packard Enterprise continually improves and changes product parts. For complete and current supported parts information, see the [**Hewlett Packard EnterprisePartSurfer website**](#).



Item	Description
1	Heatsink spare parts
2	Processor spare parts
3	DIMM spare parts HPE16GB NVDIMM spare part HPE Persistent Memory module spare parts
4	Controller spare parts
5	PCIe riser board spare parts
6	FlexibleLOM adapter spare parts Pensando DSP iLO Sideband ALOM Module spare parts
7	System board assembly spare parts
8	Expansion board spare parts ¹ Host bus adapter spare parts

Item Description

- [Converged network adapter spare parts](#)
- [InfiniBand adapter spare parts](#)
- [NIC network adapter spare parts](#)
- [SmartNIC network adapter spare part](#)
- [Smart I/O \(Pensando Distributed Services Platform\) spare parts](#)

¹ Not shown

Subtopics

[Heatsink spare parts](#)

[Processor spare parts](#)

[DIMM spare parts](#)

[HPE 16GB NVDIMM spare part](#)

[Persistent memory module spare parts](#)

[Controller spare parts](#)

[PCIe riser board spare parts](#)

[FlexibleLOM adapter spare parts](#)

[System board assembly spare parts](#)

[Expansion board spare parts](#)

Heatsink spare parts

[Customer self repair](#): **no**

Description	Spare part number
Standard heatsink	872452-001
Performance heatsink	872453-001

Processor spare parts

Subtopics

First Generation Intel Xeon Scalable Processors

Subtopics

31XX processor spare parts

41XX processor spare parts

51XX processor spare parts

61xx processor spare parts

81xx processor spare parts

31XX processor spare parts

Customer self repair: **no**

Description	Spare part number
1.7-GHz Intel Xeon-B 3104 6c 85 W processor	875709-001
1.7-GHz Intel Xeon-B 3106 8c 85 W processor	875710-001

41XX processor spare parts

Customer self repair: **no**

Description	Spare part number
1.8-GHz Intel Xeon-S 4108 8c 85 W processor	875712-001
2.1-GHz Intel Xeon-S 4110 8c 85 W processor	875711-001
2.6-GHz Intel Xeon-S 4112 4c 85 W processor	875714-001
2.2-GHz Intel Xeon-S 4114 10c 85 W processor	875713-001
2.1-GHz Intel Xeon-S 4116 12c 85 W processor	875716-001

51XX processor spare parts

Customer self repair: **no**

Description	Spare part number
2.4-GHz Intel Xeon-G 5115 10c 85 W processor	875715-001
2.0-GHz Intel Xeon-G 5117 14c 105 W processor	P10199-001
2.3-GHz Intel Xeon-G 5118 12c 105 W processor	875717-001
2.2-GHz Intel Xeon-G 5120 14c 105 W processor	875718-001
3.6-GHz Intel Xeon-G 5122 4c 105 W processor	875719-001

61xx processor spare parts

Customer self repair: **no**

Description	Spare part number
2.6-GHz Intel Xeon-G 6126 23c 125 W processor	875720-001
3.4-GHz Intel Xeon-G 6128 6c 115 W processor	875721-001
2.1-GHz Intel Xeon-G 6130 16c 125 W processor	874736-001
2.6-GHz Intel Xeon-G 6132 14c 140 W processor	875722-001
3.2-GHz Intel Xeon-G 6134 8c 130 W processor	875723-001
3.2-GHz Intel Xeon-G 6134M 8c 130 W processor	878083-001
3.0-GHz Intel Xeon-G 6136 12c 150 W processor	875724-001
3.9-GHz Intel Xeon-G 6137 8c 205 W processor	882008-001
2.0-GHz Intel Xeon-G 6138 20c 125 W processor	874735-001
2.3-GHz Intel Xeon-G 6140 18c 140 W processor	874734-001
2.3-GHz Intel Xeon-G 6140M 18c 140 W processor	878084-001
2.6-GHz Intel Xeon-G 6142 16c 150 W processor	874733-001
2.6-GHz Intel Xeon-G 6142M 16c 150 W processor	878085-001
2.8-GHz Intel Xeon-G 6143 16c 205 W processor	882169-001

Description	Spare part number
3.5-GHz Intel Xeon-G 6144 8c 150 W processor	875725-001
3.2-GHz Intel Xeon-G 6146 12c 165 W processor	875726-001
2.4-GHz Intel Xeon-G 6148 20c 150 W processor	874732-001
2.7-GHz Intel Xeon-G 6150 18c 165 W processor	874731-001
2.1-GHz Intel Xeon-G 6152 22c 140 W processor	874730-001
3.0-GHz Intel Xeon-G 6154 18c 200 W processor	875727-001

81xx processor spare parts

Customer self repair: **no**

Description	Spare part number
2.0-GHz Intel Xeon-P 8153 16c 125 W processor	875728-001
3.6-GHz Intel Xeon-P 8156 4c 105 W processor	875732-001
3.0-GHz Intel Xeon-P 8158 12c 150 W processor	875733-001
2.1-GHz Intel Xeon-P 8160 24c 150 W processor	874729-001
2.1-GHz Intel Xeon-P 8160M 24c 150 W processor	878086-001
2.0-GHz Intel Xeon-P 8164 26c 150 W processor	875729-001
2.3-GHz Intel Xeon-P 8165 24c 205 W processor	P00868-001
2.7-GHz Intel Xeon-P 8168 24c 205 W processor	875730-001
2.1-GHz Intel Xeon-P 8170 26c 165 W processor	874728-001
2.1-GHz Intel Xeon-P 8170M 26c 165 W processor	878087-001
2.1-GHz Intel Xeon-P 8176 28c 165 W processor	874727-001
2.1-GHz Intel Xeon-P 8176M 28c 165 W processor	878088-001
2.5-GHz Intel Xeon-P 8180 28c 205 W processor	875731-001
2.5-GHz Intel Xeon-P 8180M 28c 205 W processor	878089-001

Second Generation Intel Xeon Scalable Processors

Subtopics

[32XX processor spare parts](#)

[42XX processor spare parts](#)

[52XX processor spare parts](#)

[62xx processor spare parts](#)

[82xx processor spare parts](#)

32XX processor spare parts

Customer self repair: **no**

Description	Spare part number
1.9-GHz Intel Xeon-B 3204 6c 85 W processor	P11604-001
1.9-GHz Intel Xeon-B 3206R 8c 85 W processor	P19248-001

42XX processor spare parts

Customer self repair: **no**

Description	Spare part number
2.1-GHz Intel Xeon-S 4208 8c 85 W processor	P11605-001
2.2-GHz Intel Xeon-S 4210 10c 85 W processor	P11606-001
2.4-GHz Intel Xeon-S 4210R 10c 100 W processor	P19246-001
2.2-GHz Intel Xeon-S 4214 12c 85 W processor	P11607-001
2.4-GHz Intel Xeon-S 4214R 12c 100 W processor	P19245-001
2.2-GHz Intel Xeon-S 4214Y 12c 85 W processor	P11636-001
2.5-GHz Intel Xeon-S 4215 8c 85 W processor	P11608-001
3.2-GHz Intel Xeon-S 4215R 8c 130 W processor	P25089-001
2.1-GHz Intel Xeon-S 4216 16c 100 W processor	P11609-001

52XX processor spare parts

Customer self repair: **no**

Description	Spare part number
2.5-GHz Intel Xeon-G 5215 10c 85 W processor	P11610-001
2.5-GHz Intel Xeon-G 5215M 10c 85 W processor	P11626-001
2.5-GHz Intel Xeon-G 5215L 10c 85 W processor	P11631-001
3.0-GHz Intel Xeon-G 5217 8c 115 W processor	P11611-001
2.3-GHz Intel Xeon-G 5218 16c 125 W processor	P11612-001
2.3-GHz Intel Xeon-G 5218B 16c 125 W processor	P12532-001
2.3-GHz Intel Xeon-G 5218N 16c 110 W processor	P12021-001
2.1-GHz Intel Xeon-G 5218R 20c 125 W processor	P25090-001
2.2-GHz Intel Xeon-G 5220 18c 125 W processor	P11613-001
2.2-GHz Intel Xeon-G 5220R 24c 150 W processor	P19241-001
2.7-GHz Intel Xeon-G 5220S 18c 125 W processor	P11627-001
3.8-GHz Intel Xeon-G 5222 4c 105 W processor	P11632-001

62xx processor spare parts

Customer self repair: **no**

Description	Spare part number
2.9-GHz Intel Xeon-G 6208U 16c 150 W processor	P25102-001
2.1-GHz Intel Xeon-G 6209U 20c 125 W processor	P11644-001
2.5-GHz Intel Xeon-G 6210U 20c 150 W processor	P11642-001
2.4-GHz Intel Xeon-G 6212U 24c 165 W processor	P11643-001
1.8-GHz Intel Xeon-G 6222V 20c 115 W processor	P12019-001
2.7-GHz Intel Xeon-G 6226 12c 125 W processor	P12008-001
2.9-GHz Intel Xeon-G 6226R 16c 150 W processor	P25094-001
2.1-GHz Intel Xeon-G 6230 20c 125 W processor	P11614-001
2.3-GHz Intel Xeon-G 6230N 20c 125 W processor	P12022-001

Description	Spare part number
2.1-GHz Intel Xeon-G 6230R 26c 150 W processor	P25095-001
3.3-GHz Intel Xeon-G 6234 8c 130 W processor	P12009-001
2.1GHz Intel Xeon-G 6238 22c 140 W processor	P12010-001
2.1GHz Intel Xeon-G 6238L 22c 140 W processor	P12016-001
2.1GHz Intel Xeon-G 6238M 22c 140 W processor	P12014-001
2.2-GHz Intel Xeon-G 6238R 28c 165 W processor	P25096-001
2.6-GHz Intel Xeon-G 6240 18c 150 W processor	P11615-001
2.6GHz Intel Xeon-G 6240L 18c 150 W processor	P12015-001
2.4-GHz Intel Xeon-G 6240R 24c 165 W processor	P25097-001
2.6-GHz Intel Xeon-G 6240Y 18c 150 W processor	P11637-001
2.6GHz Intel Xeon-G 6240M 18c 150 W processor	P12013-001
2.8-GHz Intel Xeon-G 6242 16c 150 W processor	P11616-001
3.1-GHz Intel Xeon-G 6242R 20c 205 W processor	P25098-001
3.6-GHz Intel Xeon-G 6244 8c 150 W processor	P11617-001
3.3GHz Intel Xeon-G 6246 12c 165 W processor	P12018-001
3.4-GHz Intel Xeon-G 6246R 16c 205 W processor	P25099-001
2.5-GHz Intel Xeon-G 6248 20c 150 W processor	P11618-001
3.0-GHz Intel Xeon-G 6248R 24c 205 W processor	P25100-001
3.9-GHz Intel Xeon-G 6250 8c 185 W processor	P25092-001
3.9-GHz Intel Xeon-G 6250L 8c 185 W processor	P28864-001
2.1-GHz Intel Xeon-G 6252 24c 150 W processor	P11619-001
2.3GHz Intel Xeon-G 6252N 24c 150 W processor	P12023-001
3.1-GHz Intel Xeon-G 6254 18c 200 W processor	P11620-001
3.6-GHz Intel Xeon-G 6256 12c 205 W processor	P25093-001
2.7-GHz Intel Xeon-G 6258R 28c 205 W processor	P25101-001
1.9-GHz Intel Xeon-G 6262V 24c 135 W processor	P12020-001

82xx processor spare parts

Customer self repair: **no**

Description	Spare part number
2.2-GHz Intel Xeon-P 8253 24c 135 W processor	P12011-001
3.8-GHz Intel Xeon-P 8256 4c 105 W processor	P12012-001
2.4-GHz Intel Xeon-P 8260 24c 165 W processor	P11621-001
2.4-GHz Intel Xeon-P 8260M 24c 165 W processor	P11628-001
2.4-GHz Intel Xeon-P 8260Y 16c 165 W processor	P11638-001
2.4-GHz Intel Xeon-P 8260L 24c 165 W processor	P11633-001
2.9-GHz Intel Xeon-P 8268 24c 205 W processor	P11622-001
2.7-GHz Intel Xeon-P 8270 26c 205 W processor	P11623-001
2.2-GHz Intel Xeon-P 8276 28c 165 W processor	P11624-001
2.2-GHz Intel Xeon-P 8276M 28c 165 W processor	P11629-001
2.2-GHz Intel Xeon-P 8276L 28c 165 W processor	P11634-001
2.7-GHz Intel Xeon-P 8280 28c 205 W processor	P11625-001
2.7-GHz Intel Xeon-P 8280M 28c 205 W processor	P11630-001
2.7-GHz Intel Xeon-P 8280L 28c 205 W processor	P11635-001

DIMM spare parts

Customer self repair: **mandatory**

2666 MT/s DIMMs

Description	Spare part number
DIMM, 8GB PC4-2666V-R, 1Gx8	850879-001
DIMM, 8GB PC4-2666V-R, 512Mx8	878490-001
DIMM, 16GB PC4-2666V-R, 2Rx4	850880-001
DIMM, 16GB PC4-2666V-R, 2Rx8	868846-001
DIMM, 32GB PC4-2666V-R, 2Gx4	850881-001
DIMM, 64GB PC4-2666V-L, 2Gx4	850882-001
DIMM, 128GB PC4-2666V-L, 2Gx4	850883-001

2933 MT/s DIMMs

Description	Spare part number
DIMM, 8GB PC4-2933Y-R, 1Gx8	P06186-001
DIMM, 16GB PC4-2933Y-R, 2Gx4	P06187-001
DIMM, 16GB PC4-2933Y-R, 1Gx8	P06188-001
DIMM, 32GB PC4-2933Y-R, 2Gx4	P06189-001
DIMM, 32GB PC4-2933Y-R, 4Gx4	P39382-001
DIMM, 64GB PC4-2933Y-L, 2Gx4	P06190-001
DIMM, 64GB PC4-2933Y-R, 4Gx4	P06192-001
DIMM, 128GB PC4-2933Y-L, 2Gx4	P06191-001
DIMM, 128GB PC4-2933Y-L, 4Gx4	P16001-001

3200 MT/s DIMMs

Description	Spare part number
DIMM, 32GB PC4-2933AA-R, 4Gx4	P39381-001

HPE 16GB NVDIMM spare part

Customer self repair: **mandatory**

Description	Spare part number
NVDIMM 16GB 1Rx4 NN4-2666V-R	874540-001

Persistent memory module spare parts

Customer self repair: **mandatory**

Description	Spare part number
Intel Optane 128GB PMEM 100 for HPE	844071-001
Intel Optane 256GB PMEM 100 for HPE	844072-001
Intel Optane 512GB PMEM 100 for HPE	844073-001

Controller spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE Smart Array E208i-a SR Gen10 Controller	836259-001
HPE Smart Array E208i-a LH SR Gen10 Controller	871039-001
HPE Smart Array P408i-a SR Gen10 Controller	836260-001
HPE Smart Array P408i-a LH SR Gen10 Controller	871040-001
HPE Smart Array P816i-a SR Gen10 Controller	836261-001
HPE Smart Array E208i-p SR Gen10 Controller	836266-001
HPE Smart Array E208e-p SR Gen10 Controller	836267-001
HPE Smart Array P408i-p SR Gen10 Controller	836269-001
HPE Smart Array P408e-p SR Gen10 Controller	836270-001
HPE Smart Array P816i-p SR Gen10 Controller	836271-001
HPE Smart Array P824i-p MR Gen10 Controller	871043-001
Broadcom MegaRAID MR216i-a Tri-Mode Gen10 Plus Controller	P28348-001
Broadcom MegaRAID MR216i-p Tri-Mode Gen10 Plus Controller	P28336-001
Broadcom MegaRAID MR416i-a Tri-Mode Gen10 Plus Controller	P28335-001
Broadcom MegaRAID MR416i-p Tri-Mode Gen10 Plus Controller	P17303-001

PCIe riser board spare parts

Customer self repair: **optional**

Description	Spare part number
Primary riser board, x16 x8, GPU	875545-001
Primary riser board, x16 x8 + PCIe M.2	875546-001
Primary riser board, x16 x16 + SATA M.2	875547-001
Primary riser board, x16 x8, GPU, 2x4 NVMe ports	875548-001
Secondary riser board, 8x4/2x4 NVMe ports	875549-001

FlexibleLOM adapter spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE InfiniBand FDR/EN 40Gb 2-port, 544+ FLR-QSFP Adapter	764737-001
HPE InfiniBand FDR/EN 40Gb 2-port, 544+ QSFP Adapter	764736-001
HPE Ethernet 10/25Gb 2-port 640FLR-SFP28 Adapter	840139-001
HPE Ethernet 10/25Gb 2-port 631FLR-SFP28 Adapter	840133-001
HPE Ethernet 10/25Gb 2-port 622FLR-SFP28 Adapter	869572-001
HPE Ethernet 10Gb 2-port 537FLR-SFP+ Adapter	P12925-001
HPE Ethernet 10Gb 2-port 562FLR-SFP+ Adapter	790317-001
HPE Ethernet 10Gb 2-port 562FLR-T Adapter	840138-001
HPE FlexFabric 10Gb 4-port 536FLR-T Adapter	768082-001
HPE Ethernet 10Gb 2-port 535FLR-T Adapter	854177-001
HPE FlexFabric 10Gb 2-port 534FLR-SFP+ Adapter	701531-001

Description	Spare part number
HPE FlexFabric 10Gb 2-port 533FLR-T Adapter	701534-001
HPE Ethernet 1Gb 4-port 331FLR Adapter	789897-001
HPE Ethernet 1Gb 4-port 366FLR Adapter	669280-001

System board assembly spare parts

Customer self repair: **optional**

Description	Spare part number
System board (DL360) first-generation Intel Xeon scalable processors	875552-001
System board (DL360) first and second-generation Intel Xeon scalable processors	P11781-001
System board (DL360) without embedded LOM	P19926-001
System board blank assembly	P19929-001

Expansion board spare parts

Subtopics

Host bus adapter spare parts

Converged network adapter spare parts

InfiniBand adapter spare parts

NIC network adapter spare parts

SmartNIC network adapter spare part

Solid state NVMe/PCIe Add-In-Card spare parts

Host bus adapter spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE SN1600Q 32Gb 1p FC HBA	868140-001
HPE SN1600Q 32Gb 2p FC HBA	868141-001
HPE SN1600E 32Gb 2p FC HBA	869999-001
HPE SN1610E 32Gb 1p FC HBA	P14421-001
HPE SN1610E 32Gb 2p FC HBA	P14422-001
HPE 81Q PCI-e FC HBA	489190-001
HPE SN1200E 16Gb 1p FC HBA	870001-001
HPE SN1200E 16Gb 2p FC HBA	870002-001
HPE 82Q 8Gb Dual Port PCI-e FC HBA	489191-001
HPE StoreFabric SN1100E 4p 16Gb FC HBA	853008-001
HPE StoreFabric 84E 4-port 8Gb FC HBA	780686-001
HPE 81E 8Gb SP PCI-e FC HBA	697889-001
HPE 82E 8Gb Dual-port PCI-e FC HBA	697890-001
HPE StoreFabric 84Q 4p 8Gb FC HBA	853009-001
HPE SN1100Q 16Gb 1p FC HBA	863010-001
HPE SN1100Q 16Gb 2p FC HBA	853011-001

Converged network adapter spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE CN1100R 2P Converged Network Adapter	706801-001
HPE StoreFabric CN1100R-T 10Gb Converged Network Adapter	872605-001
HPE StoreFabric CN1200E 10Gb Converged Network Adapter	767078-001
HPE StoreFabric CN1200E-T 10Gb Converged Network Adapter	827607-001

InfiniBand adapter spare parts

Customer self repair: **mandatory**

The 940QSFP56 x16 adapter must be installed in PCIe slot 2 and the auxiliary card must be installed in PCIe slot 3.

Description	Spare part number
HPE InfiniBand HDR100/Ethernet 1-port 940QSFP56 x16 Adapter	P08356-001
HPE InfiniBand HDR100/Ethernet 100Gb 2-port 940QSFP56 x16 Adapter	P08355-001
HPE InfiniBand HDR/Ethernet 200Gb 1-port 940QSFP56 x16 Adapter	P08354-001
HPE InfiniBand HDR PCIe G3 Auxiliary card with 150 mm cable kit ¹	P10330-001
HPE InfiniBand HDR PCIe G3 Auxiliary card with 350 mm cable kit ¹	P10331-001

¹ Choose the cable length based on the distance between the adapter card slot and the auxiliary card slot.

NIC network adapter spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE Ethernet 100Gb 1-port 842QSFP28 Adapter	877697-001
HPE Ethernet 10/25Gb 2-port 640SFP28 Adapter	840140-001
HPE Ethernet 10/25Gb 2-port 631SFP28 Adapter	840130-001
HPE Ethernet 10/25Gb 2-port 621SFP28 Adapter	869570-001
HPE Ethernet 10Gb 2-port 537SFP+ Adapter	P10006-001
HPE Ethernet 10Gb 2-port 562T Adapter	840137-001
HPE Ethernet 10Gb 2-port 562SFP+ Adapter	790316-001
HPE Ethernet 10Gb 2-port 548SFP+ Adapter	P12531-001

Description	Spare part number
HPE Ethernet 10Gb 2-port 535T Adapter	815669-001
HPE Ethernet 10Gb 2-port 530SFP Adapter	656244-001
HPE Ethernet 10Gb 2-port 524SFP+ Adapter	P11585-001
HPE Ethernet 10Gb 2-port 530T Adapter	657128-001
HPE Ethernet 10Gb 2-port 521T Adapter	869573-001
HPE Ethernet 1Gb 4-port 366T Adapter	816551-001
HPE Ethernet 1Gb 4-port 331T Adapter	649871-001
HPE Ethernet 1Gb 2-port 361T Adapter	656241-002
HPE Ethernet 1Gb 2-port 332T Adapter	616012-001
HPE Ethernet 10/25 Gb 2-port SFP28 X2522 Adapter	P24737-001
HPE Ethernet 10/25Gb 2-port 642SFP28 Adapter	P25987-001
Intel E810 Ethernet 10/25Gb 2-port SFP28 Adapter for HPE	P08441-001
Intel E810 Ethernet 10/25Gb 4-port SFP28 Adapter for HPE	P08456-001

SmartNIC network adapter spare part

Customer self repair: **mandatory**

Description	Spare part number
HPE SmartNIC 25Gb 2P 691 SFP28 Adapter	P18715-001

Solid state NVMe/PCIe Add-In-Card spare parts

Customer self repair: **mandatory**

Table 1. PCIe cards

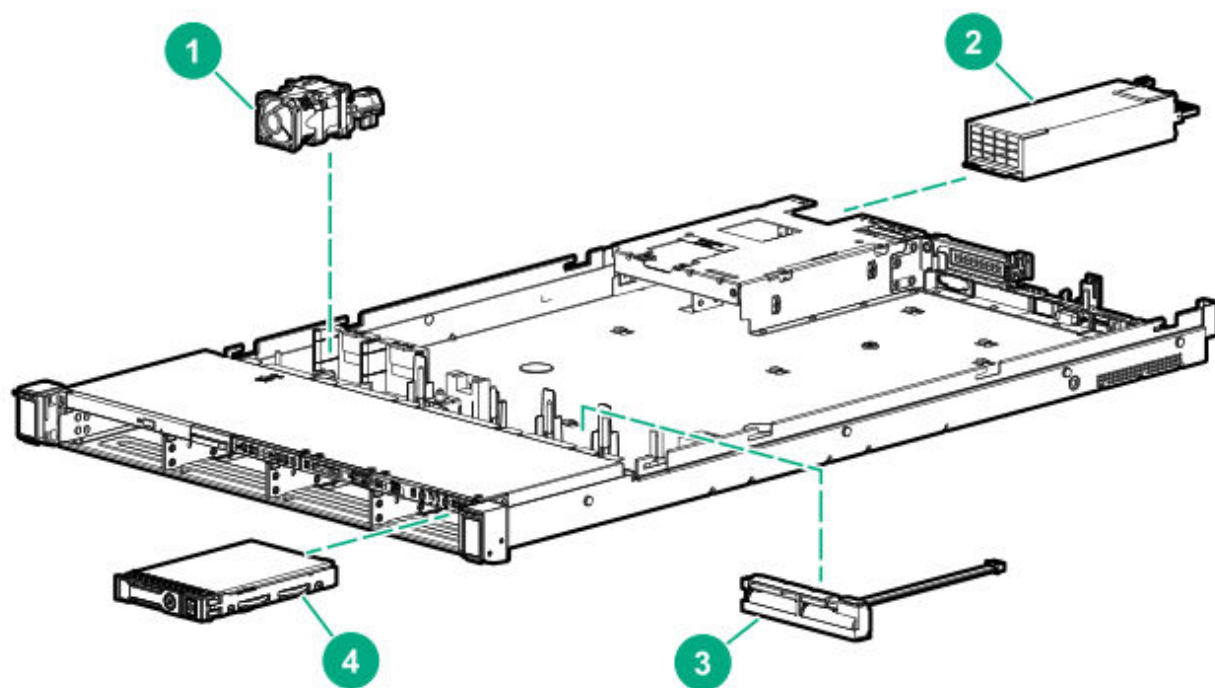
Description	Spare part number
HPE 750 GB PCIe x4 WI HH DS Card	P03580-001
HPE 1.6 TB PCIe x8 MU HH DS Card	P28069-001
HPE 3.2 TB PCIe x8 MU HH DS Card	P28070-001
HPE 6.4 TB PCIe x8 MU HH DS Card	P28071-001

Table 2. NVMe cards

Description	Spare part number
HPE 1.6 TB NVMe x8 MU HH DS Card	P10670-001
HPE 3.2 TB NVMe x8 MU HH DS Card	P10671-001
HPE 6.4 TB NVMe x8 MU HH DS Card	P10672-001

Server options

Hewlett Packard Enterprise continually improves and changes product parts. For complete and current supported parts information, see the [**Hewlett Packard Enterprise PartSurfer website**](#).



Item	Description
1	Fan spare parts
2	Power supply spare parts
3	Battery and capacitor spare parts
4	Hot-plug drive spare parts
5	Solid state NVMe drive spare parts ¹
6	Solid state M.2 drive spare parts ¹
7	Solid state SAS drive spare parts ¹
8	Solid state SATA drive spare parts ¹
9	Solid state NVMe/PCIe Add-In-Card spare parts ¹
10	microSD spare parts ¹
11	Systems Insight Display spare parts ¹
12	Power switch spare parts ¹
13	Universal Media Bay spare parts ¹
14	Drive backplane spare parts ¹
15	Cable kit spare parts ¹
16	Pensando Distributed Services Platform spare parts ¹
17	Accelerator and GPU spare parts ¹

Item	Description
18	HPE Trusted Platform Module 2.0 spare part ¹

¹ Not shown

Subtopics

[Fan spare parts](#)

[Power supply spare parts](#)

[Battery and capacitor spare parts](#)

[Hot-plug drive spare parts](#)

[Solid state NVMe drive spare parts](#)

[Solid state M.2 drive spare parts](#)

[Solid state SAS drive spare parts](#)

[Solid state SATA drive spare parts](#)

[microSD spare parts](#)

[Systems Insight Display spare parts](#)

[Power switch spare parts](#)

[Universal Media Bay spare parts](#)

[Drive backplane spare parts](#)

[Cable kit spare parts](#)

[Pensando Distributed Services Platform spare parts](#)

[HPE NS204i-p NVMe OS Boot Device spare parts](#)

[Accelerator and GPU spare parts](#)

[HPE Trusted Platform Module 2.0 spare part](#)

Fan spare parts

Customer self repair: **mandatory**

Description	Spare part number
Standard fans	875283-001
High-performance fans	875284-001

Power supply spare parts

Customer self repair: **mandatory**

Description	Spare part number
800W Flex-slot 277VDC	866727-001
800W Flex-slot 48VDC	866728-001
500W Flex-slot Platinum	866729-001
800W Flex-slot Platinum	866730-001
800W Flex-slot High Efficiency Titanium 96%	866793-001
1000W Flex-slot Titanium 96%	P44412-001
1600W Flex-slot Platinum	863373-001

Battery and capacitor spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE Smart Storage Hybrid Capacitor	P07473-001
HPE Smart Storage Battery	878643-001
System battery	319603-001

Hot-plug drive spare parts

Customer self repair: **mandatory**

LFF drives

Description	Spare part number
1TB SATA 7.2K LFF SC DS HDD	862128-001

Description	Spare part number
1TB SAS 7.2K LFF SC DS HDD	846612-001
2TB SAS 12G 7.2K LFF SC DS HDD	872744-001
2TB SAS 7.2K LFF SC HDD	819078-001
2TB SATA 7.2K 6G LFF SC DS HDD	872771-001
2TB SATA 7.2K LFF SC HDD	862126-001
3TB SATA 7.2K LFF SC DS HDD	862129-001
3TB SAS 7.2K LFF SC DS HDD	846614-001
4TB SAS 7.2K LFF SC 512e DS HDD	862141-001
4TB SAS 12G 7.2K LFF SC DS HDD	872745-001
4TB SAS 7.2K LFF SC HDD	819079-001
4TB SATA 7.2K 6G LFF SC DS HDD	872772-001
4TB SATA 7.2K LFF SC HDD	862127-001
4TB SATA 7.2K LFF SC 512e DS HDD	862139-001
6TB SATA 7.2K LFF SC DS HDD	846608-001
6TB SATA 7.2K LFF SC 512e DS HDD	862138-001
6TB SAS 7.2K LFF SC 512e DS HDD	862140-001
6TB SAS 7.2K LFF SC DS HDD	846610-001
8TB SATA 7.2K LFF SC He 512e DS HDD	861609-001
8TB SATA 7.2K LFF SC 512e DS HDD	820033-001
8TB SAS 7.2K LFF SC He 512e DS HDD	861607-001
8TB SAS 7.2K LFF SC 512e DS HDD	820032-001
10TB SAS 7.2K LFF SC He 512e DS HDD	857965-001
10TB SATA 7.2K LFF SC He 512e DS HDD	857967-001
10TB SAS 7.2K LFF DS He ST HDD	P11187-001
14TB SAS 7.2K LFF DS He SC HDD	P11188-001
14TB SATA 7.2K LFF DS He SC HDD	P11185-001
HPE 16TB SATA 7.2K LFF ISE He SC HDD	P25246-001
HPE 16TB SAS 7.2K LFF ISE He SC HDD	P25247-001

SFF drives

Description	Spare part number
300GB SAS 10k SFF SC HDD	785410-001

Description	Spare part number
300GB SAS 10K SFF SC DS HDD	872735-001
300GB SAS 15K SFF SC HDD	759546-001
300GB SAS 15K SFF SC DS HDD	870792-001
450GB SAS 15K SFF SC HDD	759547-001
600GB SAS 10K SFF SC DS HDD	872736-001
600GB SAS 10K SFF SC HDD	781577-001
600GB SAS 15K SFF SC DS HDD	870794-001
600GB SAS 15K SFF SC 512e DS HDD	870797-001
600GB SAS 15K SFF SC HDD	759548-001
600GB SAS 15K SFF SC 512e HDD	748435-001
900GB SAS 10K SFF SC HDD	785411-001
900GB SAS 15K SFF SC DS HDD	870795-001
900GB SAS 15K SFF SC 512e DS HDD	870798-001
1TB SAS 7.2K SFF SC DS HDD	832984-001
1TB SAS 7.2K SFF SC 512e DS HDD	765872-001
1TB SATA 7.2K SFF SC 512e DS HDD	765868-001
1TB SATA 7.2K SFF SC DS HDD	656108-001
1.2TB SAS 10K SFF SC HDD	781578-001
1.2TB SAS 10K SFF SC DS HDD	872737-001
1.8TB SAS 10K SFF SC 512e DS HDD	872738-001
1.8TB SAS 10K SFF SC 512e HDD	791055-001
2TB SAS 7.2K SFF SC 512e DS HDD	765873-001
2TB SATA 7.2K SFF SC 512e DS HDD	765869-001
HPE 2.4TB SAS 10K SFF BC 512e MV HDD	P30562-001
HPE 1.2TB SAS 10K SFF BC MV HDD	P30563-001
HPE 300GB SAS 10K SFF BC MV HDD	P40785-001
HPE 900GB SAS 15K SFF BC MV HDD	P40784-001
HPE 300GB SAS 15K SFF BC MV HDD	P30561-001
HPE 2TB SAS 7.2K SFF BC 512e HDD	P30575-001

Solid state NVMe drive spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE 400GB NVMe x4 WI SFF SCN SSD	765059-001
HPE 400GB NVMe x4 RI SFF SCN SSD	765067-001
HPE 400GB NVMe x4 MU SFF SCN SSD	765063-001
HPE 800GB NVMe x4 WI SFF SCN SSD	765060-001
HPE 800GB NVMe x4 MU SFF SCN SSD	765064-001
HPE 800GB NVMe x4 MU SFF SCN DS SSD	P10648-001
HPE 800GB NVMe x4 MU SFF SCN U.3 SSD	P26422-001
HPE 800GB NVMe PERF MU SFF U.3 SCN DS SSD	P20754-001
HPE 800GB NVMe MAIN MU SFF SCN U.3 SSD	P28066-001
HPE 800GB NVMe MAIN MU SFF SCN U.3 SSD	P20598-001
HPE 800GB NVMe MAIN MU SFF SC U.3 SSD	P20597-001
HPE 960GB NVMe x4 RI SFF SCN DS SSD	P10645-001
HPE 960GB NVMe x4 RI SFF SCN DS SSD	P10652-001
HPE 960GB NVMe x4 RI SFF SCN U.3 SSD	P26423-001
HPE 960GB NVMe PERF RI SFF SCN U.3 SSD	P20764-001
HPE 960GB NVMe MAIN RI SFF SCN U.3 SSD	P28068-001
HPE 960GB NVMe MAIN RI SFF SCN U.3 SSD	P20611-001
HPE 960GB NVMe MAIN RI SFF SC U.3 SSD	P20610-001
HPE 1.2TB NVMe x4 RI SFF SCN SSD	765068-001
HPE 1.6TB NVMe x4 WI SFF SCN SSD	765061-001
HPE 1.6TB NVMe x4 MU SFF SCN SSD	765065-001
HPE 1.6TB NVMe x4 MU SFF SCN DS SSD	P10470-001
HPE 1.6TB NVMe x4 MU SFF SCN DS SSD	P10649-001
HPE 1.6TB NVMe x4 MU SFF RW DS SSD	P12435-001
HPE 1.6TB NVMe x4 MU SFF SCN U.3 SSD	P26414-001
HPE 1.6TB NVMe PERF MU SFF U.3 SCN SSD	P20755-001
HPE 1.6TB NVMe MAIN MU SFF SCN U.3 SSD	P20794-001
HPE 1.6TB NVMe MAIN MU SFF SCN U.3 SSD	P20600-001
HPE 1.6TB NVMe MAIN MU SFF SC U.3 SSD	P20599-001

Description	Spare part number
HPE 1.6TB NVMe MU SCN FIPS U.3 CM6 SSD	P44777-001
HPE 1.92TB NVMe x4 RI SFF SCN DS SSD	P10466-001
HPE 1.92TB NVMe x4 RI SFF SCN DS SSD	P10646-001
HPE 1.92TB NVMe x4 RI SFF SCN DS SSD	P10653-001
HPE 1.92TB NVMe x4 RI SFF SCN U.3 SSD	P26418-001
HPE 1.92TB NVMe PERF RI SFF SCN U.3 SSD	P20765-001
HPE 1.92TB NVMe PERF RI SFF SCN U.3 SSD	P20802-001
HPE 1.92TB NVMe MAIN RI SFF SC U.3 SSD	P20612-001
HPE 1.92TB NVMe MAIN RI SFF SCN U.3 SSD	P20613-001
HPE 1.92TB NVMe RI SCN FIPS U.3 CM6 SSD	P44781-001
HPE 2TB NVMe x4 MU SFF SCN SSD	765066-001
HPE 2TB NVMe x4 RI SFF SCN SSD	765069-001
HPE 2TB NVMe x4 WI SFF SCN SSD	765062-001
HPE 3.2TB NVMe x4 MU SFF SCN DS SSD	P10471-001
HPE 3.2TB NVMe x4 MU SFF SCN DS SSD	P10650-001
HPE 3.2TB NVMe x4 MU SFF SCN U.3 SSD	P26415-001
HPE 3.2TB NVMe PERF MU SFF U.3 SCN SSD	P20756-001
HPE 3.2TB NVMe MAIN MU SFF SCN U.3 SSD	P20795-001
HPE 3.2TB NVMe MAIN MU SFF SCN U.3 SSD	P20602-001
HPE 3.2TB NVMe MAIN MU SFF SC U.3 SSD	P20601-001
HPE 3.2TB NVMe MU SCN FIPS U.3 CM6 SSD	P44779-001
HPE 3.84TB NVMe x4 RI SFF SCN DS SSD	P10467-001
HPE 3.84TB NVMe x4 RI SFF SCN DS SSD	P10647-001
HPE 3.84TB NVMe x4 RI SFF SCN DS SSD	P10654-001
HPE 3.84TB NVMe x4 RI SFF RW DS SSD	P12430-001
HPE 3.84TB NVMe x4 RI SFF SCN U.3 SSD	P26419-001
HPE 3.84TB NVMe PERF RI SFF SCN U.3 SSD	P20766-001
HPE 3.84TB NVMe PERF RI SFF SCN U.3 SSD	P20803-001
HPE 3.84TB NVMe MAIN RI SFF SCN U.3 SSD	P20614-001
HPE 3.84TB NVMe MAIN RI SFF SCN U.3 SSD	P20615-001
HPE 3.84TB NVMe MAIN RI SFF SC U.3 SSD	P20614-001
HPE 3.84TB NVMe RI SCN FIPS U.3 CM6 SSD	P44783-001
HPE 6.4TB NVMe x4 MU SFF SCN DS SSD	P10472-001

Description	Spare part number
HPE 6.4TB NVMe x4 MU SFF SCN DS SSD	P10651-001
HPE 6.4TB NVMe x4 MU SFF SCN U.3 SSD	P26416-001
HPE 6.4TB NVMe PERF MU SFF U.3 SCN SSD	P20757-001
HPE 6.4TB NVMe MAIN MU SFF SCN U.3 SSD	P20796-001
HPE 6.4TB NVMe MAIN MU SFF SCN U.3 SSD	P20604-001
HPE 6.4TB NVMe MAIN MU SFF SC U.3 SSD	P20603-001
HPE 7.68TB NVMe x4 RI SFF SCN DS SSD	P10468-001
HPE 7.68TB NVMe x4 RI SFF SCN DS SSD	P10655-001
HPE 7.68TB NVMe x4 RI SFF SCN U.3 SSD	P26420-001
HPE 7.68TB NVMe PERF RI SFF SCN U.3 SSD	P20767-001
HPE 7.68TB NVMe PERF RI SFF SCN U.3 SSD	P20804-001
HPE 7.68TB NVMe MAIN RI SFF SCN U.3 SSD	P20606-001
HPE 7.68TB NVMe MAIN RI SFF SC U.3 SSD	P20605-001
HPE 12.8TB NVMe x4 MU SFF SCN U.3 SSD	P26417-001
HPE 12.8TB NVMe MAIN MU SFF SCN U.3 SSD	P20797-001
HPE 15.36TB NVMe x4 RI SFF SCN DS SSD	P10656-001
HPE 15.36TB NVMe x4 RI SFF SCN U.3 SSD	P26421-001
HPE 15.36TB NVMe PERF RI SFF SCN U.3 SSD	P20805-001

Solid state M.2 drive spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE 120GB SATA RI uFF Dual M.2 Kit	831995-001
HPE 120GB SATA RI uFF M.2 Kit	781565-001
HPE 240GB SATA RI M.2 2280 SSD	P27211-001
HPE 340GB SATA RI uFF Dual M.2 Kit	781566-001
HPE 340GB SATA RI uFF M.2 Kit	781566-001
HPE 2x240GB SATA RI M.2 SFF SCM SSD kit	P25248-001
HPE 2x480GB SATA MU M.2 SFF SCM SSD kit	P25249-001

Description	Spare part number
HPE 480GB SATA MU M.2 2280 SSD	P27212-001
HPE 960GB SATA MU M.2 2280 SSD	P27213-001

Solid state SAS drive spare parts

Customer self repair: **mandatory**

LFF drives

Description	Spare part number
HPE 800GB SAS MU LFF LPC DS SSD	P06579-001
HPE 800GB SAS MU LFF SCC DS SSD	P06578-001
HPE 960GB SAS MU LFF SCC VS DS SSD	P10605-001
HPE 960GB SAS MU LFF LPC VS DS SSD	P10606-001
HPE 960GB SAS MU LFF LPC MV SSD	P37070-001
HPE 1.6TB SAS MU LFF LPC DS SSD	P06581-001
HPE 1.92TB SAS MU LFF SCC VS DS SSD	P10608-001
HPE 1.92TB SAS MU LFF LPC VS DS SSD	P10609-001
HPE 1.92TB SAS MU LFF SCC MV SSD	P37072-001
HPE 3.84TB SAS MU LFF LPC VS DS SSD	P10611-001

SFF drives

Description	Spare part number
HPE 400GB SAS MU SFF SC DS SSD	P09922-001
HPE 400GB SAS MU SFF SC DS SSD	P06576-001
HPE 400GB SAS MU SFF SC SSD	822784-001
HPE 400GB SAS WI SFF SC DS SSD	P09947-001
HPE 400GB SAS WI SFF SC DS SSD	P06600-001
HPE 400GB SAS WI SFF ST DS SSD	P06601-001
HPE 400GB SAS WI SFF SC SSD	P22585-001
HPE 400GB SAS WI SFF SC SSD	P37176-001

Description	Spare part number
HPE 480GB SAS RI SFF SC SSD	817047-001
HPE 800GB SAS MU SFF SC DS SSD	P09923-001
HPE 800GB SAS MU SFF SC DS SSD	P06577-001
HPE 800GB SAS MU SFF SC SSD	822786-001
HPE 800GB SAS WI SFF SC SSD	P37177-001
HPE 800GB SAS WI SFF SC DS SSD	P09948-001
HPE 800GB SAS WI SFF SC DS SSD	P06602-001
HPE 800GB SAS WI SFF ST DS SSD	P06603-001
HPE 800GB SAS MU SFF SC SSD	P20838-001
HPE 800GB SAS MU SFF SC SSD	P37171-001
HPE 800GB SAS WI SFF SC SSD	P22586-001
HPE 960GB SAS RI SFF SC DS SSD	P06596-001
HPE 960GB SAS RI SFF SC SSD	817049-001
HPE 960GB SAS RI SFF SC VS DS SSD	P10637-001
HPE 960GB SAS MU SFF SC VS DS SSD	P10604-001
HPE 960GB SAS RI SFF SC SSD	P20833-001
HPE 960GB SAS RI SFF SC SSD	P37166-001
HPE 960GB SAS RI SFF SC MV SSD	P37064-001
HPE 960GB SAS MU SFF SC MV SSD	P37068-001
HPE 1.6TB SAS MU SFF SC DS SSD	P09924-001
HPE 1.6TB SAS MU SFF SC DS SSD	P06580-001
HPE 1.6TB SAS MU SFF SC SSD	822788-001
HPE 1.6TB SAS WI SFF SC DS SSD	P09949-001
HPE 1.6TB SAS WI SFF SC DS SSD	P06604-001
HPE 1.6TB SAS MU SFF SC SSD	P20839-001
HPE 1.6TB SAS MU SFF SC SSD	P37172-001
HPE 1.6TB SAS WI SFF SC SSD	P22587-001
HPE 1.6TB SAS WI SFF SC SSD	P37178-001
HPE 1.92TB SAS RI SFF SC DS SSD	P06597-001
HPE 1.92TB SAS RI SFF SC SSD	817051-001
HPE 1.92TB SAS RI SFF SC VS DS SSD	P10638-001
HPE 1.92TB SAS MU SFF SC VS DS SSD	P10607-001
HPE 1.92TB SAS RI SFF SC SSD	P20834-001

Description	Spare part number
HPE 1.92TB SAS RI SFF SC SSD	P37167-001
HPE 1.92TB SAS RI SFF SC MV SSD	P37065-001
HPE 1.92TB SAS MU SFF SC MV SSD	P37071-001
HPE 3.2TB SAS MU SFF SC DS SSD	P09925-001
HPE 3.2TB SAS MU SFF SC DS SSD	P06582-001
HPE 3.2TB SAS MU SFF SC SSD	822790-001
HPE 3.2TB SAS WI SFF SC DS SSD	P06605-001
HPE 3.2TB SAS MU SFF SC SSD	P20840-001
HPE 3.2TB SAS MU SFF SC SSD	P37173-001
HPE 3.2TB SAS WI SFF SC SSD	P37179-001
HPE 3.84TB SAS RI SFF SC DS SSD	P06598-001
HPE 3.84TB SAS RI SFF SC SSD	817053-001
HPE 3.84TB SAS RI SFF SC VS DS SSD	P10639-001
HPE 3.84TB SAS MU SFF SC VS DS SSD	P10610-001
HPE 3.84TB SAS RI SFF SC SSD	P20835-001
HPE 3.84TB SAS RI SFF SC SSD	P37168-001
HPE 3.84TB SAS RI SFF SC MV SSD	P37066-001
HPE 3.84TB SAS MU SFF SC MV SSD	P37074-001
HPE 6.4TB SAS MU SFF SC DS SSD	P09926-001
HPE 6.4TB SAS MU SFF SC DS SSD	P06583-001
HPE 6.4TB SAS MU SFF SC SSD	P20841-001
HPE 6.4TB SAS MU SFF SC SSD	P37174-001
HPE 7.68TB SAS RI SFF SC DS SSD	P06599-001
HPE 7.68TB SAS RI SFF SC VS DS SSD	P10640-001
HPE 7.68TB SAS RI SFF SC SSD	P20836-001
HPE 7.68TB SAS RI SFF SC SSD	P37169-001
HPE 7.68TB SAS RI SFF SC MV SSD	P37067-001
HPE 15.3TB SAS RI SFF SC SSD	P20837-001
HPE 15.3TB SAS RI SFF SC SSD	P37170-001
HPE 12.8TB SAS MU SFF SC SSD	P37175-001
HPE 30.72TB SAS RI SFF SC SS	P22580-001
HPE 3.84TB SAS RI SFF BC MV SSD	P49737-001
HPE 800GB SAS WI SFF BC PM6 SSD	P41506-001

Description	Spare part number
HPE 3.2TB SAS MU SFF BC MV SSD	P49749-001
HPE 3.84TB SAS MU SFF BC VS MV SSD	P41537-001
HPE 1.92TB SAS RI SFF BC VS MV SSD	P41532-001
HPE 6.4TB SAS MU SFF BC MV SSD	P49751-001
HPE 960GB SAS MU SFF BC VS MV SSD	P41535-001
HPE 1.92TB SAS MU SFF BC VS MV SSD	P41536-001
HPE 800GB SAS MU SFF BC MV SSD	P49745-001
HPE 1.6TB SAS MU SFF BC MV SSD	P49747-001
HPE 3.84TB SAS RI SFF BC VS MV SSD	P41533-001

Solid state SATA drive spare parts

Customer self repair: **mandatory**

LFF drives

Description	Spare part number
HPE 200GB SATA WI LFF SCC SSD	805386-001
HPE 400GB SATA 6G WI LFF SCC DS SSD	872513-001
HPE 400GB SATA WI LFF SCC SSD	805388-001
HPE 480GB SATA RI LFF SCC DS SSD	P09845-001
HPE 480GB SATA 6G MU LFF SCC DS SSD	872519-001
HPE 480GB SATA 6G RI LFF SCC DS SSD	869578-001
HPE 480GB SATA RI LFF LPC SSD	P21086-001
HPE 480GB SATA MU LFF SCC SSD	P21092-001
HPE 800GB SATA 6G WI LFF SCC DS SSD	872515-001
HPE 800GB SATA WI LFF SCC SSD	805390-001
HPE 960GB SATA RI LFF SCC DS SSD	P09846-001
HPE 960GB SATA RI LFF LPC DS SSD	P09847-001
HPE 960GB SATA MU LFF SCC DS SSD	P09910-001
HPE 960GB SATA 6G MU LFF SCC DS SSD	872521-001
HPE 960GB SATA MU LFF LPC SSD	P21093-001

Description	Spare part number
HPE 1.2TB SATA WI LFF SCC SSD	805392-001
HPE 1.6TB SATA 6G WI LFF SCC DS SSD	872517-001
HPE 1.6TB SATA 6G RI LFF SCC DS SSD	869582-001
HPE 1.92TB SATA MU LFF SCC DS SSD	P09913-001
HPE 1.92TB SATA MU LFF LPC DS SSD	P09914-001
HPE 1.92TB SATA RI LFF SCC DS SSD	P09848-001
HPE 1.92TB SATA RI LFF LPC SSD	P21087-001
HPE 1.92TB SATA MU LFF SCC SSD	P21094-001
HPE 1.92TB SATA MU LFF LPC SSD	P21095-001

SFF drives

Description	Spare part number
HPE 150GB SATA 6G RI SFF SC DS SSD	869575-001
HPE 200GB SATA WI SFF SC SSD	805385-001
HPE 240GB SATA RI SFF RW DS SSD	P09844-001
HPE 240GB SATA RI SFF SC DS SSD	P08565-001
HPE 240GB SATA 6G RI SFF SC DS SSD	868924-001
HPE 240GB SATA 6G RI SFF SC DS SSD	869576-001
HPE 240GB SATA RI SFF SC MV SSD	P18481-001
HPE 240GB SATA RI SFF SC SSD	P21080-001
HPE 400GB SATA 6G WI SFF SC DS SSD	872512-001
HPE 400GB SATA WI SFF SC SSD	805387-001
HPE 480GB SATA 6G RI SFF SC DS SSD	868926-001
HPE 480GB SATA 6G MU SFF SC DS SSD	872518-001
HPE 480GB SATA MU SFF SC MV SSD	P18477-001
HPE 480GB SATA RI SFF SC MV SSD	P18482-001
HPE 480GB SATA 6G RI SFF SC DS SSD	869577-001
HPE 480GB SATA MU SFF SC DS SSD	P09907-001
HPE 480GB SATA MU SFF RW DS SSD	P09908-001
HPE 480GB SATA MU SFF SC DS SSD	P08620-001
HPE 480GB SATA RI SFF SC DS SSD	P08567-001
HPE 480GB SATA RI SFF SC SSD	P21081-001

Description	Spare part number
HPE 480GB SATA MU SFF SC SSD	P21088-001
HPE 800GB SATA 6G WI SFF SC DS SSD	872514-001
HPE 800GB SATA WI SFF SC SSD	805389-001
HPE 960GB SATA MU SFF SC DS SSD	P09909-001
HPE 960GB SATA MU SFF RW DS SSD	P09911-001
HPE 960GB SATA MU SFF SC DS SSD	P08622-001
HPE 960GB SATA RI SFF SC DS SSD	P08569-001
HPE 960GB SATA 6G RI SFF SC DS SSD	868928-001
HPE 960GB SATA 6G RI SFF SC DS SSD	869580-001
HPE 960GB SATA 6G MU SFF SC DS SSD	872520-001
HPE 960GB SATA MU SFF SC MV SSD	P18478-001
HPE 960GB SATA RI SFF SC MV SSD	P18483-001
HPE 960GB SATA RI SFF SC SSD	P21082-001
HPE 960GB SATA MU SFF SC SSD	P21089-001
HPE 1.2TB SATA WI SFF SC SSD	805391-001
HPE 1.6TB SATA 6G WI SFF SC DS SSD	872516-001
HPE 1.6TB SATA 6G RI SFF SC DS SSD	869581-001
HPE1.92TB SATA MU SFF SC DS SSD	P09912-001
HPE 1.92TB SATA RI SFF RW DS SSD	P09849-001
HPE 1.92TB SATA MU SFF RW DS SSD	P09915-001
HPE 1.92TB SATA MU SFF SC DS SSD	P08625-001
HPE 1.92TB SATA RI SFF SC DS SSD	P08572-001
HPE 1.92TB SATA 6G RI SFF SC DS SSD	868930-001
HPE 1.92TB SATA MU SFF SC MV SSD	P18479-001
HPE 1.92TB SATA 6G MU SFF SC DS SSD	872522-001
HPE 1.92TB SATA RI SFF SC MV SSD	P18484-001
HPE 1.92TB SATA RI SFF SC SSD	P21083-001
HPE 1.92TB SATA MU SFF SC SSD	P21090-001
HPE 1.92TB SATA MU SFF RW SSD	P21096-001
HPE 3.84TB SATA 6G RI SFF SC DS SSD	868932-001
HPE 3.84TB SATA RI SFF RW DS SSD	P09850-001
HPE 3.84TB SATA MU SFF SC DS SSD	P08632-001
HPE 3.84TB SATA RI SFF SC DS SSD	P08575-001

Description	Spare part number
HPE 3.84TB SATA MU SFF SC MV SSD	P18480-001
HPE 3.84TB SATA RI SFF SC MV SSD	P18485-001
HPE 3.84TB SATA RI SFF SC SSD	P21084-001
HPE 3.84TB SATA MU SFF SC DS SS	P22588-001
HPE 3.84TB SATA MU SFF SC SSD	P21091-001
HPE 7.68TB SATA RI SFF SC MV SSD	P18486-001
HPE 7.68TB SATA RI SFF SC SSD	P21085-001
HPE 960GB SATA MU SFF BC MV SSD	P41528-001
HPE 480GB SATA RI SFF BC MV SSD	P41522-001
HPE 480GB SATA MU SFF BC MV SSD	P41527-001
HPE 240GB SATA RI SFF BC MV SSD	P41521-001
HPE 960GB SATA RI SFF BC MV SSD	P41523-001
HPE 3.84TB SATA MU SFF BC MV SSD	P41530-001
HPE 1.92TB SATA MU SFF BC MV SSD	P41529-001
HPE 1.92TB SATA RI SFF BC MV SSD	P41524-001
HPE 3.84TB SATA RI SFF BC MV SSD	P41525-001

microSD spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE 32 GB microSD RAID 1 USB Boot Drive	P23103-001

Systems Insight Display spare parts

Customer self repair: **mandatory**

Description	Spare part number
SFF Systems Insight Display	875563-001
LFF Systems Insight Display	875564-001

Power switch spare parts

Customer self repair: **mandatory**

Description	Spare part number
SFF Power/UID power switch	875565-001
LFF Power/UID power switch	P25160-001

Universal Media Bay spare parts

Customer self repair: **mandatory**

Description	Spare part number
SFF display port/USB/Optical disk drive blank	875560-001
LFF display port/USB Option	875562-001

Drive backplane spare parts

Customer self repair: **mandatory**

Description	Spare part number
2 SFF drive cage only for 10SFF	875553-001
2 SFF SAS/SATA drive backplane plus cage assembly	875554-001

Description	Spare part number
2 SFF NVMe drive backplane plus cage assembly	875555-001
10 SFF NVMe/SAS/SATA drive backplane	875556-001
2 SFF SATA UFF drive backplane	875557-001
1 SFF rear drive backplane plus cage assembly	875558-001
4 LFF drive backplane	775402-001
8 SFF drive backplane	780428-001
PCA 1U 8SFF 12Gx1SAS,UBM6 BP	P62068-001
PCA,8SFF x1 SAS/SATA UBM2 E BP	P31227-001

Cable kit spare parts

Customer self repair: **mandatory**

Description	Spare part number
10SFF P824i-P cables kit	P03217-001
Controller backup power cable (short)	878645-001
Controller backup power cable (long)	878646-001
2SFF/8SFF cables kit	875566-001
10SFF cables kit	875567-001
4LFF backplane power cable	875568-001
1SFF rear backplane SATA	875569-001
4LFF Mini-SAS cable kit	875572-001
SFF/2SFF Mini-SAS cable kit	875573-001
10SFF NVMe cable kit	875574-001
Intrusion Detection switch cable	875570-001
Serial port cable	875571-001
CA,2/4/8SFF S/S BP, Tmode AROC/P1/P2	P56045-001

Pensando Distributed Services Platform spare parts

Subtopics

[Pensando DSP iLO Sideband ALOM Module spare parts](#)

[Smart I/O \(Pensando Distributed Services Platform\) spare parts](#)

Pensando DSP iLO Sideband ALOM Module spare parts

Customer self repair: **optional**

Description	Spare part number
Pensando Distributed Services Platform for HPE iLO Sideband Management ALOM Module	P27683-001

Smart I/O (Pensando Distributed Services Platform) spare parts

Customer self repair: **optional**

Description	Spare part number
Pensando Distributed Services Platform DSC-25 10/25G 2-port SFP28 card ¹	P27682-001

¹ To use with the accompanying Pensando DSP iLO Sideband ALOM Module, the Smart I/O card must be installed on slot 3 of the primary riser.

HPE NS204i-p NVMe OS Boot Device spare parts

Customer self repair: **mandatory**

Description	Spare part number
HPE NS204i-p NVMe OS Boot Device	P14379-001
SPS-DRV SSD 480GB NVMe RI M.2 SV SMSUNG	P69616-001*

* SSDs with the part number P13688-004 are End of Life and no longer available.



IMPORTANT

Do not mix drives. Configurations with different SSD manufacturers are not supported. If you are replacing P13688-004 SSDs, you must replace both drives. For more information on removing and replacing two boot drives, see the [HPE NS204i Boot Device User Guide](#).

Accelerator and GPU spare parts

Customer self repair: **mandatory**

Description	Spare part number
NVIDIA Quadro P2000 GPU module	871969-001
NVIDIA Quadro P4000 GPU module	871970-001
NVIDIA Quadro RTX 4000 module	P11603-001
HPE NVIDIA Quadro P2200 Graphics Accelerator	P16021-001
HPE NVIDIA Tesla T4 16GB Accelerator	P09571-001
Intel Arria 10 GX FPGA Accelerator	P02425-001
HPE DL360 Gen10 2P FH GPU v2 Enable Kit	P26841-001

HPE Trusted Platform Module 2.0 spare part

Customer self repair: **no**

Description	Spare part number
HPE Trusted Platform Module 2.0 Gen 10 kit, TAA	872159-001

Customer self repair

Hewlett Packard Enterprise products are designed with many Customer Self Repair (CSR) parts to minimize repair time and allow for greater flexibility in performing defective parts replacement. If during the diagnosis period Hewlett Packard Enterprise (or Hewlett Packard Enterprise service providers or service partners) identifies that the repair can be accomplished by the use of a CSR part, Hewlett Packard Enterprise will ship that part directly to you for replacement. There are two categories of CSR parts:

- **Mandatory**—Parts for which customer self repair is mandatory. If you request Hewlett Packard Enterprise to replace these parts, you will be charged for the travel and labor costs of this service.
- **Optional**—Parts for which customer self repair is optional. These parts are also designed for customer self repair. If, however, you require that Hewlett Packard Enterprise replace them for you, there may or may not be additional charges, depending on the type of warranty service designated for your product.



NOTE

Some Hewlett Packard Enterprise parts are not designed for customer self repair. In order to satisfy the customer warranty, Hewlett Packard Enterprise requires that an authorized service provider replace the part. These parts are identified as "No" in the Illustrated Parts Catalog.

Based on availability and where geography permits, CSR parts will be shipped for next business day delivery. Same day or four-hour delivery may be offered at an additional charge where geography permits. If assistance is required, you can call the Hewlett Packard Enterprise Support Center and a technician will help you over the telephone. Hewlett Packard Enterprise specifies in the materials shipped with a replacement CSR part whether a defective part must be returned to Hewlett Packard Enterprise. In cases where it is required to return the defective part to Hewlett Packard Enterprise, you must ship the defective part back to Hewlett Packard Enterprise within a defined period of time, normally five (5) business days. The defective part must be returned with the associated documentation in the provided shipping material. Failure to return the defective part may result in Hewlett Packard Enterprise billing you for the replacement. With a customer self repair, Hewlett Packard Enterprise will pay all shipping and part return costs and determine the courier/carrier to be used.

For more information about the Hewlett Packard Enterprise CSR program, contact your local service provider.

Parts only warranty service

Your Hewlett Packard Enterprise Limited Warranty may include a parts only warranty service. Under the terms of parts only warranty service, Hewlett Packard Enterprise will provide replacement parts free of charge.

For parts only warranty service, CSR part replacement is mandatory. If you request Hewlett Packard Enterprise to replace these parts, you will be charged for the travel and labor costs of this service.

Réparation par le client (CSR)

Les produits Hewlett Packard Enterprise comportent de nombreuses pièces CSR (Customer Self Repair = réparation par le client) afin de minimiser les délais de réparation et faciliter le remplacement des pièces défectueuses. Si pendant la période de diagnostic, Hewlett Packard Enterprise (ou ses partenaires ou mainteneurs agréés) détermine que la réparation peut être effectuée à l'aide d'une pièce CSR, Hewlett Packard Enterprise vous l'envoie directement. Il existe deux catégories de pièces CSR :

- **Obligatoire**—Pièces pour lesquelles la réparation par le client est obligatoire. Si vous demandez à Hewlett Packard Enterprise de remplacer ces pièces, les coûts de déplacement et main d'œuvre du service vous seront facturés.
- **Facultatif**—Pièces pour lesquelles la réparation par le client est facultative. Ces pièces sont également conçues pour permettre au client d'effectuer lui-même la réparation. Toutefois, si vous demandez à Hewlett Packard Enterprise de remplacer ces pièces, l'intervention peut ou non vous être facturée, selon le type de garantie applicable à votre produit.

REMARQUE: Certaines pièces Hewlett Packard Enterprise ne sont pas conçues pour permettre au client d'effectuer lui-même la réparation. Pour que la garantie puisse s'appliquer, Hewlett Packard Enterprise exige que le remplacement de la pièce soit effectué par un Mainteneur Agréé. Ces pièces sont identifiées par la mention "Non" dans le Catalogue illustré.

Les pièces CSR sont livrées le jour ouvré suivant, dans la limite des stocks disponibles et selon votre situation géographique. Si votre situation géographique le permet et que vous demandez une livraison le jour même ou dans les 4 heures, celle-ci vous sera facturée. Pour toute assistance, appelez le Centre d'assistance Hewlett Packard Enterprise pour qu'un technicien vous aide au téléphone. Dans les documents envoyés avec la pièce de rechange CSR, Hewlett Packard Enterprise précise s'il est nécessaire de lui retourner la pièce défectueuse. Si c'est le cas, vous devez le faire dans le délai indiqué, généralement cinq (5) jours ouvrés. La pièce et sa documentation doivent être retournées dans l'emballage fourni. Si vous ne retournez pas la pièce défectueuse, Hewlett Packard Enterprise se réserve le droit de vous facturer les coûts de remplacement. Dans le cas d'une pièce CSR, Hewlett Packard Enterprise supporte l'ensemble des frais d'expédition et de retour, et détermine la société de courses ou le transporteur à utiliser.

Pour plus d'informations sur le programme CSR de Hewlett Packard Enterprise, contactez votre Mainteneur Agréé local.

Service de garantie "pièces seules"

Votre garantie limitée Hewlett Packard Enterprise peut inclure un service de garantie "pièces seules". Dans ce cas, les pièces de rechange fournies par Hewlett Packard Enterprise ne sont pas facturées.

Dans le cadre de ce service, la réparation des pièces CSR par le client est obligatoire. Si vous demandez à Hewlett Packard Enterprise de remplacer ces pièces, les coûts de déplacement et main d'œuvre du service vous seront facturés.

Riparazione da parte del cliente

Per abbreviare i tempi di riparazione e garantire una maggiore flessibilità nella sostituzione di parti difettose, i prodotti Hewlett Packard Enterprise sono realizzati con numerosi componenti che possono essere riparati direttamente dal cliente (CSR, Customer Self Repair). Se in fase di diagnostica Hewlett Packard Enterprise (o un centro di servizi o di assistenza Hewlett Packard Enterprise) identifica il guasto come riparabile mediante un ricambio CSR, Hewlett Packard Enterprise lo spedisce direttamente al cliente per la sostituzione. Vi sono due categorie di parti CSR:

- **Obbligatorie**—Parti che devono essere necessariamente riparate dal cliente. Se il cliente ne affida la riparazione ad Hewlett Packard Enterprise, deve sostenere le spese di spedizione e di manodopera per il servizio.
- **Opzionali**—Parti la cui riparazione da parte del cliente è facoltativa. Si tratta comunque di componenti progettati per questo scopo. Se tuttavia il cliente ne richiede la sostituzione ad Hewlett Packard Enterprise, potrebbe dover sostenere spese aggiuntive a seconda del tipo di garanzia previsto per il prodotto.

NOTA: alcuni componenti Hewlett Packard Enterprise non sono progettati per la riparazione da parte del cliente. Per rispettare la garanzia, Hewlett Packard Enterprise richiede che queste parti siano sostituite da un centro di assistenza autorizzato. Tali parti sono identificate da un "No" nel Catalogo illustrato dei componenti.

In base alla disponibilità e alla località geografica, le parti CSR vengono spedite con consegna entro il giorno lavorativo seguente. La consegna nel giorno stesso o entro quattro ore è offerta con un supplemento di costo solo in alcune zone. In caso di necessità si può richiedere l'assistenza telefonica di un addetto del centro di supporto tecnico Hewlett Packard Enterprise. Nel materiale fornito con una parte di ricambio CSR, Hewlett Packard Enterprise specifica se il cliente deve restituire dei componenti. Qualora sia richiesta la resa ad Hewlett Packard Enterprise del componente difettoso, lo si deve spedire ad Hewlett Packard Enterprise entro un determinato periodo di tempo, generalmente cinque (5) giorni lavorativi. Il componente difettoso deve essere restituito con la documentazione associata nell'imballo di spedizione fornito. La mancata restituzione del componente può comportare la fatturazione del ricambio da parte di Hewlett Packard Enterprise. Nel caso di riparazione da parte del cliente, Hewlett Packard Enterprise sostiene tutte le spese di spedizione e resa e sceglie il corriere/vettore da utilizzare.

Per ulteriori informazioni sul programma CSR di Hewlett Packard Enterprise, contattare il centro di assistenza di zona.

Servizio di garanzia per i soli componenti

La garanzia limitata Hewlett Packard Enterprise può includere un servizio di garanzia per i soli componenti. Nei termini di garanzia del servizio per i soli componenti, Hewlett Packard Enterprise fornirà gratuitamente le parti di ricambio.

Per il servizio di garanzia per i soli componenti è obbligatoria la formula CSR che prevede la riparazione da parte del cliente. Se il cliente invece richiede la sostituzione ad Hewlett Packard Enterprise dovrà sostenere le spese di spedizione e di manodopera per il servizio.

Customer Self Repair

Hewlett Packard Enterprise Produkte enthalten viele CSR-Teile (Customer Self Repair), um Reparaturzeiten zu minimieren und höhere Flexibilität beim Austausch defekter Bauteile zu ermöglichen. Wenn Hewlett Packard Enterprise (oder ein Hewlett Packard Enterprise Servicepartner) bei der Diagnose feststellt, dass das Produkt mithilfe eines CSR-Teils repariert werden kann, sendet Ihnen Hewlett Packard Enterprise dieses Bauteil zum Austausch direkt zu. CSR-Teile werden in zwei Kategorien unterteilt:

- **Zwingend**—Teile, für die das Customer Self Repair-Verfahren zwingend vorgegeben ist. Wenn Sie den Austausch dieser Teile von Hewlett Packard Enterprise vornehmen lassen, werden Ihnen die Anfahrt- und Arbeitskosten für diesen Service berechnet.
- **Optional**—Teile, für die das Customer Self Repair-Verfahren optional ist. Diese Teile sind auch für Customer Self Repair ausgelegt. Wenn Sie jedoch den Austausch dieser Teile von Hewlett Packard Enterprise vornehmen lassen möchten, können bei diesem Service je nach den für Ihr Produkt vorgesehenen Garantiebedingungen zusätzliche Kosten anfallen.

HINWEIS: Einige Hewlett Packard Enterprise Teile sind nicht für Customer Self Repair ausgelegt. Um den Garantieanspruch des Kunden zu erfüllen, muss das Teil von einem Hewlett Packard Enterprise Servicepartner ersetzt werden. Im illustrierten Teilekatalog sind diese Teile mit „No“ bzw. „Nein“ gekennzeichnet.

CSR-Teile werden abhängig von der Verfügbarkeit und vom Lieferziel am folgenden Geschäftstag geliefert. Für bestimmte Standorte ist eine Lieferung am selben Tag oder innerhalb von vier Stunden gegen einen Aufpreis verfügbar. Wenn Sie Hilfe benötigen, können Sie das Hewlett Packard Enterprise Support Center anrufen und sich von einem Mitarbeiter per Telefon helfen lassen. Den Materialien von Hewlett Packard Enterprise, die mit einem CSR-Ersatzteil geliefert werden, können Sie entnehmen, ob das defekte Teil an Hewlett Packard Enterprise zurückgeschickt werden muss. Wenn es erforderlich ist, das defekte Teil an Hewlett Packard Enterprise zurückzuschicken, müssen Sie dies innerhalb eines vorgegebenen Zeitraums tun, in der Regel innerhalb von fünf (5) Geschäftstagen. Das defekte Teil muss mit der zugehörigen Dokumentation in der Verpackung zurückgeschickt werden, die im Lieferumfang enthalten ist. Wenn Sie das defekte Teil nicht zurückschicken, kann Hewlett Packard Enterprise Ihnen das Ersatzteil in Rechnung stellen. Im Falle von Customer Self Repair kommt Hewlett Packard Enterprise für alle Kosten für die Lieferung und Rücksendung auf und bestimmt den Kurier-/Frachtdienst.

Weitere Informationen über das Hewlett Packard Enterprise Customer Self Repair Programm erhalten Sie von Ihrem Servicepartner vor Ort.

Parts-only Warranty Service (Garantieservice ausschließlich für Teile)

Ihre Hewlett Packard Enterprise Garantie umfasst möglicherweise einen Parts-only Warranty Service (Garantieservice ausschließlich für Teile). Gemäß den Bestimmungen des Parts-only Warranty Service stellt Hewlett Packard Enterprise Ersatzteile kostenlos zur Verfügung.

Für den Parts-only Warranty Service ist das CSR-Verfahren zwingend vorgegeben. Wenn Sie den Austausch dieser Teile von Hewlett Packard Enterprise vornehmen lassen, werden Ihnen die Anfahrt- und Arbeitskosten für diesen Service berechnet.

Reparaciones del propio cliente

Los productos de Hewlett Packard Enterprise incluyen muchos componentes que el propio usuario puede reemplazar (Customer Self Repair, CSR) para minimizar el tiempo de reparación y ofrecer una

mayor flexibilidad a la hora de realizar sustituciones de componentes defectuosos. Si, durante la fase de diagnóstico, Hewlett Packard Enterprise (o los proveedores o socios de servicio de Hewlett Packard Enterprise) identifica que una reparación puede llevarse a cabo mediante el uso de un componente CSR, Hewlett Packard Enterprise le enviará dicho componente directamente para que realice su sustitución. Los componentes CSR se clasifican en dos categorías:

- **Obligatorio**—Componentes cuya reparación por parte del usuario es obligatoria. Si solicita a Hewlett Packard Enterprise que realice la sustitución de estos componentes, tendrá que hacerse cargo de los gastos de desplazamiento y de mano de obra de dicho servicio.
- **Opcional**—Componentes cuya reparación por parte del usuario es opcional. Estos componentes también están diseñados para que puedan ser reparados por el usuario. Sin embargo, si precisa que Hewlett Packard Enterprise realice su sustitución, puede o no conllevar costes adicionales, dependiendo del tipo de servicio de garantía correspondiente al producto.

NOTA: Algunos componentes de Hewlett Packard Enterprise no están diseñados para que puedan ser reparados por el usuario. Para que el usuario haga valer su garantía, Hewlett Packard Enterprise pone como condición que un proveedor de servicios autorizado realice la sustitución de estos componentes. Dichos componentes se identifican con la palabra "No" en el catálogo ilustrado de componentes.

Según la disponibilidad y la situación geográfica, los componentes CSR se enviarán para que lleguen a su destino al siguiente día laborable. Si la situación geográfica lo permite, se puede solicitar la entrega en el mismo día o en cuatro horas con un coste adicional. Si precisa asistencia técnica, puede llamar al Centro de asistencia técnica de Hewlett Packard Enterprise y recibirá ayuda telefónica por parte de un técnico. Con el envío de materiales para la sustitución de componentes CSR, Hewlett Packard Enterprise especificará si los componentes defectuosos deberán devolverse a Hewlett Packard Enterprise. En aquellos casos en los que sea necesario devolver algún componente a Hewlett Packard Enterprise, deberá hacerlo en el periodo de tiempo especificado, normalmente cinco días laborables. Los componentes defectuosos deberán devolverse con toda la documentación relacionada y con el embalaje de envío. Si no enviara el componente defectuoso requerido, Hewlett Packard Enterprise podrá cobrarle por el de sustitución. En el caso de todas sustituciones que lleve a cabo el cliente, Hewlett Packard Enterprise se hará cargo de todos los gastos de envío y devolución de componentes y escogerá la empresa de transporte que se utilice para dicho servicio.

Para obtener más información acerca del programa de Reparaciones del propio cliente de Hewlett Packard Enterprise, póngase en contacto con su proveedor de servicios local.

Servicio de garantía exclusivo de componentes

La garantía limitada de Hewlett Packard Enterprise puede que incluya un servicio de garantía exclusivo de componentes. Según las condiciones de este servicio exclusivo de componentes, Hewlett Packard Enterprise le facilitará los componentes de repuesto sin cargo adicional alguno.

Para este servicio de garantía exclusivo de componentes, es obligatoria la sustitución de componentes por parte del usuario (CSR). Si solicita a Hewlett Packard Enterprise que realice la sustitución de estos componentes, tendrá que hacerse cargo de los gastos de desplazamiento y de mano de obra de dicho servicio.

Customer Self Repair

Veel onderdelen in Hewlett Packard Enterprise producten zijn door de klant zelf te repareren, waardoor de reparatieduur tot een minimum beperkt kan blijven en de flexibiliteit in het vervangen van defecte onderdelen groter is. Deze onderdelen worden CSR-onderdelen (Customer Self Repair) genoemd. Als

Hewlett Packard Enterprise (of een Hewlett Packard Enterprise Service Partner) bij de diagnose vaststelt dat de reparatie kan worden uitgevoerd met een CSR-onderdeel, verzendt Hewlett Packard Enterprise dat onderdeel rechtstreeks naar u, zodat u het defecte onderdeel daarmee kunt vervangen. Er zijn twee categorieën CSR-onderdelen:

- **Verplicht**—Onderdelen waarvoor reparatie door de klant verplicht is. Als u Hewlett Packard Enterprise verzoekt deze onderdelen voor u te vervangen, worden u voor deze service reiskosten en arbeidsloon in rekening gebracht.
- **Optioneel**—Onderdelen waarvoor reparatie door de klant optioneel is. Ook deze onderdelen zijn ontworpen voor reparatie door de klant. Als u echter Hewlett Packard Enterprise verzoekt deze onderdelen voor u te vervangen, kunnen daarvoor extra kosten in rekening worden gebracht, afhankelijk van het type garanteservice voor het product.

OPMERKING: Sommige Hewlett Packard Enterprise onderdelen zijn niet ontwikkeld voor reparatie door de klant. In verband met de garantievoorwaarden moet het onderdeel door een geautoriseerde Service Partner worden vervangen. Deze onderdelen worden in de geïllustreerde onderdelencatalogus aangemerkt met "Nee".

Afhankelijk van de leverbaarheid en de locatie worden CSR-onderdelen verzonden voor levering op de eerstvolgende werkdag. Levering op dezelfde dag of binnen vier uur kan tegen meerkosten worden aangeboden, indien dit mogelijk is gezien de locatie. Indien assistentie is gewenst, belt u het Hewlett Packard Enterprise Support Center om via de telefoon ondersteuning van een technicus te ontvangen. Hewlett Packard Enterprise vermeldt in de documentatie bij het vervangende CSR-onderdeel of het defecte onderdeel aan Hewlett Packard Enterprise moet worden geretourneerd. Als het defecte onderdeel aan Hewlett Packard Enterprise moet worden teruggezonden, moet u het defecte onderdeel binnen een bepaalde periode, gewoonlijk vijf (5) werkdagen, retourneren aan Hewlett Packard Enterprise. Het defecte onderdeel moet met de bijbehorende documentatie worden geretourneerd in het meegeleverde verpakkingsmateriaal. Als u het defecte onderdeel niet terugzendt, kan Hewlett Packard Enterprise u voor het vervangende onderdeel kosten in rekening brengen. Bij reparatie door de klant betaalt Hewlett Packard Enterprise alle verzendkosten voor het vervangende en geretourneerde onderdeel en kiest Hewlett Packard Enterprise zelf welke koerier/transportonderneming hiervoor wordt gebruikt.

Neem contact op met een Service Partner voor meer informatie over het Customer Self Repair programma van Hewlett Packard Enterprise.

Garanteservice "Parts Only"

Het is mogelijk dat de Hewlett Packard Enterprise garantie alleen de garantieservice "Parts Only" omvat. Volgens de bepalingen van de Parts Only garantieservice zal Hewlett Packard Enterprise kosteloos vervangende onderdelen ter beschikking stellen.

Voor de Parts Only garantieservice is vervanging door CSR-onderdelen verplicht. Als u Hewlett Packard Enterprise verzoekt deze onderdelen voor u te vervangen, worden u voor deze service reiskosten en arbeidsloon in rekening gebracht.

Reparo feito pelo cliente

Os produtos da Hewlett Packard Enterprise são projetados com muitas peças para reparo feito pelo cliente (CSR) de modo a minimizar o tempo de reparo e permitir maior flexibilidade na substituição de peças com defeito. Se, durante o período de diagnóstico, a Hewlett Packard Enterprise (ou fornecedores/parceiros da

Hewlett Packard Enterprise) concluir que o reparo pode ser efetuado pelo uso de uma peça CSR, a Hewlett Packard Enterprise enviará a peça diretamente ao cliente. Há duas categorias de peças CSR:

- **Obrigatória**—Peças cujo reparo feito pelo cliente é obrigatório. Se desejar que a Hewlett Packard Enterprise substitua essas peças, serão cobradas as despesas de transporte e mão-de-obra do serviço.
- **Opcional**—Peças cujo reparo feito pelo cliente é opcional. Essas peças também são projetadas para o reparo feito pelo cliente. No entanto, se desejar que a Hewlett Packard Enterprise as substitua, pode haver ou não a cobrança de taxa adicional, dependendo do tipo de serviço de garantia destinado ao produto.

OBSERVAÇÃO: Algumas peças da Hewlett Packard Enterprise não são projetadas para o reparo feito pelo cliente. A fim de cumprir a garantia do cliente, a Hewlett Packard Enterprise exige que um técnico autorizado substitua a peça. Essas peças estão identificadas com a marca "No" (Não), no catálogo de peças ilustrado.

Conforme a disponibilidade e o local geográfico, as peças CSR serão enviadas no primeiro dia útil após o pedido. Onde as condições geográficas permitirem, a entrega no mesmo dia ou em quatro horas pode ser feita mediante uma taxa adicional. Se precisar de auxílio, entre em contato com o Centro de suporte técnico da Hewlett Packard Enterprise para que um técnico o ajude por telefone. A Hewlett Packard Enterprise especifica nos materiais fornecidos com a peça CSR de reposição se a peça com defeito deve ser devolvida à Hewlett Packard Enterprise. Nos casos em que isso for necessário, é preciso enviar a peça com defeito à Hewlett Packard Enterprise, você deverá enviar a peça com defeito de volta para a Hewlett Packard Enterprise dentro do período de tempo definido, normalmente em 5 (cinco) dias úteis. A peça com defeito deve ser enviada com a documentação correspondente no material de transporte fornecido. Caso não o faça, a Hewlett Packard Enterprise poderá cobrar a reposição. Para as peças de reparo feito pelo cliente, a Hewlett Packard Enterprise paga todas as despesas de transporte e de devolução da peça e determina a transportadora/serviço postal a ser utilizado.

Para obter mais informações sobre o programa de reparo feito pelo cliente da Hewlett Packard Enterprise, entre em contato com o fornecedor de serviços local.

Serviço de garantia apenas para peças

A garantia limitada da Hewlett Packard Enterprise pode incluir um serviço de garantia apenas para peças. Segundo os termos do serviço de garantia apenas para peças, a Hewlett Packard Enterprise fornece as peças de reposição sem cobrar nenhuma taxa.

No caso desse serviço, a substituição de peças CSR é obrigatória. Se desejar que a Hewlett Packard Enterprise substitua essas peças, serão cobradas as despesas de transporte e mão-de-obra do serviço.

カスタマーセルフリペア

修理時間を短縮し、故障部品の交換における高い柔軟性を確保するために、Hewlett Packard Enterprise製品には多数のカスタマーセルフリペア（CSR）部品があります。診断の際に、CSR部品を使用すれば修理ができるとHewlett Packard Enterprise（Hewlett Packard EnterpriseまたはHewlett Packard Enterprise正規保守代理店）が判断した場合、Hewlett Packard Enterpriseはその部品を直接、お客様に発送し、お客様に交換していただきます。CSR部品には以下の2種類があります。

- **必須** - カスタマーセルフリペアが必須の部品。当該部品について、もしもお客様がHewlett Packard Enterpriseに交換作業を依頼される場合には、その修理サービスに関する交通費および人件費がお客様に請求されます。
- **任意** - カスタマーセルフリペアが任意である部品。この部品もカスタマーセルフリペア用です。当該部品について、もしもお客様がHewlett Packard Enterpriseに交換作業を依頼される場合には、お買い上げの製品に適用される保証サービス内容の範囲内においては、別途費用を負担していただくことなく保証サービスを受けることができます。

注：Hewlett Packard Enterprise製品の一部の部品は、カスタマーセルフリペアの対象外です。製品の保証を継続するためには、Hewlett Packard EnterpriseまたはHewlett Packard Enterprise正規保守代理店による交換作業が必須となります。部品カタログには、当該部品がカスタマーセルフリペア除外品である旨が記載されています。

部品供給が可能な場合、地域によっては、CSR部品を翌営業日に届くように発送します。また、地域によっては、追加費用を負担いただくことにより同日または4時間以内に届くように発送することも可能な場合があります。サポートが必要なときは、Hewlett Packard Enterpriseサポートセンターに電話していただければ、技術者が電話でアドバイスします。交換用のCSR部品または同梱物には、故障部品をHewlett Packard Enterpriseに返送する必要があるかどうかが表示されています。故障部品をHewlett Packard Enterpriseに返送する必要がある場合は、指定期限内（通常は5営業日以内）に故障部品をHewlett Packard Enterpriseに返送してください。故障部品を返送する場合は、届いた時の梱包箱に関連書類とともに入れてください。故障部品を返送しない場合、Hewlett Packard Enterpriseから部品費用が請求されます。カスタマーセルフリペアの際には、Hewlett Packard Enterpriseは送料および部品返送費を全額負担し、使用する宅配便会社や運送会社を指定します。

部品のみ保証サービス

Hewlett Packard Enterprise保証サービスには、部品のみ保証サービスが適用される場合があります。このサービスでは、交換部品は無償で提供されます。

部品のみ保証サービスにおいては、CSR部品をお客様により交換作業していただくことが必須となります。当該部品について、もしもお客様がHewlett Packard Enterpriseに交換作業を依頼される場合には、その修理サービスに関する交通費および人件費がお客様のご負担となります。

客户自行维修

Hewlett Packard Enterprise 产品提供许多客户自行维修 (CSR) 部件，以尽可能缩短维修时间和在更换缺陷部件方面提供更大的灵活性。如果在诊断期间 Hewlett Packard Enterprise（或 Hewlett Packard Enterprise 服务提供商或服务合作伙伴）确定可以通过使用 CSR 部件完成维修，Hewlett Packard Enterprise 将直接把该部件发送给您进行更换。有两类 CSR 部件：

- **强制性的** — 要求客户必须自行维修的部件。如果您请求 Hewlett Packard Enterprise 更换这些部件，则必须为该服务支付差旅费和人工费用。
- **可选的** — 客户可以选择是否自行维修的部件。这些部件也是为客户自行维修设计的。不过，如果您要求 Hewlett Packard Enterprise 为您更换这些部件，则根据您的产品指定的保修服务类型，Hewlett Packard Enterprise 可能收取或不再收取任何附加费用。

注：某些 Hewlett Packard Enterprise 部件的设计并未考虑客户自行维修。为了满足客户保修的需要，Hewlett Packard Enterprise 要求授权服务提供商更换相关部件。这些部件在部件图解目录中标记为“否”。

CSR 部件将在下一个工作日发运（取决于备货情况和允许的地理范围）。在允许的地理范围内，可在当天或四小时内发运，但要收取额外费用。如果需要帮助，您可以致电 Hewlett Packard Enterprise 技术支持中心，将会有技术人员通过电话为您提供帮助。Hewlett Packard Enterprise 会在随更换的 CSR 部件发运的材料中指明是否必须将有缺陷的部件返还给 Hewlett Packard Enterprise。如果要求您将有缺陷的部件返还给 Hewlett Packard Enterprise，那么您必须在规定的期限内（通常是五 (5) 个工作日）将缺陷部件发给 Hewlett Packard Enterprise。有缺陷的部件必须随所提供的发运材料中的相关文件一起返还。如果未能送还有缺陷的部件，Hewlett Packard Enterprise 可能会要求您支付更换费用。客户自行维修时，Hewlett Packard Enterprise 将承担所有相关运输和部件返回费用，并指定快递商/承运商。

有关 Hewlett Packard Enterprise 客户自行维修计划的详细信息，请与您当地的服务提供商联系。

仅部件保修服务

您的 Hewlett Packard Enterprise 有限保修服务可能涉及仅部件保修服务。根据仅部件保修服务条款的规定，Hewlett Packard Enterprise 将免费提供更换的部件。

仅部件保修服务要求进行 CSR 部件更换。如果您请求 Hewlett Packard Enterprise 更换这些部件，则必须为该服务支付差旅费和人工费用。

客戶自行維修

Hewlett Packard Enterprise 產品設計了許多「客戶自行維修」(CSR) 的零件以減少維修時間，並且使得更換瑕疵零件時能有更大的彈性。如果在診斷期間，Hewlett Packard Enterprise (或 Hewlett Packard Enterprise 服務供應商或維修夥伴) 辨認出此項維修工作可以藉由使用 CSR 零件來完成，則 Hewlett Packard Enterprise 將直接寄送該零件給您作更換。CSR 零件分為兩種類別：

- **強制的** — 客戶自行維修所使用的零件是強制性的。如果您要求 Hewlett Packard Enterprise 更換這些零件，Hewlett Packard Enterprise 將會向您收取此服務所需的外出費用與勞動成本。
- **選購的** — 客戶自行維修所使用的零件是選購的。這些零件也設計用於客戶自行維修之用。不過，如果您要求 Hewlett Packard Enterprise 為您更換，則可能需要也可能不需要負擔額外的費用，端視針對此產品指定的保固服務類型而定。

備註：某些 Hewlett Packard Enterprise 零件沒有消費者可自行維修的設計。為符合客戶保固，Hewlett Packard Enterprise 需要授權的服務供應商更換零件。這些零件在圖示的零件目錄中，被標示為「否」。

基於材料取得及環境允許的情況下，CSR 零件將於下一個工作日以快遞寄送。在環境的允許下當天或四小時內送達，則可能需要額外的費用。若您需要協助，可致電 Hewlett Packard Enterprise 支援中心，會有一位技術人員透過電話來協助您。不論損壞的零件是否必須退回，Hewlett Packard Enterprise 皆會在與 CSR 替換零件一起運送的材料中註明。若要將損壞的零件退回 Hewlett Packard Enterprise，您必須在指定的一段時間內 (通常為五 (5) 個工作天)，將損壞的零件寄回 Hewlett Packard Enterprise。損壞的零件必須與寄送資料中隨附的相關技術文件一併退還。如果無法退還損壞的零件，Hewlett Packard Enterprise 可能要向您收取替換費用。針對客戶自行維修情形，Hewlett Packard Enterprise 將負責所有運費及零件退還費用，並指定使用何家快遞/貨運公司。

如需 Hewlett Packard Enterprise 的 CSR 方案詳細資訊，請連絡您當地的服務供應商。

僅限零件的保固服務

您的「Hewlett Packard Enterprise 有限保固」可能包含僅限零件的保固服務。在僅限零件的保固服務情況下，Hewlett Packard Enterprise 將免費提供替換零件。

針對僅限零件的保固服務，CSR 零件替換是強制性的。如果您要求 Hewlett Packard Enterprise 更換這些零件，Hewlett Packard Enterprise 將會向您收取此服務所需的外出費用與勞動成本。

고객 셀프 수리

Hewlett Packard Enterprise 제품은 수리 시간을 최소화하고 결함이 있는 부품 교체 시 더욱 융통성을 발휘할 수 있도록 하기 위해 고객 셀프 수리(CSR) 부품을 다량 사용하여 설계되었습니다. 진단 기간 동안 Hewlett Packard Enterprise(또는 Hewlett Packard Enterprise 서비스 공급업체 또는 서비스 협력업체)에서 CSR 부품을 사용하여 수리가 가능하다고 판단되면 Hewlett Packard Enterprise는 해당 부품을 바로 사용자에게 보내어 사용자가 교체할 수 있도록 합니다. CSR 부품에는 두 가지 종류가 있습니다.

- 필수 - 고객 셀프 수리가 의무 사항인 필수 부품. 사용자가 Hewlett Packard Enterprise에 이 부품의 교체를 요청할 경우 이 서비스에 대한 출장비 및 작업비가 청구됩니다.
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참고: 일부 Hewlett Packard Enterprise 제품은 고객 셀프 수리가 불가능하도록 설계되었습니다. Hewlett Packard Enterprise는 만족스러운 고객 보증을 위해 공인 서비스 제공업체를 통해 부품을 교체하도록 하고 있습니다. 이러한 부품들은 Illustrated Parts Catalog에 "No"라고 표시되어 있습니다.

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Removal and replacement procedures

Subtopics

Required tools

Safety considerations

Preparation procedures

Removing and replacing the bezel

Removing and replacing the cable management arm

Removing and replacing the easy install rails

Removing and replacing the ball bearing rails

Removing the hard drive blank

Removing a SAS/SATA drive

Removing a hot-plug SAS/SATA basic drive

Removing and replacing an NVMe drive

Removing and replacing an M.2 SSD

Removing and replacing a uFF drive

Removing and replacing a power supply blank

Removing and replacing the AC power supply

Removing and replacing a fan

Removing and replacing a fan blank

Removing and replacing a PCIe blank

Removing and replacing a riser cage

Removing and replacing an expansion board

Removing and replacing the Pensando DSP DSC-25 2p SFP28 card with the iLO Sideband ALOM Module

Removing and replacing the HPE NS204i-p NVMe OS Boot Device option

Removing and replacing a 940QSFP56 x16 adapter and auxiliary card

Removing and replacing a primary riser card

Removing and replacing the serial port

Removing and replacing the intrusion detection switch

Removing and replacing a GPU in the primary riser

Removing and replacing a GPU in the secondary riser

Removing and replacing the FlexibleLOM

Removing and replacing a storage controller

Removing and replacing a DIMM

Removing and replacing an NVDIMM

Removing and replacing a persistent memory module

Removing and replacing the system battery

Removing and replacing the Smart Storage Battery or Hybrid Capacitor

Removing and replacing a microSD card

Removing and replacing a processor heatsink assembly

Removing and replacing the system board

Removing and replacing the LFF optical disk drive/display port/USB

Removing and replacing the SFF optical disk drive/display port/USB

Removing and replacing the LFF power switch module

Removing and replacing the Systems Insight Display power module

Removing and replacing the rear SFF drive cage

Removing and replacing the 2 SFF drive cage

Removing and replacing the 4 LFF drive backplane

Removing and replacing the 8 SFF drive backplane

Removing and replacing the 10 SFF drive backplane

HPE Trusted Platform Module 2.0 Gen10 Option

Required tools

The following tools might be required to perform some procedures:

- T-10/T-15/T-30 Torx screwdriver
- HPE Insight Diagnostics software

Safety considerations

Before performing service procedures, review all the safety information.

Subtopics

Electrostatic discharge

Server warnings and cautions

Rack warnings

Electrostatic discharge

Be aware of the precautions you must follow when setting up the system or handling components. A discharge of static electricity from a finger or other conductor may damage system boards or other static-sensitive devices. This type of damage may reduce the life expectancy of the system or component.

To prevent electrostatic damage:

- Avoid hand contact by transporting and storing products in static-safe containers.
- Keep electrostatic-sensitive parts in their containers until they arrive at static-free workstations.
- Place parts on a grounded surface before removing them from their containers.
- Avoid touching pins, leads, or circuitry.
- Always be properly grounded when touching a static-sensitive component or assembly. Use one or more of the following methods when handling or installing electrostatic-sensitive parts:
 - Use a wrist strap connected by a ground cord to a grounded workstation or computer chassis. Wrist straps are flexible straps with a minimum of 1 megohm ± 10 percent resistance in the ground cords. To provide proper ground, wear the strap snug against the skin.
 - Use heel straps, toe straps, or boot straps at standing workstations. Wear the straps on both feet when standing on conductive floors or dissipating floor mats.

- Use conductive field service tools.
- Use a portable field service kit with a folding static-dissipating work mat.

If you do not have any of the suggested equipment for proper grounding, have an authorized reseller install the part.

For more information on static electricity or assistance with product installation, contact an authorized reseller.

Server warnings and cautions



WARNING

This server is heavy. To reduce the risk of personal injury or damage to the equipment:

- Observe local occupational health and safety requirements and guidelines for manual material handling.
- Get help to lift and stabilize the product during installation or removal, especially when the product is not fastened to the rails. Hewlett Packard Enterprise recommends that a minimum of two people are required for all rack server installations. If the server is installed higher than chest level, a third person may be required to help align the server.
- Use caution when installing the server in or removing the server from the rack; it is unstable when not fastened to the rails.



WARNING

To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.



WARNING

To reduce the risk of personal injury, electric shock, or damage to the equipment, remove the power cord to remove power from the server. The front panel Power On/Standby button does not completely shut off system power. Portions of the power supply and some internal circuitry remain active until AC/DC power is removed.



WARNING

To reduce the risk of fire or burns after removing the energy pack:

- Do not disassemble, crush, or puncture the energy pack.
- Do not short external contacts.
- Do not dispose of the energy pack in fire or water.

After power is disconnected, battery voltage might still be present for 1s to 160s.

AVERTISSEMENT: Pour réduire les risques d'incendie ou de brûlures après le retrait du module batterie :

- N'essayez pas de démonter, d'écraser ou de percer le module batterie.

Rack warnings



WARNING

To reduce the risk of personal injury or damage to the equipment, be sure that:

- The leveling jacks are extended to the floor.
- The full weight of the rack rests on the leveling jacks.
- The stabilizing feet are attached to the rack if it is a single-rack installation.
- The racks are coupled together in multiple-rack installations.
- Only one component is extended at a time. A rack may become unstable if more than one component is extended for any reason.



WARNING

To reduce the risk of personal injury or equipment damage when unloading a rack:

- At least two people are needed to safely unload the rack from the pallet. An empty 42U rack can weigh as much as 115 kg (253 lb), can stand more than 2.1 m (7 ft) tall, and might become unstable when being moved on its casters.
- Never stand in front of the rack when it is rolling down the ramp from the pallet. Always handle the rack from both sides.



WARNING

To reduce the risk of personal injury or damage to the equipment, adequately stabilize the rack before extending a component outside the rack. Extend only one component at a time. A rack may become unstable if more than one component is extended.



WARNING

When installing a server in a telco rack, be sure that the rack frame is adequately secured at the top and bottom to the building structure.

Preparation procedures

To access some components and perform certain service procedures, you must perform one or more of the following procedures:

- Extend the server from the rack.

If you are performing service procedures in a Hewlett Packard Enterprise, Compaq branded, Telco, or third-party rack cabinet, you can use the locking feature of the rack rails to support the server and gain access to internal components.

For more information about Telco rack solutions, refer to the [RackSolutions.com website](http://RackSolutions.com).

- Power down the server.

If you must remove a server from a rack or a non-hot-plug component from a server, power down the server.

- Remove the server from the rack.

If the rack environment, cabling configuration, or the server location in the rack creates awkward conditions, remove the server from the rack.

Subtopics

Power up the server

Power down the server

Extend the server from the rack

Remove the server from the rack

Remove the access panel

Install the access panel

Release the cable management arm

Power up the server

About this task

To power up the server, use one of the following methods:

- Press the Power On/Standby button.
- Use the virtual power button through iLO.

Power down the server

Before powering down the server for any upgrade or maintenance procedures, perform a backup of critical server data and programs.



IMPORTANT

When the server is in standby mode, auxiliary power is still being provided to the system.

To power down the server, use one of the following methods:

- Press and release the Power On/Standby button.
This method initiates a controlled shutdown of applications and the OS before the server enters standby mode.
- Press and hold the Power On/Standby button for more than 4 seconds to force the server to enter standby mode.
This method forces the server to enter standby mode without properly exiting applications and the OS. If an application stops responding, you can use this method to force a shutdown.
- Use a virtual power button selection through iLO.
This method initiates a controlled remote shutdown of applications and the OS before the server enters standby mode.

Before proceeding, verify that the server is in standby mode by observing that the system power LED is amber.

Extend the server from the rack

About this task



NOTE

If the cable management arm option is installed, you can extend the server without powering down the server or disconnecting peripheral cables and power cords. These steps are only necessary with the standard cable management solution.

Procedure

1. Power down the server.
2. Disconnect all peripheral cables and power cords.

3. Loosen the front panel thumbscrews.
4. Extend the server on the rack rails until the server rail-release latches engage.

**WARNING**

To reduce the risk of personal injury or equipment damage, be sure that the rack is adequately stabilized before extending a component from the rack.

**WARNING**

To reduce the risk of personal injury, be careful when pressing the server rail-release latches and sliding the server into the rack. The sliding rails could pinch your fingers.

5. After the installation or maintenance procedure, slide the server into the rack:
 - a. Slide the server fully into the rack.
 - b. Secure the server by tightening the thumbscrews.
6. Connect the peripheral cables and power cords.

Remove the server from the rack

Prerequisites

Before you perform this procedure, review the:

- [Rack warnings](#)
- [Server warnings and cautions](#)

About this task

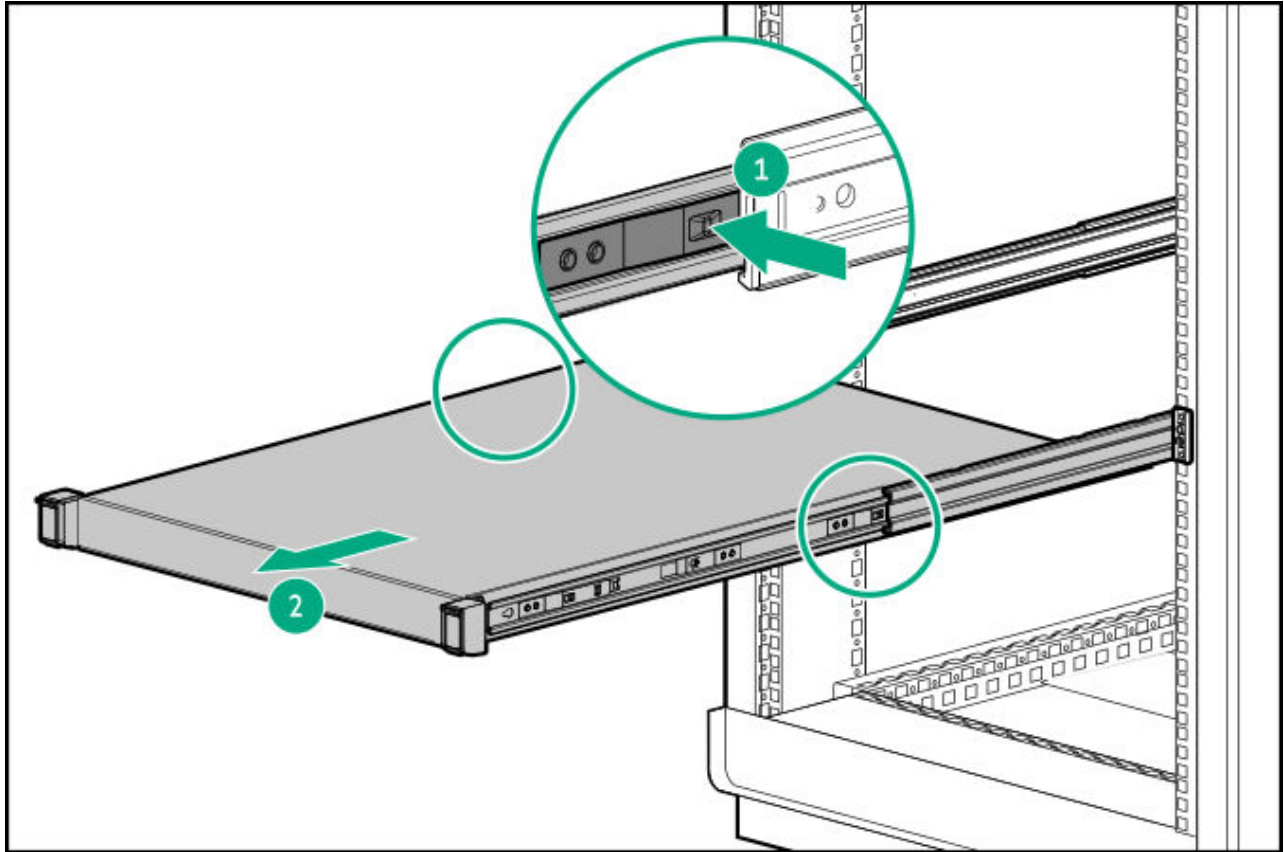
**WARNING**

To reduce the risk of personal injury or equipment damage, be sure that the rack is adequately stabilized before extending anything from the rack.

Procedure

1. [Power down the server.](#)

2. Release the cable management arm.
3. Disconnect the cabling from the rear panel.
4. Extend the server from the rack.
5. Remove the server from the rack:



6. Place the server on a sturdy, level surface.

Remove the access panel

About this task



WARNING

To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.



CAUTION

Do not operate the server for long periods with the access panel open or removed. Operating the server in this manner results in improper airflow and improper cooling that can lead to thermal damage.

Procedure

1. Power down the server.
2. Extend the server from the rack.
3. Open or unlock the locking latch, slide the access panel to the rear of the chassis, and remove the access panel.

Install the access panel

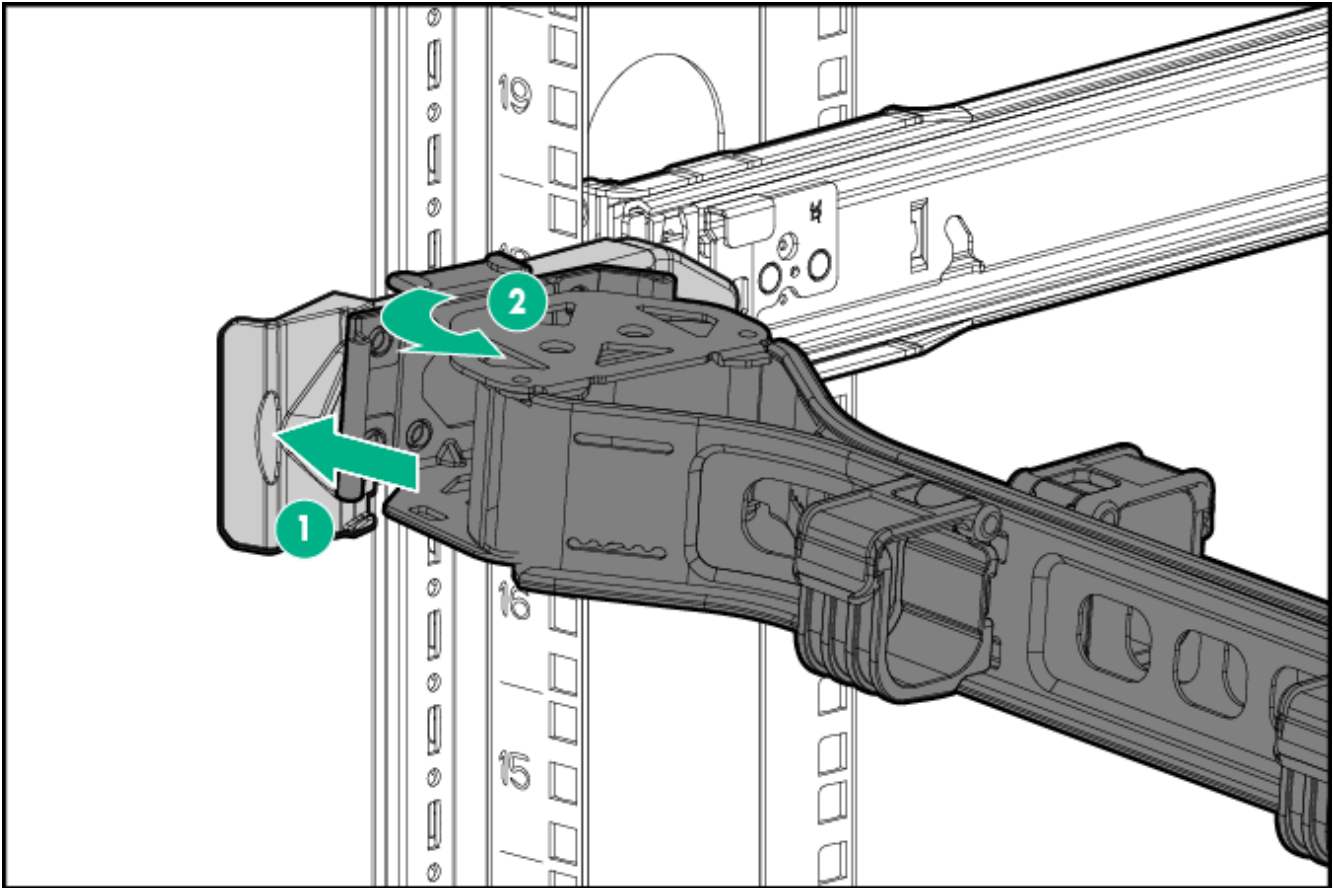
Procedure

1. Place the access panel on top of the server with the latch open.
Allow the panel to extend past the rear of the server approximately 1.25 cm (0.5 in).
2. Push down on the latch.
The access panel slides to a closed position.
3. Tighten the security screw on the latch, if needed.

Release the cable management arm

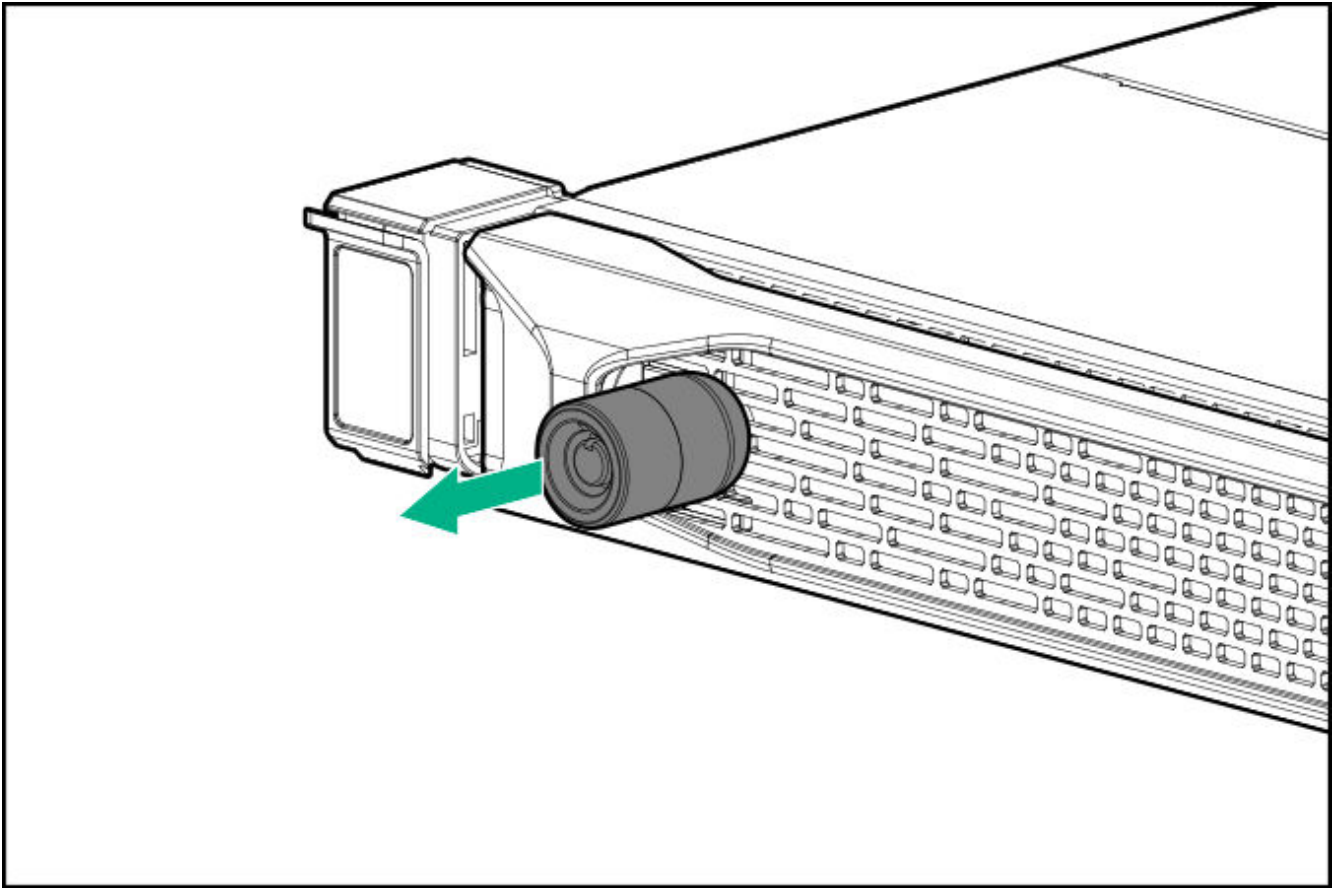
About this task

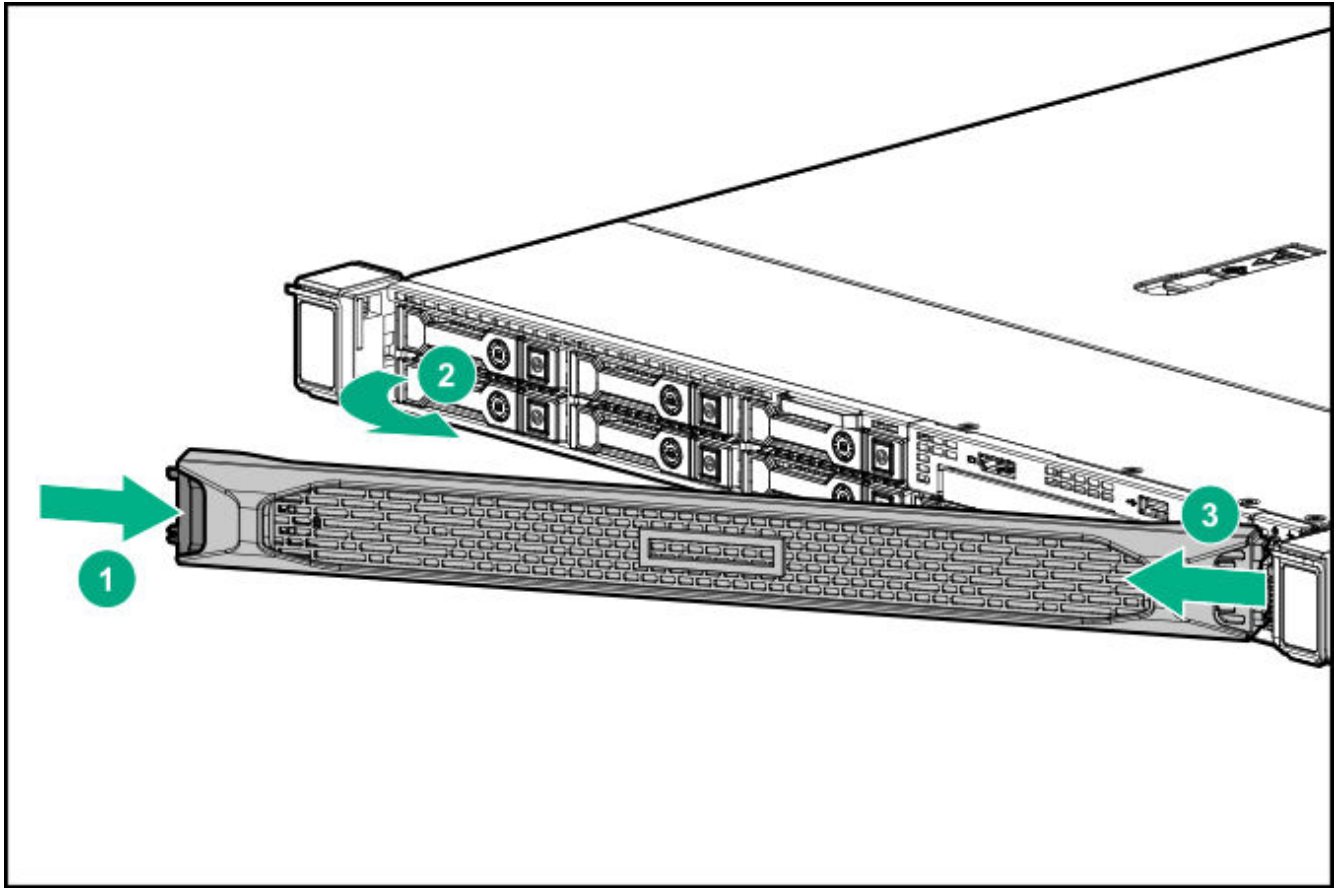
Release the cable management arm and then swing the arm away from the rack.



Removing and replacing the bezel

About this task





Removing and replacing the cable management arm

About this task

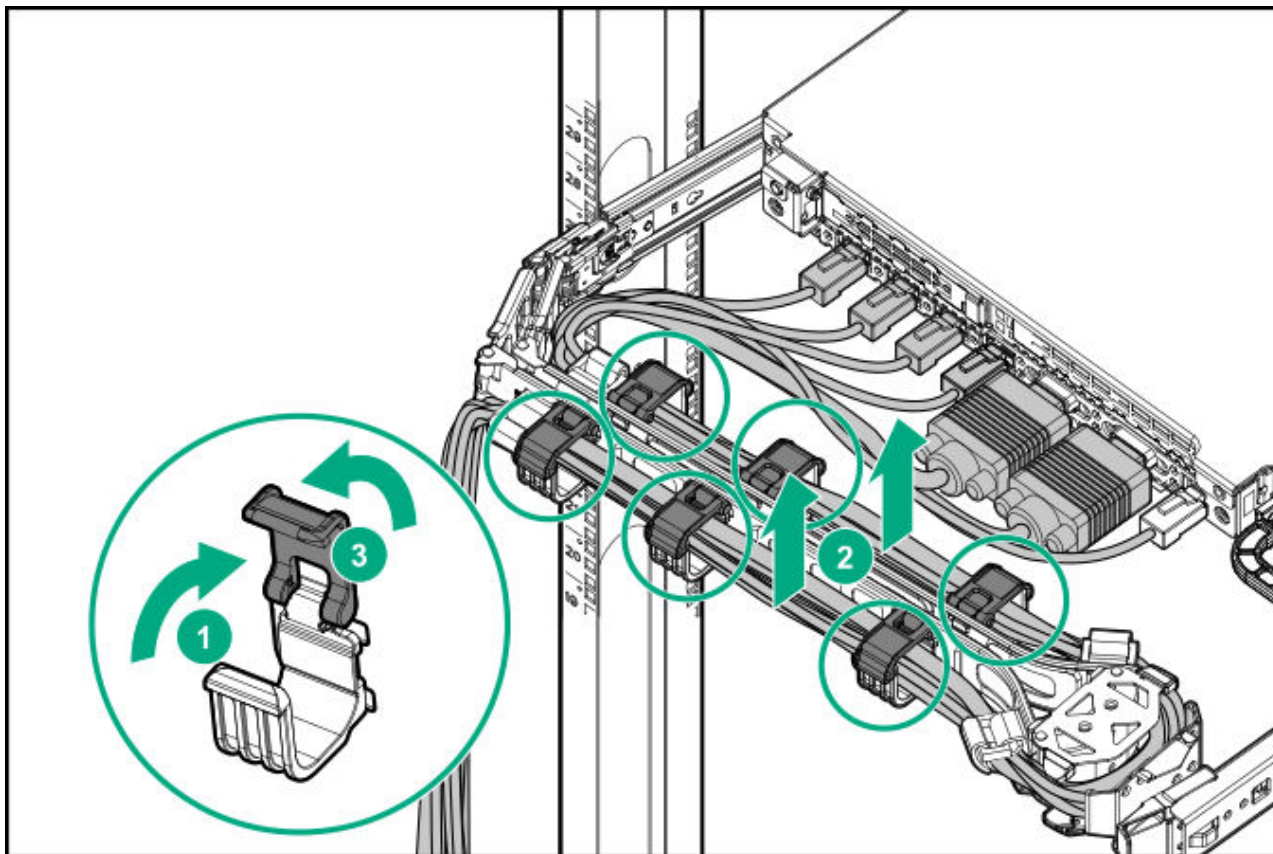


CAUTION

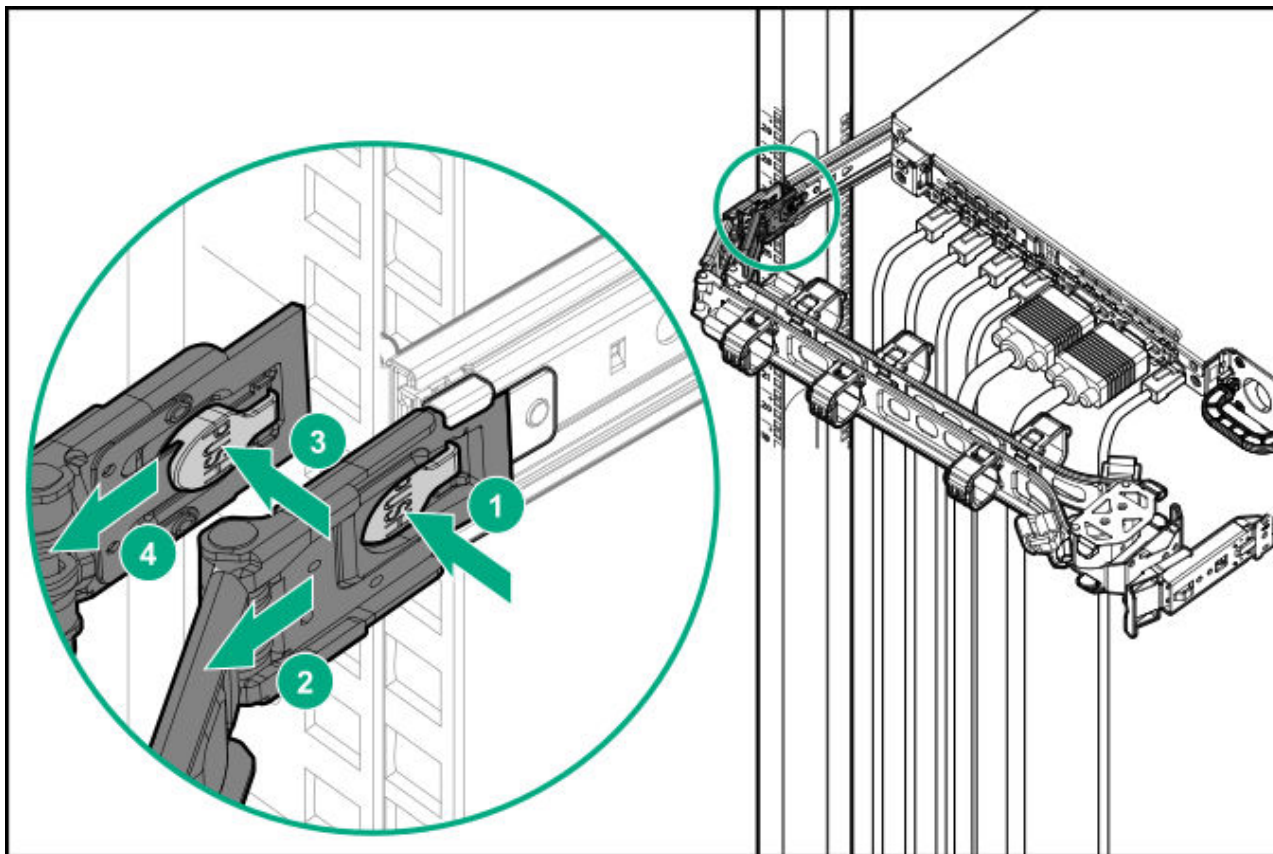
Support the CMA during the removal/replacement procedure. Do not allow the CMA to hang by its own weight during the procedure.

Procedure

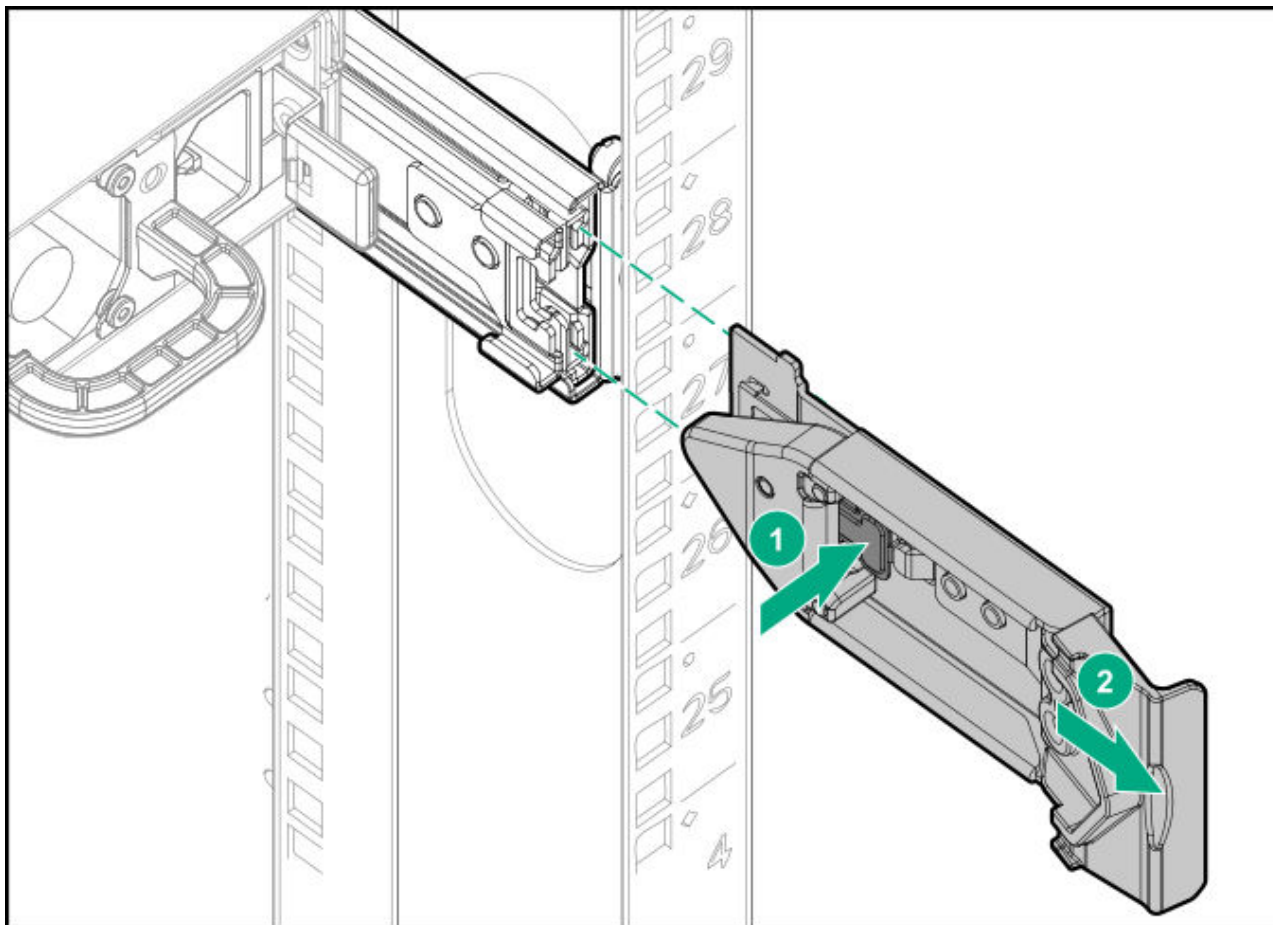
1. Extend the server from the rack.
2. Open the cable clamps and remove the cables routed through the CMA.



3. Release the CMA.



4. Remove the CMA latch.



Results

To replace the component, reverse the removal procedure.

Removing and replacing the easy install rails

Prerequisites

Before you begin, be sure that you have appropriate screwdriver for the screws in the rack.

About this task

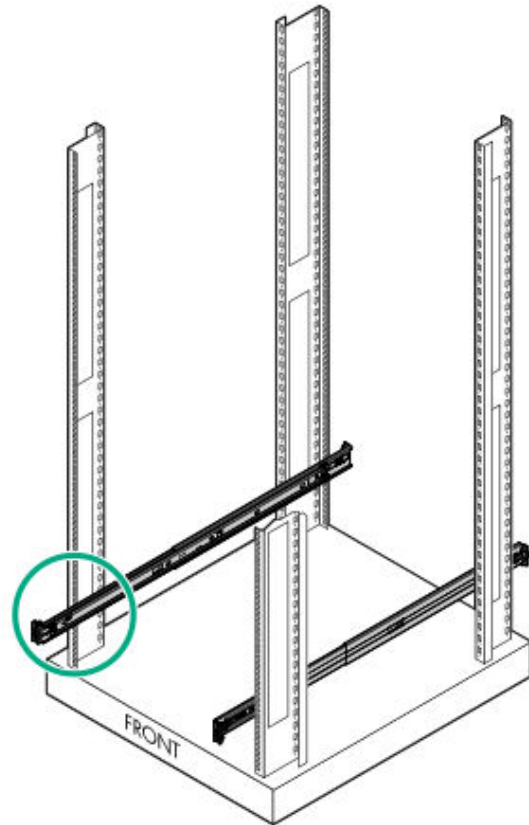
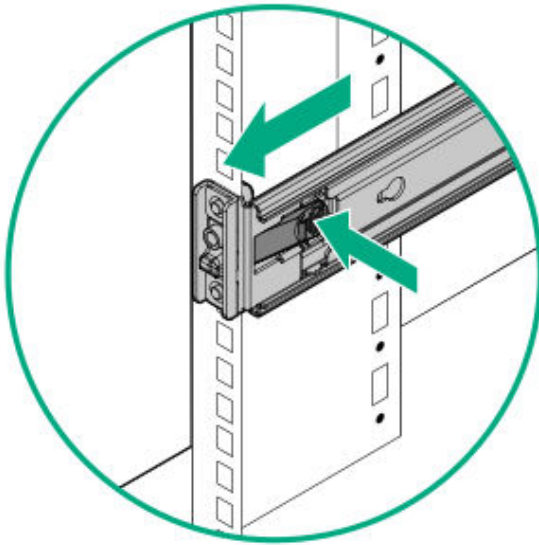


WARNING

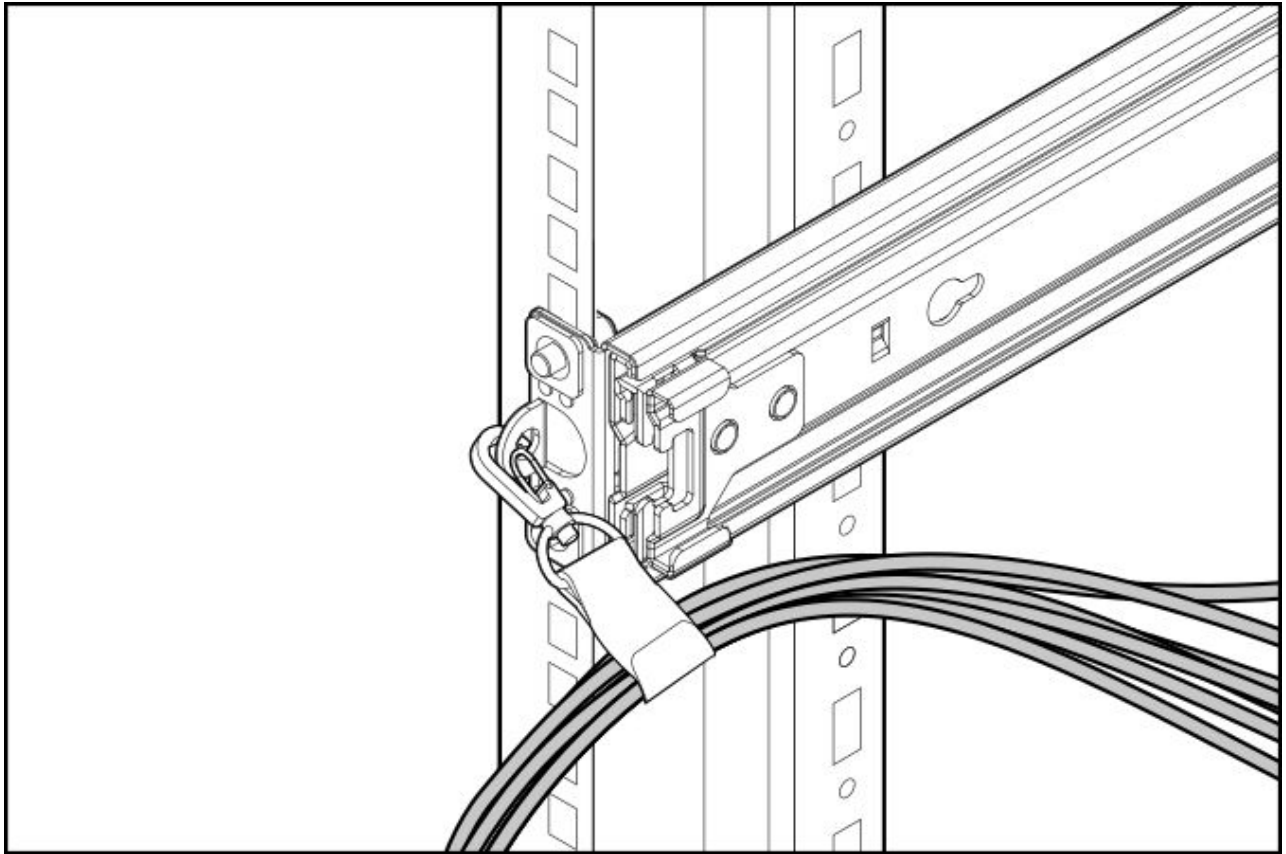
To reduce the risk of personal injury or damage to the equipment, you must adequately support the enclosure during installation and removal.

Procedure

1. Power down the server.
2. Extend the server from the rack.
3. Disconnect the cabling.
4. Remove the server from the rack.
5. Place the server on a sturdy, level surface.
6. Remove the front of the rail.



7. Secure the cables.



Results

To replace the component, reverse the removal procedure.

Removing and replacing the ball bearing rails

Prerequisites

Before you begin, be sure that you have the following items:

- Appropriate screwdriver for the screws in the rack.
- Optional installation tool (695539-001), if the rails were installed using it.

About this task

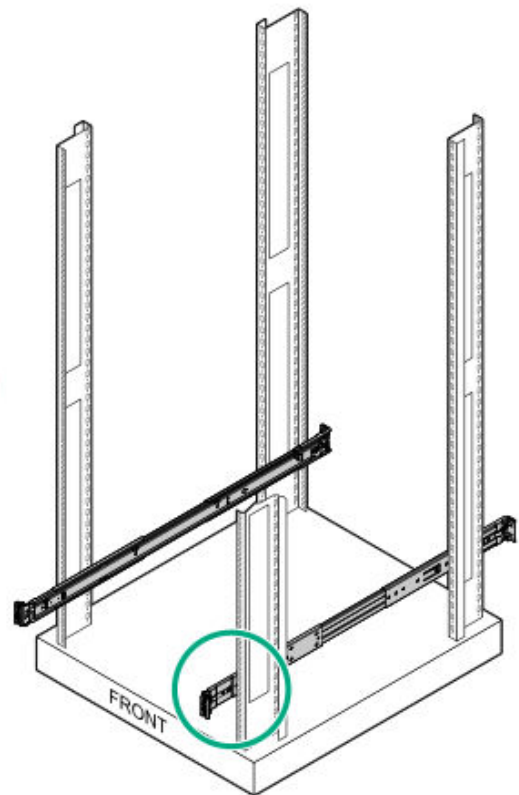
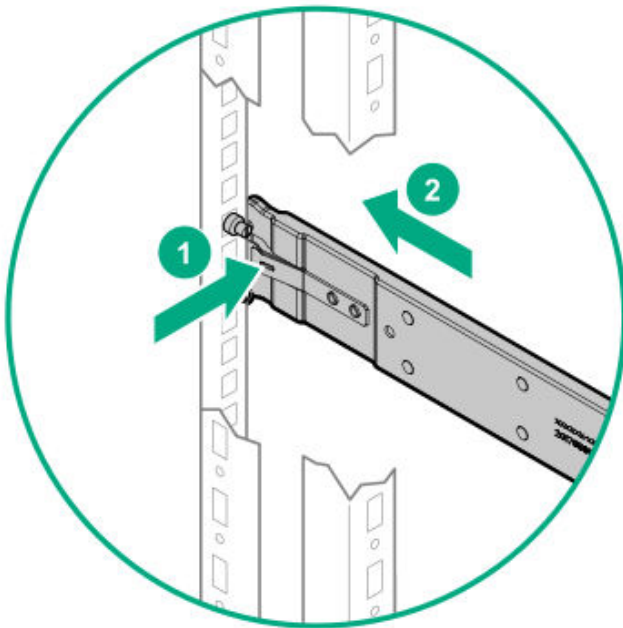


WARNING

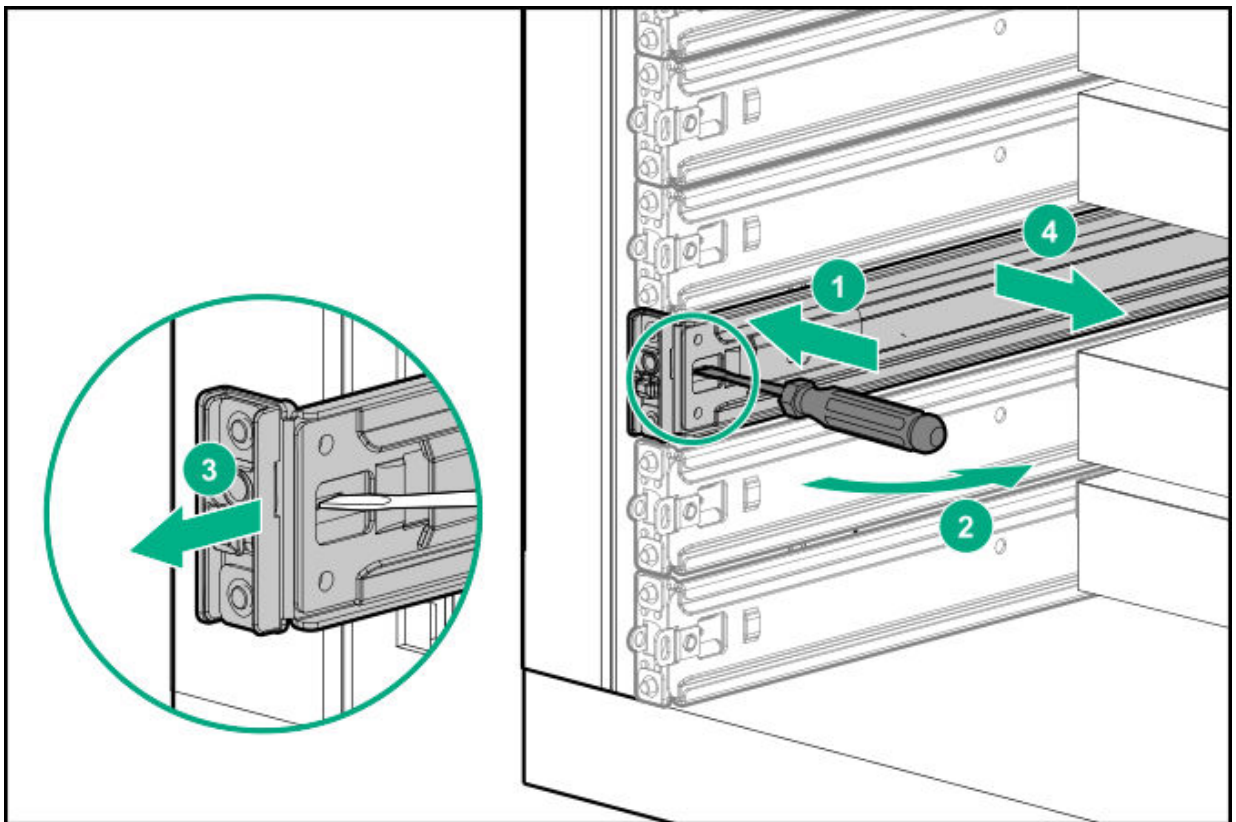
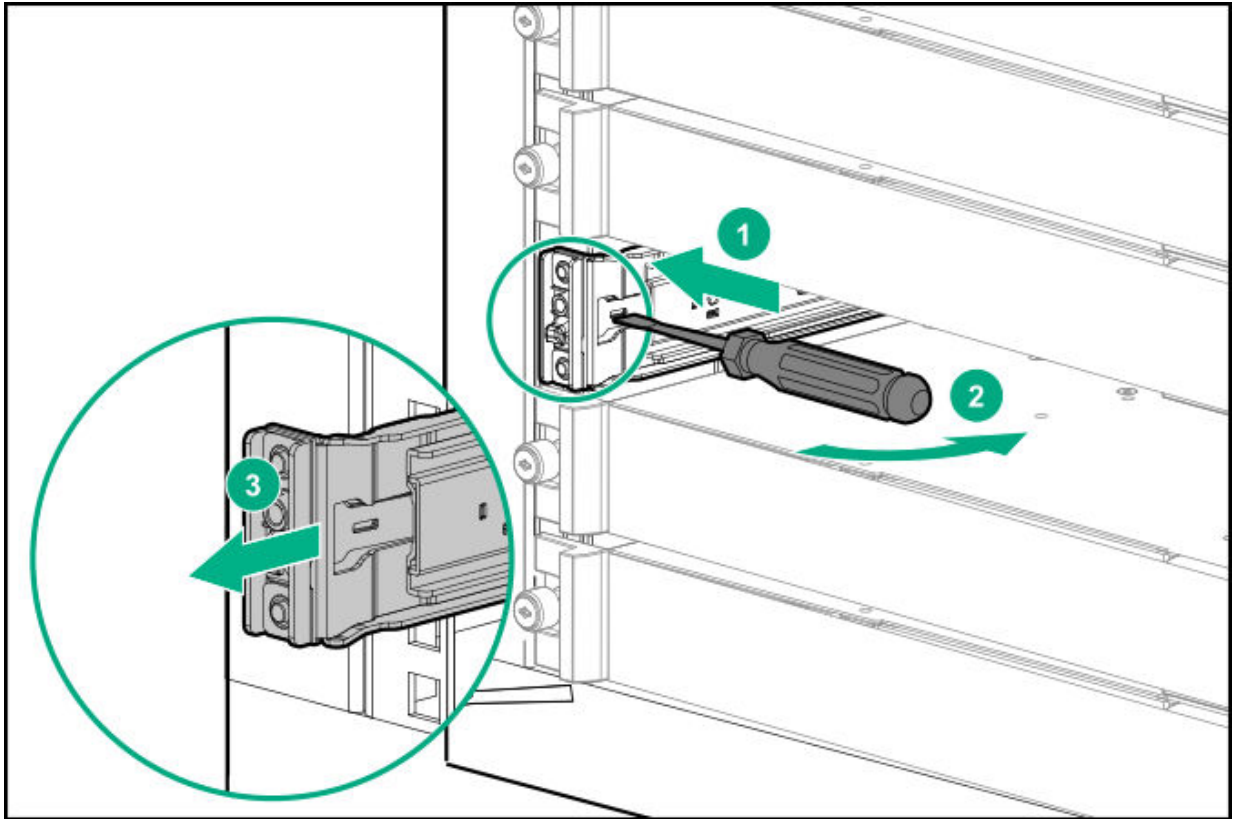
To reduce the risk of personal injury or damage to the equipment, you must adequately support the enclosure during installation and removal.

Procedure

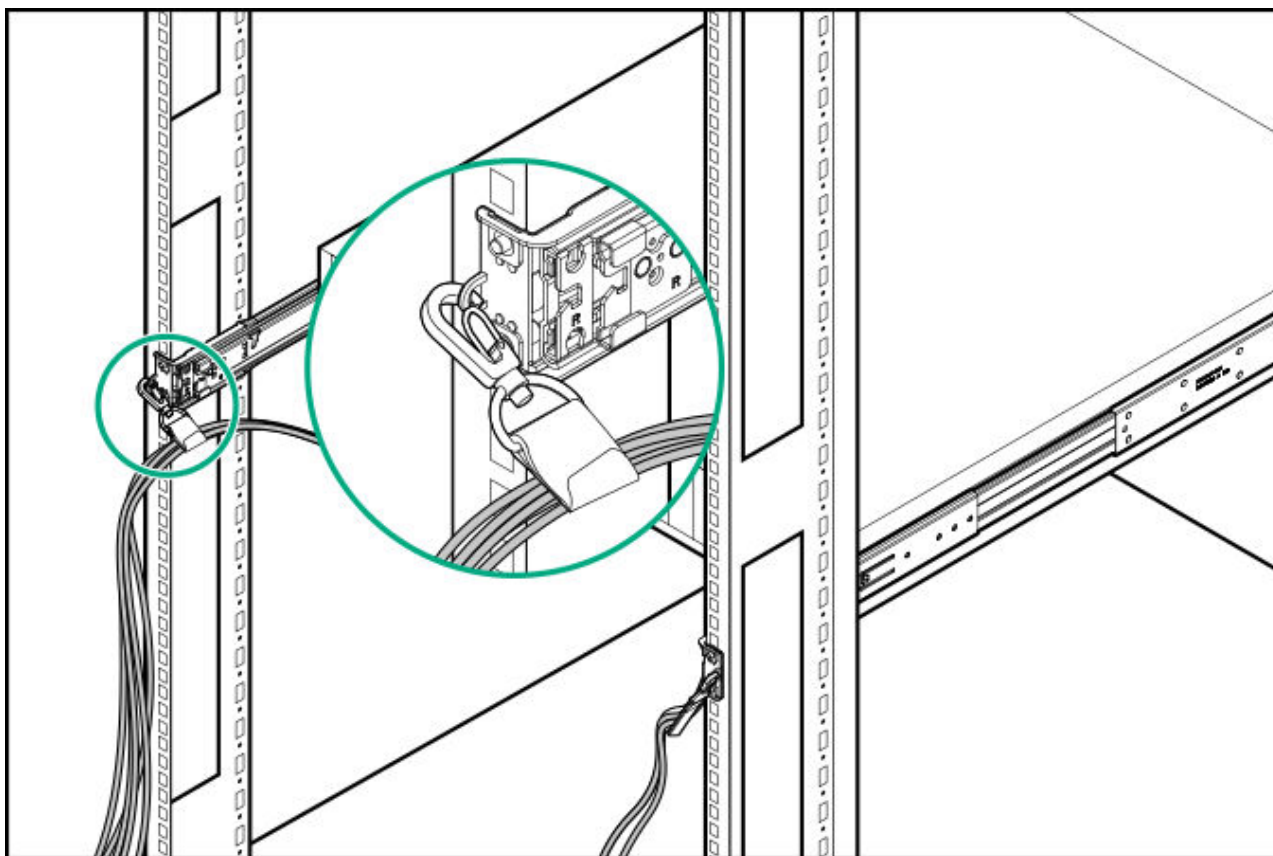
1. Power down the server.
2. Extend the server from the rack.
3. Disconnect the cabling.
4. Remove the server from the rack.
5. Place the server on a sturdy, level surface.
6. Do one of the following:
 - (Optional) Remove the rail without a tool.



- (Optional) Remove the rail with the tool.



7. Secure the cables.



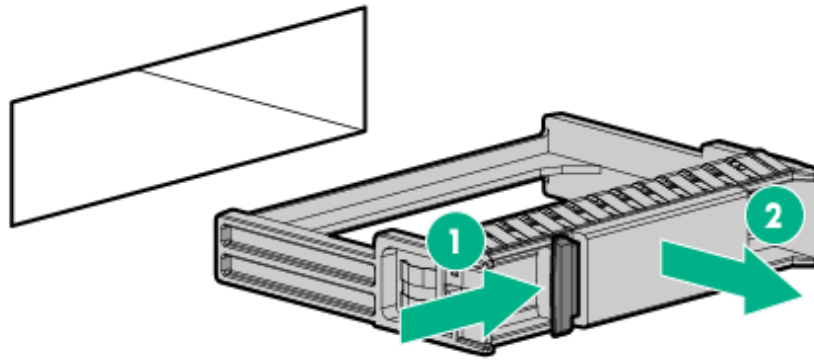
Results

To replace the component, reverse the removal procedure.

Removing the hard drive blank

About this task

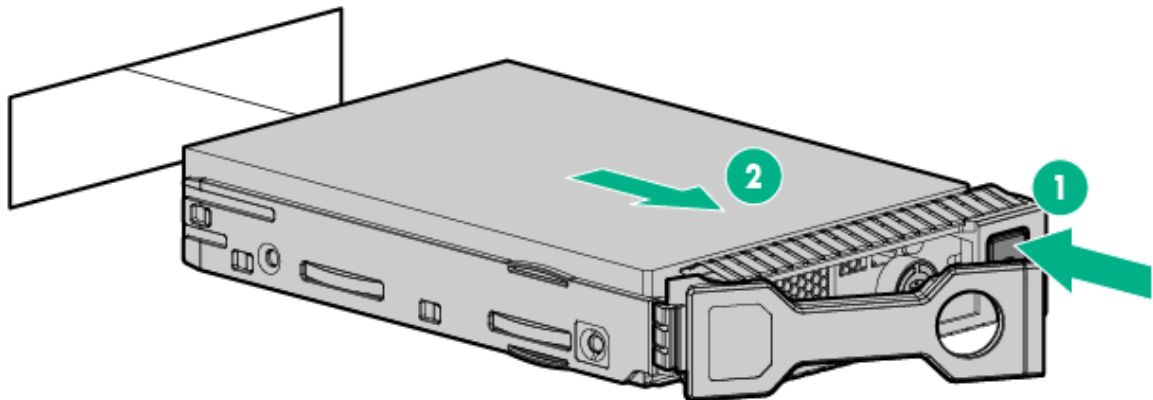
Remove the component as indicated.



Removing a SAS/SATA drive

Procedure

1. Back up all server data.
2. Observe the LED status of the drive and determine if it can be removed.
3. Remove the drive.



Results

To replace the component, reverse the removal procedure.

Removing a hot-plug SAS/SATA basic drive



CAUTION

For proper cooling, do not operate the server without the access panel, baffles, expansion slot covers, or blanks installed. If the server supports hot-plug components, minimize the amount of time the access panel is open.

1. Determine the status of the drive from the [hot-plug drive LED definitions](#).
2. Back up all server data on the drive.
3. Remove the drive.

Figure 1. SFF drive

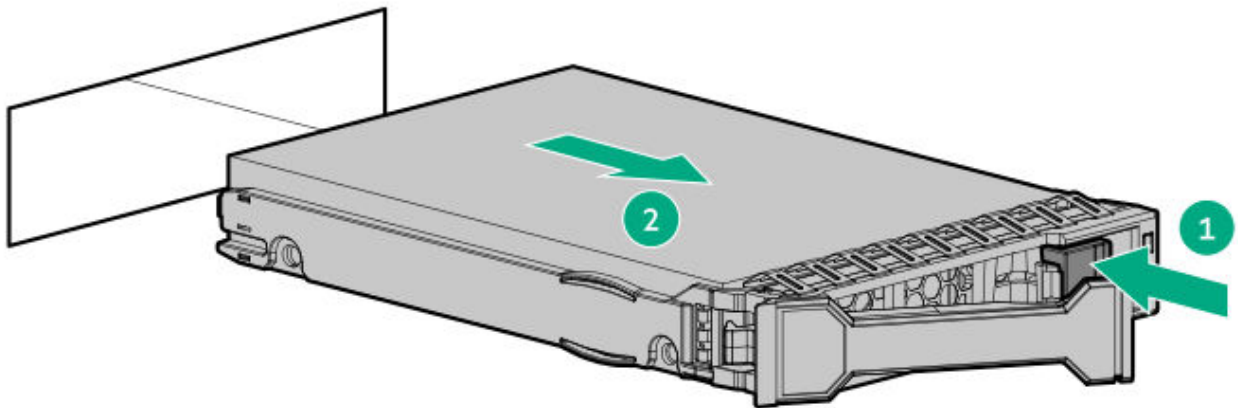
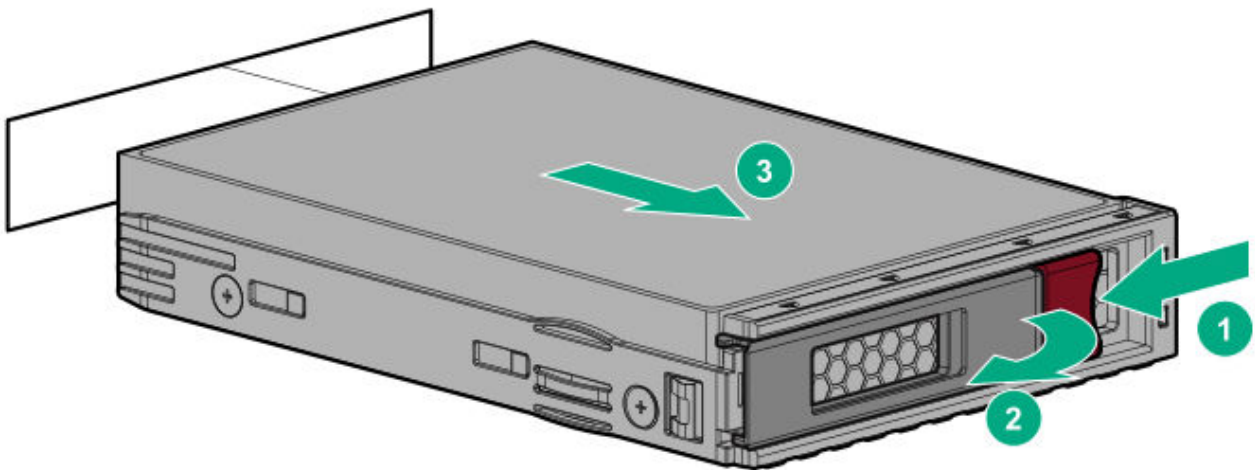


Figure 2. LFF drive



Removing and replacing an NVMe drive

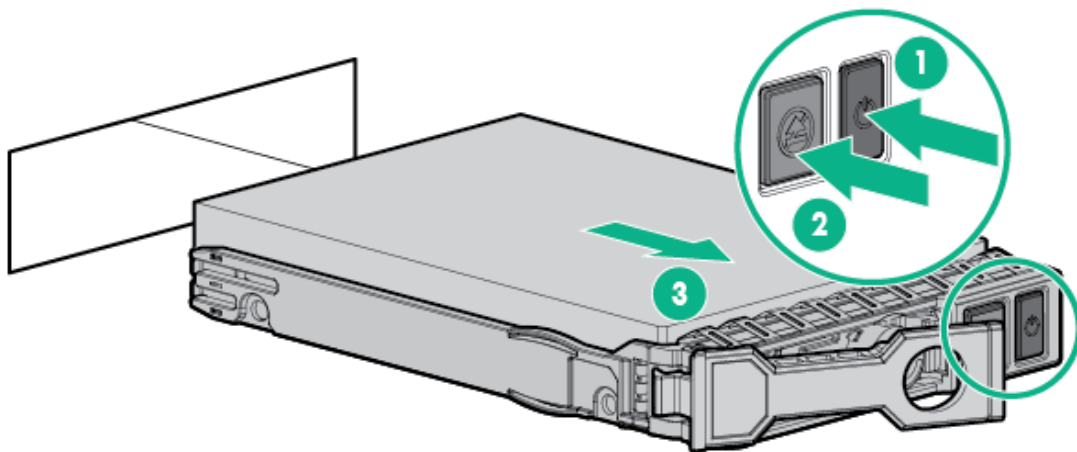
About this task

An NVMe SSD is a PCIe BUS device. Devices attached to a PCIe bus cannot be removed without allowing the device and the bus to complete and cease signal/traffic flow.

Procedure

1. Back up all server data.
2. Observe the LED status of the drive and determine if it can be removed.
3. Remove the drive:
 - a. Push the Power button.

The Do Not Remove button illuminates and flashes.
 - b. Wait until the flashing stops and the Do Not Remove button is no longer illuminated.
 - c. Push the Do Not Remove button and then remove the drive.



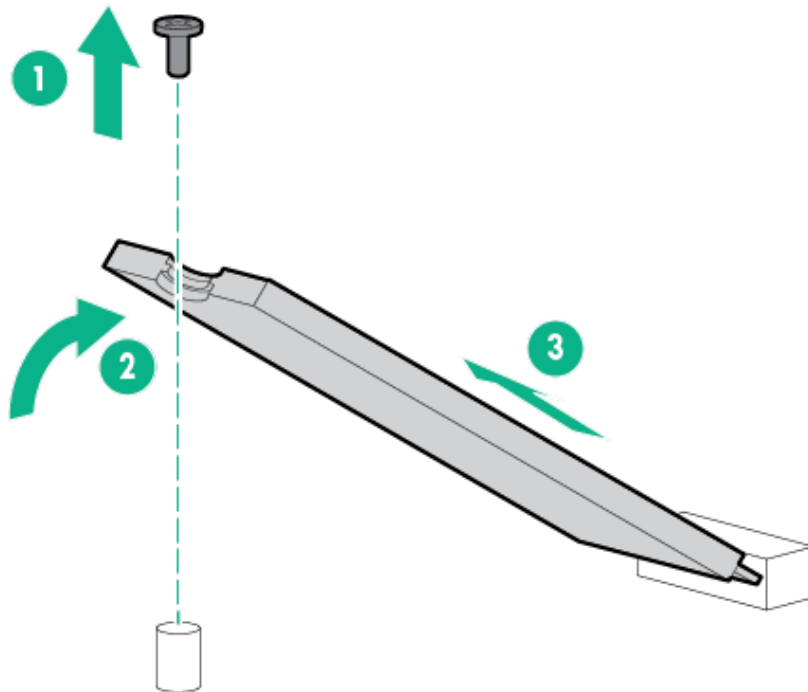
Removing and replacing an M.2 SSD

About this task

This procedure is for drives on the system board, riser, or expansion card. Do not use this procedure on uFF drives.

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Remove the primary riser cage.
6. Remove the M.2 SSD.



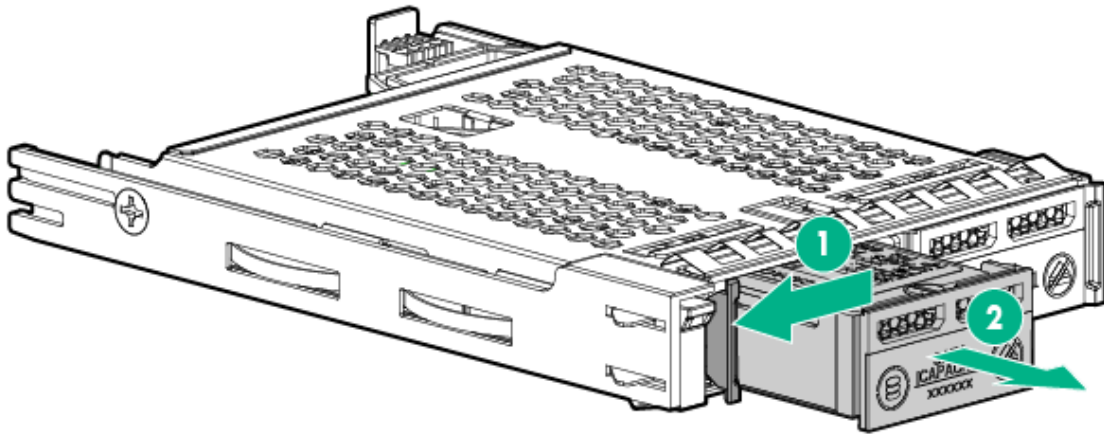
Results

To replace the component, reverse the removal procedure.

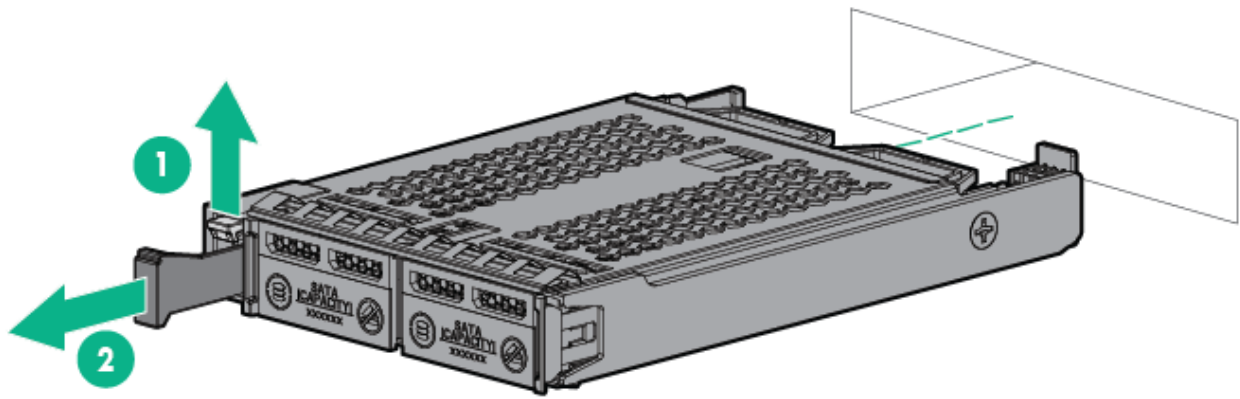
Removing and replacing a uFF drive

Procedure

1. Back up all server data.
2. Observe the LED status of the drive and determine if it can be removed.
3. Remove the drive.



To remove the drive carrier:



Results

To replace the component, reverse the removal procedure.

Removing and replacing a power supply blank

About this task



WARNING

To reduce the risk of personal injury from hot surfaces, allow the power supply or power supply blank to cool before touching it.

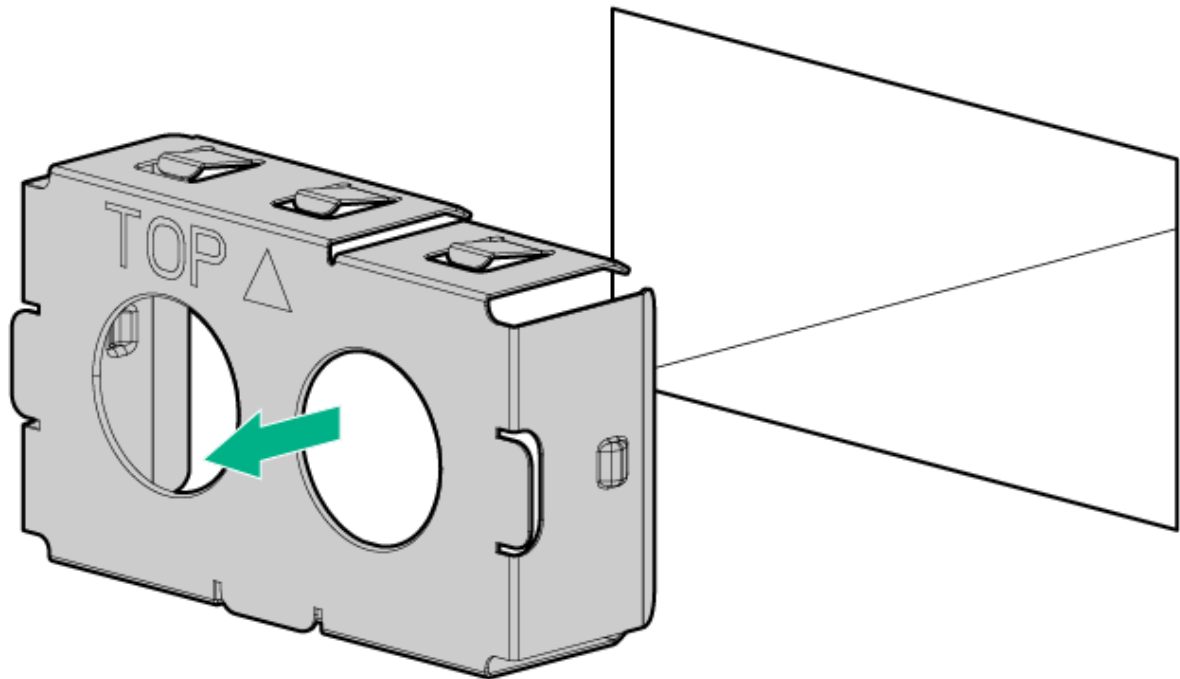


CAUTION

To prevent improper cooling and thermal damage, do not operate the server unless all bays are populated with either a component or a blank.

Procedure

1. Remove the power supply blank.



2. Immediately install the replacement power supply blank spare.

Removing and replacing the AC power supply

About this task



CAUTION

To prevent improper cooling and thermal damage, do not operate the server unless all bays are populated with either a component or a blank.

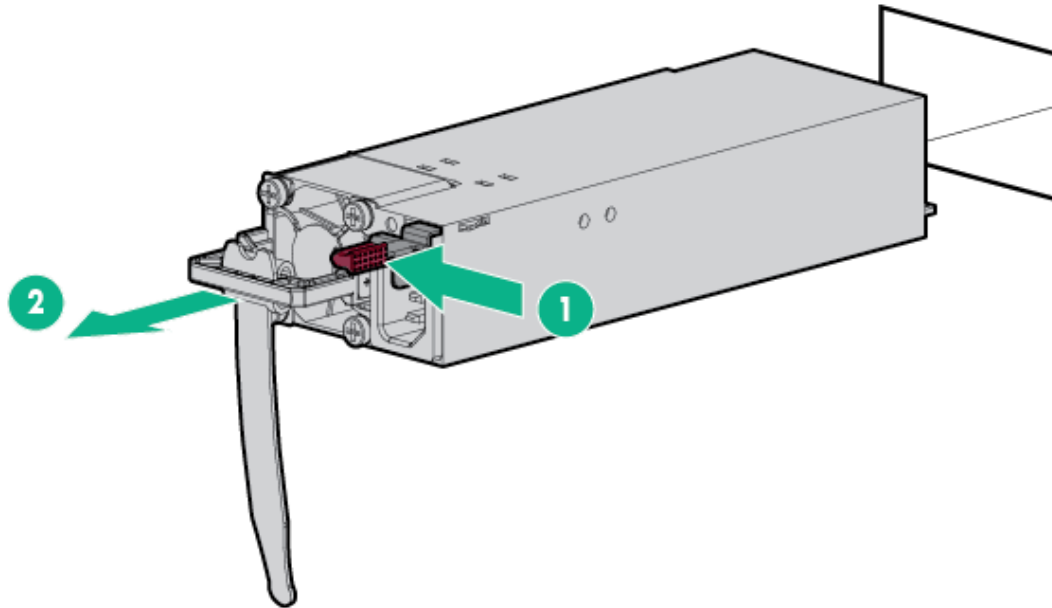
Procedure

1. Power down the server ([Power down the server](#)).
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Access the product rear panel ([Release the cable management arm](#)).

4. Remove the power supply.

**WARNING**

To reduce the risk of personal injury from hot surfaces, allow the power supply or power supply blank to cool before touching it.

**Results**

To replace the component, reverse the removal procedure.

Removing and replacing a fan

About this task



CAUTION

Do not operate the server for long periods with the access panel open or removed. Operating the server in this manner results in improper airflow and improper cooling that can lead to thermal damage.



IMPORTANT

For optimum cooling, install fans in all primary fan locations.

Procedure

1. Extend the server from the rack.
2. Remove the access panel.
3. Remove the fan.



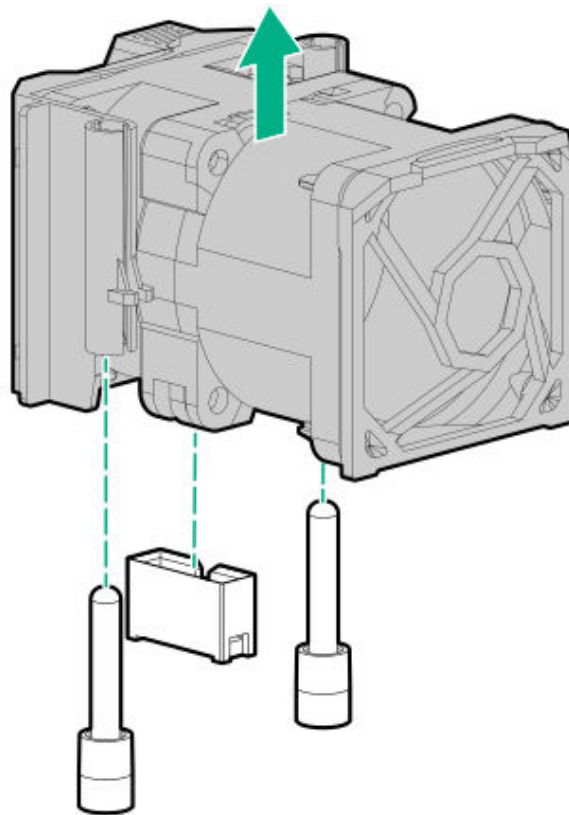
IMPORTANT

When a hot-plug fan is removed, the other fans in the server will increase speed to compensate.



CAUTION

To avoid server shutdown, a fan must be replaced within 60 seconds of being removed.



Results

To replace the component, reverse the removal procedure.

Removing and replacing a fan blank

About this task



CAUTION

Do not operate the server for long periods with the access panel open or removed. Operating the server in this manner results in improper airflow and improper cooling that can lead to thermal damage.

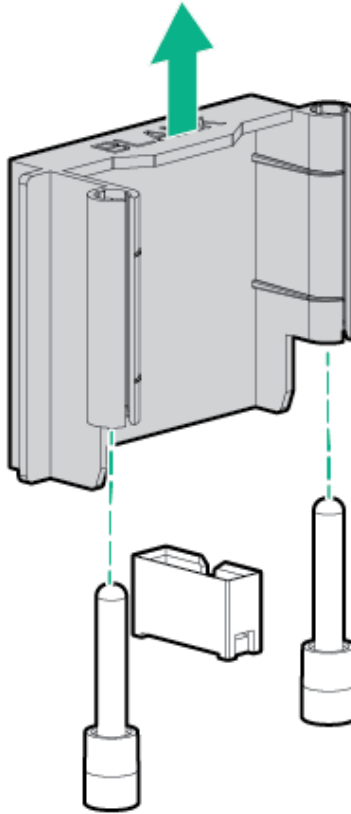


IMPORTANT

For optimum cooling, install fans in all primary fan locations.

Procedure

1. Extend the server from the rack.
2. Remove the access panel.
3. Remove the fan blank.



Results

To replace the component, reverse the removal procedure.

Removing and replacing a PCIe blank

About this task



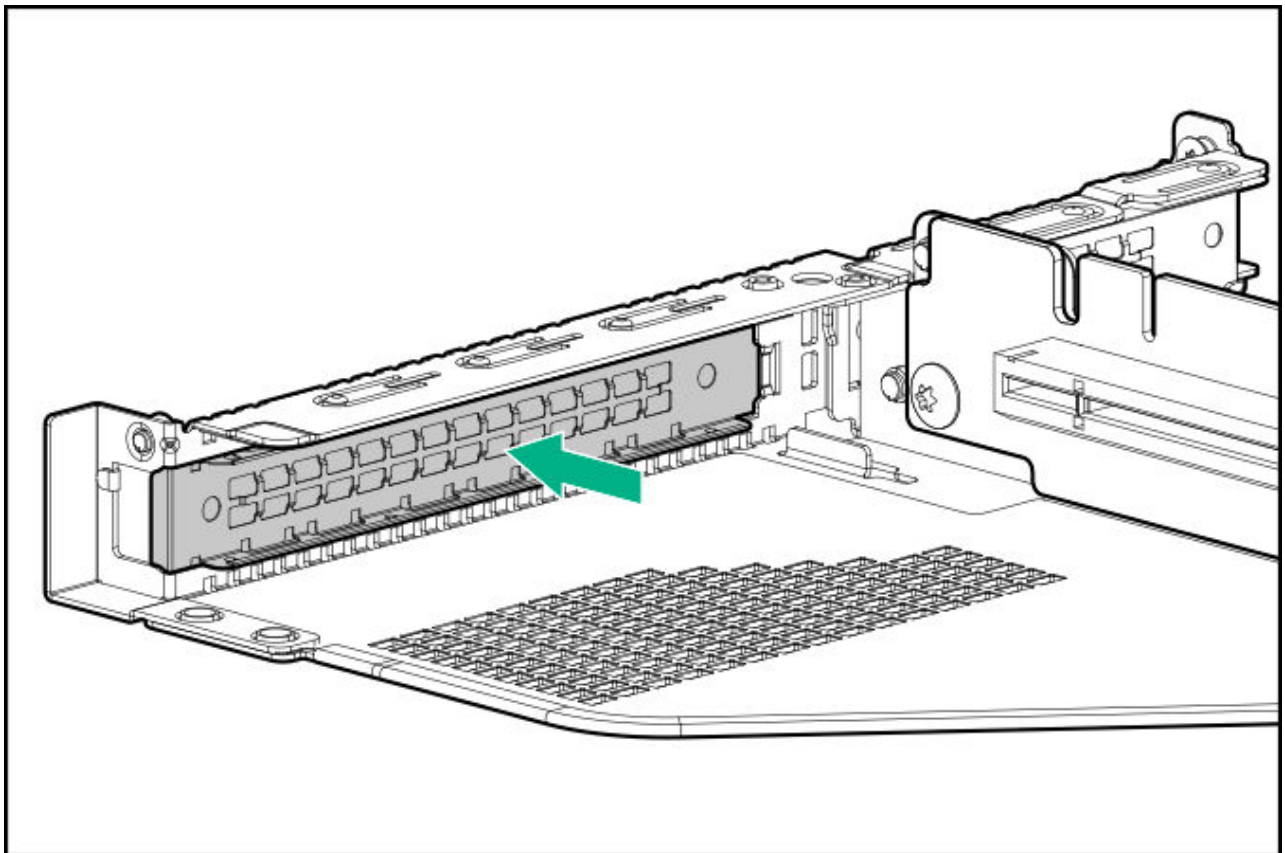
CAUTION

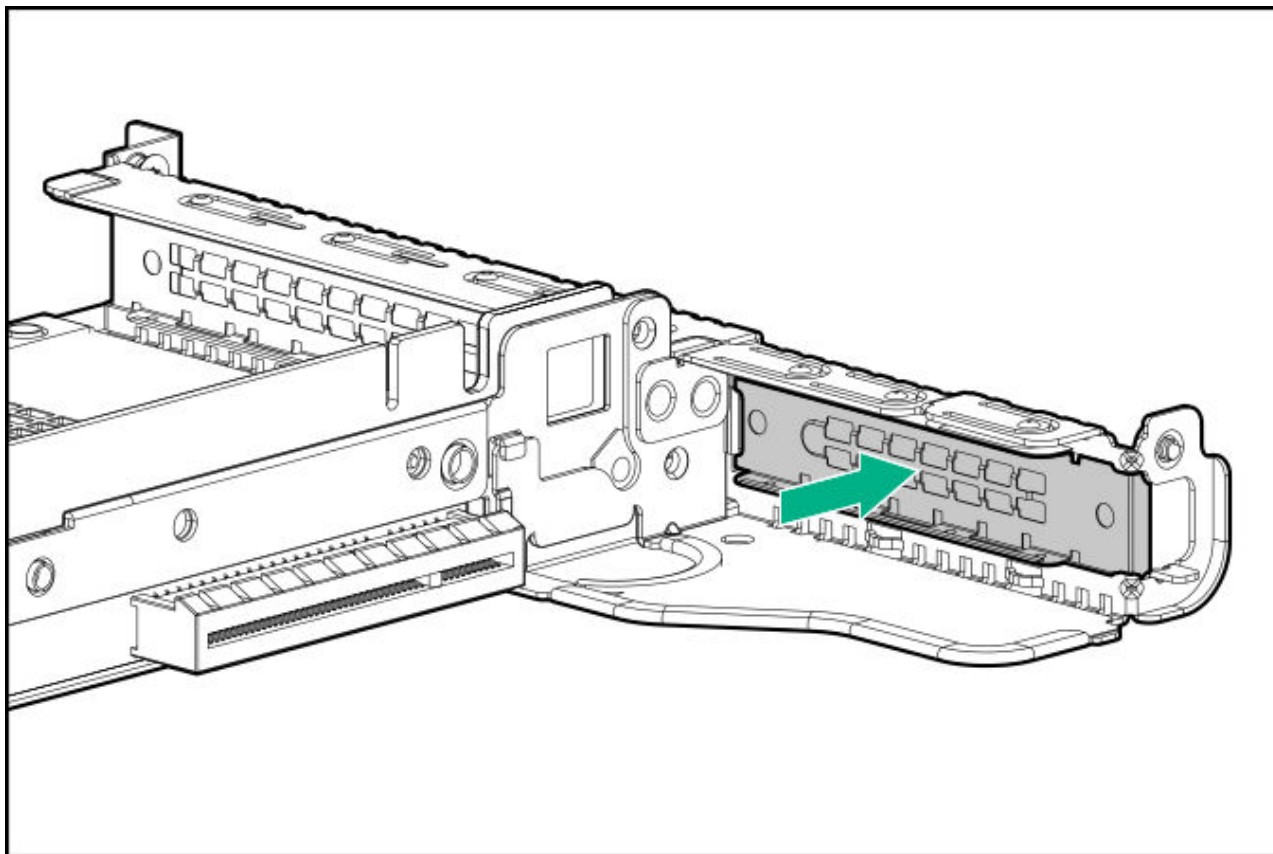
To prevent damage to the server or expansion boards, power down the server and remove all AC power cords before removing or installing the PCI riser cage.

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Remove the riser cage.
6. Remove the riser blank.

To remove blanks from the primary riser:

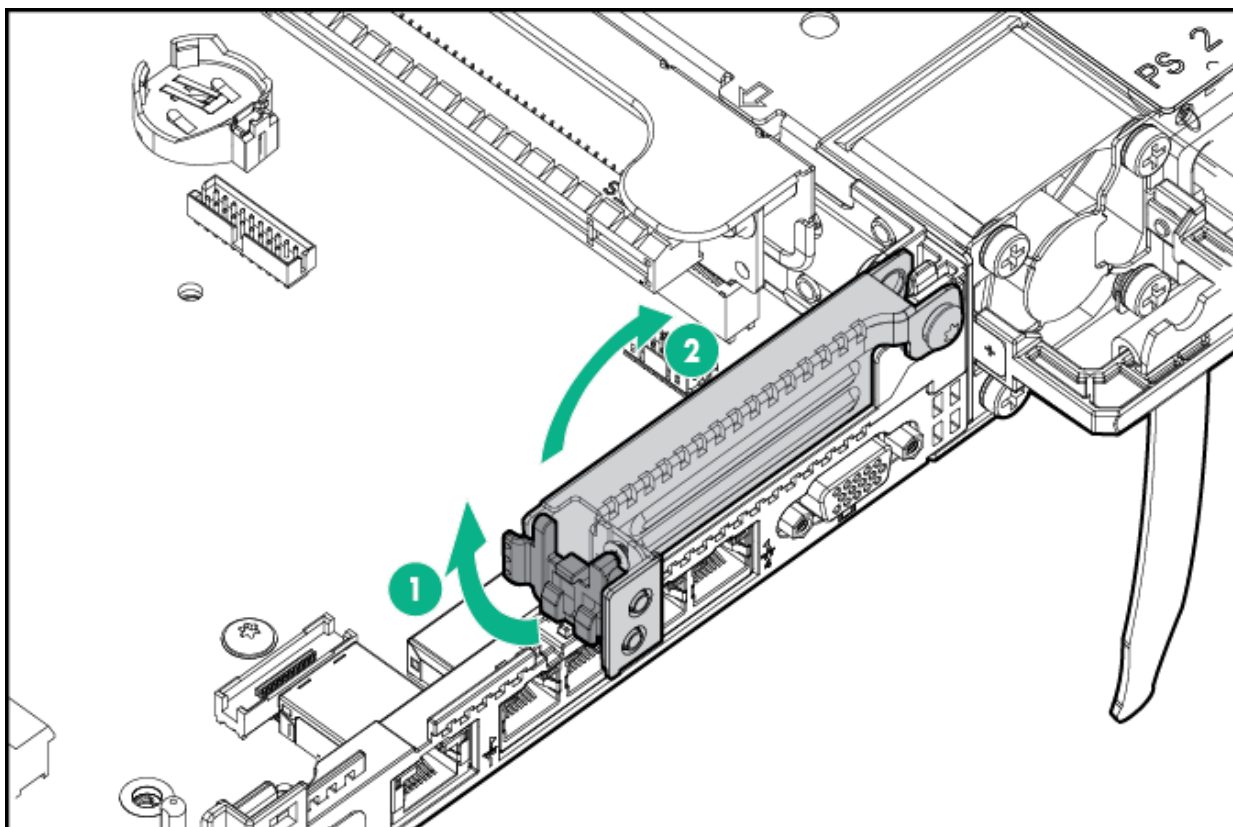




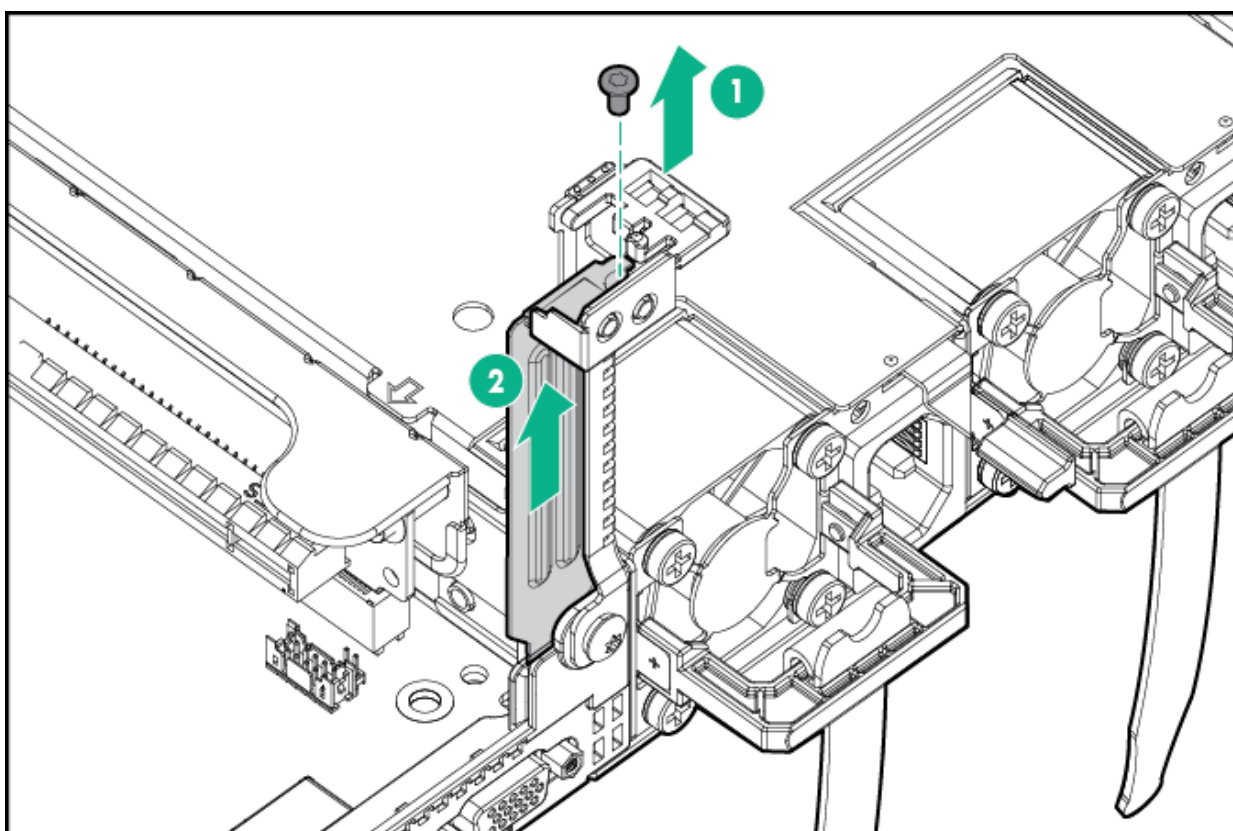
To remove blanks from the secondary riser:

- a. Lift the rear wall latch.

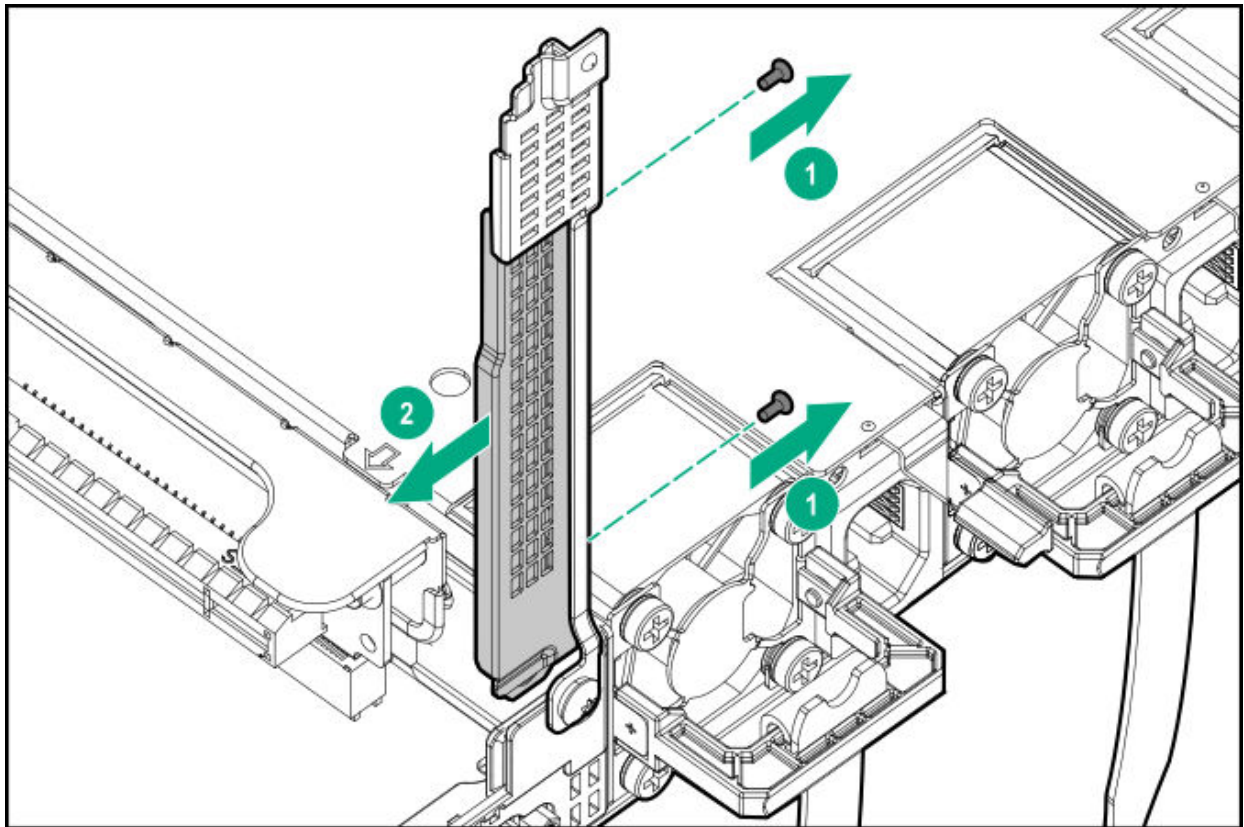
Half-height



b. Remove the riser blank.



Full-height



Results

To replace the component, reverse the removal procedure.

Removing and replacing a riser cage

About this task



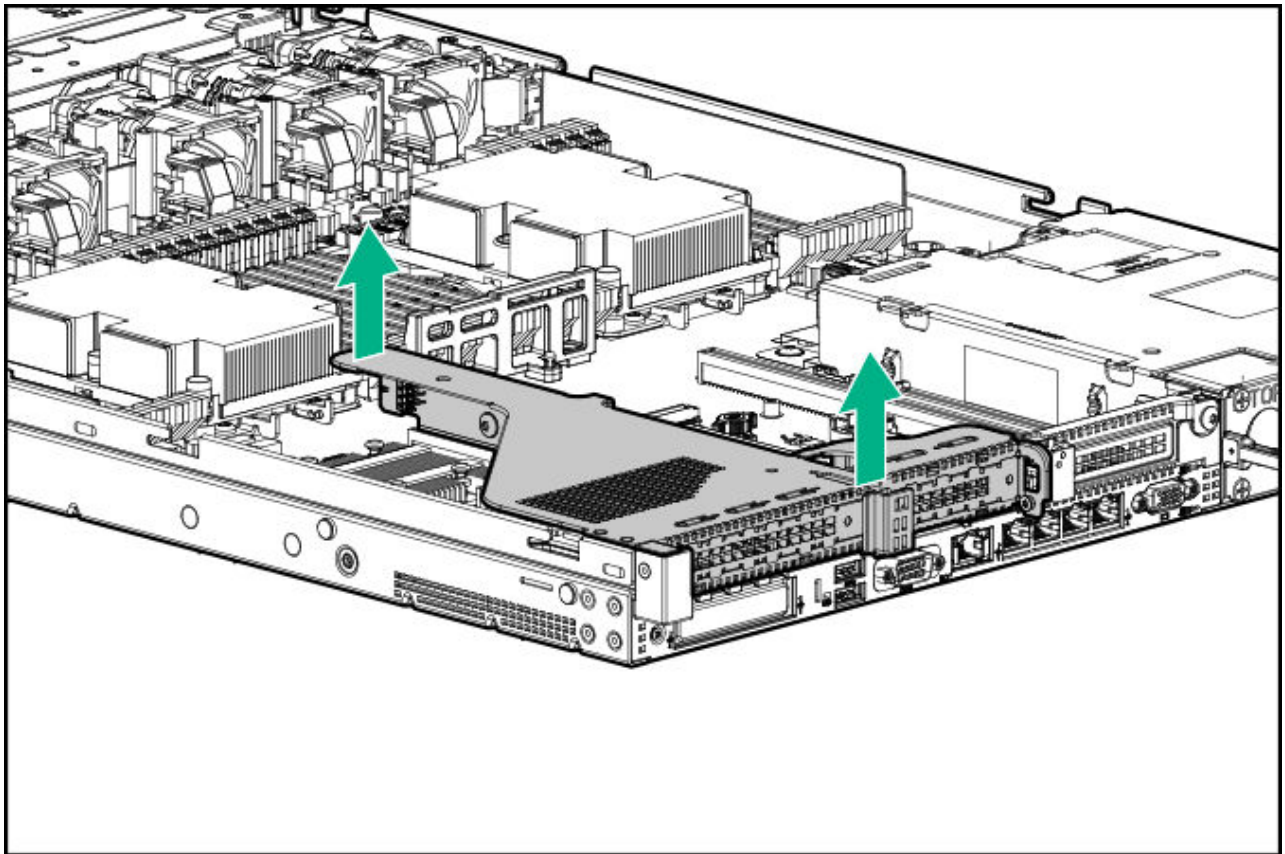
CAUTION

To prevent damage to the server or expansion boards, power down the server and remove all AC power cords before removing or installing the PCI riser cage.

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Remove the riser cage.

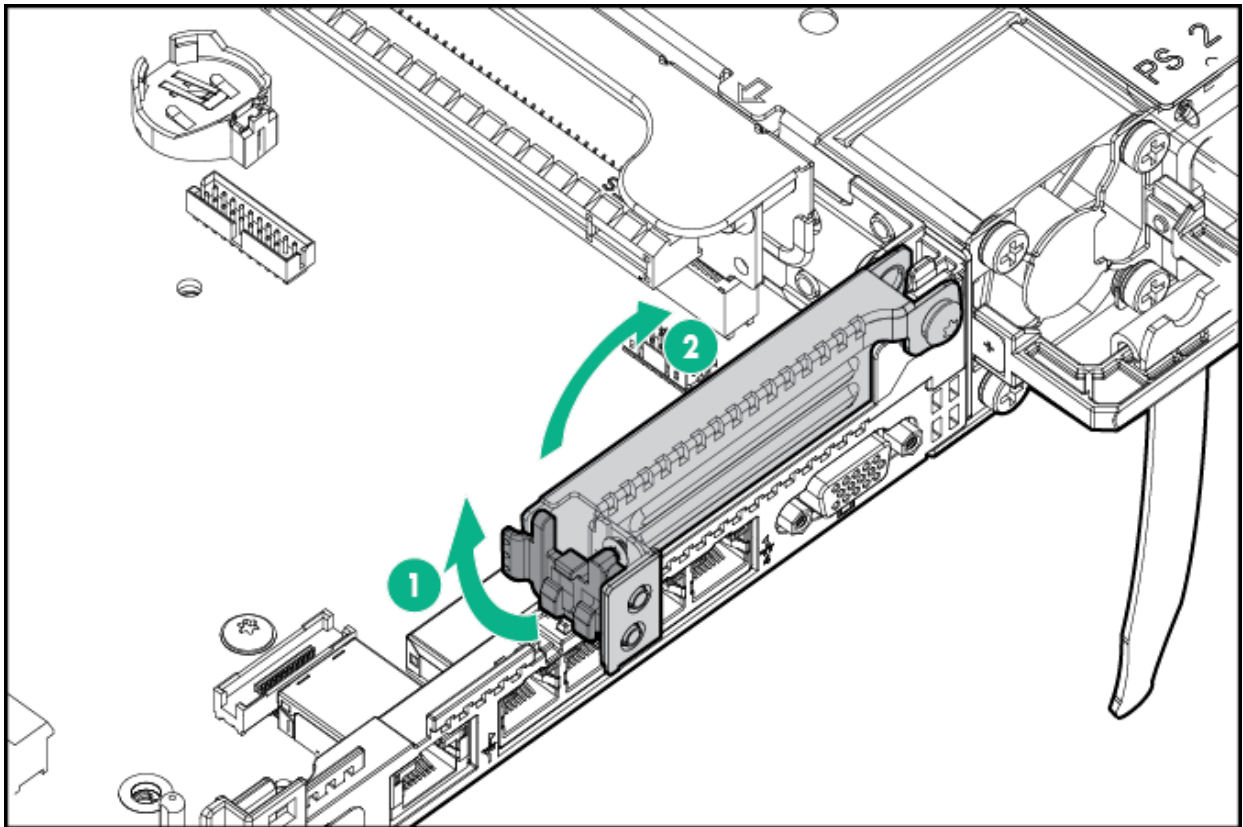
To remove the primary riser cage:



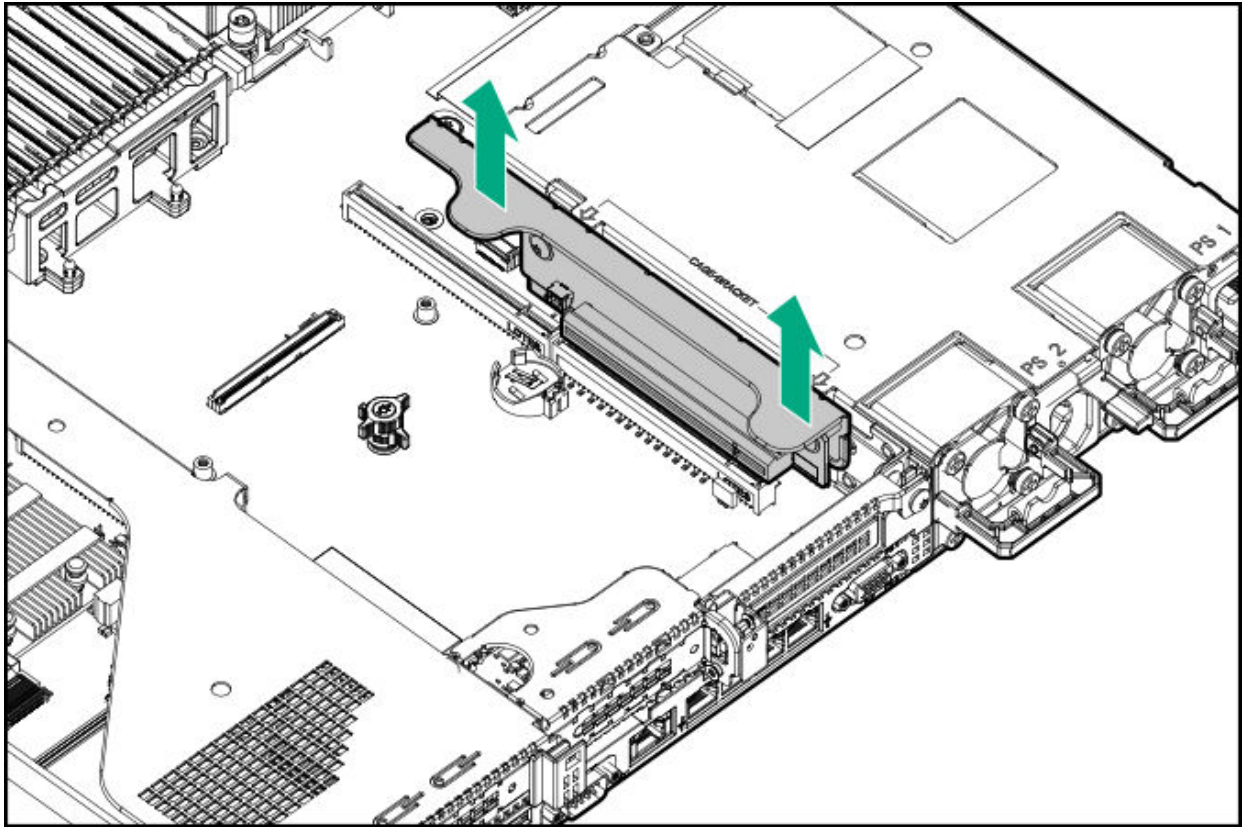
To remove the secondary riser cage:

- a. Remove the primary riser cage.

- b. Open the secondary rear wall.



- c. Remove any expansion boards.
- d. Remove the riser cage.



Results

To replace the component, reverse the removal procedure.

Removing and replacing an expansion board

About this task

Use these instructions to remove and replace expansion boards such as Smart I/O cards (such as the Pensando DSP DSC-25 2p SFP28 card), solid state NVMe/PCIe Add-In-Cards, HBAs, CNAs, InfiniBand adapters, and accelerators.



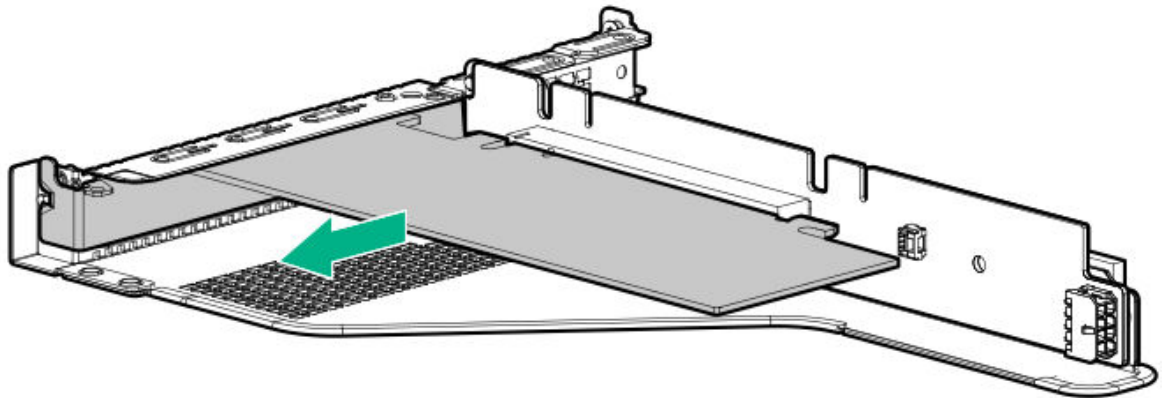
WARNING

To reduce the risk of personal injury, electric shock, or damage to the equipment, remove power from the server by removing the power cord. The front panel Power On/Standby button does not shut off system power. Portions of the power supply and some internal circuitry remain active until AC power is removed.

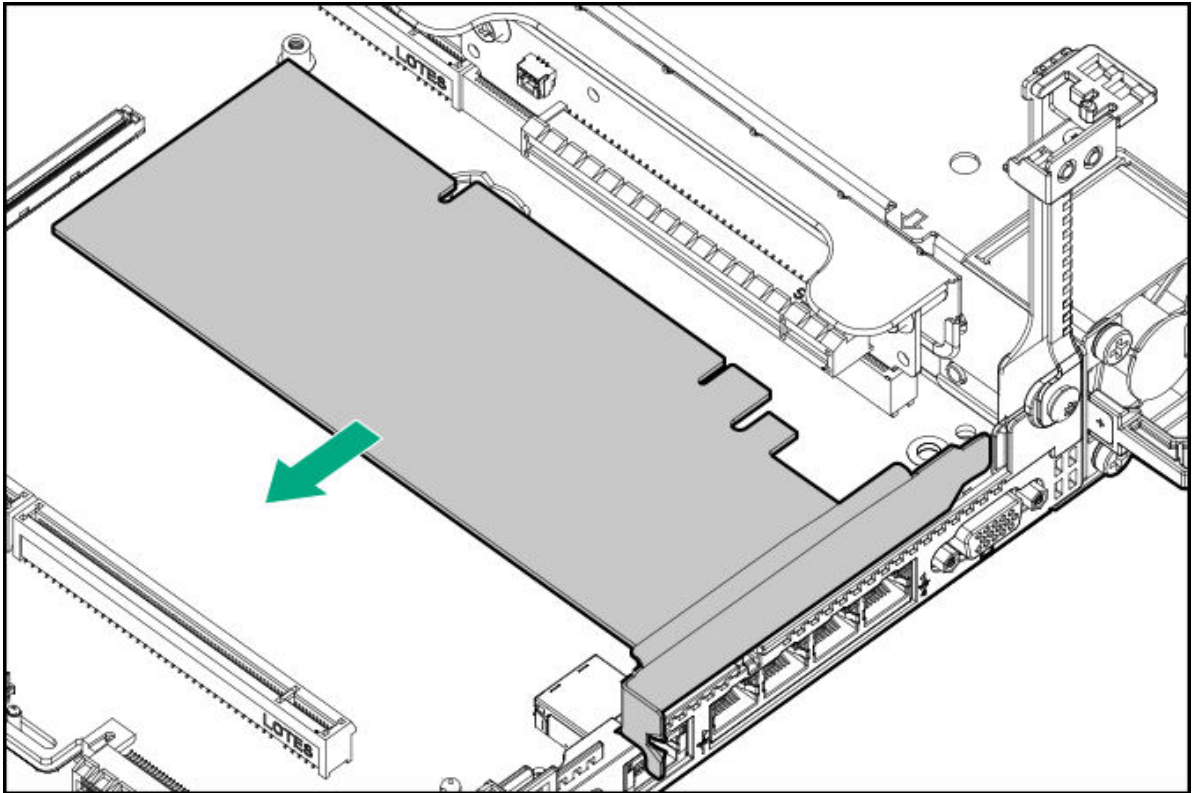
Procedure

1. Power down the server.

2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Remove the riser cage.
6. If a retaining screw is installed, remove the screw.
7. Do one of the following:
 - Remove the expansion board from the primary riser cage:
 - a. Remove the expansion board.



- Remove the expansion board from the secondary riser cage:
 - a. Lift the secondary rear wall.
 - b. Remove the expansion card.



Results

To replace the component, reverse the removal procedure.

Removing and replacing the Pensando DSP DSC-25 2p SFP28 card with the iLO Sideband ALOM Module

Subtopics

[Decommissioning the Pensando DSP DSC-25 2p SFP28 card](#)

[Removing the Pensando DSP DSC-25 2p SFP28 card with the iLO Sideband ALOM Module](#)

[Replacing the Pensando DSP DSC-25 2p SFP28 card with the iLO Sideband ALOM Module](#)

Decommissioning the Pensando DSP DSC-25 2p SFP28 card

About this task

Procedure

Before removing the DSC-25 card from the server, you must first decommission the card using the Pensando Policy and Services Manager (PSM).

For more information, see Pensando Distributed Services Card for Single-Wire Management (DSC-25-SWM) Removal/Replacement Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/support/dsc25-replacement>).

Removing the Pensando DSP DSC-25 2p SFP28 card with the iLO Sideband ALOM Module

About this task

Pensando Distributed Services Platform for HPE iLO Sideband Management ALOM Module requires that the accompanying Smart I/O card (such as the Pensando DSP DSC-25 2p SFP28 card) be installed on slot 1 of the primary PCIe riser cage.

Procedure

1. Observe the following alerts:



WARNING

To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.



CAUTION

Electrostatic discharge (ESD) can damage electronic components. Be sure that you are properly grounded (earthed) before beginning any installation procedure.



CAUTION

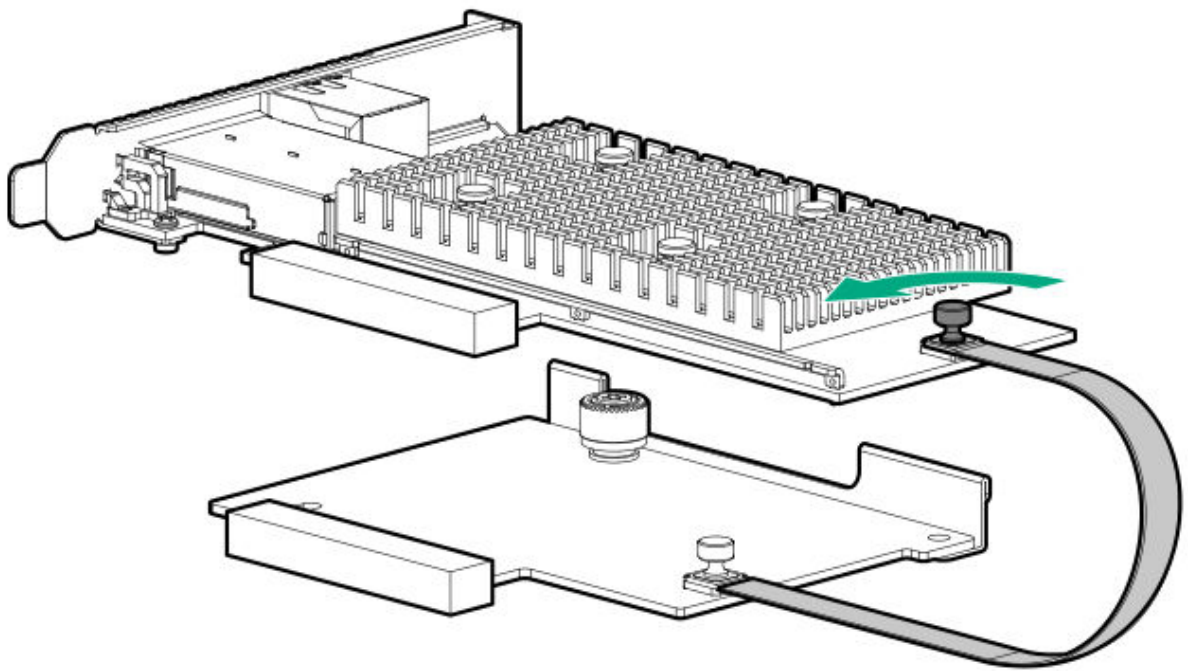
To prevent improper cooling and thermal damage, do not operate the server unless all PCI slots have either an expansion slot cover or an expansion board installed.

2. Power down the server.
3. Remove all power:

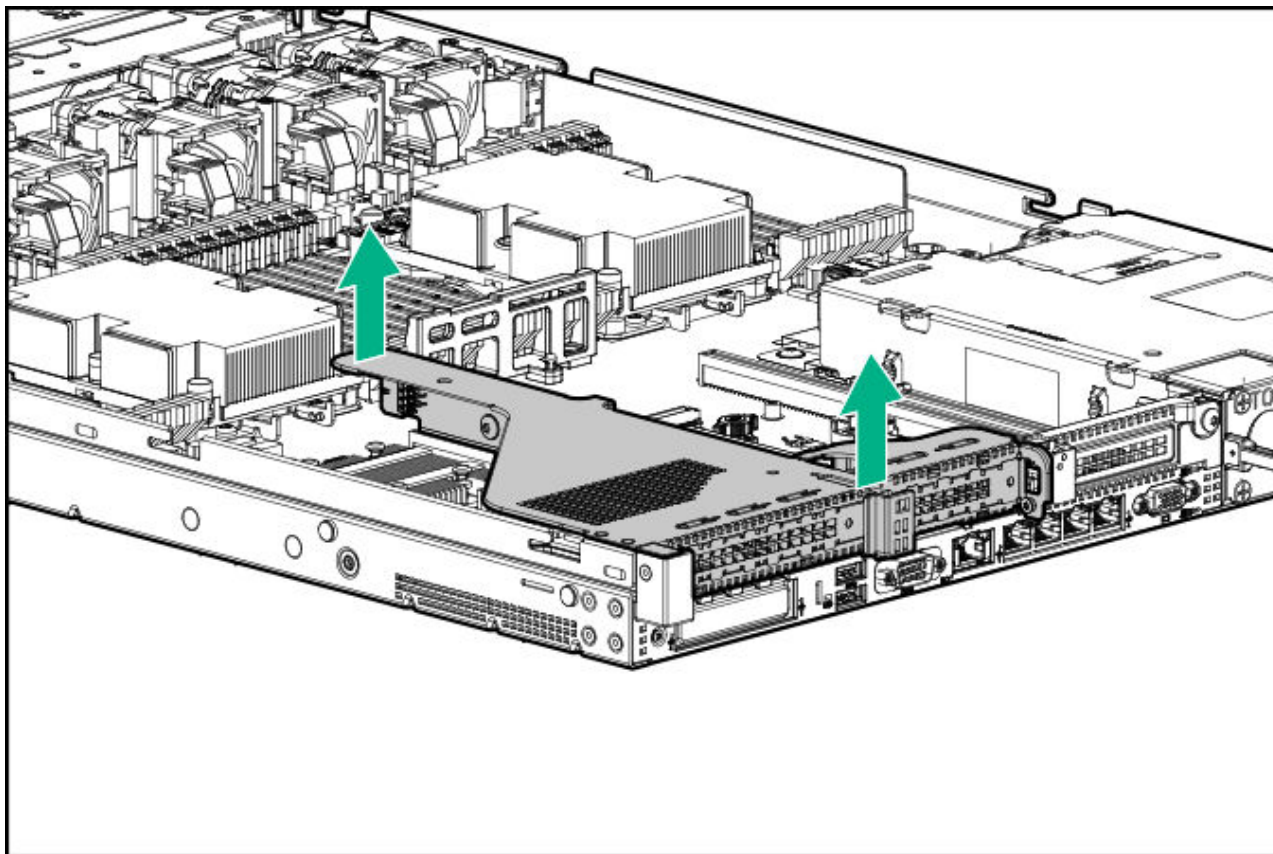
- a. Disconnect each power cord from the power source.
- b. Disconnect each power cord from the server.
4. Extend the server from the rack.
5. Remove the server from the rack.
6. Remove the access panel.
7. Disconnect the auxiliary cable from the DSC-25 card before removing the PCIe riser cage.

**NOTE**

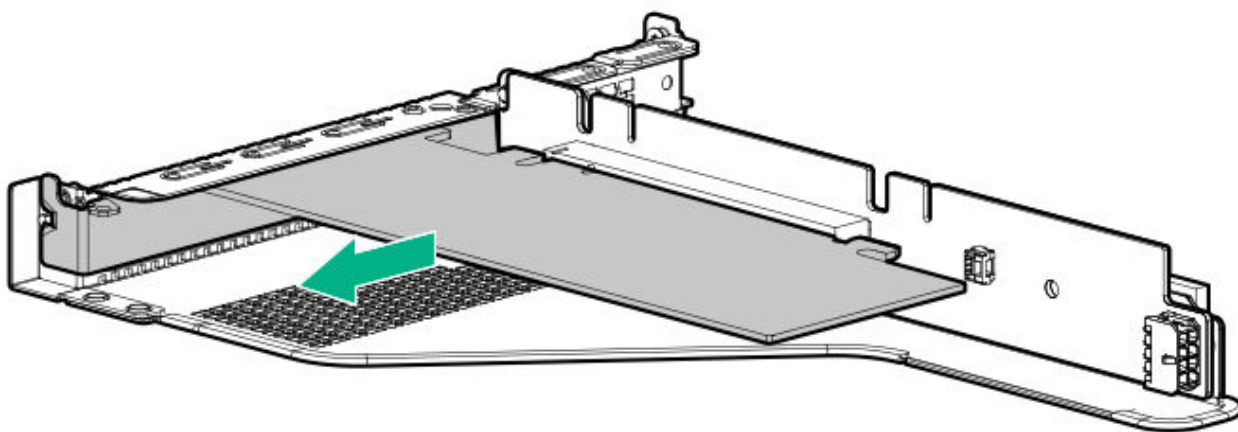
The PCIe riser cage is not shown in this illustration for clarity purposes.



8. Remove the primary PCIe riser cage.



9. Remove the DSC-25 card from the slot.

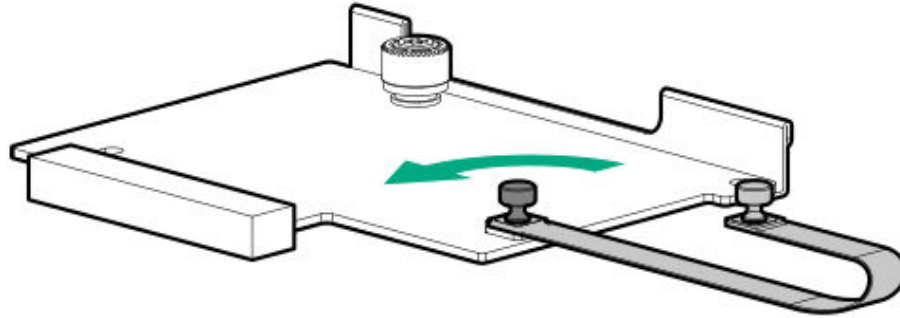


- .0. Disconnect the auxiliary cable from the ALOM module using the thumb screw.

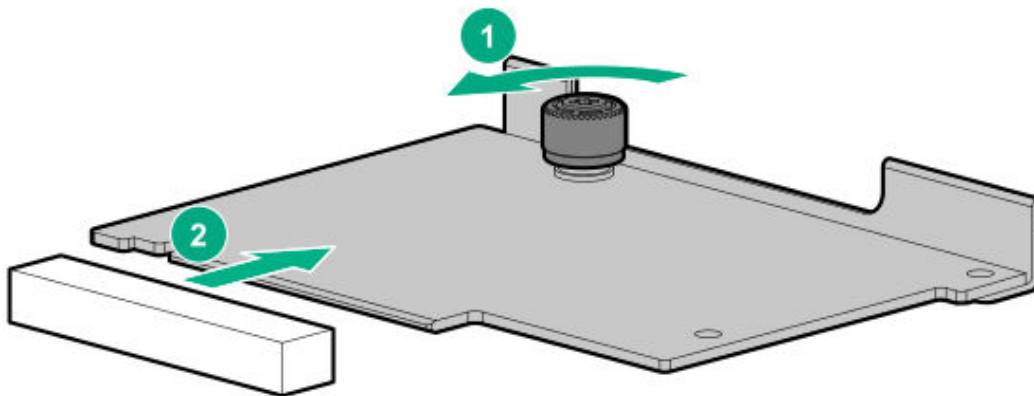


NOTE

Be sure not to twist the cable.



- .1. Remove the Pensando DSP iLO Sideband ALOM Module from the FlexibleLOM adapter connector on the system board.



Replacing the Pensando DSP DSC-25 2p SFP28 card with the iLO Sideband ALOM Module

Prerequisites

Before installing this option, be sure you have the following:

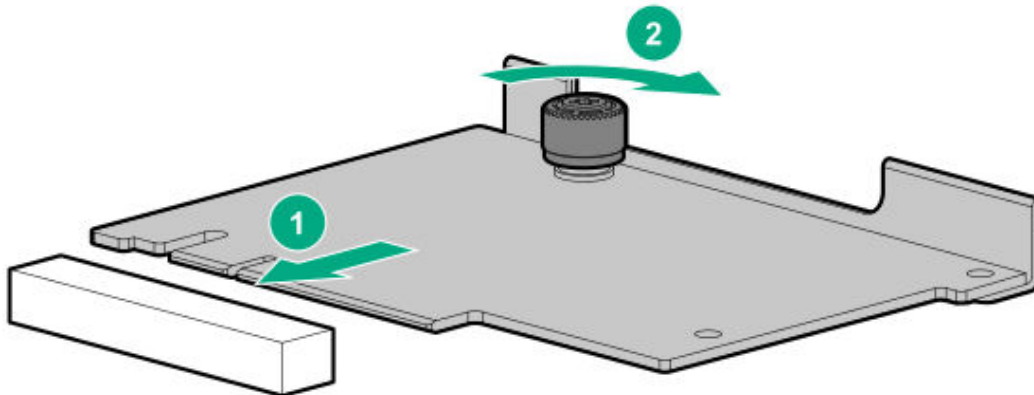
- The components included with the hardware option kit.
- T-10 Torx screwdriver.

About this task

Pensando DSP iLO Sideband ALOM Module requires that the accompanying Smart I/O card (such as the Pensando DSP DSC-25 2p SFP28 card) be installed on slot 1 of the primary PCIe riser cage.

Procedure

1. Install the Pensando DSP iLO Sideband ALOM Module into the FlexibleLOM adapter connector on the system board.



2. Remove the shipping cap from the auxiliary cable.
3. Connect the auxiliary cable to the iLO Sideband ALOM Module:
 - a. Align the connector keying posts to the ALOM module.

- b. Secure the auxiliary cable to the ALOM module using the thumbscrew.



CAUTION

Do not over torque the screw.



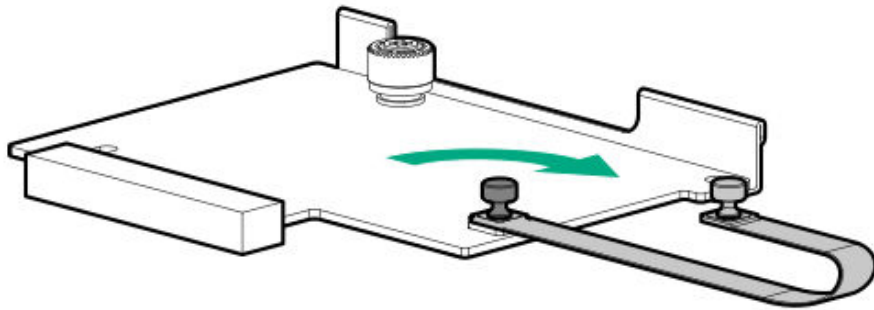
CAUTION

Do not use a screwdriver to tighten the screw.

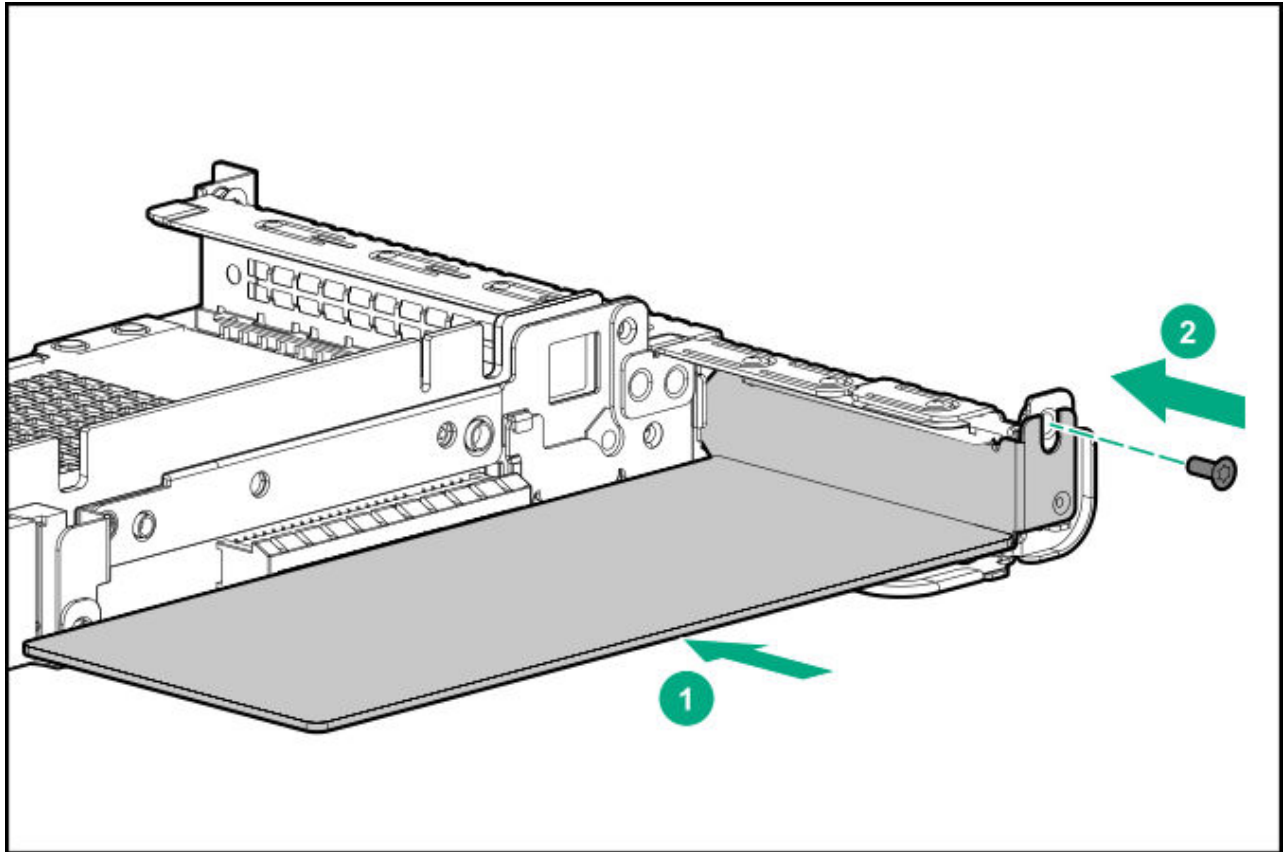


CAUTION

Do not twist the cable service loop during installation.



4. Install the DSC-25 card into the PCIe slot.

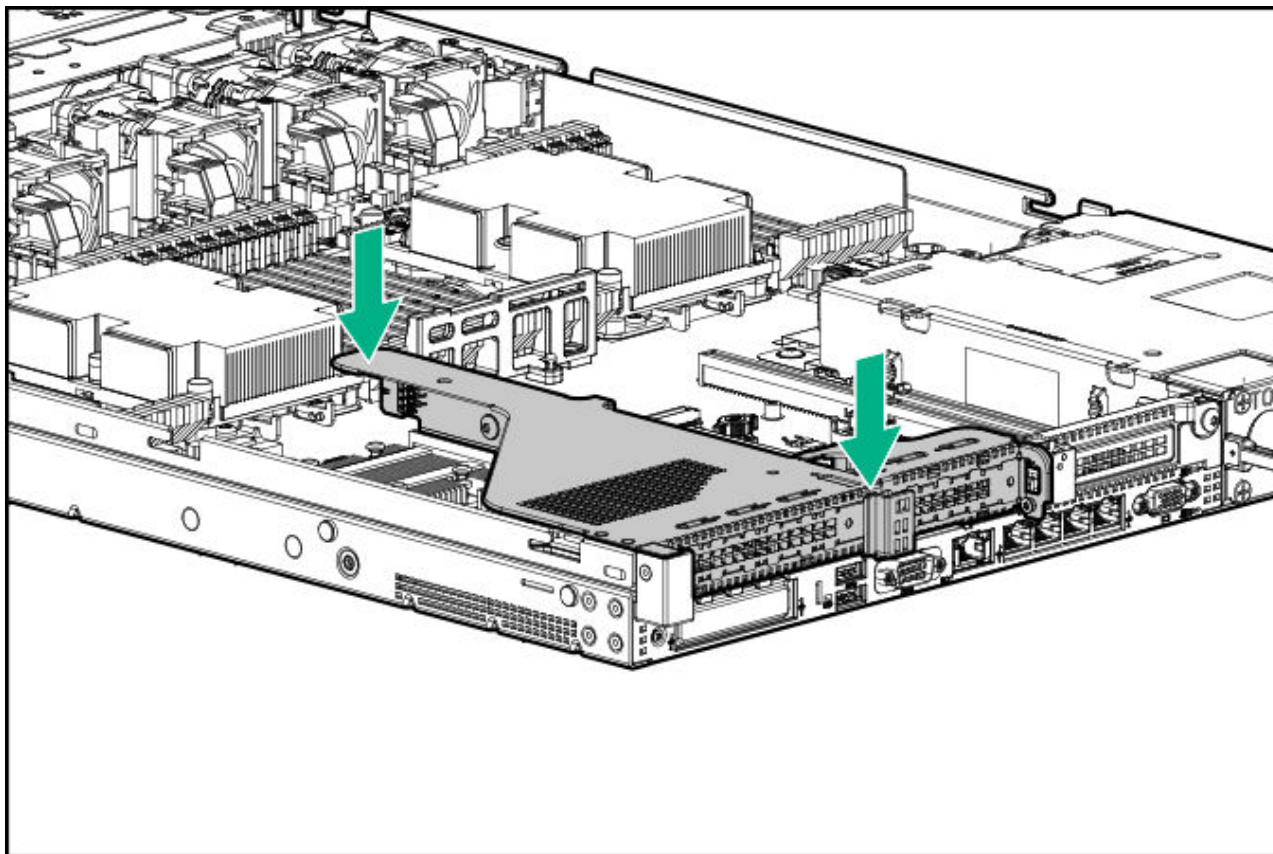


5. Install the PCIe riser cage into the server.



CAUTION

Do not twist the cable service loop during installation.



6. Remove the shipping cap from the auxiliary cable.
7. Connect the auxiliary cable to the DSC-25 card:
 - a. Align the connector keying posts to the DSC-25 card.

- b. Secure the auxiliary cable to the DSC-25 card using the thumbscrew.



CAUTION

Do not over torque the screw.



CAUTION

Do not use a screwdriver to tighten the screw.



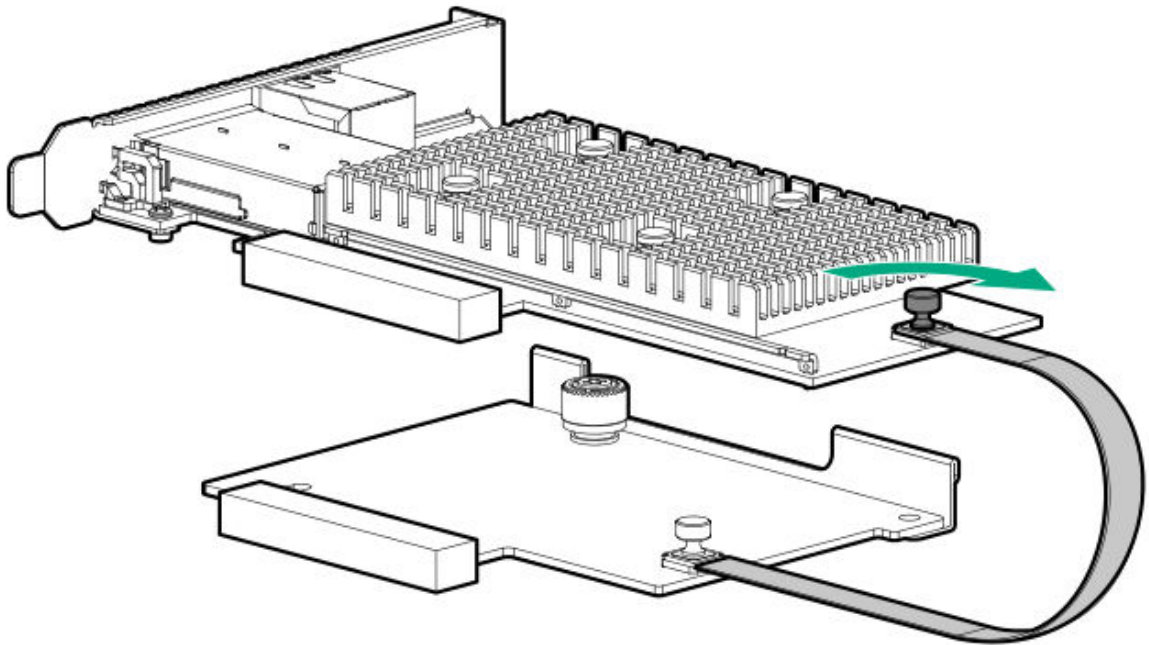
CAUTION

Do not twist the cable service loop during installation.



NOTE

The PCIe riser cage is not shown in this illustration for clarity purposes.



8. Install the access panel.
9. Slide the server into the rack.
10. Connect each power cord to the server.

- .1. Connect each power cord to the power source.
- .2. Power up the server.
- .3. Admit the new DSC-25 card to the Pensando Policy and Services Manager (PSM).

For more information, see Pensando Distributed Services Card for Single-Wire Management (DSC-25-SWM) Removal/Replacement Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/support/dsc25-replacement>).

Removing and replacing the HPE NS204i-p NVMe OS Boot Device option

Subtopics

[Removing and replacing the boot device](#)

[Removing and replacing a boot device drive](#)

Removing and replacing the boot device

Prerequisites

Before installing this option, be sure you have the following:

- The components included with the hardware option kit.
- T-10 Torx screwdriver.

Procedure

1. Observe the following alerts:



CAUTION

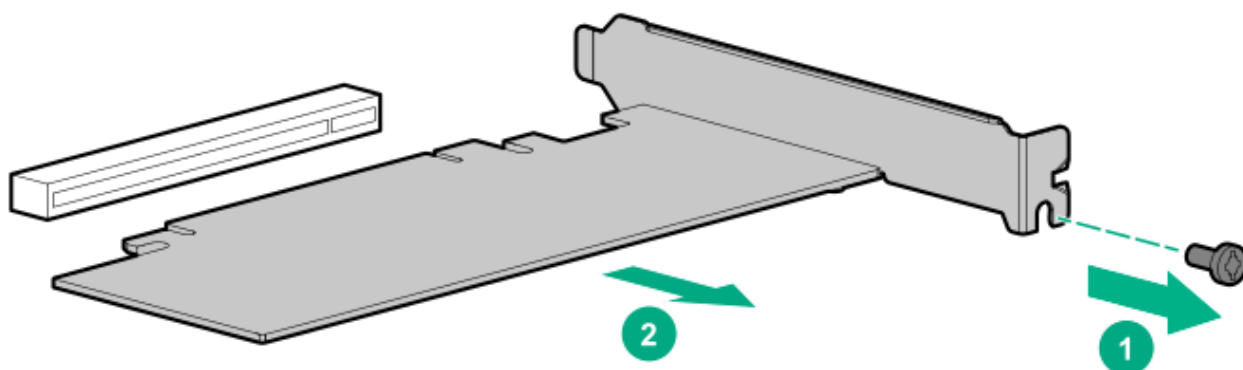
The boot device and the drives installed on the boot device are not hot-pluggable. To remove the boot device, or a drive from the boot device, you must first power down the server.



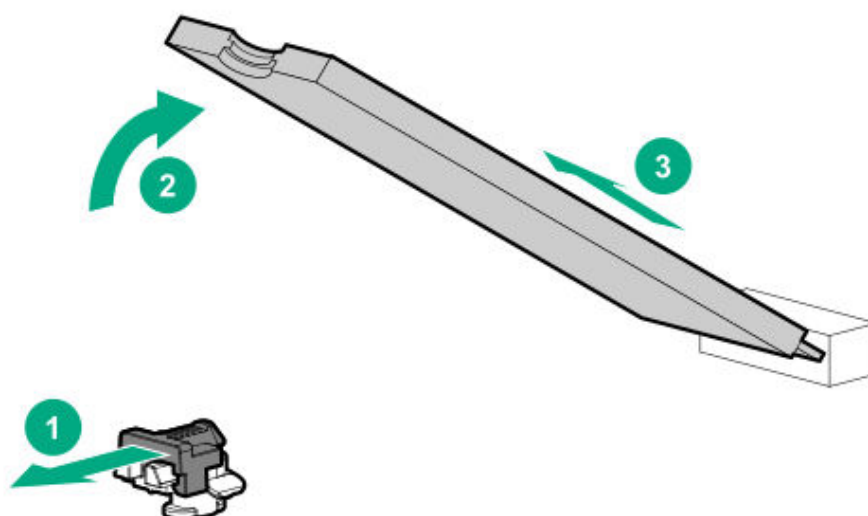
WARNING

To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.

2. Back up all server data.
3. Power down the server.
4. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
5. Do one of the following:
 - Extend the server from the rack.
 - Remove the server from the rack.
6. Place the server on a flat, level work surface.
7. Remove the access panel.
8. Remove the PCIe riser cage, if necessary.
9. Remove the boot device.



10. Remove the drives from the boot device.



Retain these drives for installation onto the replacement boot device.

Results

To replace the component, reverse the removal procedure.

Removing and replacing a boot device drive

Prerequisites

Review the [boot device drive bay LEDs](#) to determine the failed drive.

About this task

The boot device supports two physical drive sizes:

- 22110 model (110mm)
- 22080 model (80mm)

Two drives of the same physical size, or one of each size (80mm and 110mm), can be installed at the same time. Depending on the replacement drive model, you might be required to relocate the M.2 drive retaining latches on the boot device.

Procedure

1. Observe the following alerts:



CAUTION

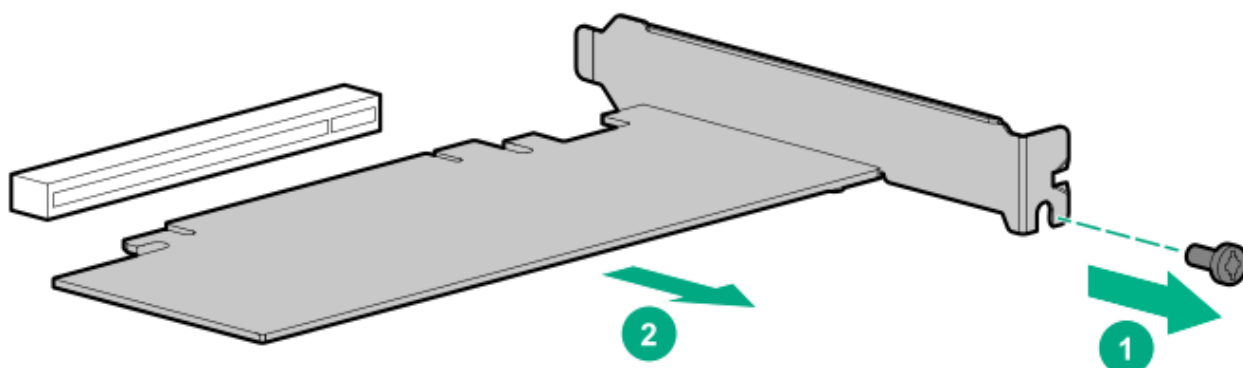
The boot device and the drives installed on the boot device are not hot-pluggable. To remove the boot device, or a drive from the boot device, you must first power down the server.



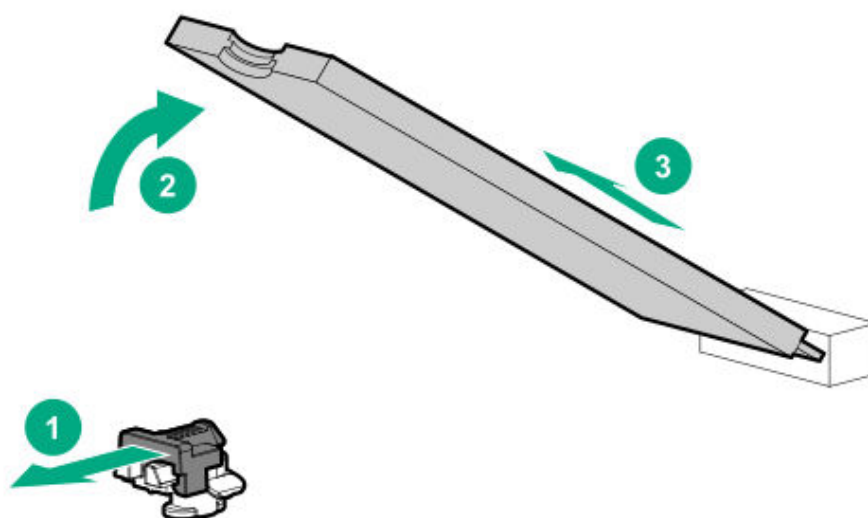
WARNING

To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.

2. Back up all server data.
3. Power down the server.
4. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
5. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
6. Place the server on a flat, level work surface.
7. Remove the access panel.
8. Remove the riser cage.
9. Remove the boot device.



.0. Remove the failed drive from the boot device.



Results

To replace the component, reverse the removal procedure.

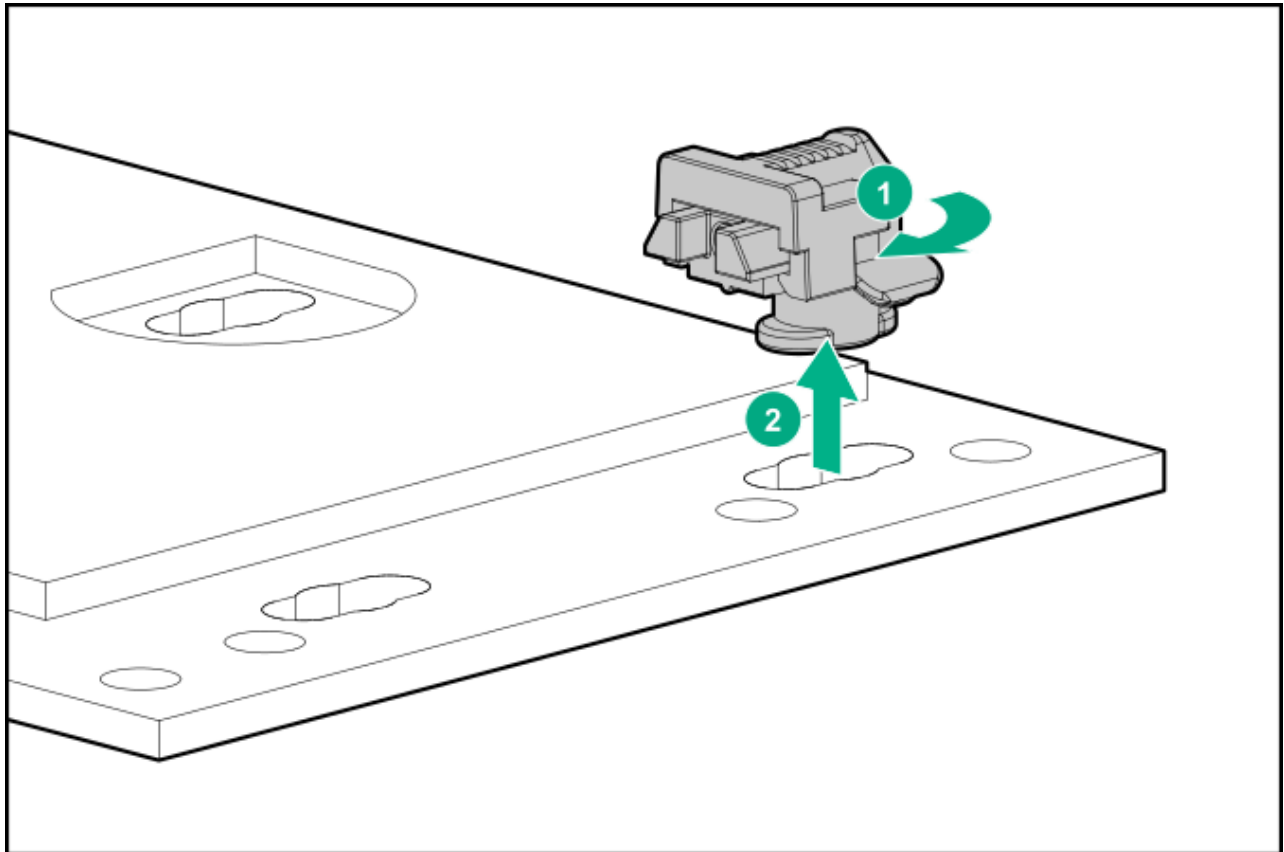
Subtopics

Relocating the M.2 drive retaining latches

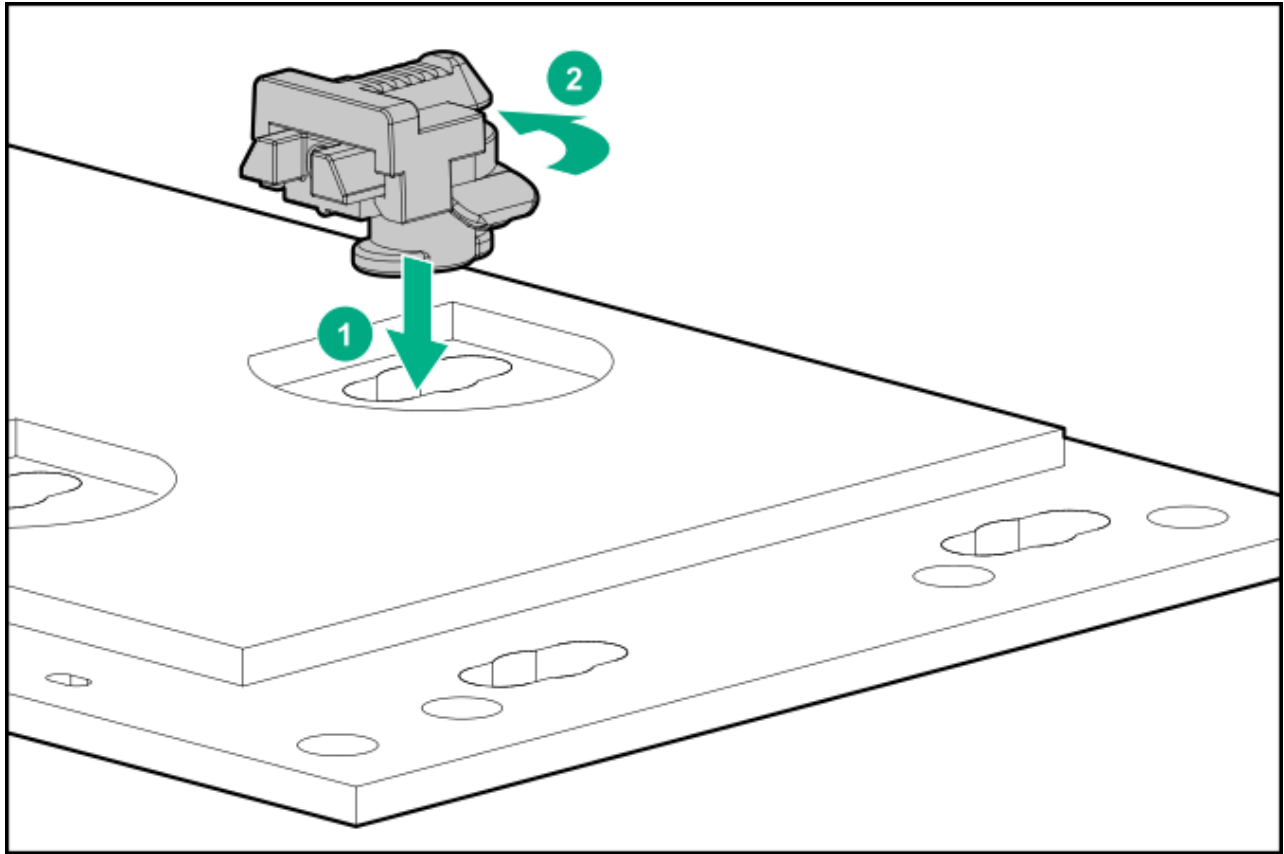
Relocating the M.2 drive retaining latches

Procedure

1. If your spare kit includes an 80 mm replacement drive, remove the retaining latch from its current location.



2. Install the retaining latch at the 80 mm drive location.



Removing and replacing a 940QSFP56 x16 adapter and auxiliary card

Procedure

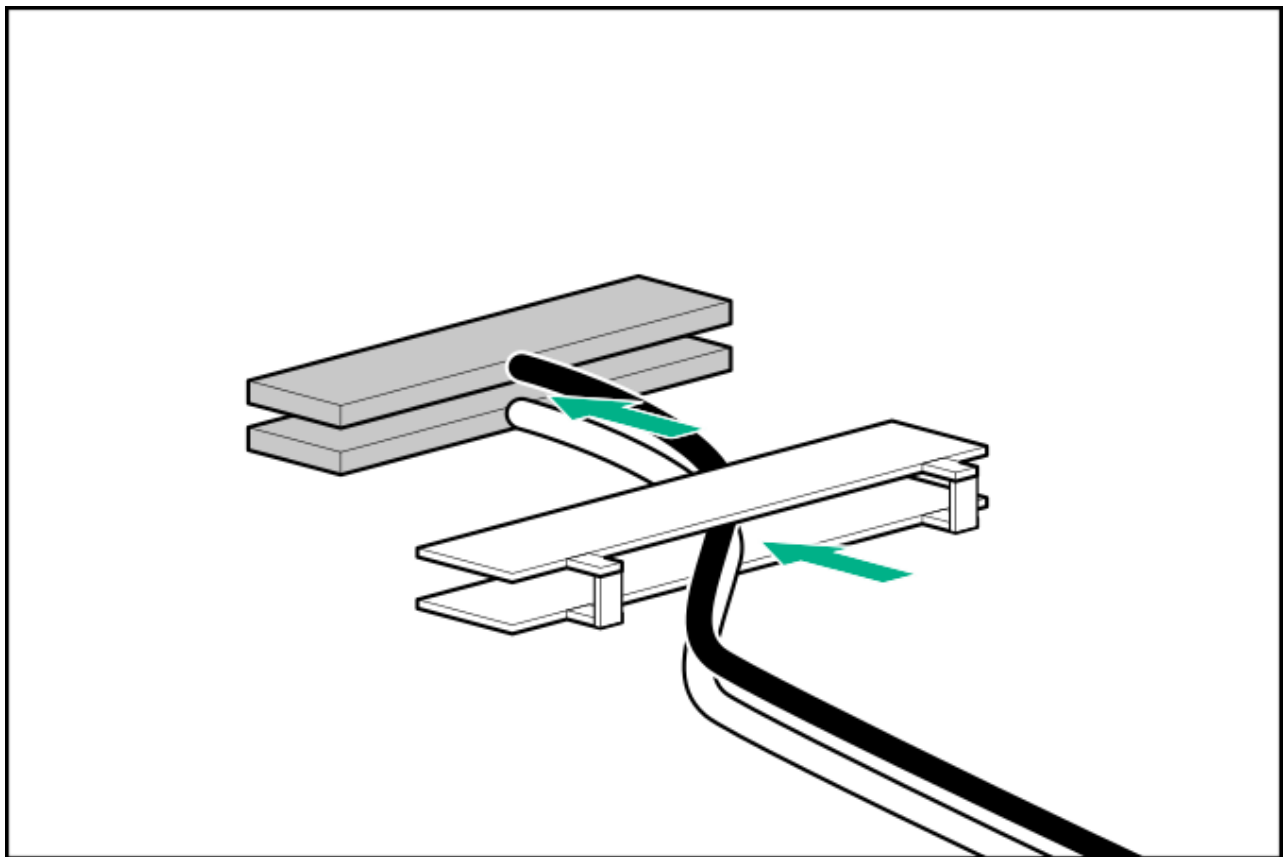
1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.

5. Remove the riser cage ([Removing and replacing a riser cage](#)).
6. Remove the auxiliary card from slot 3 ([Removing and replacing an expansion board](#)).
7. Disconnect the cables to the auxiliary card:
 - a. Open the retention clip and slide it away from the connectors.
 - b. Gently pull up the top corner of the cable latch door.
 - c. Disconnect the card cables.
8. Remove 940QSFP56 x16 adapter from slot 2 ([Removing and replacing an expansion board](#)).
9. Disconnect the cables to the 940QSFP56 x16 adapter.

Results

To replace the component:

1. Thread the adapter end of the auxiliary card cables through one of the retention clips provided with the installation kit.



2. Open the cable latch door, connect the auxiliary card cables to the 940QSFP56 x16 adapter ports so that the golden side of the cable connector is on the top, and then close the cable latch door.

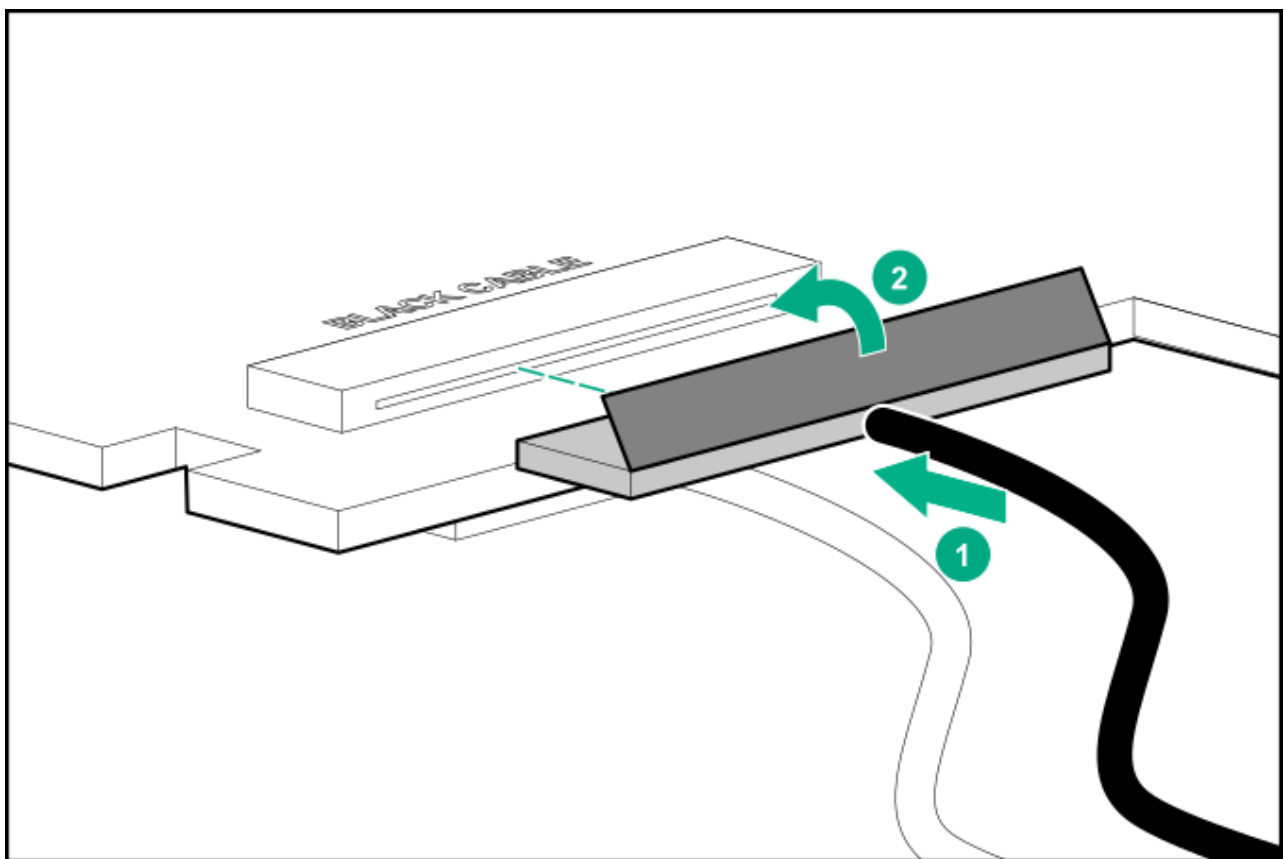
The white and black cables from the auxiliary card connect to the expansion board ports labeled WHITE CABLE and BLACK CABLE, respectively.

The cable latch door must be open when connecting the cables.

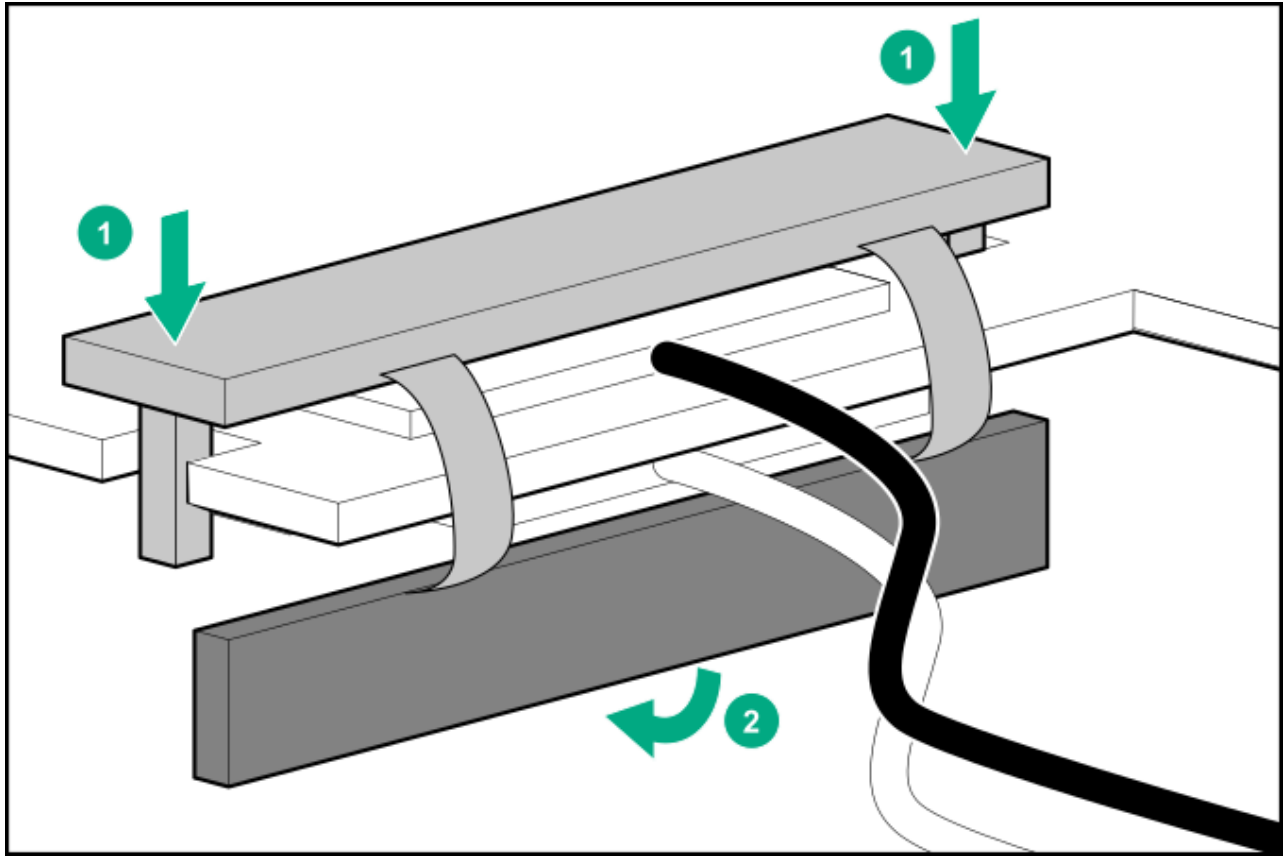


CAUTION

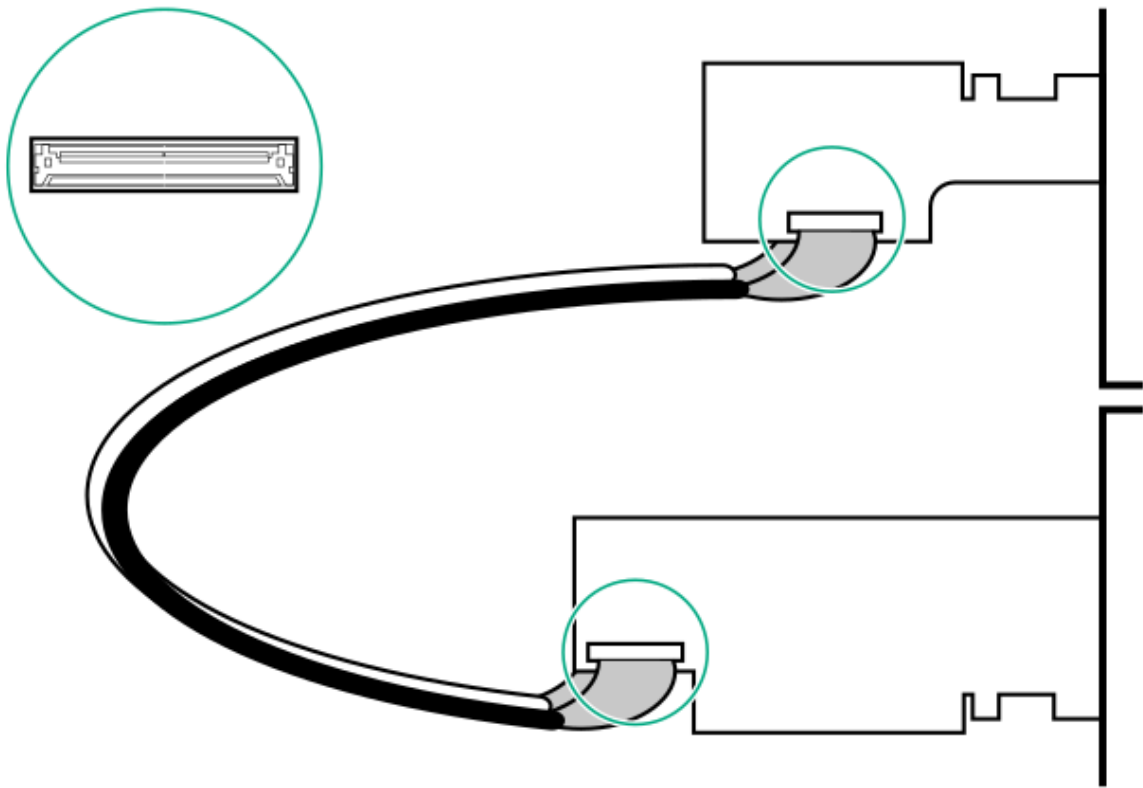
The connector pins are fragile and easily damaged. To avoid damaging the connector pins, do not use excessive force when connecting the cables.



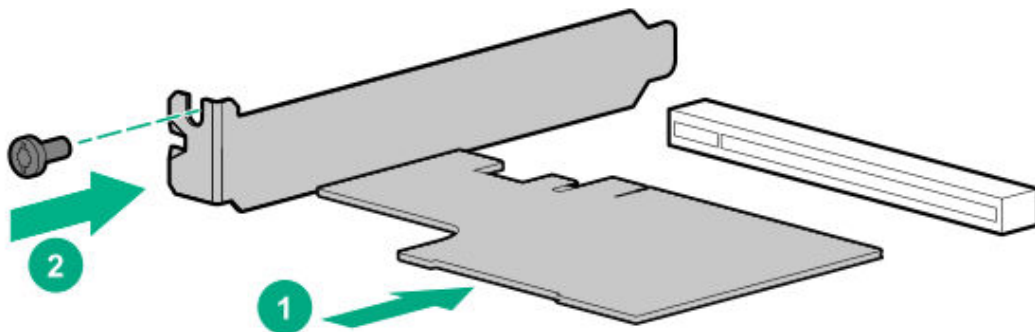
3. Install the retention clip.



4. Thread the auxiliary cables through the second retention clip.
5. Connect the auxiliary cables to the auxiliary card.
6. Confirm that the orientation of the cables is correct.

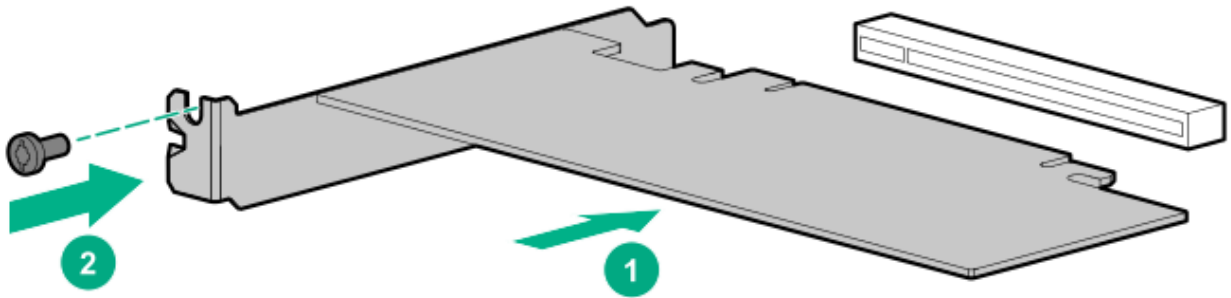


7. Install the auxiliary card into expansion slot 3.



8. Install the retaining screw, if removed.

9. Install the 940QSFP56 x16 adapter into expansion slot 2.



- .0. Install the retaining screw, if removed.
- .1. Install the riser cage.
- .2. Install the access panel.
- .3. Connect each power cord to the server.
- .4. Connect each power cord to the power source.
- .5. Power up the server.

Removing and replacing a primary riser card

About this task



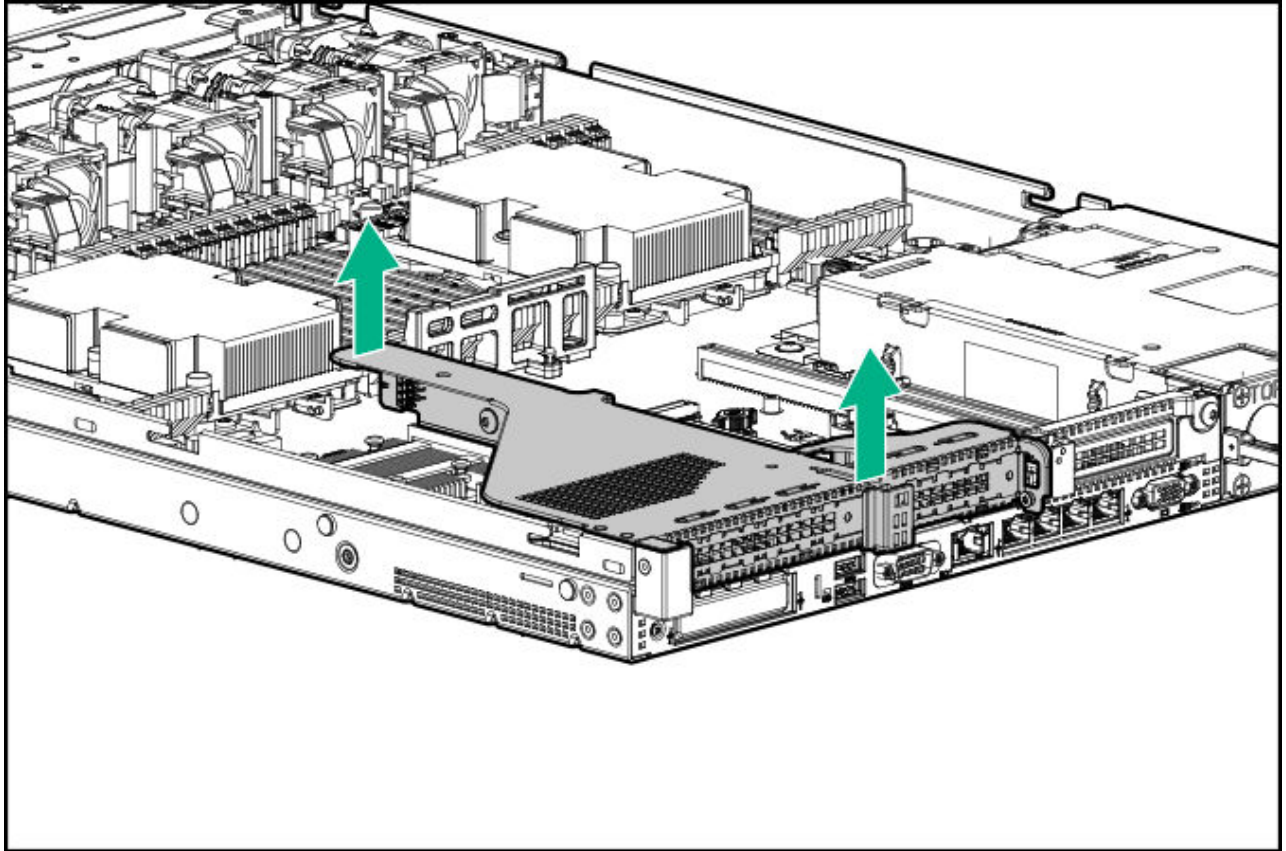
CAUTION

To prevent damage to the server or expansion boards, power down the server and remove all AC power cords before removing or installing the PCI riser cage.

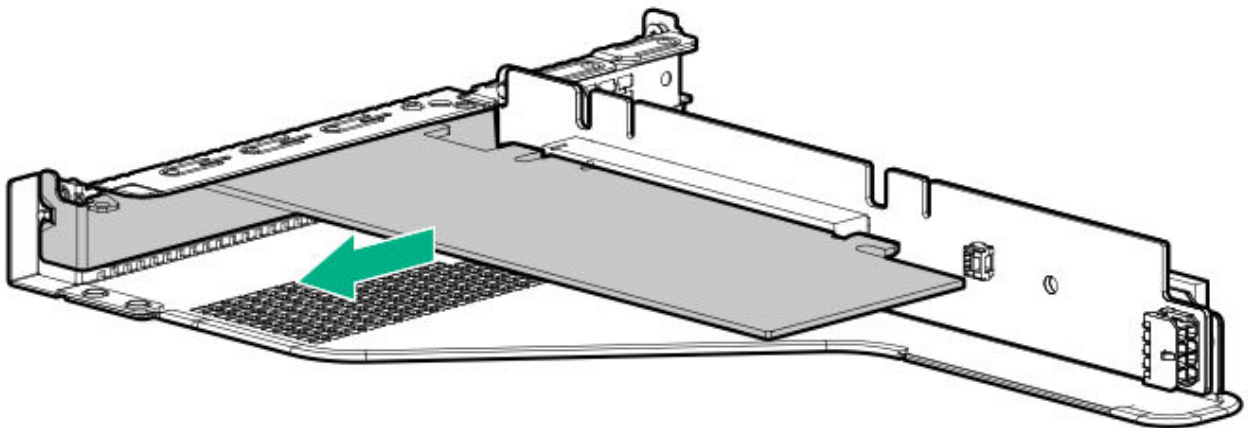
Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.

- b. Remove the server from the rack.
- 4. Remove the access panel.
- 5. Remove the riser cage.

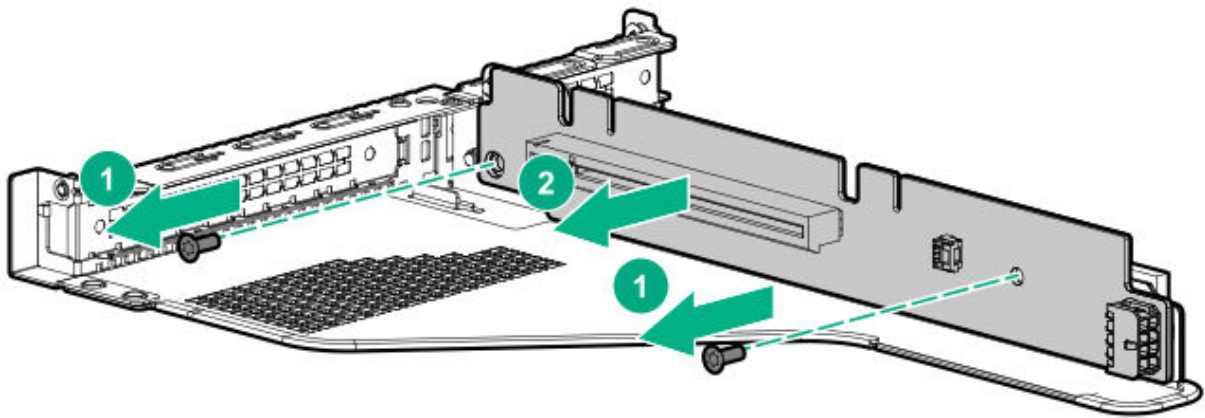


- 6. Remove all expansion boards.



7. Remove the riser.

Secondary risers are spared with the riser cage.



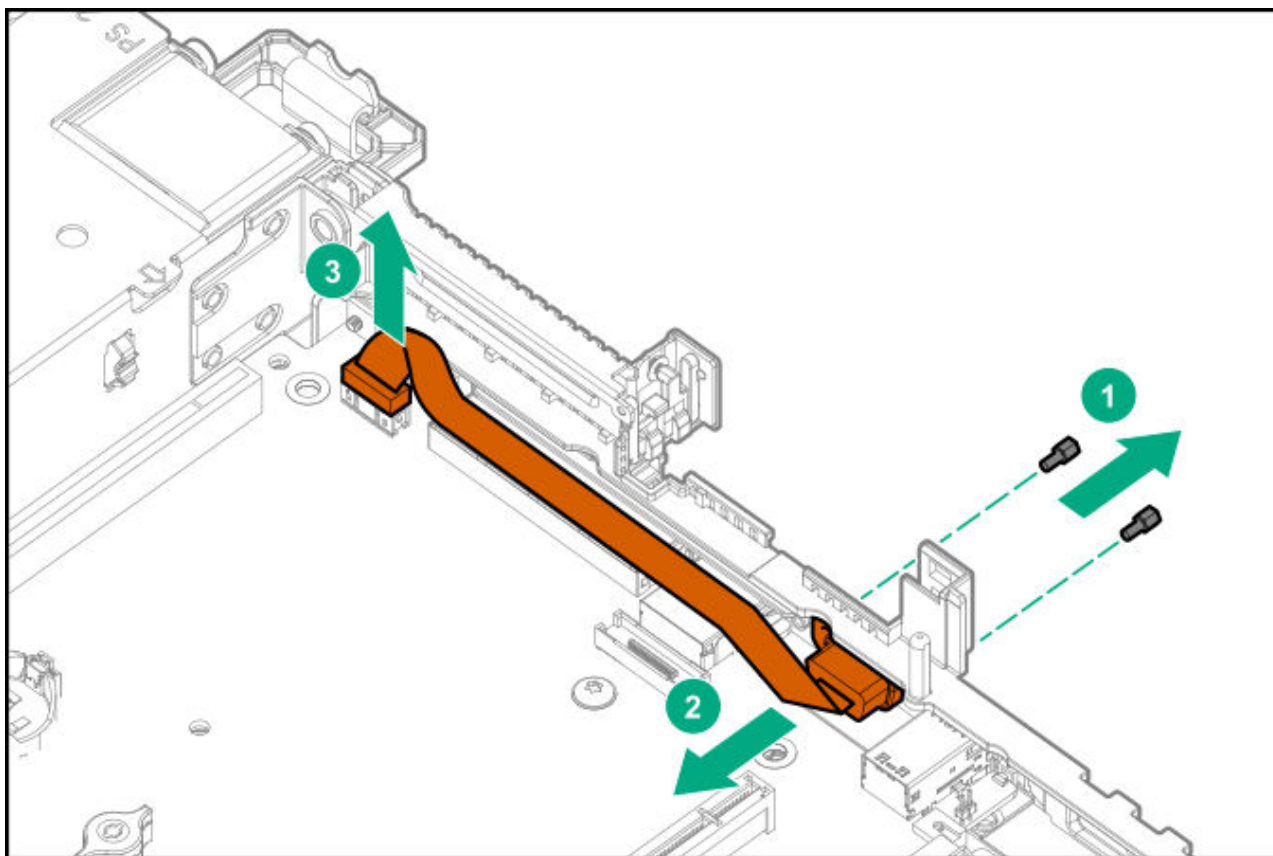
Results

To replace the component, reverse the removal procedure.

Removing and replacing the serial port

Procedure

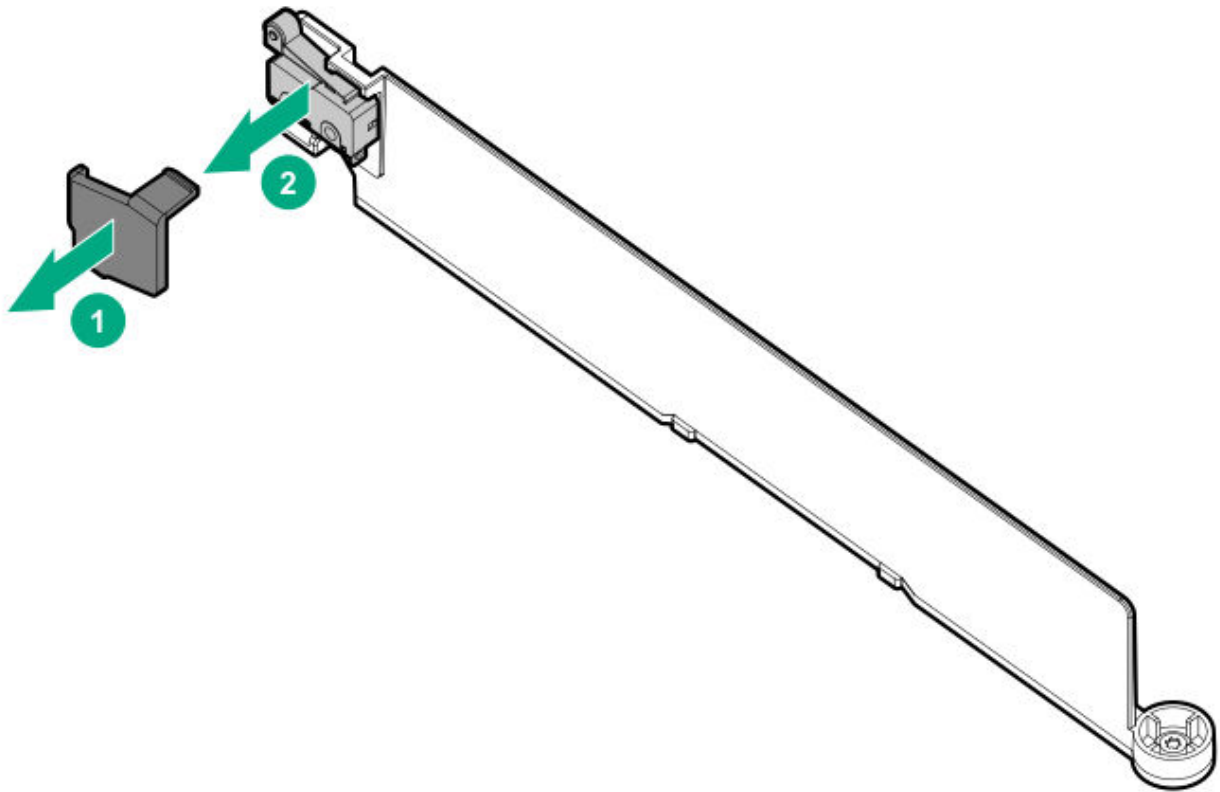
1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Remove all riser cages.
6. Remove the switch and then disconnect the cable.



Removing and replacing the intrusion detection switch

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Disconnect the cable and remove the switch.



Results

To replace the component, reverse the removal procedure.

Removing and replacing a GPU in the primary riser

Procedure

1. Observe the following warning:

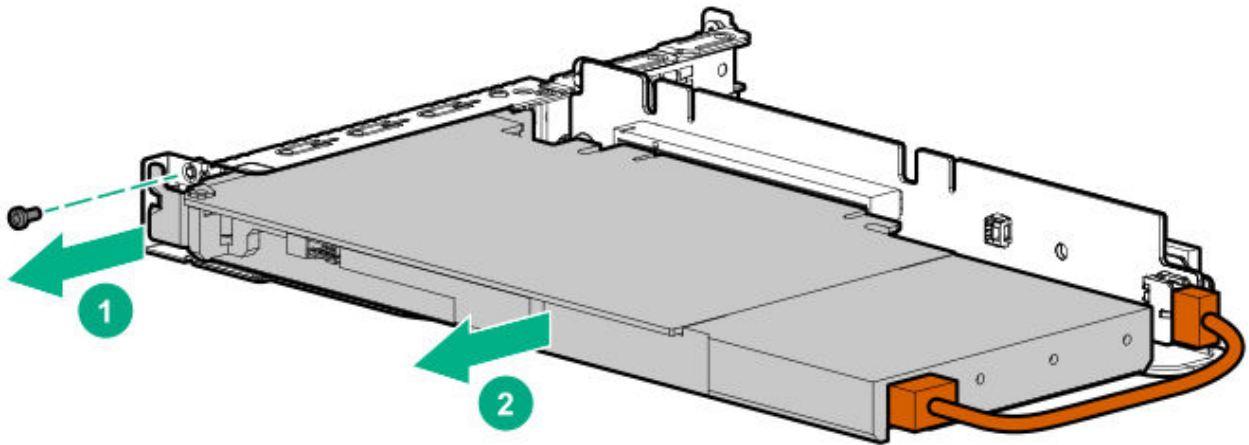


WARNING

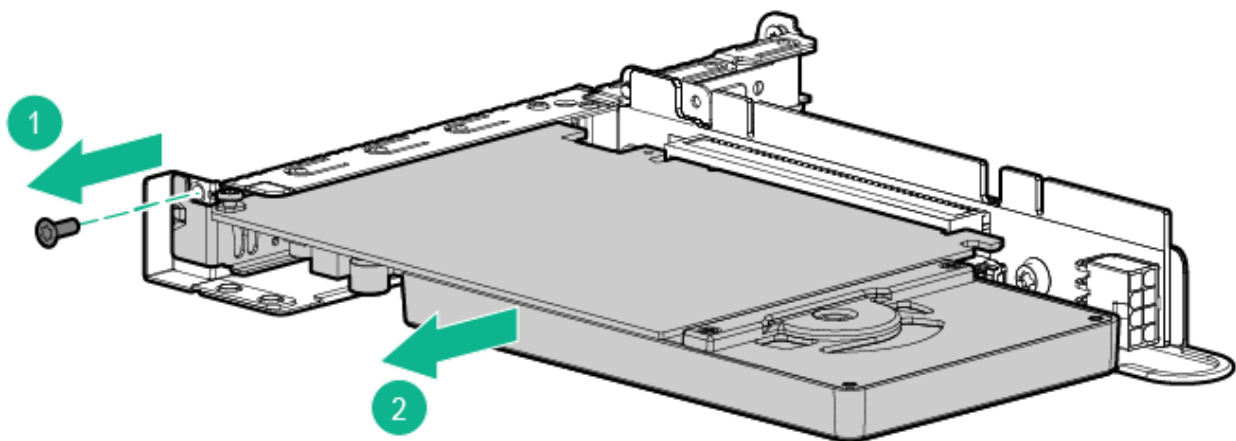
To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.

2. Power down the server.
3. Remove all power:

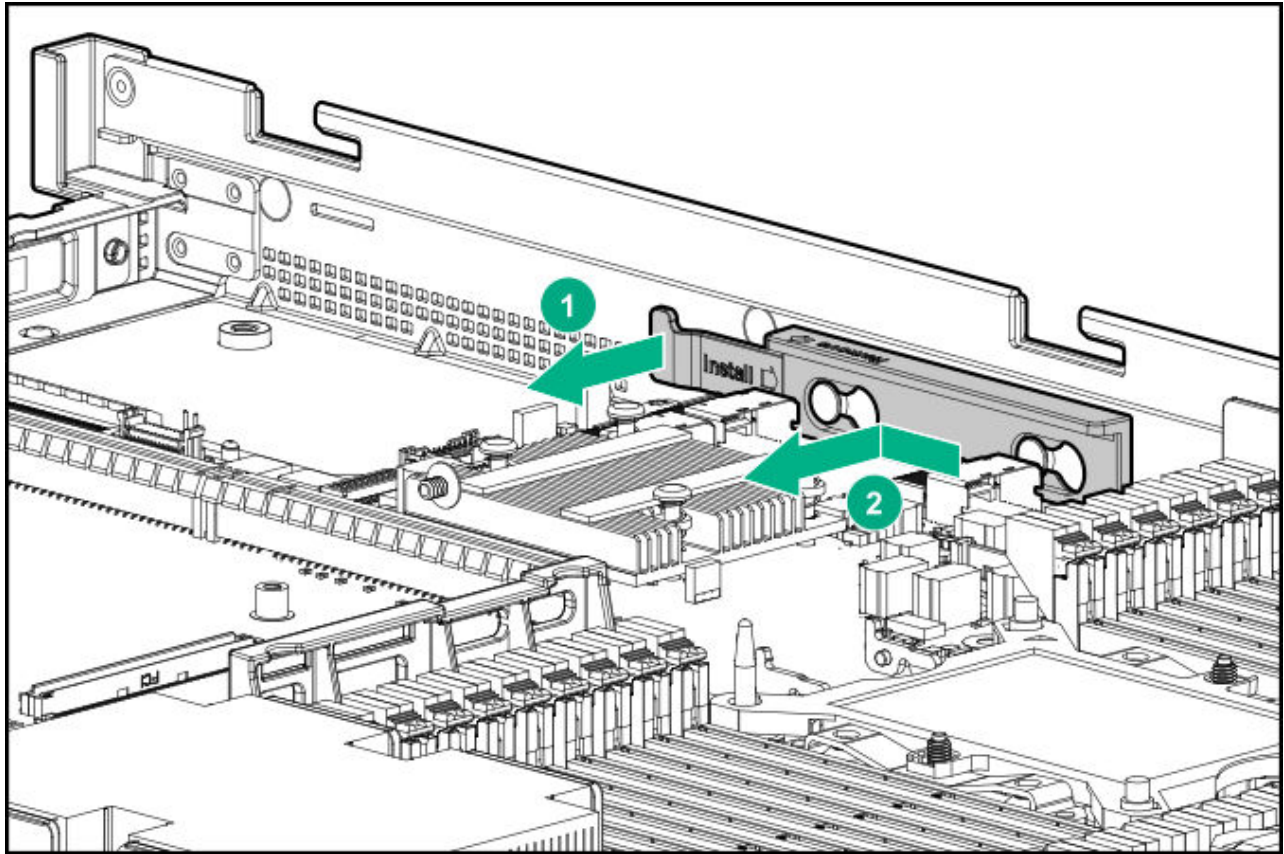
- a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
4. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
5. Remove the access panel.
6. Remove the riser cage.
7. If needed, disconnect the GPU power cable and then remove the GPU from the riser.



If there is no power cable, remove the GPU:



8. If replacing the system board, remove the GPU support bracket.



Results

To replace the component, reverse the removal procedure.

Removing and replacing a GPU in the secondary riser

About this task

This procedure applies to 3/4 length GPUs. Some GPUs might not have a power cable or a support bracket. Be sure to reuse the power cable or bracket from the existing GPU, if needed.

Procedure

1. Observe the following warning:

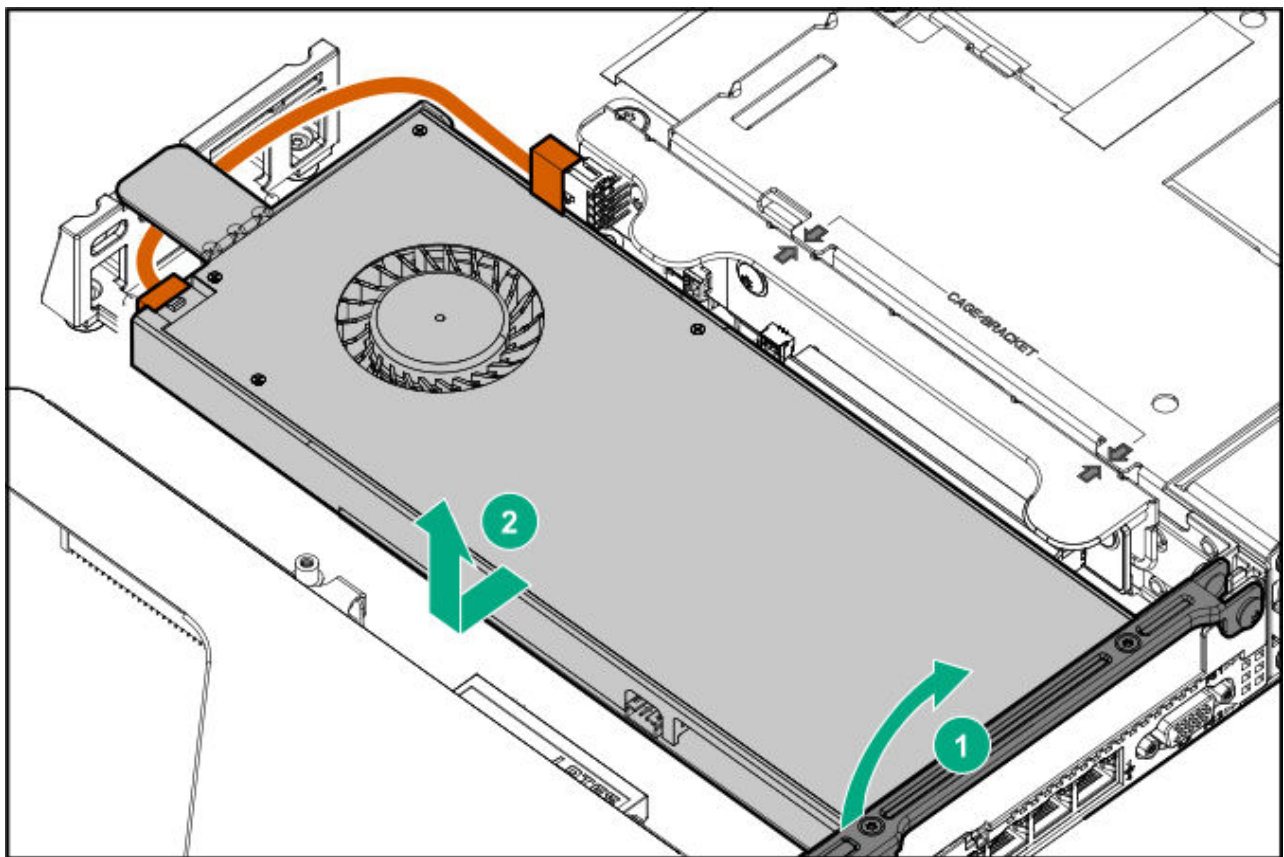


WARNING

To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.

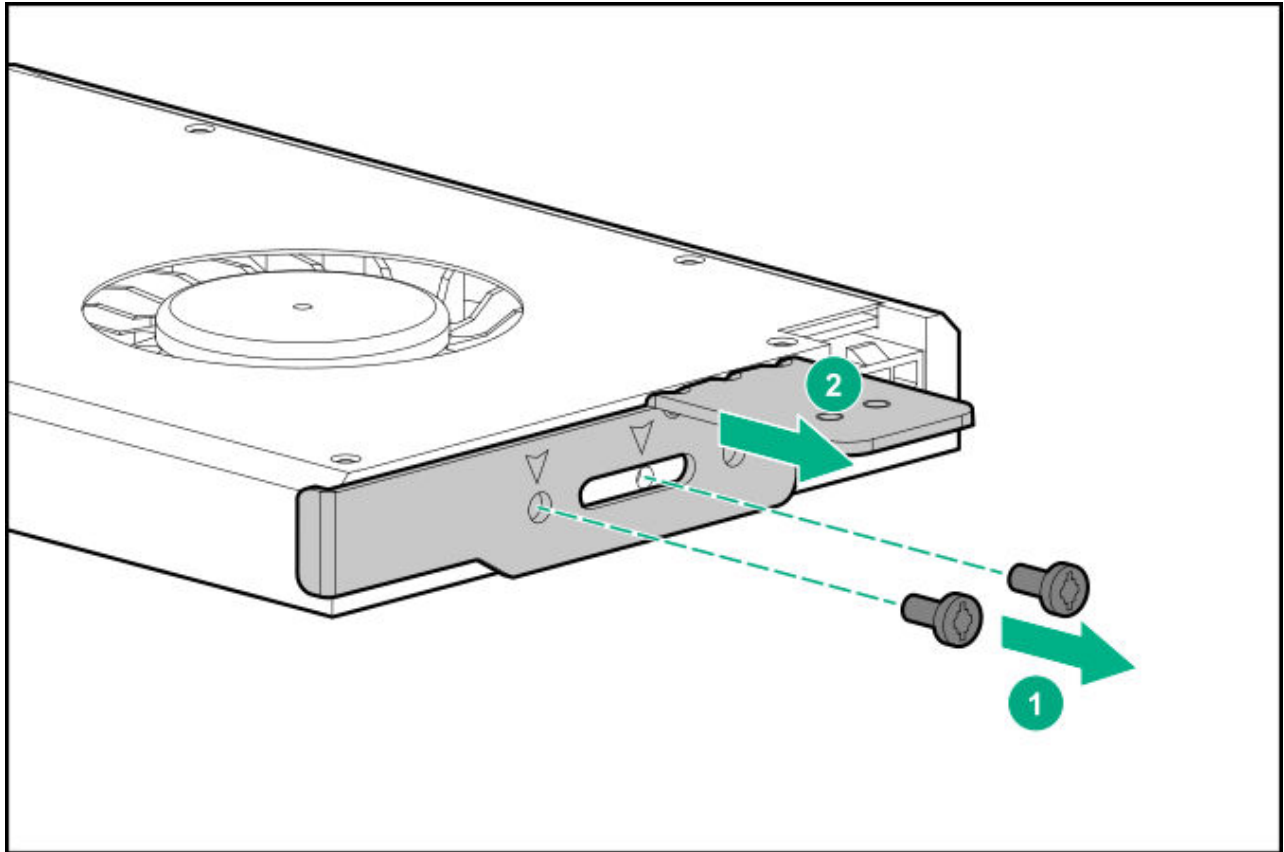
2. Power down the server.

3. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
4. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
5. Remove the access panel.
6. Remove the primary riser cage.
7. Disconnect the GPU power cable from the riser, and then remove the GPU from the riser.



8. If needed, remove the bracket from the existing GPU, and then install it on the new GPU.

Your bracket might look different than the one shown.



Results

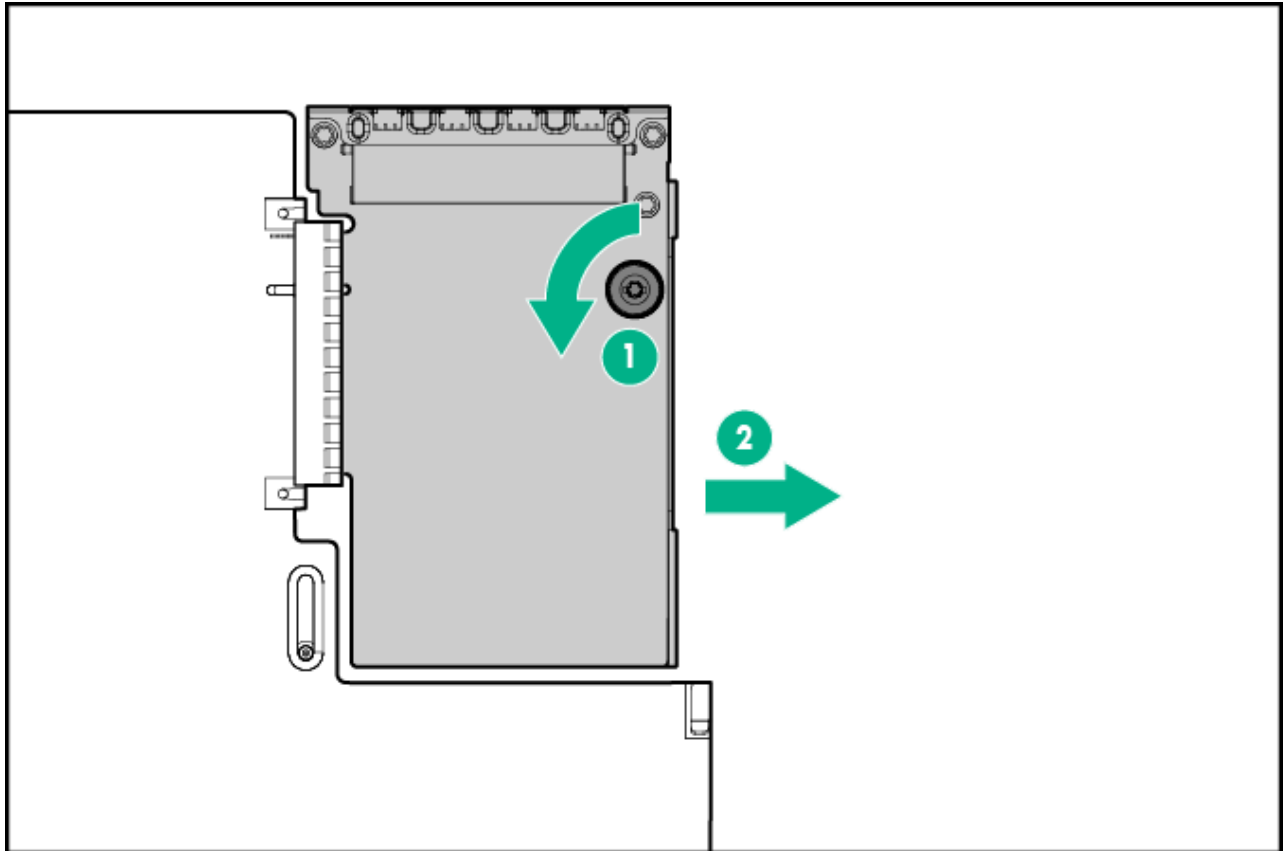
To replace the component, reverse the removal procedure.

Removing and replacing the FlexibleLOM

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Disconnect the LAN segment cables.
4. Do one of the following:
 - Extend the server from the rack.

- Remove the server from the rack.
5. Remove the access panel.
 6. Remove the FlexibleLOM.



Results

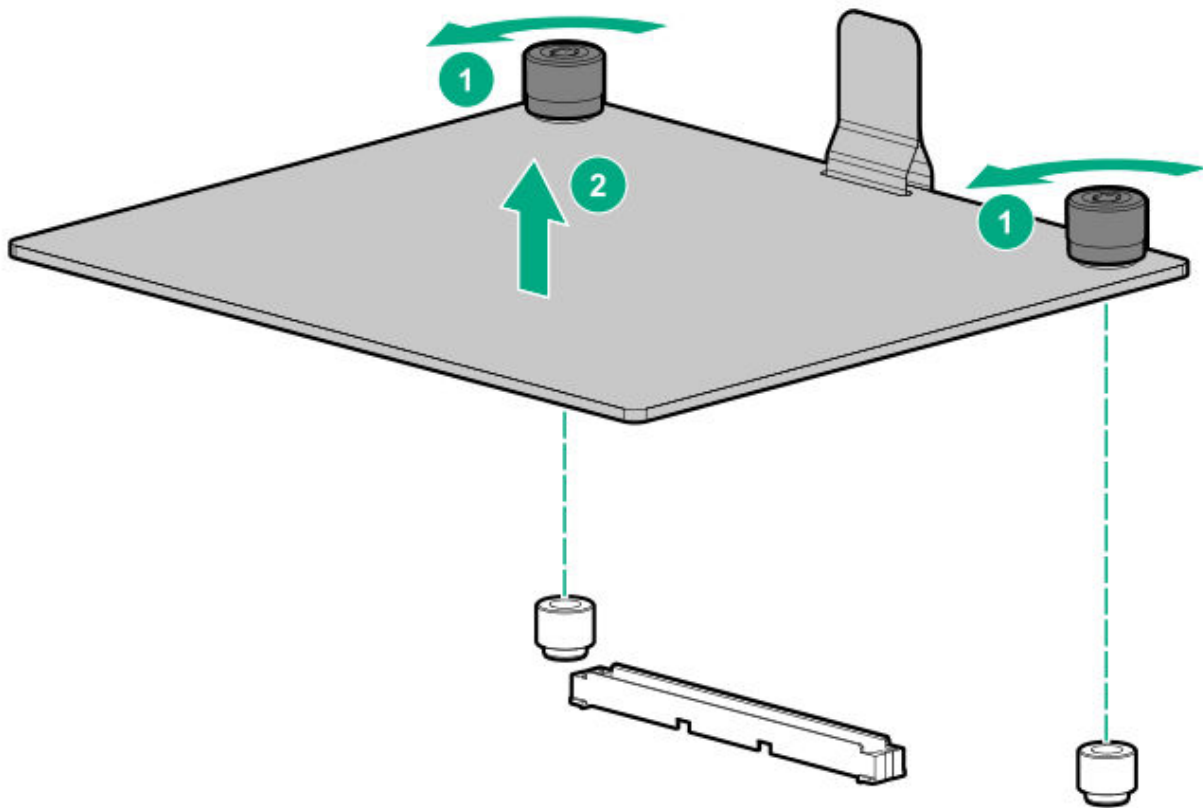
To replace the component, reverse the removal procedure.

Removing and replacing a storage controller

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.

- b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Disconnect any cables from the controller
Note the port numbers before disconnecting the cables.
6. Remove the controller.



Results

To replace the component, reverse the removal procedure.

Removing and replacing a DIMM

Prerequisites

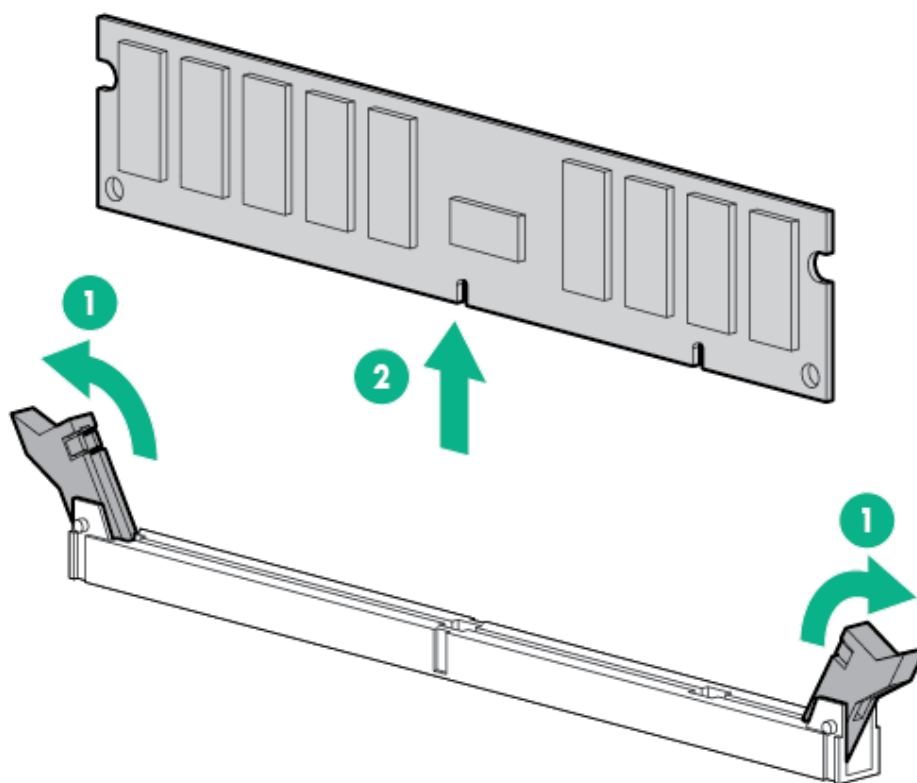
Before replacing memory, read the memory configuration and population guidelines in the server user guide.

About this task

To identify DIMMs installed in the server, see "[DIMM slot locations](#)."

Procedure

1. [Power down the server](#).
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. [Extend the server from the rack](#).
 - b. [Remove the server from the rack](#).
4. [Remove the access panel](#).
5. Remove the DIMM.



Results

To replace the component, reverse the removal procedure.

Subtopics

DIMM-processor compatibility

DIMM-processor compatibility

The installed processor determines the type of DIMM that is supported in the server:

- First-generation Intel Xeon Scalable processors support DDR4-2666 DIMMs.
- Second-generation Intel Xeon Scalable processors support DDR4-2933 DIMMs.

Mixing DIMM types is not supported. Install only the supported DDR4-2666 or DDR4-2933 DIMMs in the server.

Removing and replacing an NVDIMM

Prerequisites

Before replacing memory, read the memory configuration and population guidelines in the server user guide.

About this task



CAUTION

Do not remove an NVDIMM when any LEDs on any NVDIMM in the system are illuminated. Removing an NVDIMM when an LED is illuminated might cause a loss of data.



CAUTION

Electrostatic discharge can damage electronic components. Be sure you are properly grounded before beginning this procedure.



CAUTION

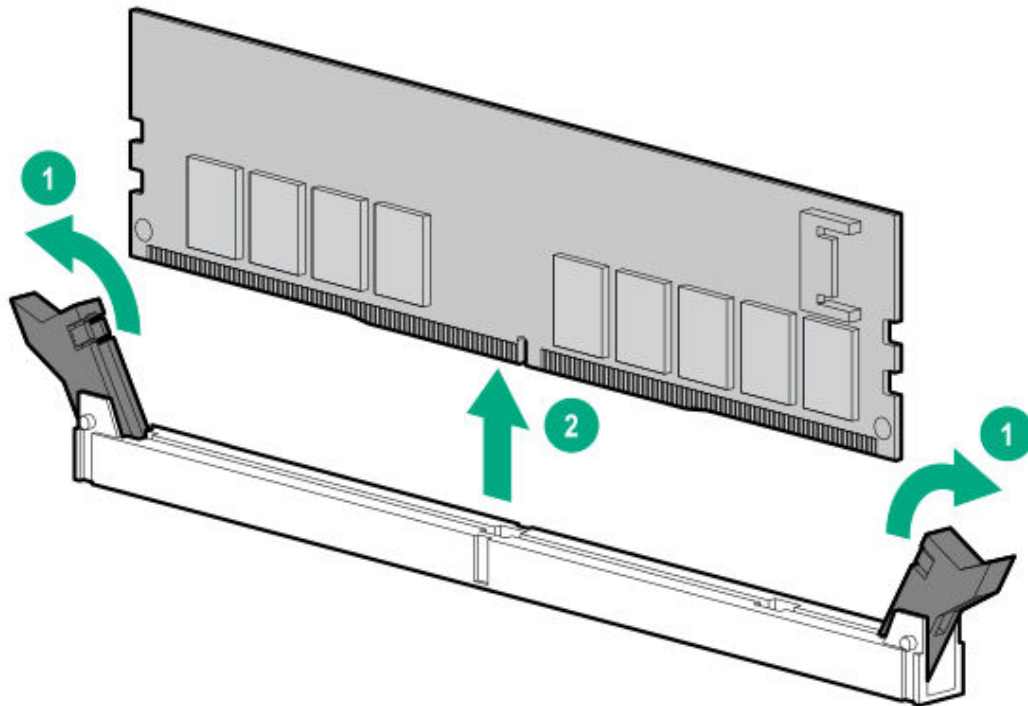
Failure to properly handle DIMMs can cause damage to DIMM components and the system board connector.

To identify NVDIMMs installed in the server, see "[NVDIMM identification](#)"

Procedure

1. Observe [NVDIMM relocation guidelines](#).
2. [Power down the server](#).
3. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
4. Do one of the following:
 - a. [Extend the server from the rack](#).
 - b. [Remove the server from the rack](#).
5. [Remove the access panel](#).
6. Observe the NVDIMM LEDs. Do not remove an NVDIMM when any NVDIMM LED in the system is illuminated.

7. Remove the NVDIMM-N.



Results

To replace the component, reverse the removal procedure.

Subtopics

[NVDIMM-processor compatibility](#)

[DIMM and NVDIMM population information](#)

[NVDIMM sanitization](#)

[NVDIMM relocation guidelines](#)

[Recovering restored data from an NVDIMM-N DRAM](#)

[Configuring the server for NVDIMMs](#)

NVDIMM-processor compatibility

HPE 16GB NVDIMMs are only supported in servers with first-generation Intel Xeon Scalable processors installed.

DIMM and NVDIMM population information

For specific DIMM and NVDIMM population information, see the DIMM population guidelines on the Hewlett Packard Enterprise website (<https://www.hpe.com/docs/memory-population-rules>).

NVDIMM sanitization

Media sanitization is defined by NIST SP800-88 Guidelines for Media Sanitization (Rev 1, Dec 2014) as "a general term referring to the actions taken to render data written on media unrecoverable by both ordinary and extraordinary means."

The specification defines the following levels:

- Clear: Overwrite user-addressable storage space using standard write commands; might not sanitize data in areas not currently user-addressable (such as bad blocks and overprovisioned areas)
- Purge: Overwrite or erase all storage space that might have been used to store data using dedicated device sanitize commands, such that data retrieval is "infeasible using state-of-the-art laboratory techniques"
- Destroy: Ensure that data retrieval is "infeasible using state-of-the-art laboratory techniques" and render the media unable to store data (such as disintegrate, pulverize, melt, incinerate, or shred)

The NVDIMM-N Sanitize options are intended to meet the Purge level.

For more information on sanitization for NVDIMMs, see the following sections in the HPE 16GB NVDIMM User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/nvdimm-docs>):

- NVDIMM sanitization policies
- NVDIMM sanitization guidelines
- Setting the NVDIMM-N Sanitize/Erase on the Next Reboot Policy

NIST SP800-88 Guidelines for Media Sanitization (Rev 1, Dec 2014) is available for download from the NIST website (<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-88r1.pdf>).

NVDIMM relocation guidelines

Requirements for relocating NVDIMMs or a set of NVDIMMs when the data must be preserved

- The destination server hardware must match the original server hardware configuration.

- All System Utilities settings in the destination server must match the original System Utilities settings in the original server.
- If NVDIMM-Ns are used with NVDIMM Interleaving ON mode in the original server, do the following:
 - Install the NVDIMMs in the same DIMM slots in the destination server.
 - Install the entire NVDIMM set (all the NVDIMM-Ns on the processor) on the destination server.

This guideline would apply when replacing a system board due to system failure.

If any of the requirements cannot be met during NVDIMM relocation, do the following:

- Manually back up the NVDIMM-N data before relocating NVDIMM-Ns to another server.
- Relocate the NVDIMM-Ns to another server.
- Sanitize all NVDIMM-Ns on the new server before using them.

Requirements for relocating NVDIMMs or a set of NVDIMMs when the data does not have to be preserved

If data on the NVDIMM-N or set of NVDIMM-Ns does not have to be preserved, then

- Move the NVDIMM-Ns to the new location and sanitize all NVDIMM-Ns after installing them to the new location. For more information, see [NVDIMM sanitization](#).
- Observe all DIMM and NVDIMM population guidelines. For more information, see [DIMM and NVDIMM population information](#).
- Observe the process for removing an NVDIMM.
- Observe the process for installing an NVDIMM.
- Review and configure the system settings for NVDIMMs. For more information, see [Configuring the server for NVDIMMs](#).

Recovering restored data from an NVDIMM-N DRAM

About this task



CAUTION

Do not remove an NVDIMM when any LEDs on any NVDIMM in the system are illuminated. Removing an NVDIMM when an LED is illuminated might cause a loss of data.



CAUTION

Electrostatic discharge can damage electronic components. Be sure you are properly grounded before beginning this procedure.



CAUTION

Failure to properly handle DIMMs can damage the DIMM components and the system board connector. For more information, see the DIMM handling guidelines in the troubleshooting guide for your product on the Hewlett Packard Enterprise website:

- HPE ProLiant Gen10 (<https://www.hpe.com/info/gen10-troubleshooting>)
- HPE Synergy (<https://www.hpe.com/info/synergy-troubleshooting>)

When the NVDIMM-N DRAM contains the only copy of restored data, perform the following procedure to recover the information:

Procedure

1. Copy the data from the NVDIMM to some other storage device (such as SSD, HDD, or another NVDIMM) as soon as possible (before cold reset or power loss).
2. Power down the server.
3. Extend or remove the server.
4. Remove the access panel.
5. Remove all components necessary to access the server DIMM slots and the HPE Smart Storage Battery.

For more information, see the server maintenance and service guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/enterprise-docs>).

6. Observe the NVDIMM LEDs. Do not remove an NVDIMM when any NVDIMM LED in the system is illuminated.

7. Remove the NVDIMM-N.
8. Install a replacement NVDIMM-N.
9. Install any components removed to access the DIMM slots and the HPE Smart Storage Battery.
0. Install the access panel.
1. Install the server in the rack.
2. Power up the server.
3. Sanitize the replacement NVDIMM.
4. Copy the data from the storage device to the NVDIMM-N.

Configuring the server for NVDIMMs

After installing NVDIMMs, configure the server for NVDIMMs. For information on configuring settings for NVDIMMs, see the HPE 16GB NVDIMM User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/nvdimm-docs>).

The server can be configured for NVDIMMs using either of the following:

- UEFI System Utilities—Use System Utilities through the Remote Console to configure the server for NVDIMM memory options by pressing the **F9** key during POST. For more information about UEFI System Utilities, see the Hewlett Packard Enterprise website (<https://www.hpe.com/info/uefi/docs>).
- iLO RESTful API for HPE iLO 5—For more information about configuring the system for NVDIMMs, see <https://hewlettpackard.github.io/ilo-rest-api-docs/ilo5/>.

Removing and replacing a persistent memory module

About this task

For specific population and configuration information, see the memory population guidelines on the Hewlett Packard Enterprise website (<https://www.hpe.com/docs/memory-population-rules>).

To identify the persistent memory modules installed in the server, see [Intel Optane persistent memory 100 series for HPE label identification](#).

Procedure

1. Observe the following alerts:



CAUTION

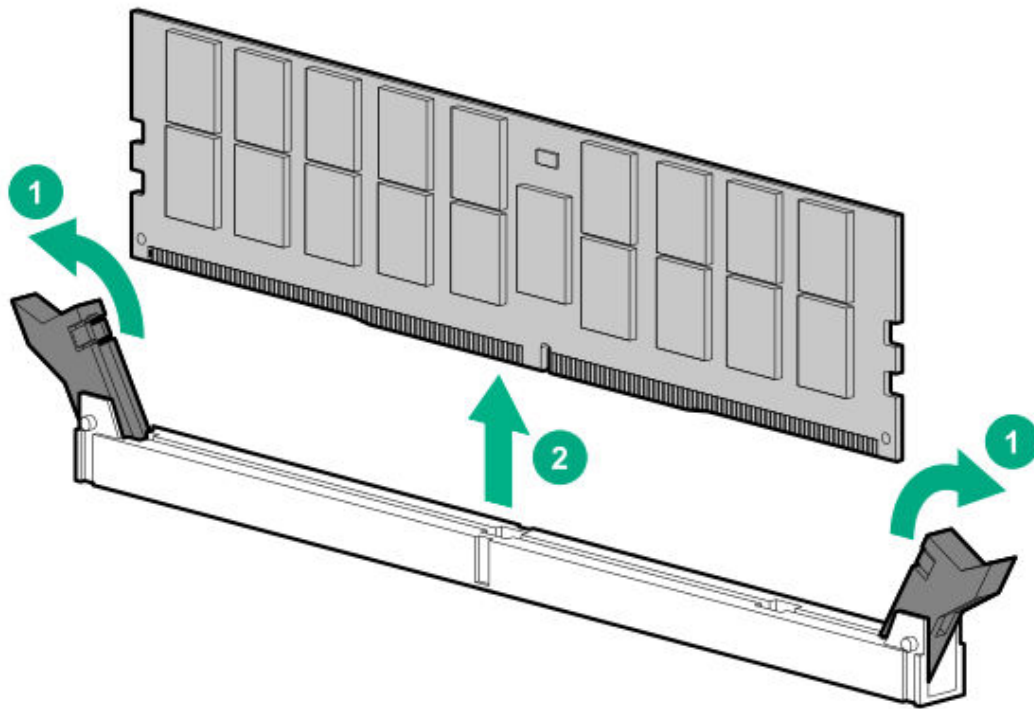
Electrostatic discharge can damage electronic components. Be sure you are properly grounded before beginning this procedure.



CAUTION

Failure to properly handle persistent memory modules can damage the component and the system board connector.

2. Power down the server:
 - a. Shut down the OS as directed by the OS documentation.
 - b. To place the server in standby mode, press the Power On/Standby button. When the server enters standby power mode, the system power LED changes to amber.
 - c. Disconnect the power cords (rack and tower servers).
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Place the server on a flat, level work surface.
5. Remove the access panel.
6. Remove all components necessary to access the DIMM slots.
7. Remove the persistent memory module.



8. Install the new persistent memory module, and review the persistent memory configuration of the server.

For more information, see [Configuring the server for Intel Optane persistent memory 100 series for HPE](#).

9. If you are relocating the persistent memory module to or from another server, see the [Persistent memory module relocation guidelines](#).

Subtopics

[**Persistent memory module-processor compatibility**](#)

[**Configuring the server for Intel Optane persistent memory 100 series for HPE**](#)

[**Persistent memory module relocation guidelines**](#)

[**Persistent memory module sanitization**](#)

Persistent memory module-processor compatibility

Intel Optane persistent memory 100 series for HPE is supported only in servers with second-generation Intel Xeon Scalable processors installed.

Configuring the server for Intel Optane persistent memory 100 series for HPE

After installing persistent memory modules, configure the server for Intel Optane persistent memory for HPE.



IMPORTANT

Always follow recommendations from your software application provider for high-availability best practices to ensure maximum uptime and data protection.

A number of configuration tools are available, including:

- UEFI System Utilities—Configure the server by using the UEFI System Utilities through the Remote Console. To access the UEFI System Utilities, press **F9** during POST.
- iLO RESTful API—Use the iLO RESTful API through tools such as the RESTful Interface Tool (ilorest) or other third-party tools.

For more information, see the Intel Optane persistent memory 100 series for HPE User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/persistentmemory-docs>).

Persistent memory module relocation guidelines

Observe the relocation guidelines when doing the following:

- When relocating persistent memory modules to another DIMM slot on the server.
- When relocating persistent memory modules to another server.
- When reinstalling persistent memory modules after replacing the server system board.



IMPORTANT

When data must be preserved, Hewlett Packard Enterprise strongly recommends that you perform a manual backup of all user data on the persistent memory modules before changing the goal configuration or performing relocation procedures.

Requirements for relocating persistent memory modules or a set of persistent memory modules when the data must be preserved

- The destination server hardware must match the original server hardware configuration.

- All System Utilities settings in the destination server must match the original System Utilities settings in the original server.
- If persistent memory modules are used with **Persistent Memory Interleaving** set to Enabled in the original server, do the following:
 - Install the persistent memory modules in the same DIMM slots in the destination server.
 - Install the entire interleaved set (all the DIMMs and persistent memory modules on the processor) on the destination server.

If any of the requirements cannot be met during relocation, do the following:

- Manually back up the persistent memory data before relocating persistent memory modules to another server.
- Relocate the persistent memory modules to another server.
- Sanitize all persistent memory modules on the new server before using them.

Requirements for relocating encrypted persistent memory modules or a set of persistent memory modules when the data must be preserved

- If persistent memory modules are encrypted with local key management, either manually retrieve the persistent memory module passwords from the server (user-generated passwords only) or export a password file to a USB key.

Hewlett Packard Enterprise recommends exporting the password file to a USB key.

- Follow the requirements for relocating persistent memory modules or a set of persistent memory modules when the data must be preserved.
- Do one of the following:
 - If persistent memory modules are encrypted with local key management, either manually enter the persistent memory module passwords in the System Utilities or import the password file from the USB key.
 - If persistent memory modules are encrypted with remote key management, enroll the HPE iLO in the key management server to provide access to the data on the persistent memory modules.

For more information, see the Intel Optane persistent memory 100 series for HPE User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/persistentmemory-docs>).

Requirements for relocating persistent memory modules or a set of persistent memory modules when the data does not have to be preserved

- Move the persistent memory modules to the new location and sanitize all persistent memory modules after installing them to the new location.

- Observe the DIMM and persistent memory module population guidelines.
- Observe the process for removing a persistent memory module.
- Observe the process for installing a persistent memory module.
- Review and configure the system settings for Intel Optane persistent memory for HPE.

For more information, see the Intel Optane persistent memory 100 series for HPE User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/persistentmemory-docs>).

Persistent memory module sanitization

Media sanitization is defined by NIST SP800-88 Guidelines for Media Sanitization (Rev 1, Dec 2014) as "a general term referring to the actions taken to render data written on media unrecoverable by both ordinary and extraordinary means."

The specification defines the following levels:

- Clear: Overwrite user-addressable storage space using standard write commands; might not sanitize data in areas not currently user-addressable (such as bad blocks and over-provisioned areas).
- Purge: Overwrite or erase all storage space that might have been used to store data using dedicated device sanitize commands, such that data retrieval is "infeasible using state-of-the-art laboratory techniques."
- Destroy: Ensure that data retrieval is "infeasible using state-of-the-art laboratory techniques" and render the media unable to store data (such as disintegrate, pulverize, melt, incinerate, or shred).

Intel Optane persistent memory for HPE supports the purge level using a cryptographic erase technique and an overwrite technique.

HPE ProLiant and HPE Synergy Gen10 server products support sanitizing persistent memory modules during POST. Use the HPE RESTful Interface Tool or UEFI System Utilities to schedule sanitization on the next boot.

For more information, see the sanitization policies and guidelines in the Intel Optane persistent memory 100 series for HPE User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/persistentmemory-docs>).

NIST SP800-88 Guidelines for Media Sanitization (Rev 1, Dec 2014) is available for download from the NIST website (<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-88r1.pdf>).

Removing and replacing the system battery

About this task

The system battery provides power to the real-time clock. If the server no longer automatically displays the correct date and time, you might need to replace the system battery.



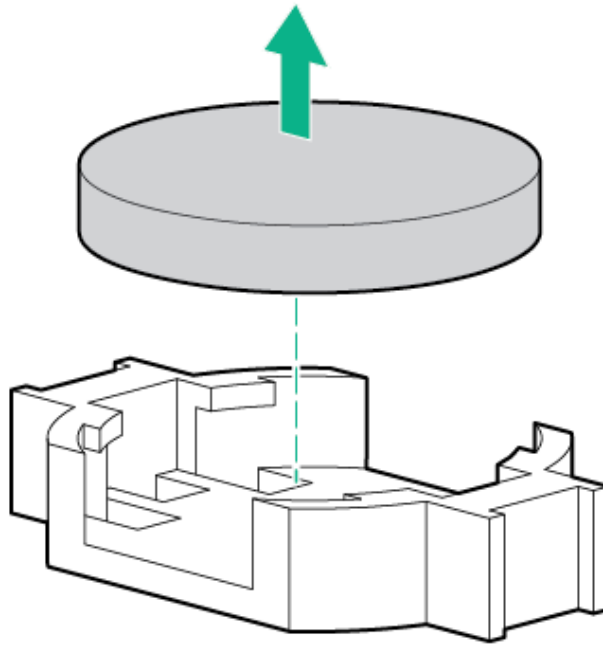
WARNING

The computer contains an internal lithium manganese dioxide, a vanadium pentoxide, or an alkaline battery pack. A risk of fire and burns exists if the battery pack is not properly handled. To reduce the risk of personal injury:

- Do not attempt to recharge the battery.
- Do not expose the battery to temperatures higher than 60°C (140°F).
- Do not disassemble, crush, puncture, short external contacts, or dispose of in fire or water.
- Replace only with the spare designated for this product.

Procedure

1. Power down the server ([Power down the server](#)).
2. Remove all power:
 - Disconnect each power cord from the power source.
 - Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack ([Extend the server from the rack](#)).
 - b. Remove the server from the rack ([Remove the server from the rack](#)).
4. Remove the access panel ([Remove the access panel](#)).
5. Locate the battery ([System board components](#)).
6. Remove the battery.



7. To replace the component, reverse the removal procedure.
8. Properly dispose of the old battery.

For more information about battery replacement or proper disposal, contact an authorized reseller or an authorized service provider.

Removing and replacing the Smart Storage Battery or Hybrid Capacitor

About this task



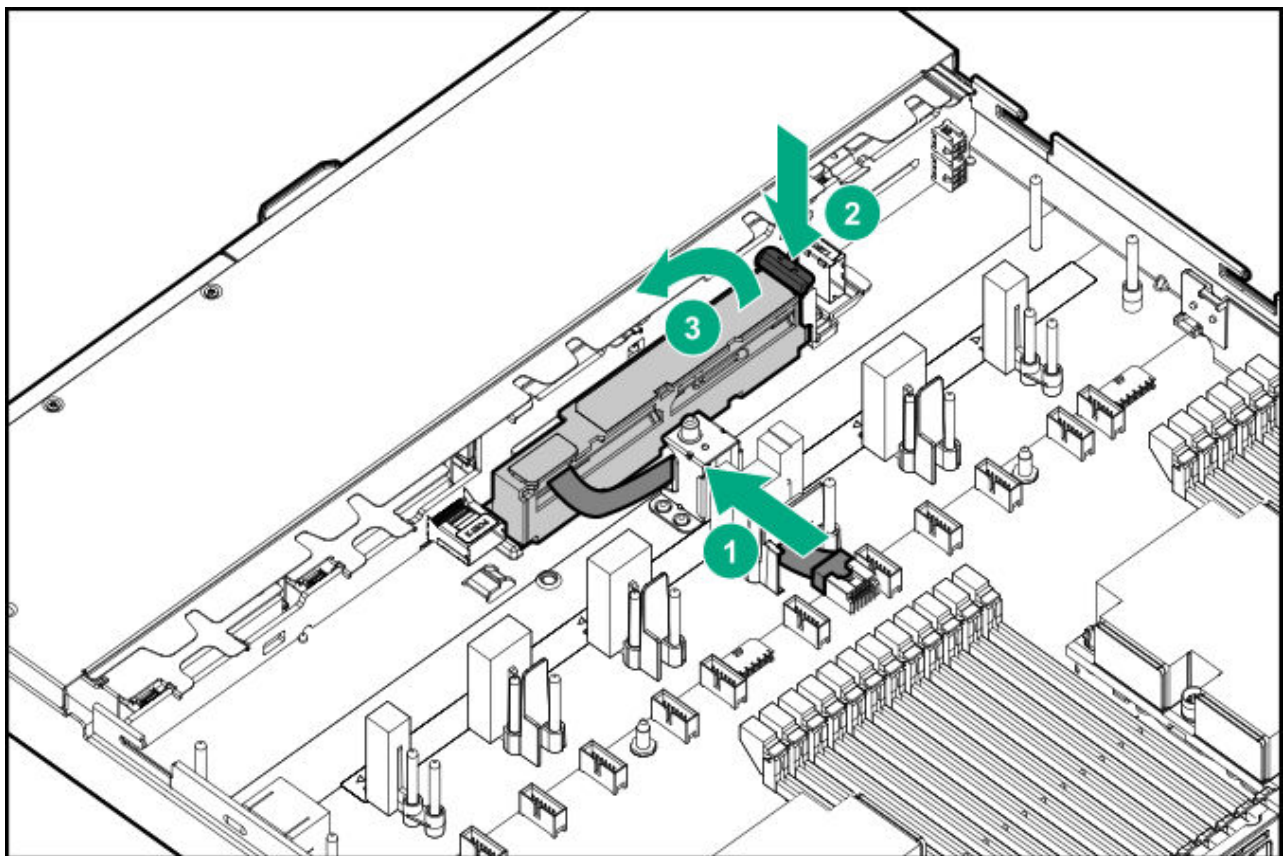
NOTE

System ROM and firmware messages might display "energy pack" in place of "Smart Storage Battery." Energy pack refers to both HPE Smart Storage batteries and HPE Smart Storage Hybrid capacitors.

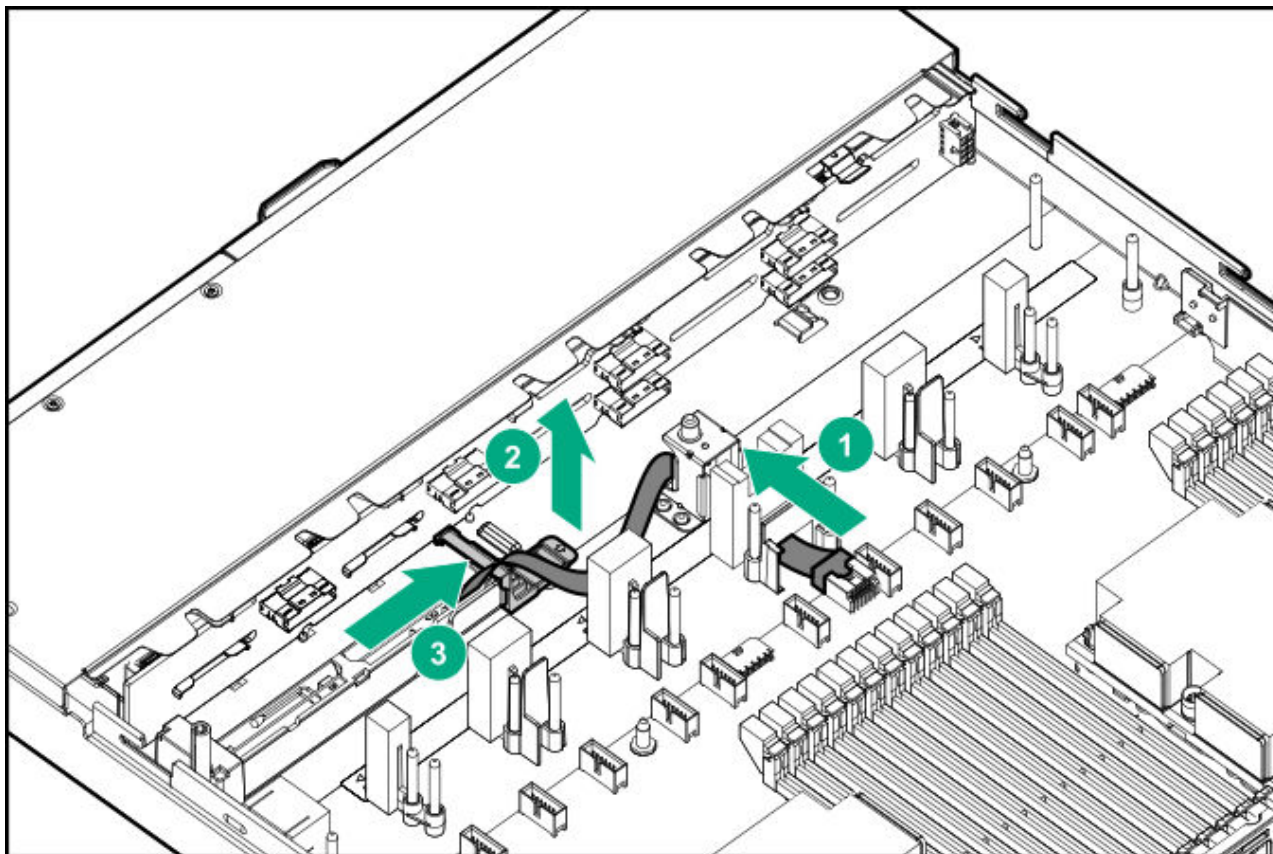
Procedure

1. Power down the server.

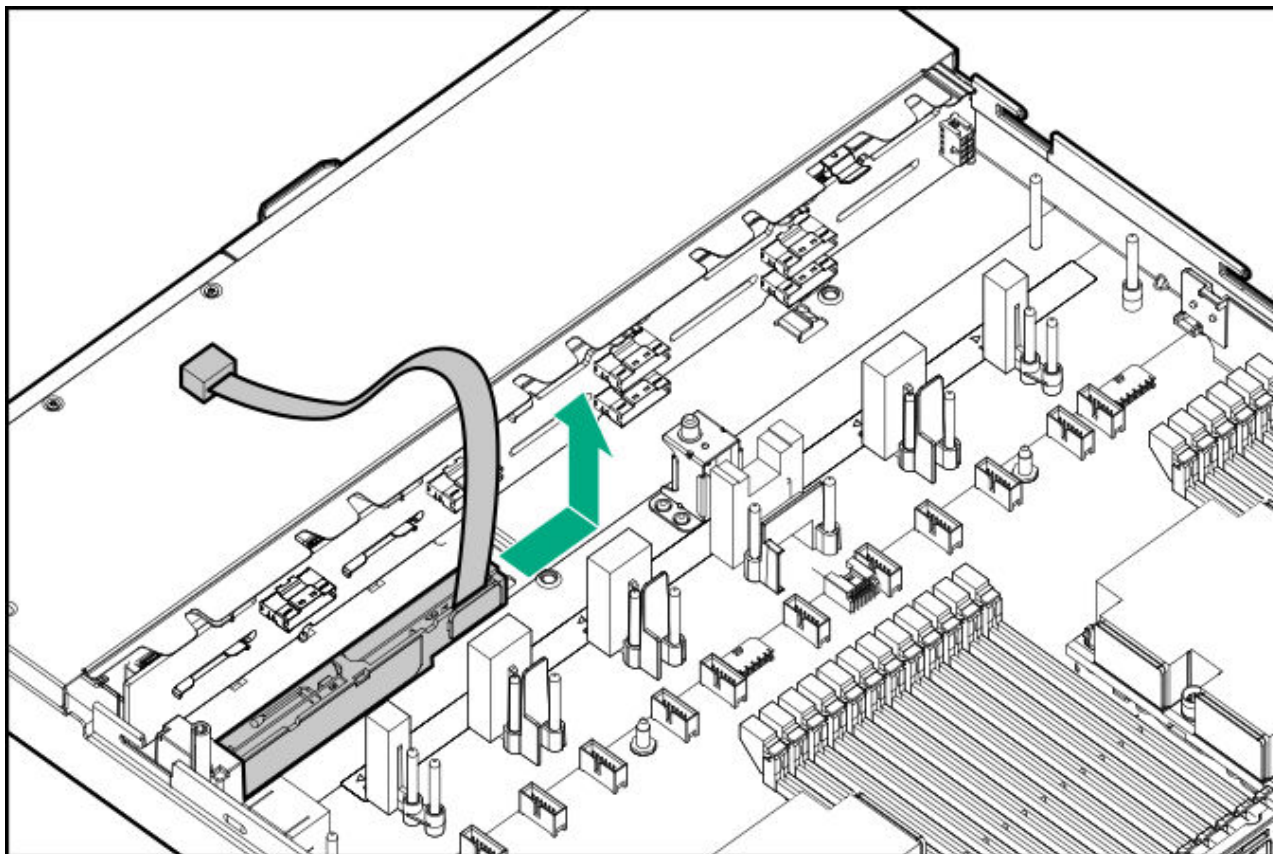
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Disconnect the cable and then remove the battery or capacitor pack.



To remove the battery or capacitor pack from the 10NVMe configuration:



When removing the cable, remove the cable and cable extender.



Results

To replace the component, reverse the removal procedure.

Subtopics

HPE Smart Storage Battery

HPE Smart Storage Battery

The HPE Smart Storage Battery supports the following devices:

- HPE Smart Array SR controllers
- HPE Smart Array MR controllers

- NVDIMMs



IMPORTANT

To support NVDIMMs, the HPE Smart Storage Battery must be installed.

A single 96W battery can support up to 24 devices.

After the battery is installed, it might take up to two hours to charge. Controller features requiring backup power are not re-enabled until the battery is capable of supporting the backup power.

This server supports the 96W HPE Smart Storage Battery with the 145mm cable.

Removing and replacing a microSD card

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Remove the microSD card.

For information on the location of the microSD card slot, see System board components.

Results

To replace the component, reverse the removal procedure.

Removing and replacing a processor heatsink assembly

About this task



IMPORTANT

Existing HPE ProLiant and HPE Synergy Gen10 server products containing first-generation Intel Xeon Scalable processors may not be upgraded to second-generation Intel Xeon Scalable processors at this time.

For more information, see the product QuickSpecs on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/qs>).

Procedure

1. Observe the following alerts:



CAUTION

To avoid damage to the processor or system board, only authorized personnel should attempt to replace or install the processor in this server.



CAUTION

If installing a processor with a faster speed, update the system ROM before installing the processor.

To download firmware and view installation instructions, see the **Hewlett Packard Enterprise Support Center website**.



CAUTION

To prevent possible server malfunction and damage to the equipment, multiprocessor configurations must contain processors with the same part number.



CAUTION

THE CONTACTS ARE VERY FRAGILE AND EASILY DAMAGED. To avoid damage to the socket or processor, do not touch the contacts.



CAUTION

When handling the heatsink, always hold it along the top and bottom of the fins. Holding it from the sides can damage the fins.

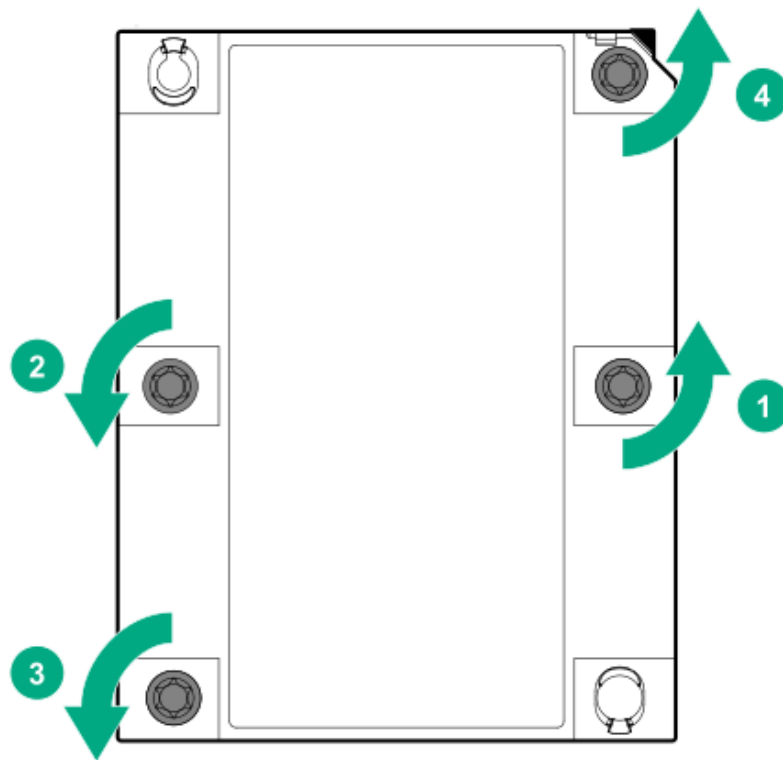


CAUTION

Observe the label on the heatsink. Tightening or loosening the screws in the wrong order can damage the heatsink.

2. Power down the server.
3. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.

4. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
5. Remove the access panel.
6. Remove the processor heatsink assembly:
 - a. Allow the heatsink to cool.
 - b. Loosen the heatsink nuts in the order specified by the label on the heatsink.



- c. Lift the processor heatsink assembly and move it away from the system board.
- d. Turn the assembly over and place it on a work surface with the processor facing up.
- e. Install the dust cover.

Results

To replace the component, reverse the removal procedure.

Removing and replacing the system board

Prerequisites

If persistent memory modules are installed in the server and are encrypted with local key management, do one of the following:

- Manually retrieve the persistent memory module passwords from the server (user-generated passwords only)
- Export a password file to a USB key.

Hewlett Packard Enterprise recommends exporting the password file to a USB key.

For more information, see the Intel Optane persistent memory 100 series for HPE User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/persistentmemory-docs>).

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove all power supplies.
5. Remove the access panel.
6. Take note of port numbers and cables connections to risers and the system board.
7. Disconnect and remove all cables that are connected to risers or the system board.
8. Remove all riser cages.
9. Remove the GPU support bracket.
- .0. Remove the controller.
- .1. Remove all fans.
- .2. Remove all fan blanks.
- .3. Remove the FlexibleLOM.

.4. Remove all DIMMs.

Be sure to note the DIMM slot locations in which each DIMM is installed. These components must be installed in the same locations on the new system board.

.5. Do the following:

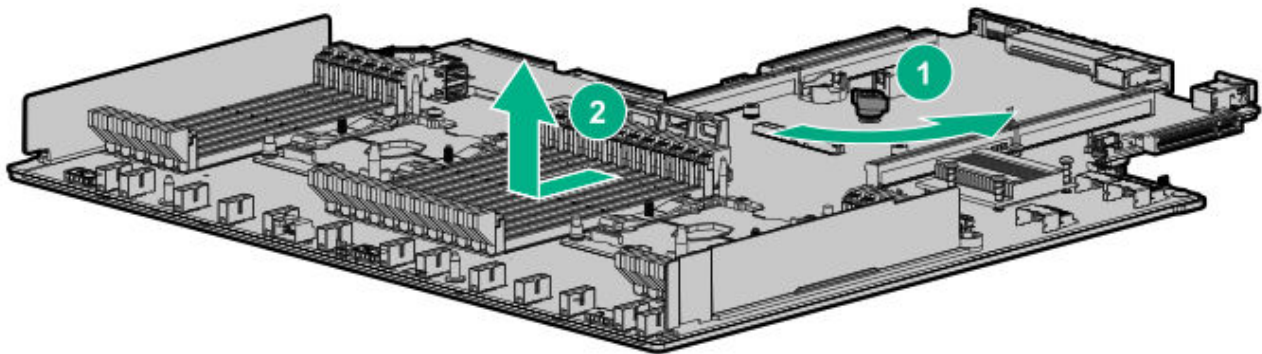
- a. Observe guidelines and procedures for NVDIMM relocation.
- b. Remove all NVDIMMs.

.6. Remove all DCPMMs.

Be sure to note the DIMM slot locations in which each persistent memory module is installed. These components must be installed in the same locations on the new system board.

.7. Remove the processor heatsink assembly.

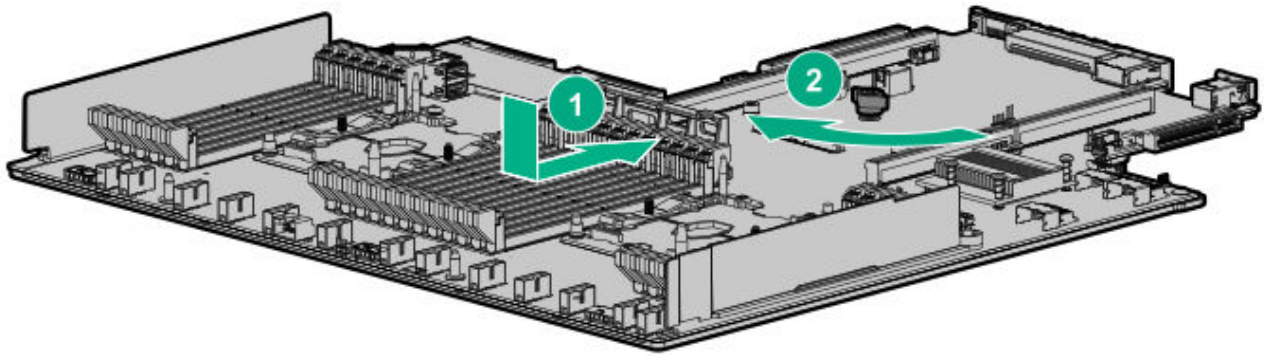
.8. Loosen the system board thumbscrew, and then remove the system board.



Results

To replace the component:

1. Install the spare system board.



2. Install all components removed from the failed system board.
Be sure to install the DIMMs and persistent memory modules in the same DIMM slots as the failed system board.
3. Install the access panel.
4. Install the power supplies.
5. Power up the server.
6. Review the persistent memory configuration of the server.
For more information, see [Configuring the server for Intel Optane persistent memory 100 series for HPE](#).
7. Do one of the following:
 - If persistent memory modules are encrypted with local key management, either manually enter the persistent memory module passwords in the System Utilities or import the password file from the USB key.
 - If persistent memory modules are encrypted with remote key management, enroll the server iLO in the key management server to provide access to the data on the persistent memory modules.

For more information, see the Intel Optane persistent memory 100 series for HPE User Guide on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/persistentmemory-docs>).

After you replace the system board, you must re-enter the server serial number and the product ID:

1. During the server startup sequence, press the **F9** key to access UEFI System Utilities.
2. Select **System Configuration > BIOS/Platform Configuration (RBSU) > Advanced Options > Advanced System ROM Options > Serial Number**, and then press the **Enter** key.
3. Enter the serial number and press the **Enter** key. The following message appears:

Only qualified service personnel must modify the serial number. This value must always match the serial number on the chassis.

4. To clear the warning, press the **Enter** key.
5. Enter the serial number and press the **Enter** key.
6. From the **System Utilities** screen, select **System Configuration > BIOS/Platform Configuration (RBSU) > Date and Time**.
7. Select a setting, and then complete your entry:
 - **Date (mm-dd-yyyy)**—Enter the date in a month-day-year (mm-dd-yyyy) format.
 - **Time (hh:mm:ss)**—Enter the time in a 24-hour format (hh:mm:ss) format.
 - **Time Zone**—Select your current time zone for the system.
 - **Daylight Savings Time:**
 - **Enabled**—Adjusts the local time displayed by one hour for Daylight Savings Time.
 - **Disabled**—Does not adjust the local time displayed for Daylight Savings Time.
 - **Time Format**
 - **Coordinated Universal Time (UTC)**—Calculates the time stored in the hardware Real Time Clock (RTC) from the associated **Time Zone** setting.
 - **Local Time**—Removes the use of the **Time Zone** setting.

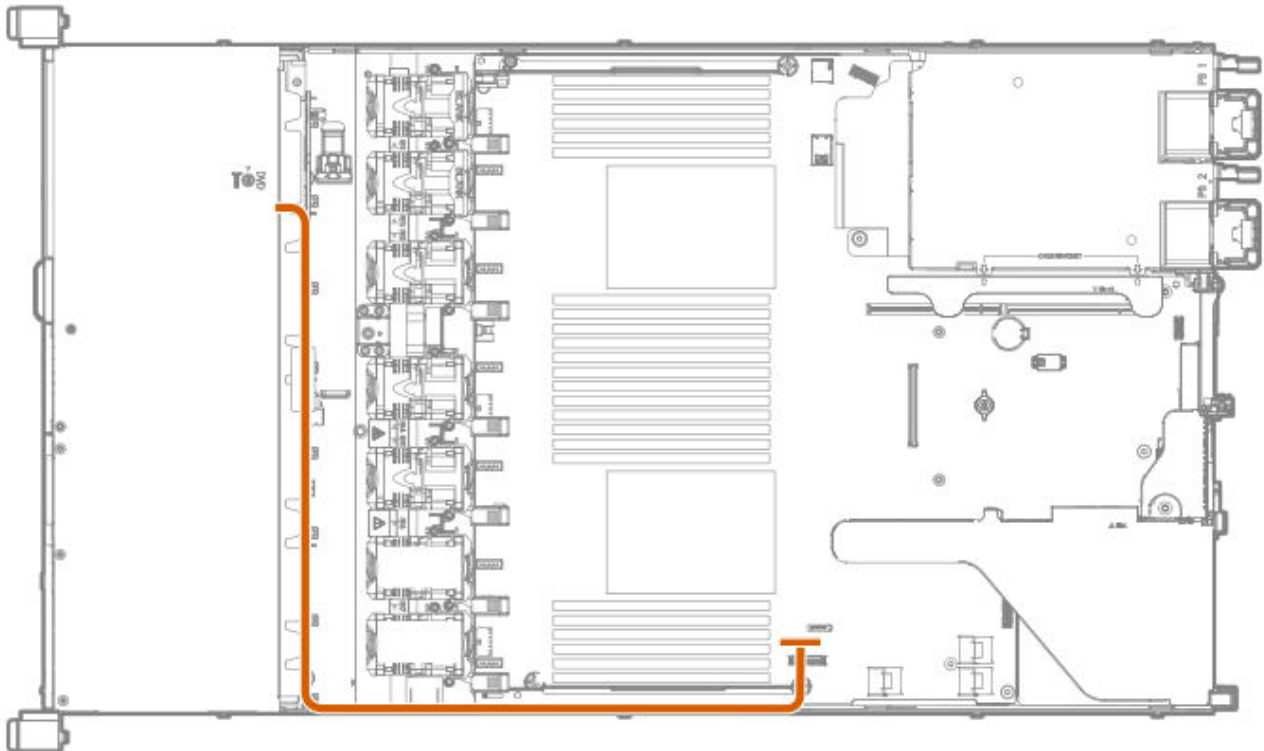
This option is useful for addressing interaction issues between Windows operating systems set in Legacy BIOS boot mode.
8. Save your settings.
9. Select Product ID. The following warning appears:

Warning: The Product ID must ONLY be modified by qualified service personnel. This value must always match the Product ID on the chassis.
10. Enter the product ID and press the **Enter** key.
11. To confirm exiting System Utilities, press the **F10** key.
12. The server automatically reboots.

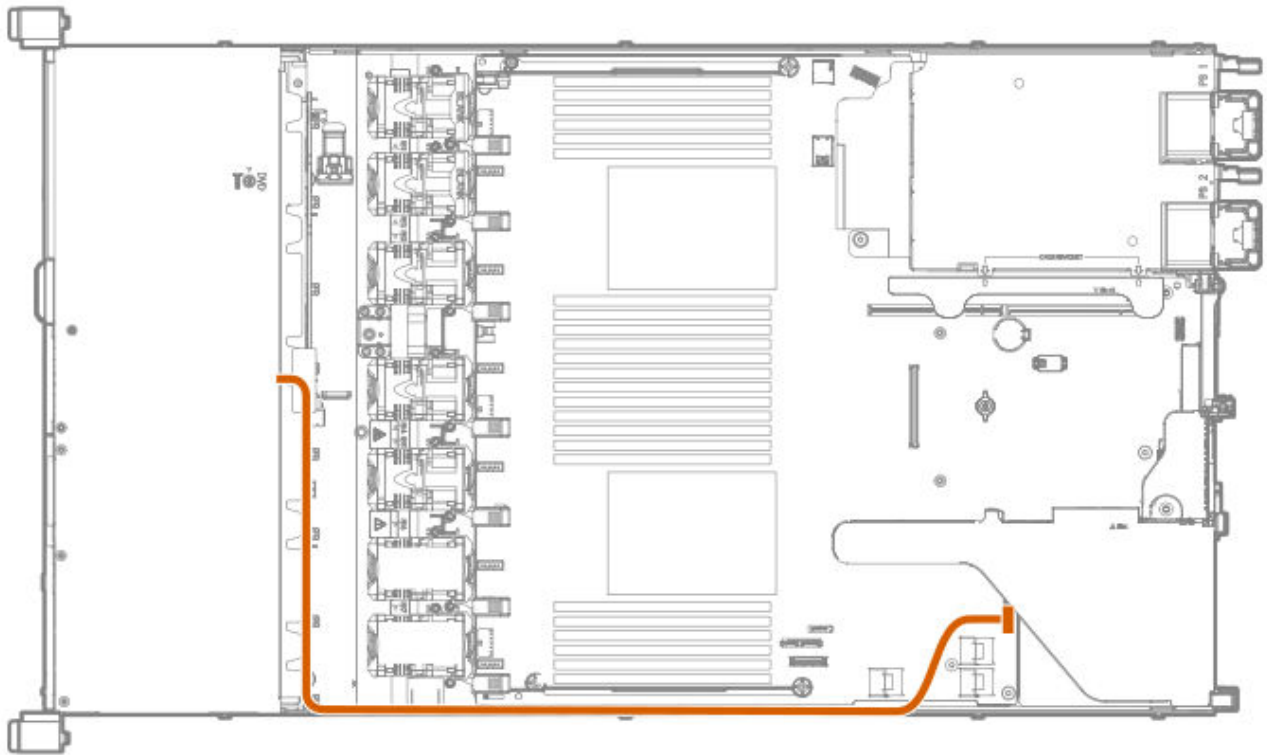
Removing and replacing the LFF optical disk drive/display port/USB

Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Disconnect the optical drive cable from the system board.

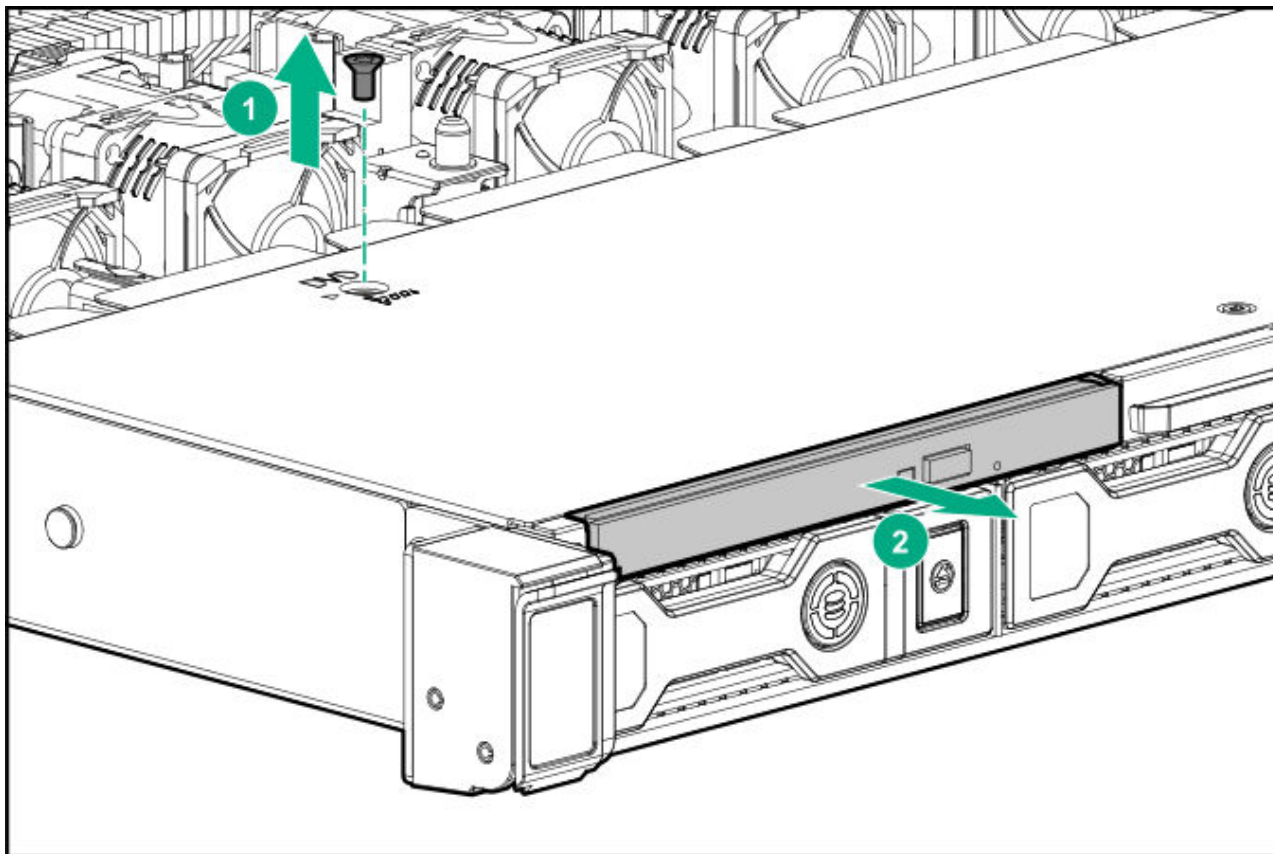


6. Disconnect the display port/USB cable from the system board.

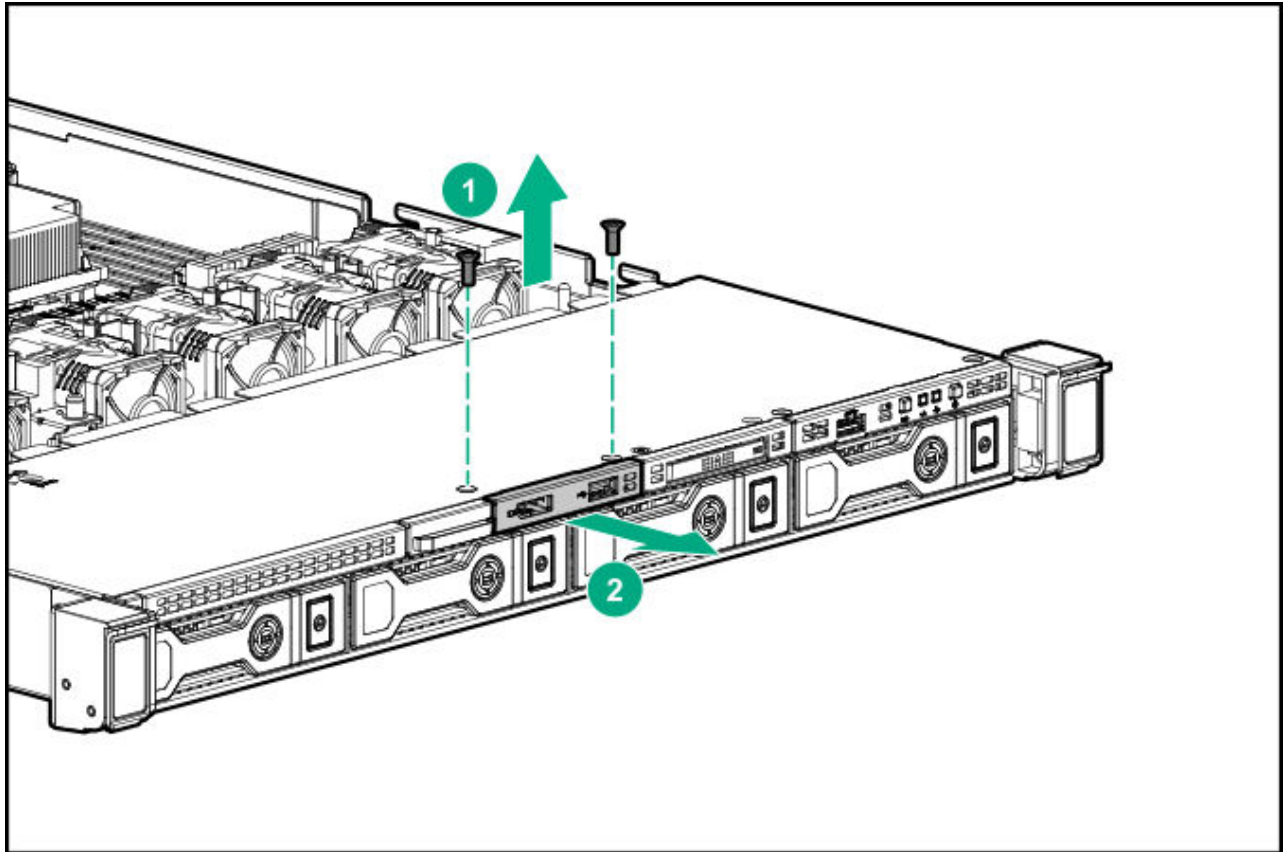


7. Remove the component.

Optical disk drive



Display port/USB



Results

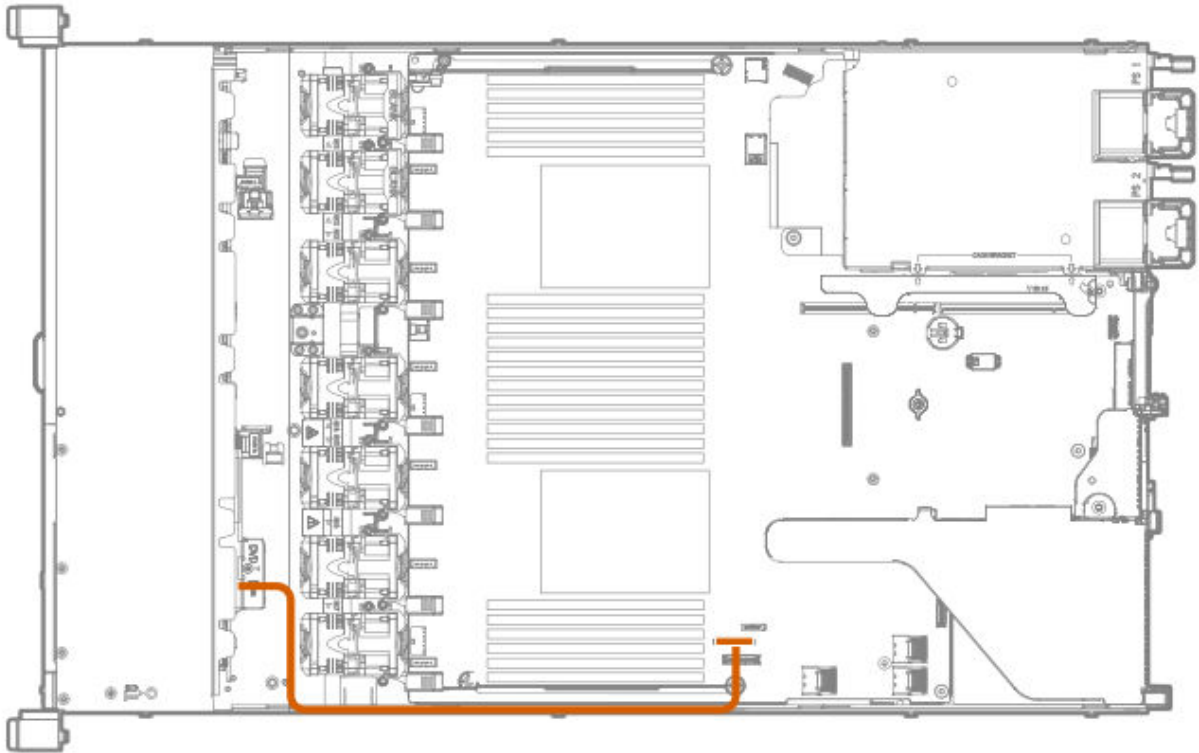
To replace the component, reverse the removal procedure.

Removing and replacing the SFF optical disk drive/display port/USB

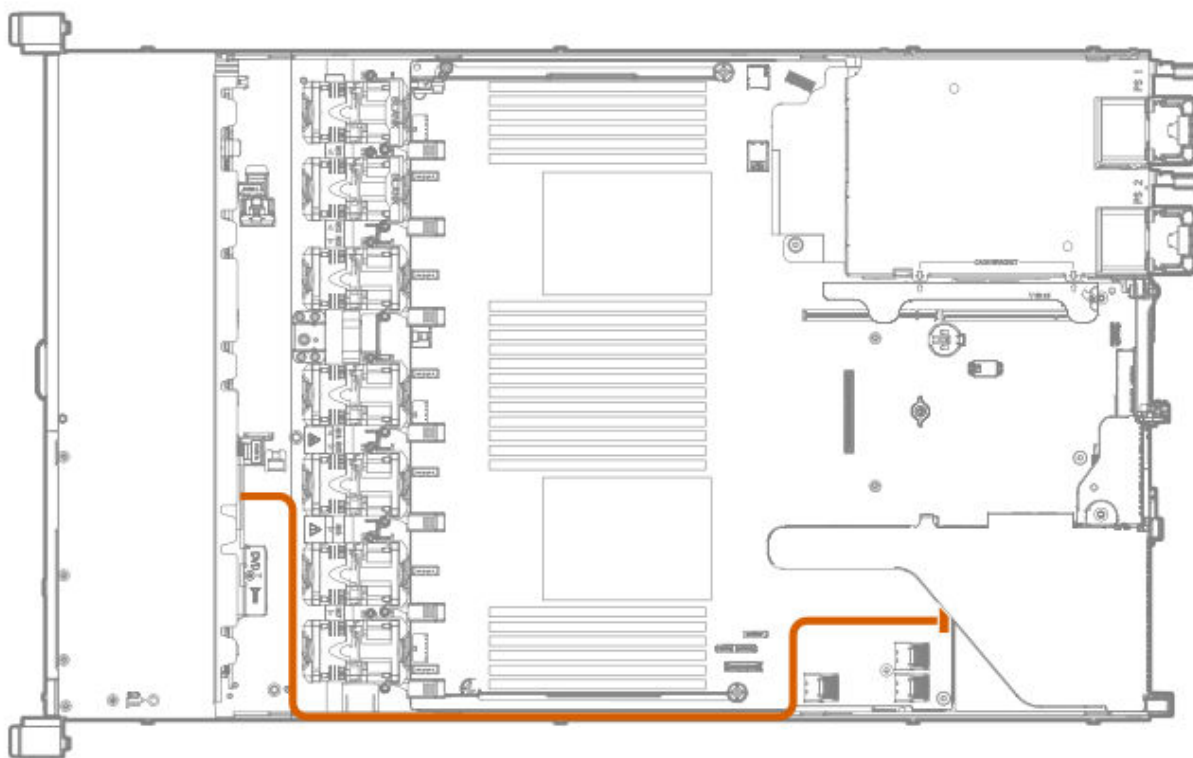
Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.

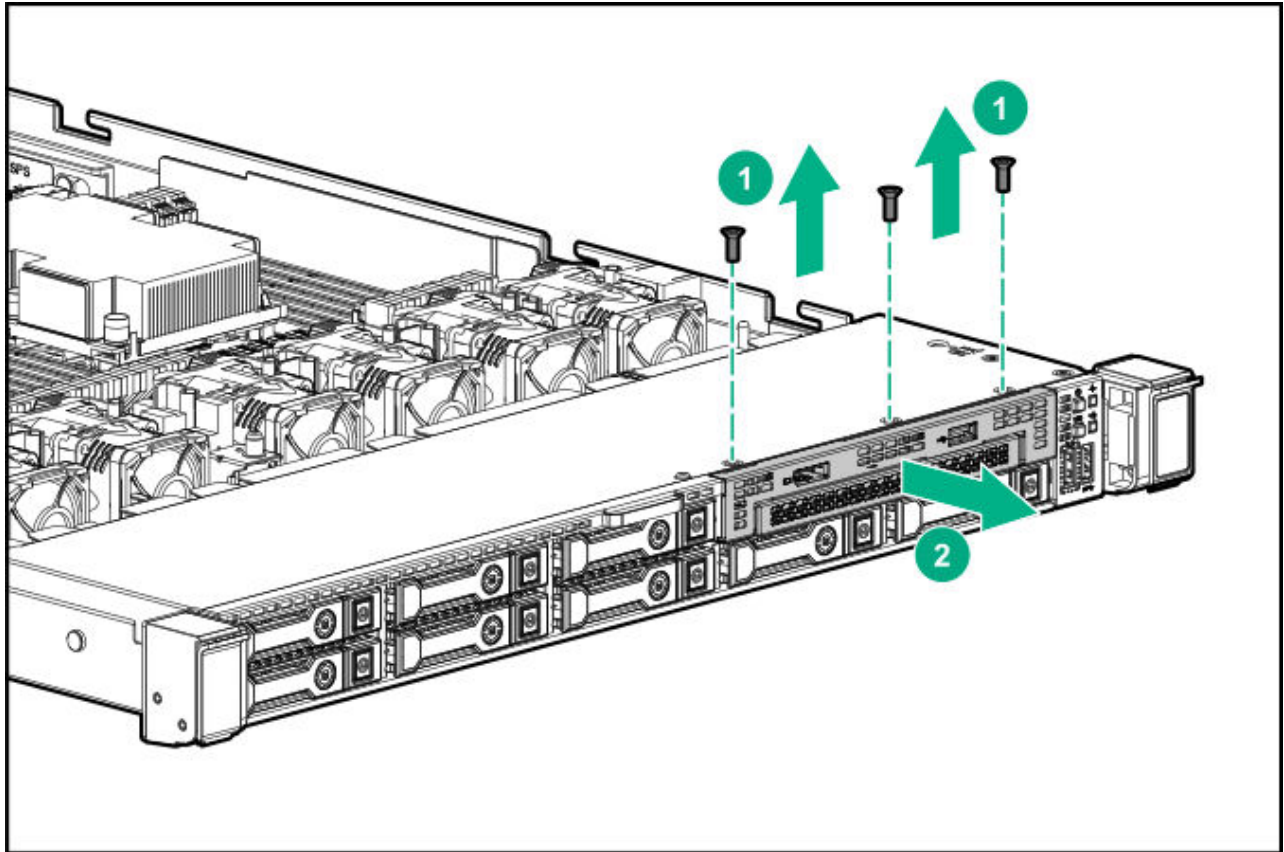
4. Remove the access panel.
5. If needed, disconnect the optical disk drive cable from the system board.



6. Disconnect the display port/USB cable from the system board.



7. Remove the component.



Results

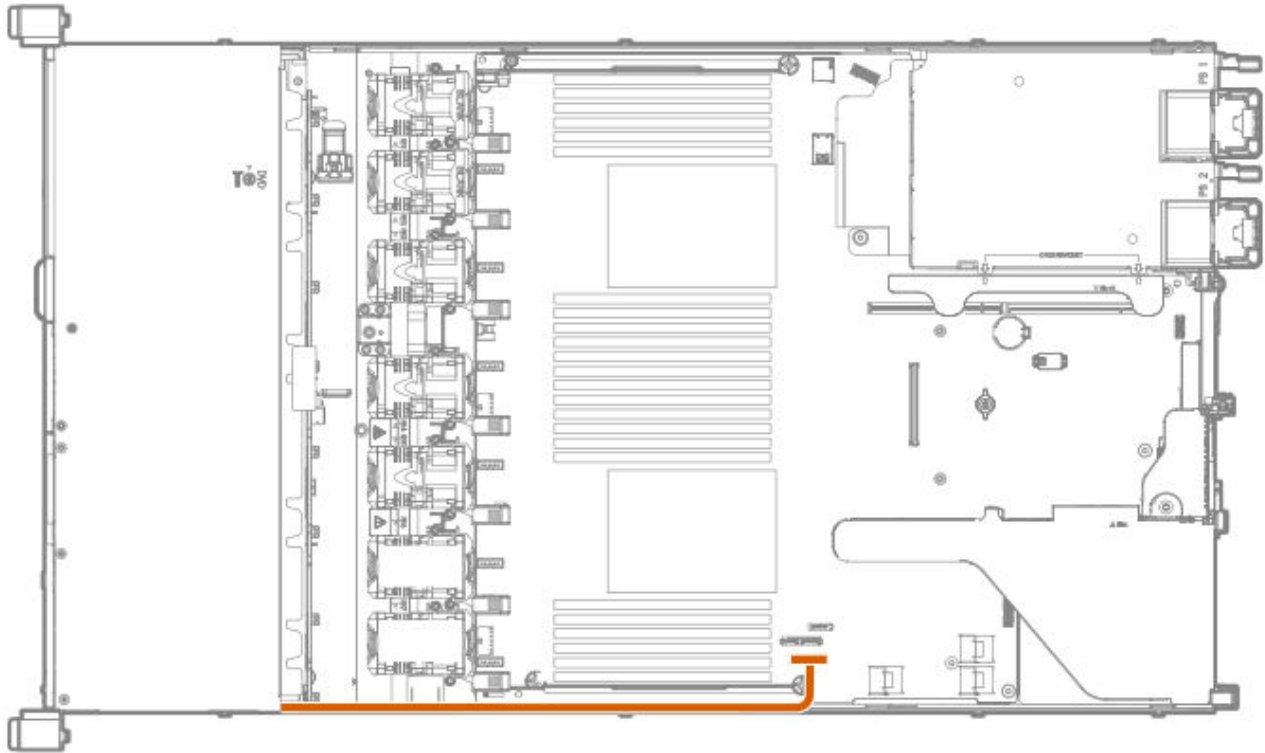
To replace the component, reverse the removal procedure.

Removing and replacing the LFF power switch module

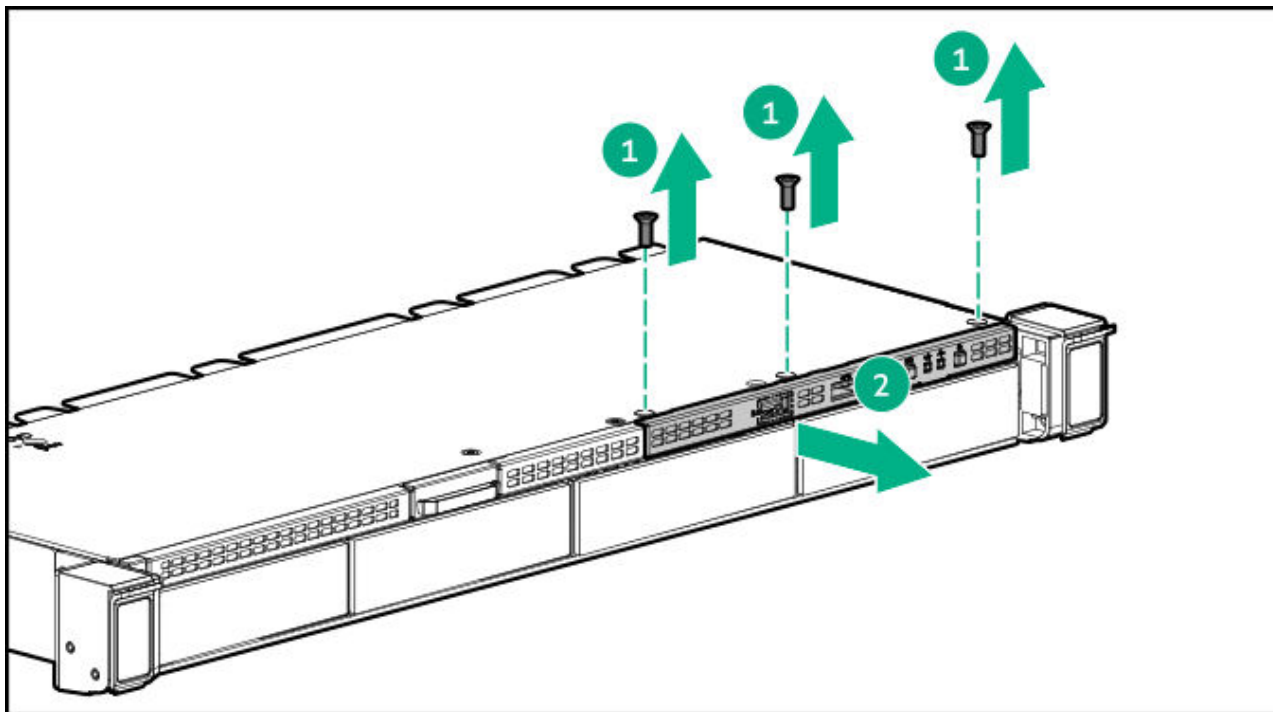
Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.

4. Remove the access panel.
5. Disconnect the cable from the system board.



6. Remove the power module.



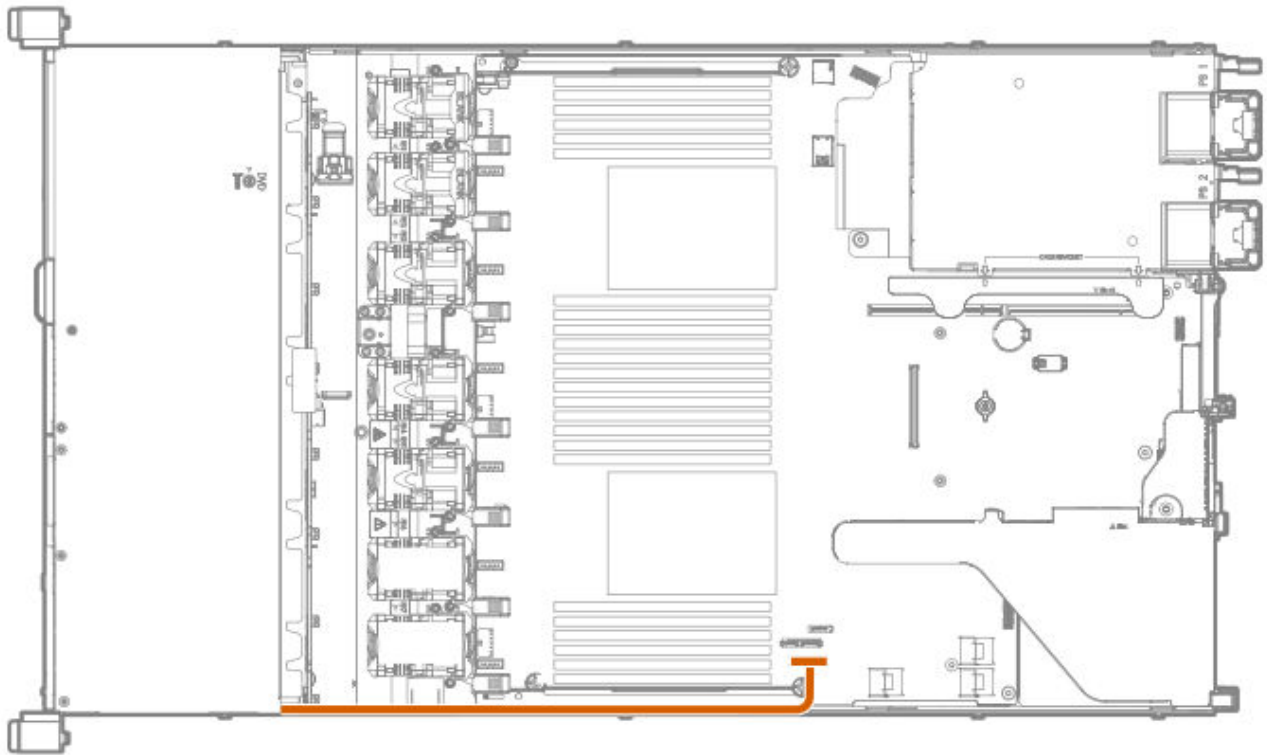
Results

To replace the component, reverse the removal procedure.

Removing and replacing the Systems Insight Display power module

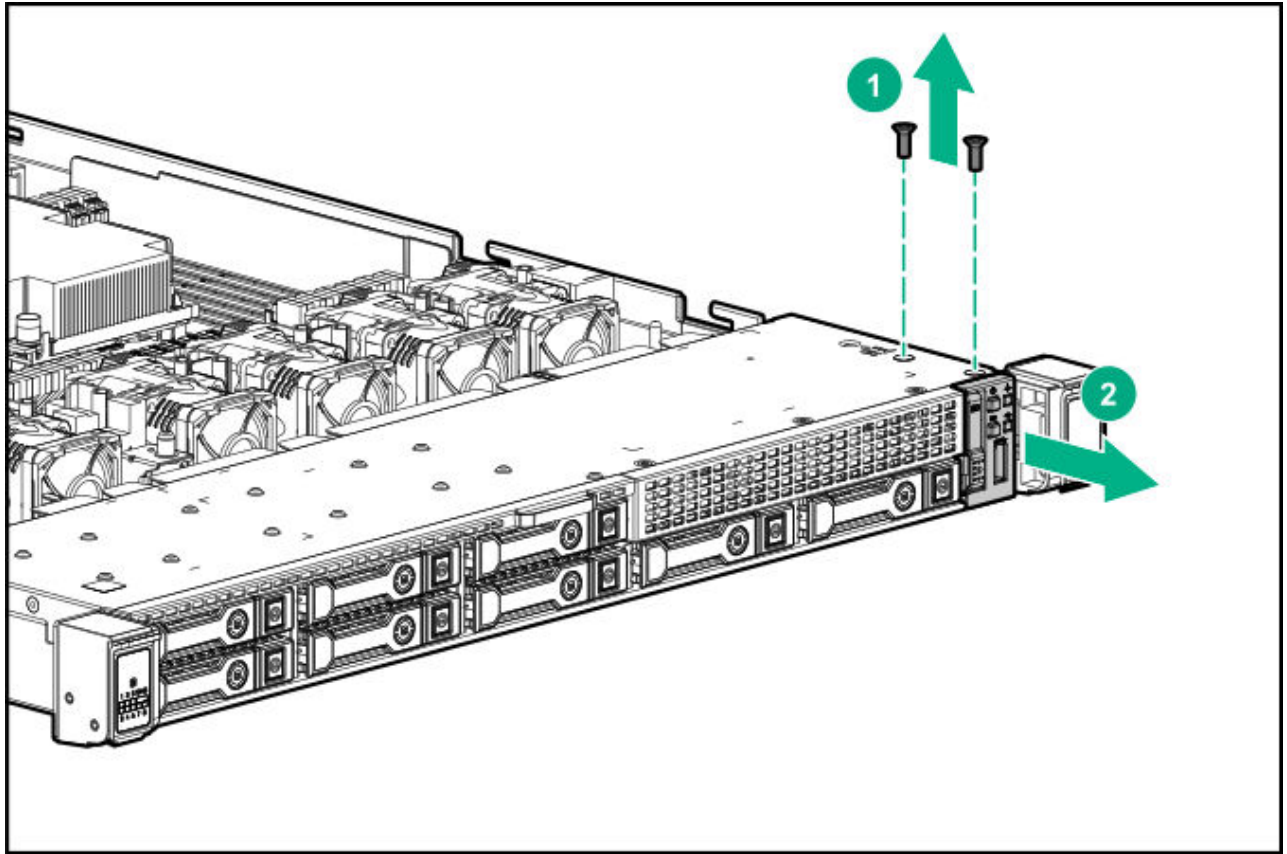
Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.
4. Remove the access panel.
5. Disconnect the cable from the system board.

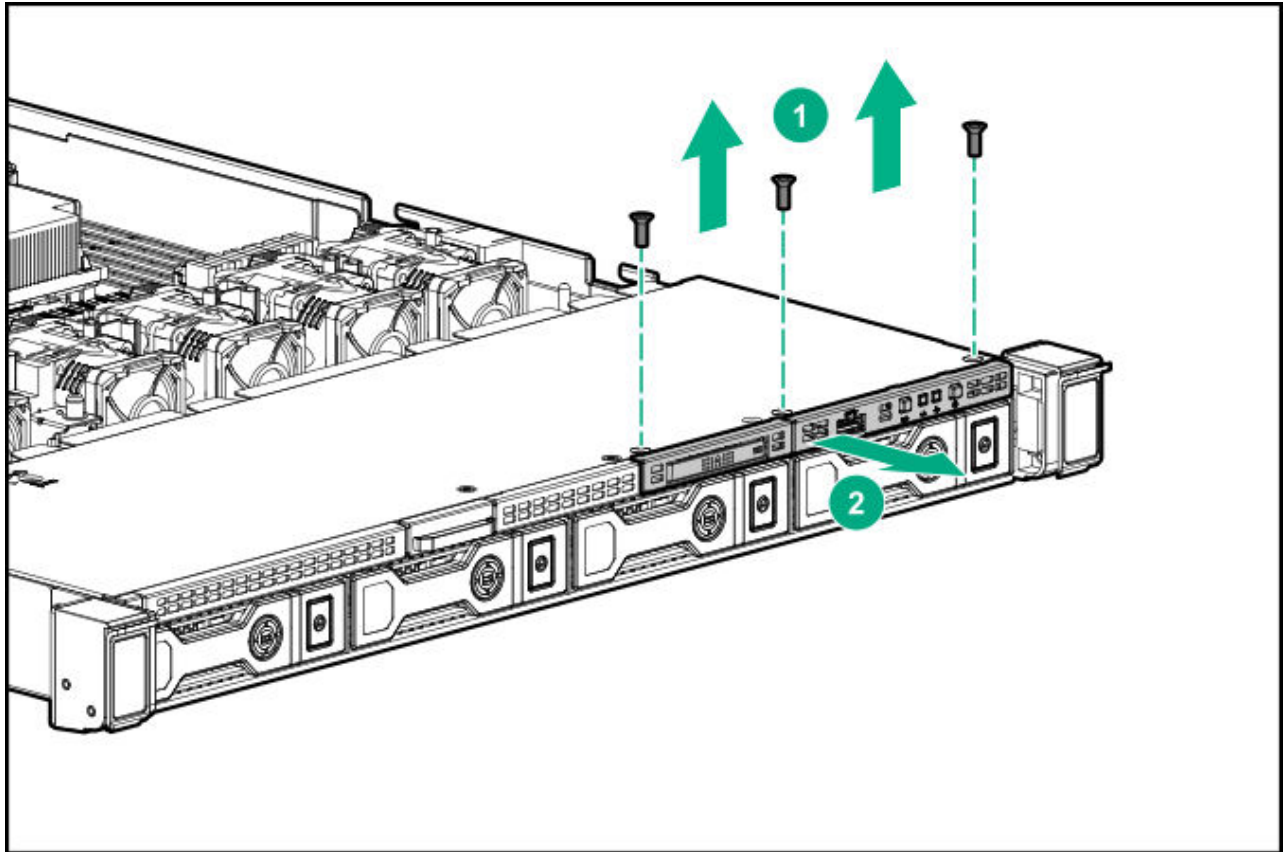


6. Remove the component.

SFF



LFF



Results

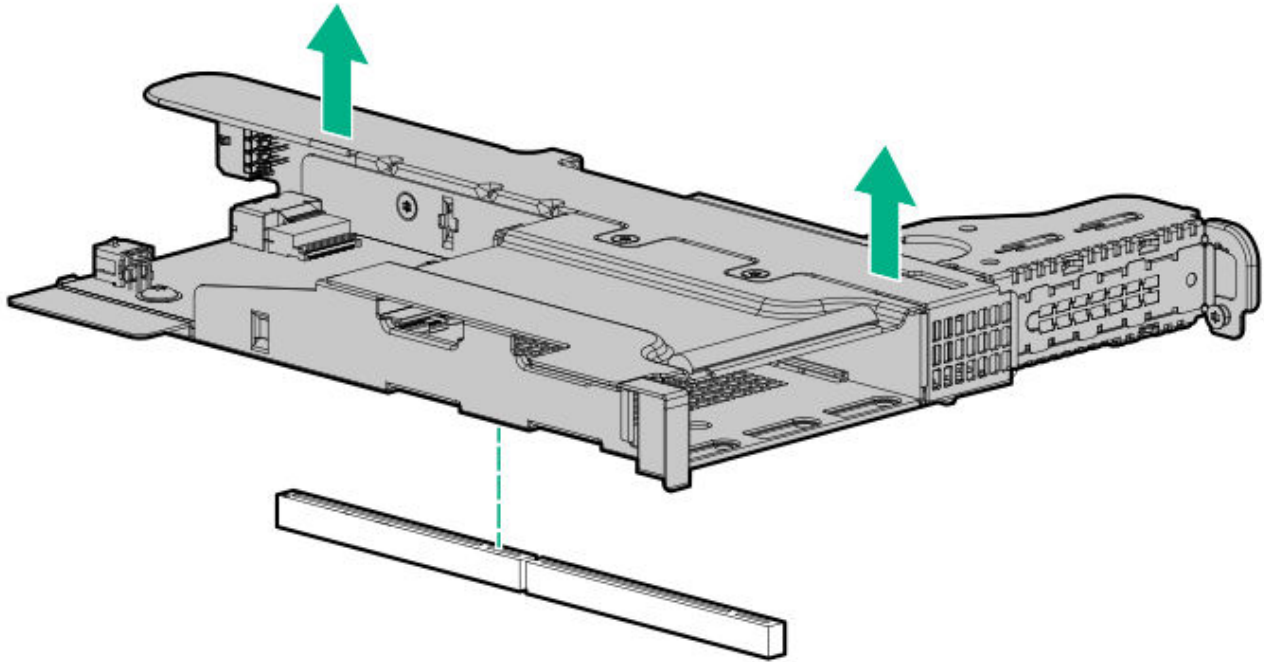
To replace the component, reverse the removal procedure.

Removing and replacing the rear SFF drive cage

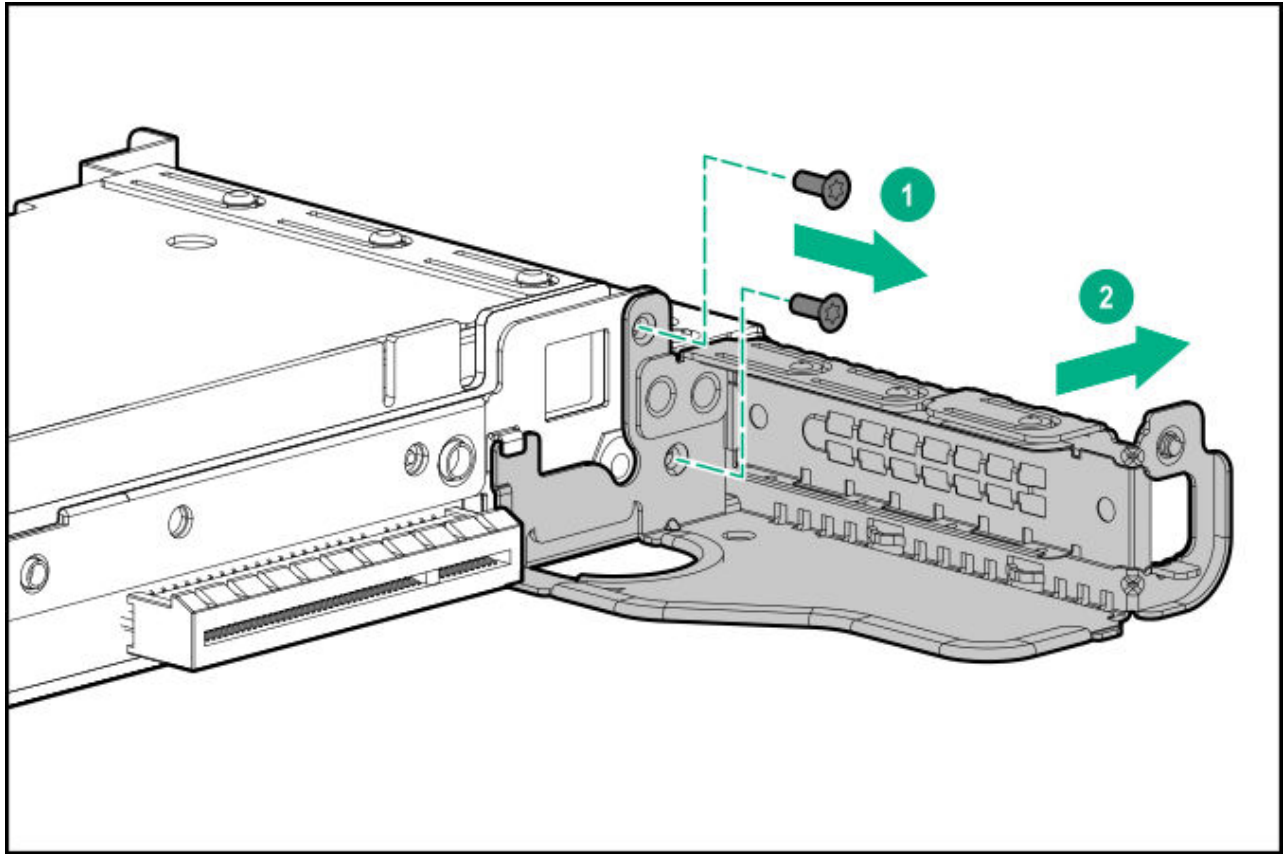
Procedure

1. Power down the server.
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack.
 - b. Remove the server from the rack.

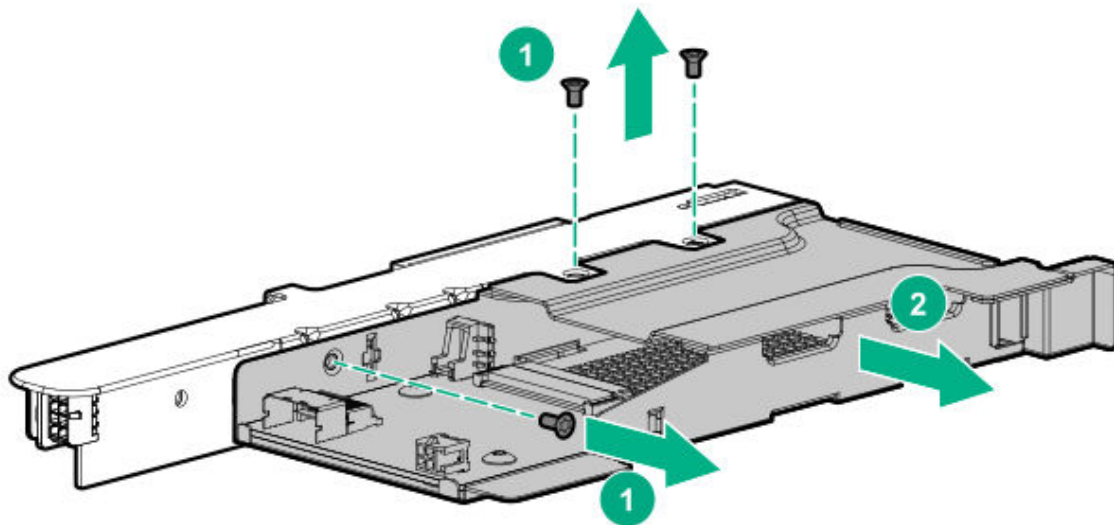
4. Remove the access panel.
5. Remove the drive or drive blank from the drive cage.
6. Disconnect all cables from the drive backplane.
7. Remove the drive cage.



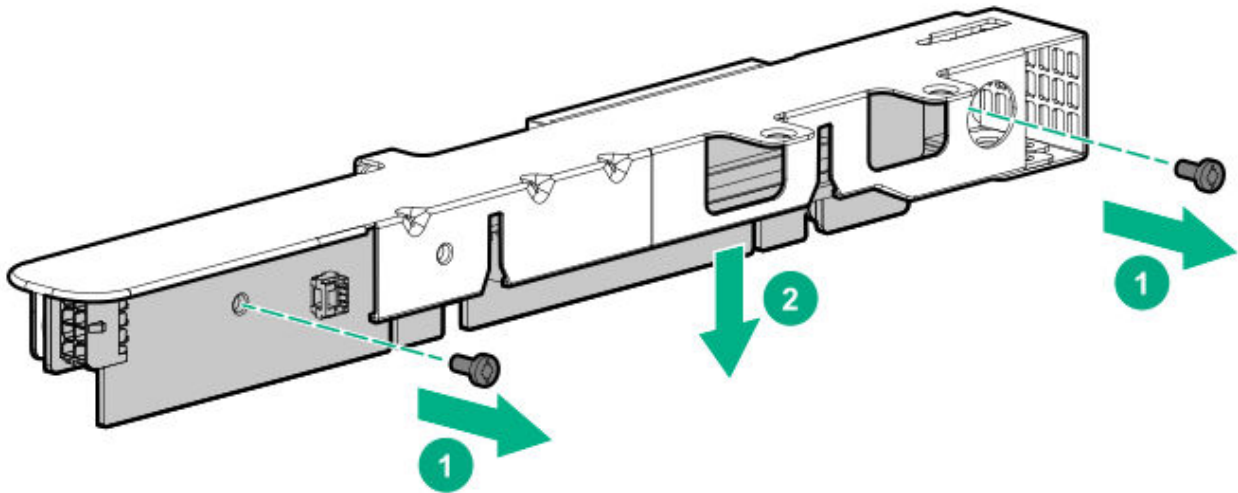
8. Remove the slot 2 bracket from the riser cage.



9. Remove the drive cage from the riser cage.



10. Remove the riser from the riser cage.



Results

To replace the component, reverse the removal procedure.

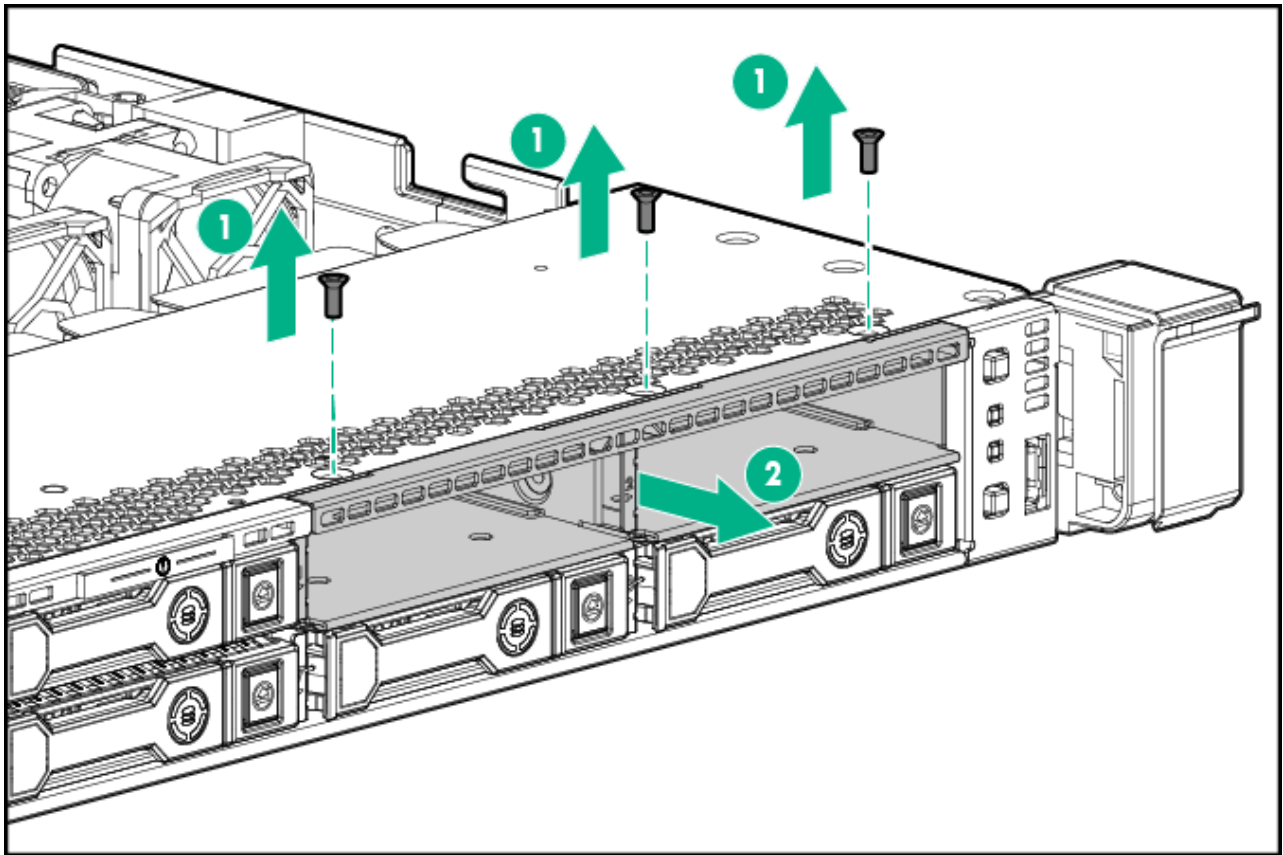
Removing and replacing the 2 SFF drive cage

Procedure

1. Power down the server ([Power down the server](#)).
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack ([Extend the server from the rack](#)).
 - b. Remove the server from the rack ([Remove the server from the rack](#)).
4. Remove the access panel ([Remove the access panel](#)).
5. Remove drives and drive blanks.

Note where each drive is located.
6. Disconnect all cables from the drive backplane.

7. Remove the drive cage.



Results

To replace the component, reverse the removal procedure.

Removing and replacing the 4 LFF drive backplane

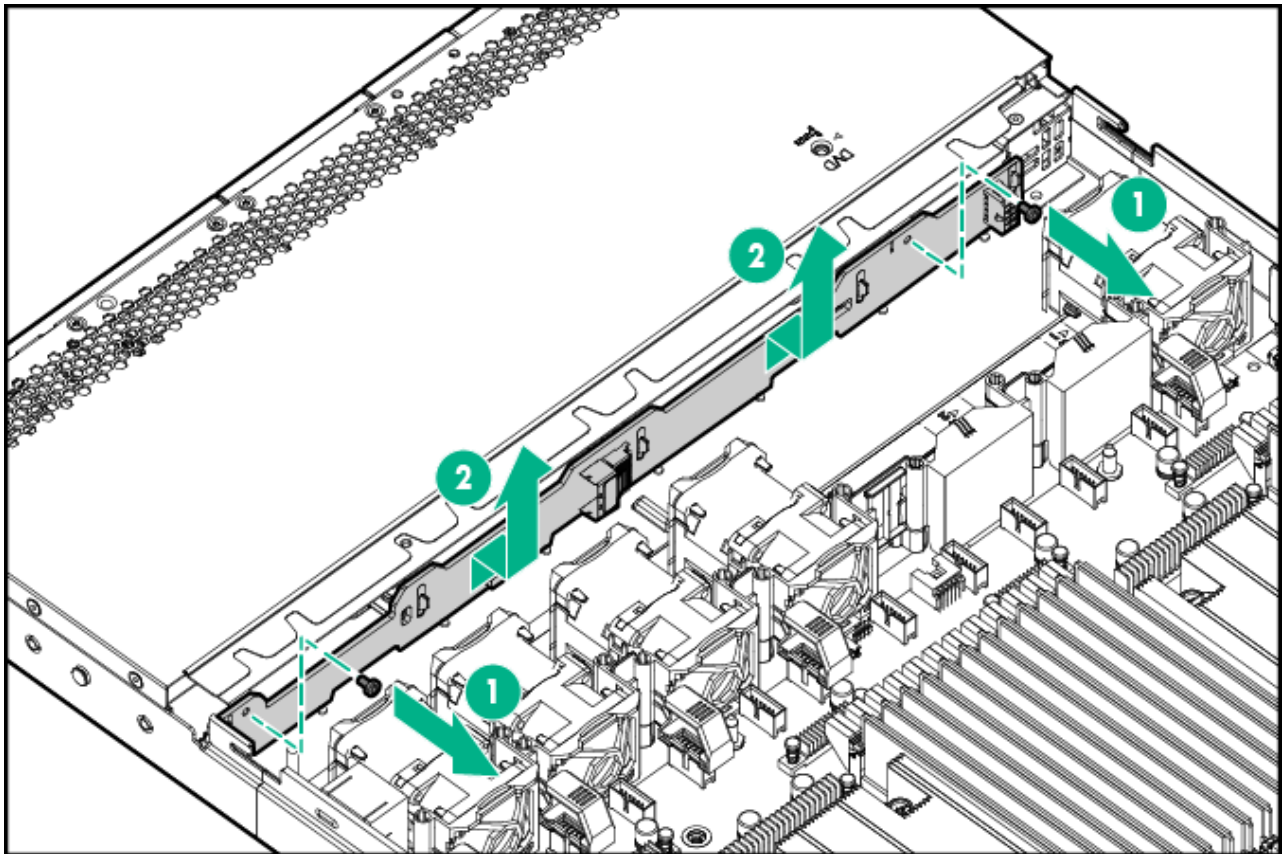
Procedure

1. Power down the server ([Power down the server](#)).
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:

- a. Extend the server from the rack ([Extend the server from the rack](#)).
- b. Remove the server from the rack ([Remove the server from the rack](#)).
4. Remove the access panel ([Remove the access panel](#)).
5. Remove all drives and drive blanks.

Note where each drive is located.

6. If installed, remove the Smart Storage Battery ([Removing and replacing the Smart Storage Battery or Hybrid Capacitor](#)).
7. Disconnect all cables from the drive backplane.
8. Remove the drive backplane.



Results

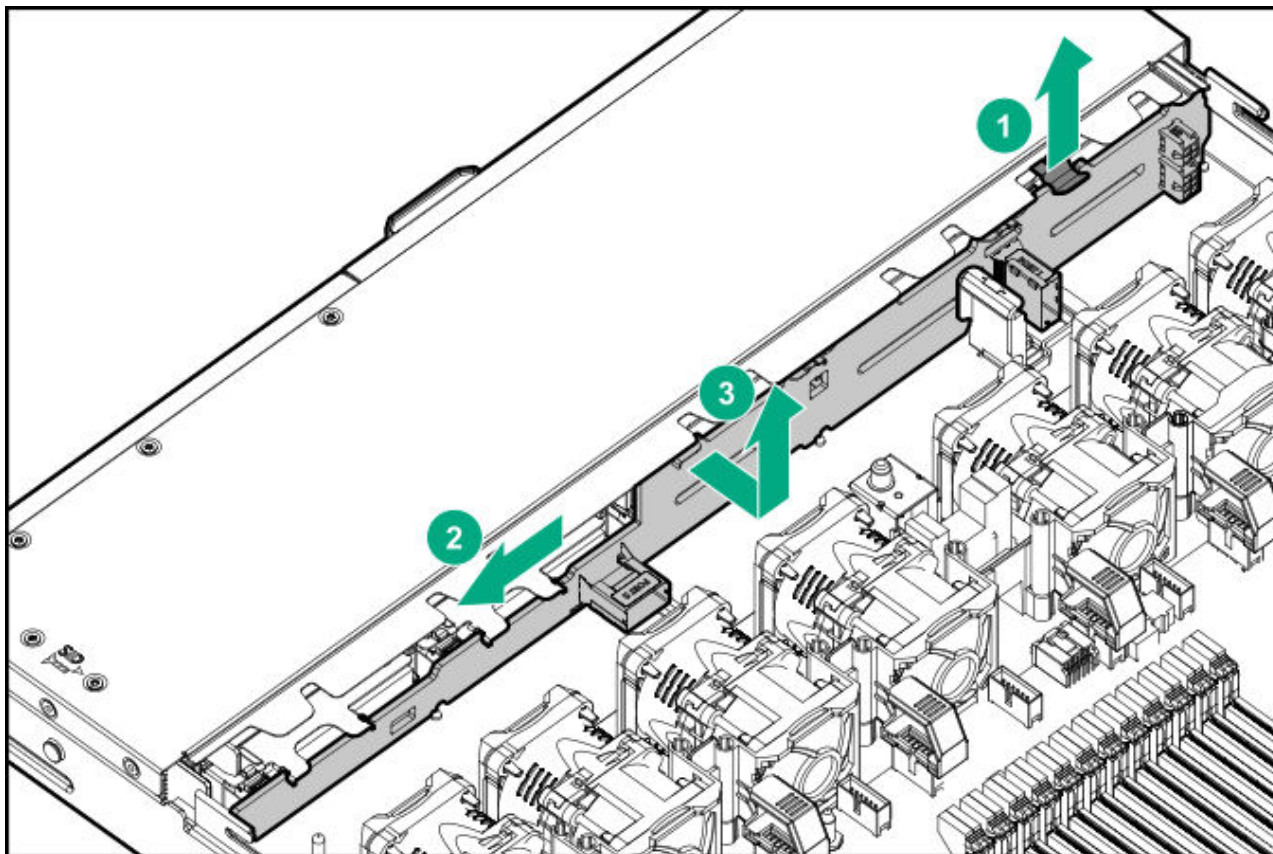
To replace the component, reverse the removal procedure.

Removing and replacing the 8 SFF drive backplane

Procedure

1. Power down the server ([Power down the server](#)).
2. Remove all power:
 - a. Disconnect each power cord from the power source.
 - b. Disconnect each power cord from the server.
3. Do one of the following:
 - a. Extend the server from the rack ([Extend the server from the rack](#)).
 - b. Remove the server from the rack ([Remove the server from the rack](#)).
4. Remove the access panel ([Remove the access panel](#)).
5. Remove all drives and drive blanks.

Note where each drive is located.
6. If installed, remove the Smart Storage Battery ([Removing and replacing the Smart Storage Battery or Hybrid Capacitor](#)).
7. Disconnect and remove all cables from the drive backplane.
8. Remove the drive backplane.



Results

To replace the component, reverse the removal procedure.

Removing and replacing the 10 SFF drive backplane

About this task



CAUTION

To prevent damage to electrical components, take the appropriate antistatic precautions before beginning any system installation. Improper grounding can cause electrostatic discharge.

Procedure

1. Back up all server data.
2. Power down the server ([Power down the server](#)).
3. Do one of the following:

- Extend the server from the rack ([Extend the server from the rack](#)).
- Remove the server from the rack ([Remove the server from the rack](#)).

4. Remove the access panel ([Remove the access panel](#)).

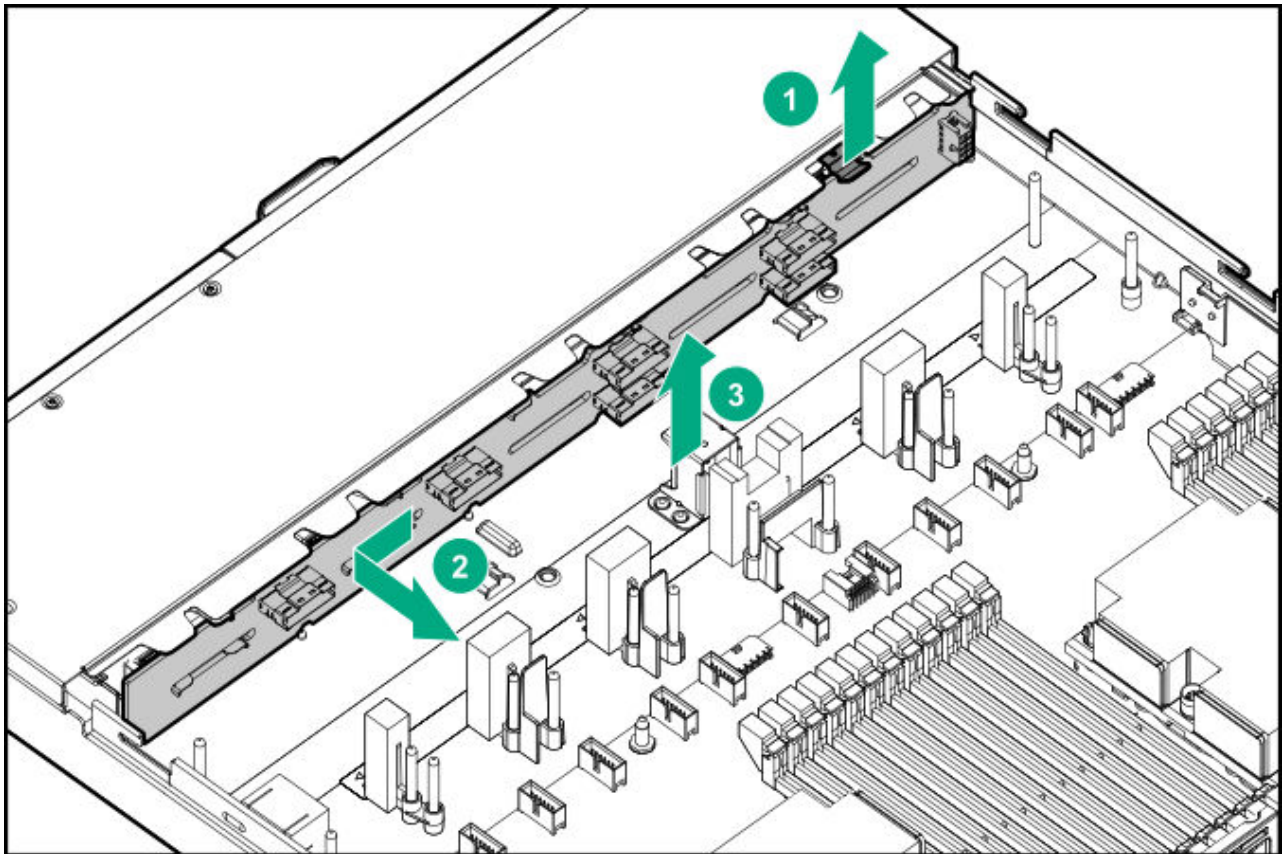
5. Remove all drives and drive blanks.

Note where each drive is located.

6. If installed, remove the Smart Storage Battery ([Removing and replacing the Smart Storage Battery or Hybrid Capacitor](#)).

7. Disconnect all cables from the drive backplane.

8. Remove the drive backplane.



Results

To replace the component, reverse the removal procedure.

HPE Trusted Platform Module 2.0 Gen10 Option

The HPE Trusted Platform Module 2.0 Gen10 Option is not a customer-removable part.



CAUTION

If the TPM is removed from the original server and powered up on a different server, data stored in the TPM including keys will be erased.

If you suspect a TPM board failure, leave the TPM installed and remove the system board. Contact a Hewlett Packard Enterprise authorized service provider for a replacement system board and TPM board.

Troubleshooting

Subtopics

Troubleshooting resources

Troubleshooting resources

Troubleshooting resources are available for HPE Gen10 and Gen10 Plus server products in the following documents:

- Troubleshooting Guide for HPE ProLiant Gen10 and Gen10 Plus servers provides procedures for resolving common problems and comprehensive courses of action for fault isolation and identification, issue resolution, and software maintenance.
- Error Message Guide for HPE ProLiant Gen10 servers and HPE Synergy provides a list of error messages and information to assist with interpreting and resolving error messages.
- Error Message Guide for HPE ProLiant Gen10 Plus servers and HPE Synergy provides a list of error messages and information to assist with interpreting and resolving error messages.
- Integrated Management Log Messages and Troubleshooting Guide for HPE ProLiant Gen10 and Gen10 Plus servers and HPE Synergy provides IML messages and associated troubleshooting information to resolve critical and cautionary IML events.

To access troubleshooting resources for your product, see the Hewlett Packard Enterprise Information Library:

- For Gen10 servers, see <https://www.hpe.com/info/gen10-troubleshooting>.
- For Gen10 Plus servers, see <https://www.hpe.com/info/gen10plus-troubleshooting>.

Diagnostic tools

Subtopics

Product QuickSpecs

UEFI System Utilities

Intelligent Provisioning

HPE Insight Remote Support

HPE InfoSight for servers

USB support

HPE Smart Storage Administrator

Product QuickSpecs

For more information about product features, specifications, options, configurations, and compatibility, see the product QuickSpecs on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/qs>).

UEFI System Utilities

The UEFI System Utilities is embedded in the system ROM. Its features enable you to perform a wide range of configuration activities, including:

- Configuring system devices and installed options.
- Enabling and disabling system features.
- Displaying system information.
- Selecting the primary boot controller or partition.
- Configuring memory options.
- Launching other preboot environments.

HPE servers with UEFI can provide:

- Support for boot partitions larger than 2.2 TB. Such configurations could previously only be used for boot drives when using RAID solutions.
- Secure Boot that enables the system firmware, option card firmware, operating systems, and software collaborate to enhance platform security.

- UEFI Graphical User Interface (GUI)
- An Embedded UEFI Shell that provides a preboot environment for running scripts and tools.
- Boot support for option cards that only support a UEFI option ROM.

Subtopics

Selecting the boot mode

Secure Boot

Launching the Embedded UEFI Shell

Selecting the boot mode

About this task

This server provides two **Boot Mode** configurations: UEFI Mode and Legacy BIOS Mode. Certain boot options require that you select a specific boot mode. By default, the boot mode is set to **UEFI Mode**. The system must boot in **UEFI Mode** to use certain options, including:

- Secure Boot, UEFI Optimized Boot, Generic USB Boot, IPv6 PXE Boot, iSCSI Boot, and Boot from URL
- Fibre Channel/FCoE Scan Policy



NOTE

The boot mode you use must match the operating system installation. If not, changing the boot mode can impact the ability of the server to boot to the installed operating system.

Prerequisite

When booting to **UEFI Mode**, leave **UEFI Optimized Boot** enabled.

Procedure

1. From the **System Utilities** screen, select **System Configuration > BIOS/Platform Configuration (RBSU) > Boot Options > Boot Mode**.
2. Select a setting.
 - **UEFI Mode** (default)—Configures the system to boot to a UEFI compatible operating system.
 - **Legacy BIOS Mode**—Configures the system to boot to a traditional operating system in Legacy BIOS compatibility mode.
3. Save your setting.

4. Reboot the server.

Secure Boot

Secure Boot is a server security feature that is implemented in the BIOS and does not require special hardware. Secure Boot ensures that each component launched during the boot process is digitally signed and that the signature is validated against a set of trusted certificates embedded in the UEFI BIOS. Secure Boot validates the software identity of the following components in the boot process:

- UEFI drivers loaded from PCIe cards
- UEFI drivers loaded from mass storage devices
- Preboot UEFI Shell applications
- OS UEFI boot loaders

When Secure Boot is enabled:

- Firmware components and operating systems with boot loaders must have an appropriate digital signature to execute during the boot process.
- Operating systems must support Secure Boot and have an EFI boot loader signed with one of the authorized keys to boot. For more information about supported operating systems, see <https://www.hpe.com/servers/ossupport>.

You can customize the certificates embedded in the UEFI BIOS by adding or removing your own certificates, either from a management console directly attached to the server, or by remotely connecting to the server using the iLO Remote Console.

You can configure Secure Boot:

- Using the **System Utilities** options described in the following sections.
- Using the iLO RESTful API to clear and restore certificates. For more information, see the Hewlett Packard Enterprise website (<https://www.hpe.com/info/redfish>).
- Using the `secboot` command in the Embedded UEFI Shell to display Secure Boot databases, keys, and security reports.

Launching the Embedded UEFI Shell

Prerequisites

- **Embedded UEFI Shell** is set to **Enabled**.

About this task

Use the **Embedded UEFI Shell** option to launch the Embedded UEFI Shell. The Embedded UEFI Shell is a preboot command-line environment for scripting and running UEFI applications, including UEFI boot loaders. The Shell also provides CLI-based commands you can use to obtain system information, and to configure and update the system BIOS.

Procedure

1. From the **System Utilities** screen, select **Embedded Applications > Embedded UEFI Shell**.

The **Embedded UEFI Shell** screen appears.

2. Press any key to acknowledge that you are physically present.

This step ensures that certain features, such as disabling **Secure Boot** or managing the **Secure Boot** certificates using third-party UEFI tools, are not restricted.

3. If an administrator password is set, enter it at the prompt and press **Enter**.

The `Shell>` prompt appears.

4. Enter the commands required to complete your task.
5. Enter the `exit` command to exit the Shell.

Intelligent Provisioning

Intelligent Provisioning is a single-server deployment tool embedded in ProLiant servers and HPE Synergy compute modules. Intelligent Provisioning simplifies server setup, providing a reliable and consistent way to deploy servers.



NOTE

- Rapid Setup Software is only available on the ProLiant 300 Series servers or below. When you launch F10 mode from the POST screen, you are prompted to select whether you want to enter the Intelligent Provisioning or HPE Rapid Setup Software mode.
- After you have selected a mode, you must reprovision the server to change the mode that launches when you boot to F10.

Intelligent Provisioning prepares the system for installing original, licensed vendor media and Hewlett Packard Enterprise-branded versions of OS software. Intelligent Provisioning also prepares the system to integrate optimized server support software from the Service Pack for ProLiant (SPP). SPP is a comprehensive systems software and firmware solution for ProLiant servers, server blades, their enclosures, and HPE Synergy compute modules. These components are preloaded with a basic set of firmware and OS components that are installed along with Intelligent Provisioning.



IMPORTANT

HPE ProLiant DX/XL servers do not support operating system installation with Intelligent Provisioning, but they do support the maintenance features. For more information, see "Performing Maintenance" in the Intelligent Provisioning user guide and online help.

After the server is running, you can update the firmware to install additional components. You can also update any components that have been outdated since the server was manufactured.

To access Intelligent Provisioning:

- Press **F10** from the POST screen and enter either Intelligent Provisioning or HPE Rapid Setup Software.
- From the iLO web interface using **Lifecycle Management**. **Lifecycle Management** allows you to access Intelligent Provisioning without rebooting your server.

Subtopics

Intelligent Provisioning operation

Intelligent Provisioning operation



NOTE

Intelligent Provisioning 3.40 and later requires iLO firmware version 2.10 or later.

Intelligent Provisioning includes the following components:

- Critical boot drivers
- Active Health System (AHS)
- Erase Utility
- Deployment Settings



IMPORTANT

- Although your server is preloaded with firmware and drivers, Hewlett Packard Enterprise recommends updating the firmware upon initial setup. Also, downloading and updating the latest version of Intelligent Provisioning ensures the latest supported features are available.
- For ProLiant servers, firmware is updated using the Intelligent Provisioning Firmware Update utility.
- Do not update firmware if the version you are currently running is required for compatibility.



NOTE

Intelligent Provisioning does not function within multihomed configurations. A multihomed host is one that is connected to two or more networks or has two or more IP addresses.

Intelligent Provisioning provides installation help for the following operating systems:

- Microsoft Windows Server
- Red Hat Enterprise Linux
- SUSE Linux Enterprise Server
- VMware ESXi/vSphere Custom Image
- ClearOS

Not all versions of an OS are supported. For information about specific versions of a supported operating system, see the OS Support Matrix on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/ossupport>).

HPE Insight Remote Support

Hewlett Packard Enterprise strongly recommends that you register your device for remote support to enable enhanced delivery of your Hewlett Packard Enterprise warranty, HPE support services, or Hewlett Packard Enterprise contractual support agreement. Insight Remote Support supplements your monitoring continuously to ensure maximum system availability by providing intelligent event diagnosis, and automatic, secure submission of hardware event notifications to Hewlett Packard Enterprise, which will initiate a fast and accurate resolution, based on your product's service level. Notifications can be sent to your authorized Hewlett Packard Enterprise Channel Partner for onsite service, if configured and available in your country.

For more information, see Insight Remote Support and Insight Online Setup Guide for ProLiant Servers and BladeSystem c-Class Enclosures on the [Hewlett Packard Enterprise website](#). Insight Remote Support is available as part of Hewlett Packard Enterprise Warranty, HPE support services, or Hewlett Packard Enterprise contractual support agreement.

HPE InfoSight for servers

The HPE InfoSight portal is a secure web interface hosted by HPE that allows you to monitor supported devices through a graphical interface.

HPE InfoSight for servers:

- Combines the machine learning and predictive analytics of HPE InfoSight with the health and performance monitoring of Active Health System (AHS) and HPE iLO to optimize performance and predict and prevent problems
- Provides automatic collection and analysis of the sensor and telemetry data from AHS to derive insights from the behaviors of the install base to provide recommendations to resolve problems and improve performance

For more information on getting started and using HPE InfoSight for servers, go to: <https://www.hpe.com/info/infosight-servers-docs>.

USB support

Hewlett Packard Enterprise Gen10 and Gen10 Plus servers support all USB operating speeds depending on the device that is connected to the server.

Subtopics

External USB functionality

External USB functionality

Hewlett Packard Enterprise provides external USB support to enable local connection of USB devices for server administration, configuration, and diagnostic procedures.

For additional security, external USB functionality can be disabled through USB options in UEFI System Utilities.

HPE Smart Storage Administrator

HPE SSA is the main tool for configuring arrays on HPE Smart Array SR controllers. It exists in three interface formats: the HPE SSA GUI, the HPE SSA CLI, and HPE SSA Scripting. All formats provide support for configuration tasks. Some of the advanced tasks are available in only one format.

The diagnostic features in HPE SSA are also available in the standalone software HPE Smart Storage Administrator Diagnostics Utility CLI.

During the initial provisioning of the server or compute module, an array is required to be configured before the operating system can be installed. You can configure the array using SSA.

HPE SSA is accessible both offline (either through HPE Intelligent Provisioning or as a standalone bootable ISO image) and online:

- Accessing HPE SSA in the offline environment



IMPORTANT

If you are updating an existing server in an offline environment, obtain the latest version of HPE SSA through Service Pack for ProLiant before performing configuration procedures.

Using one of multiple methods, you can run HPE SSA before launching the host operating system. In offline mode, users can configure or maintain detected and supported devices, such as optional Smart

Array controllers and integrated Smart Array controllers. Some HPE SSA features are only available in the offline environment, such as setting the boot controller and boot volume.

- Accessing HPE SSA in the online environment
This method requires an administrator to download the HPE SSA executables and install them. You can run HPE SSA online after launching the host operating system.

For more information, see HPE Smart Array SR Gen10 Configuration Guide at the [**Hewlett Packard Enterprise website**](#).

Component identification

Subtopics

[**Front panel components**](#)

[**Front panel LEDs and buttons**](#)

[**Systems Insight Display LEDs**](#)

[**Systems Insight Display combined LED descriptions**](#)

[**Rear panel components**](#)

[**Rear panel LEDs**](#)

[**System board components**](#)

[**Processor, heatsink, and socket components**](#)

[**Device numbers**](#)

[**Hot-plug drive LED definitions**](#)

[**Smart Carrier NVMe \(SCN\) drive LED definitions**](#)

[**HPE Basic Drive LED definitions**](#)

[**uFF drive components and LEDs**](#)

[**Hot-plug fans**](#)

[**HPE Smart Array P824i-p MR Gen10 Controller**](#)

[**HPE NS204i-p NVMe OS Boot Device components**](#)

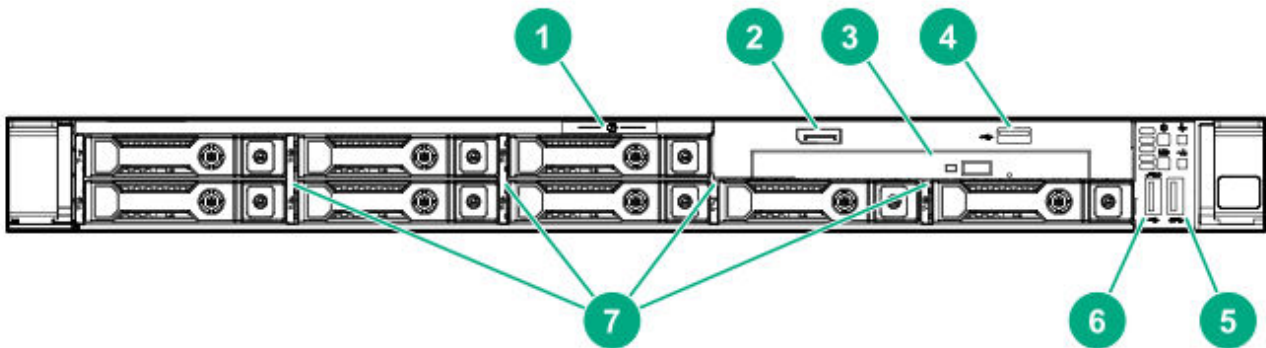
[**HPE NS204i-p NVMe OS Boot Device LED definitions**](#)

[**HPE InfiniBand HDR/Ethernet 940QSFP 56x16 adapter LEDs**](#)

[**DSC-25 2-port SFP28 card ports and LEDs**](#)

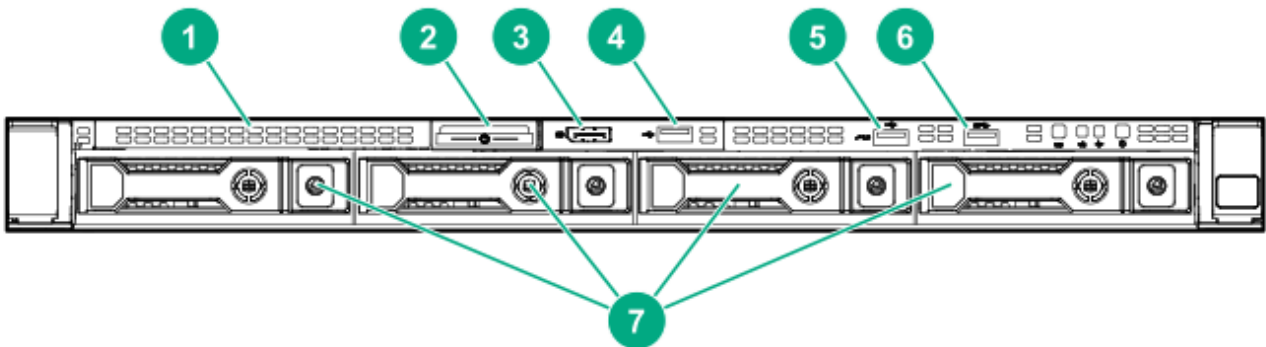
Front panel components

8 SFF



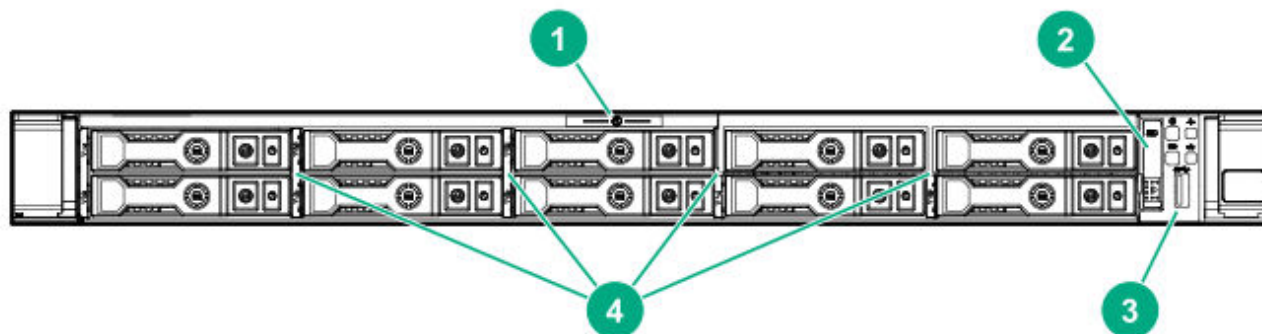
Item	Description
1	Serial label pull tab
2	Display port (optional)
3	Optical drive (optional)
4	USB 2.0 port (optional)
5	USB 3.0 port
6	iLO Service Port The operating system does not recognize this port as a USB port.
7	SAS/SATA drive bays

4 LFF



Item	Description
1	Optical drive blank (optional)
2	Serial label pull tab
3	Display port (optional)
4	USB 2.0 port (optional)
5	iLO Service Port The operating system does not recognize this port as a USB port.
6	USB 3.0 port
7	SAS/SATA drive bays

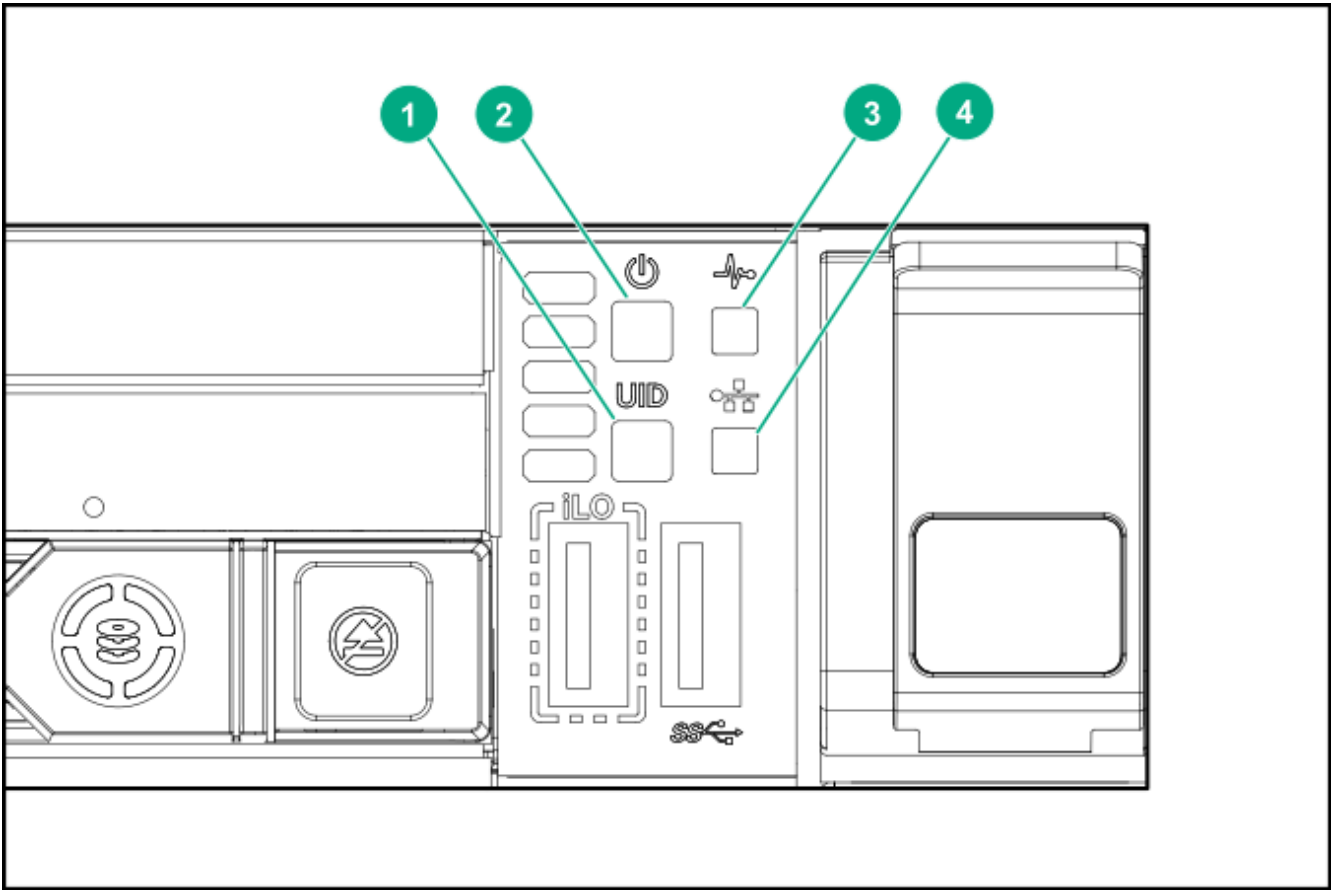
10 SFF NVMe/SAS Combo



Item	Description
1	Serial label pull tab
2	Systems Insight Display (optional)
3	USB 3.0 port
4	SAS/SATA/NVMe drive bays When the 10 SFF NVMe/SAS backplane option is installed, NVMe drives must be installed in bays 9 and 10. The other bays support a mix of NVMe and SAS drives.

Front panel LEDs and buttons

8 SFF/10 SFF

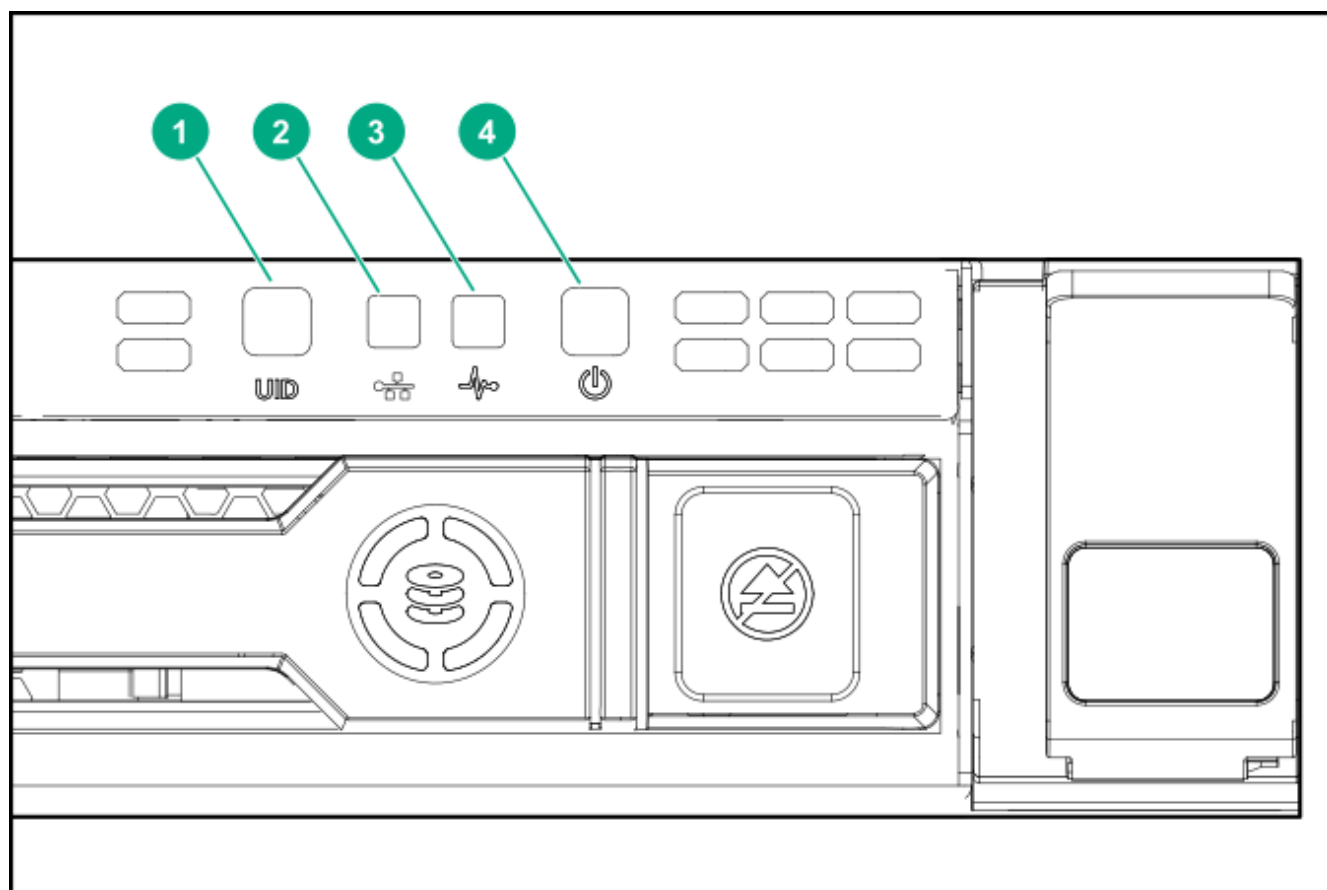


Item	Description	Status
1	UID button/LED ¹	Solid blue = Activated Flashing blue: • 1 Hz = Remote management or firmware upgrade in progress • 4 Hz = iLO manual reboot sequence initiated • 8 Hz = iLO manual reboot sequence in progress Off = Deactivated
2	Power On/Standby button and system power LED ¹	Solid green = System on Flashing green = Performing power on sequence Solid amber = System in standby Off = No power present ²

Item	Description	Status
3	Health LED ¹	Solid green = Normal Flashing green = iLO is rebooting. Flashing amber = System degraded Flashing red = System critical ³
4	NIC status LED ¹	Solid green = Link to network Flashing green = Network active Off = No network activity

- ¹ When all four LEDs described in this table flash simultaneously, a power fault has occurred.
- ² Facility power is not present, power cord is not attached, no power supplies are installed, power supply failure has occurred, or the power button cable is disconnected.
- ³ If the health LED indicates a degraded or critical state, review the system IML or use iLO to review the system health status.

4 LFF



Item	Description	Status
1	UID button/LED ¹ _—	Solid blue = Activated. Flashing blue: • 1 Hz = Remote management or firmware upgrade in progress. • 4 Hz = iLO manual reboot sequence initiated. • 8 Hz = iLO manual reboot sequence in progress. Off = Deactivated.
2	NIC status LED ¹	Solid green = Link to network. Flashing green = Network active. Off = No network activity.
3	Health LED ¹	Solid green = Normal. Flashing green = iLO is rebooting. Flashing amber = System degraded. Flashing red = System critical. ²_—
4	Power On/Standby button and system power LED ¹	Solid green = System on. Flashing green = Performing power on sequence. Solid amber = System in standby. Off = No power present. ³_—

¹ When all four LEDs described in this table flash simultaneously, a power fault has occurred.

² To identify components in a degraded or critical state, see the Systems Insight Display LEDs, check iLO/BIOS logs, and reference the server troubleshooting guide.

³ Facility power is not present, power cord is not attached, no power supplies are installed, power supply failure has occurred, or the power button cable is disconnected.

Subtopics

UID button functionality

Front panel LED power fault codes

UID button functionality

The UID button can be used to display the Server Health Summary when the server will not power on. For more information, see the latest HPE iLO 5 User Guide on the [Hewlett Packard Enterprise website](#).

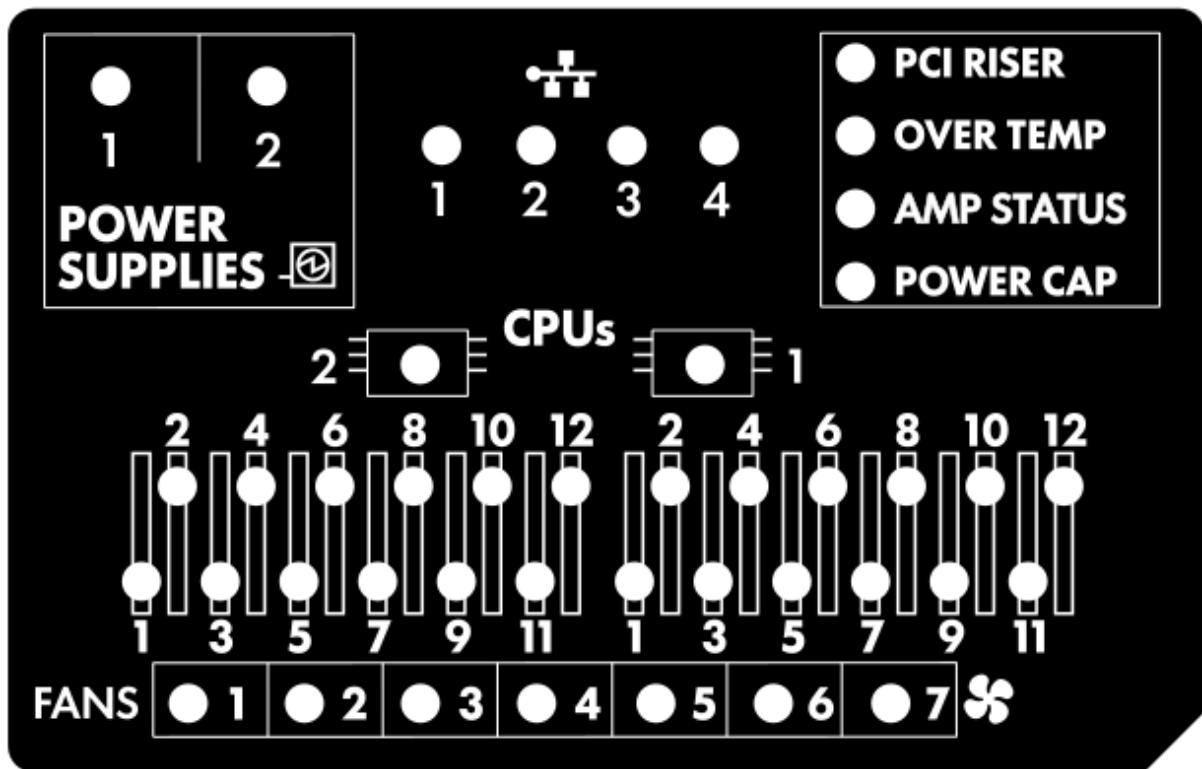
Front panel LED power fault codes

The following table provides a list of power fault codes, and the subsystems that are affected. Not all power faults are used by all servers.

Subsystem	LED behavior
System board	1 flash
Processor	2 flashes
Memory	3 flashes
Riser board PCIe slots	4 flashes
FlexibleLOM	5 flashes
Storage controller	6 flashes
System board PCIe slots	7 flashes
Power backplane	8 flashes
Storage backplane	9 flashes
Power supply	10 flashes
PCIe expansion cards installed in riser board	11 flashes
Chassis	12 flashes
GPU card	13 flashes

Systems Insight Display LEDs

The Systems Insight Display LEDs represent the system board layout. The display enables diagnosis with the access panel installed.



Description	Status
Processor LEDs	Off = Normal Amber = Failed processor
DIMM LEDs	Off = Normal Amber = Failed DIMM or configuration issue
Fan LEDs	Off = Normal Amber = Failed fan or missing fan
NIC LEDs <u>1</u>	Off = No link to network Solid green = Network link Flashing green = Network link with activity If power is off, the front panel LED is not active. For status, see Rear panel LEDs .
Power supply LEDs	Off = Normal

Description	Status
	Solid amber = Power subsystem degraded, power supply failure, or input power lost.
PCI riser LED	Off = Normal Amber = Incorrectly installed PCI riser cage
Over temp LED	Off = Normal Amber = High system temperature detected
Amp Status LED	Off = AMP modes disabled Solid green = AMP mode enabled Solid amber = Failover Flashing amber = Invalid configuration
Power cap LED	Off = System is in standby, or no cap is set. Solid green = Power cap applied

- ¹ For Networking Choice (NC) server models, the embedded NIC ports are not equipped on the server. Therefore, the NIC LEDs on the Systems Insight Display will flash based on the FlexibleLOM network port activity. In the case of a dual-port FlexibleLOM, only NIC LED 1 and 2 will illuminate to correspond with the activity of the respective network ports.

When the health LED on the front panel illuminates either amber or red, the server is experiencing a health event. For more information on the combination of these LEDs, see [Systems Insight Display combined LED descriptions](#).

Systems Insight Display combined LED descriptions

The combined illumination of the following LEDs indicates a system condition:

- Systems Insight Display LEDs
- System power LED
- Health LED

Systems Insight Display LED and Health LED System power Status

LED color	LED	System power LED	Status
Processor (amber)	Red	Amber	One or more of the following conditions might exist: • Processor in socket X has failed. • Processor X is not installed in the socket. • Processor X is unsupported. • ROM detects a failed processor during POST.
Processor (amber)	Amber	Green	Processor in socket X is in a pre-failure condition.
DIMM (amber)	Red	Green	One or more DIMMs have failed.
DIMM (amber)	Amber	Green	DIMM in slot X is in a pre-failure condition.
Over temp (amber)	Amber	Green	The Health Driver has detected a cautionary temperature level.
Over temp (amber)	Red	Amber	The server has detected a hardware critical temperature level.
PCI riser (amber)	Red	Green	The PCI riser cage is not seated properly.
Fan (amber)	Amber	Green	One fan has failed or has been removed.
Fan (amber)	Red	Green	Two or more fans have failed or been removed.
Power supply (amber)	Red	Amber	One or more of the following conditions might exist: • Only one power supply is installed and that power supply is in standby. • Power supply fault. • System board fault.
Power supply (amber)	Amber	Green	One or more of the following conditions might exist: • Redundant power supply is installed and only one power supply is functional. • AC power cord is not plugged into redundant power supply. • Redundant power supply fault.

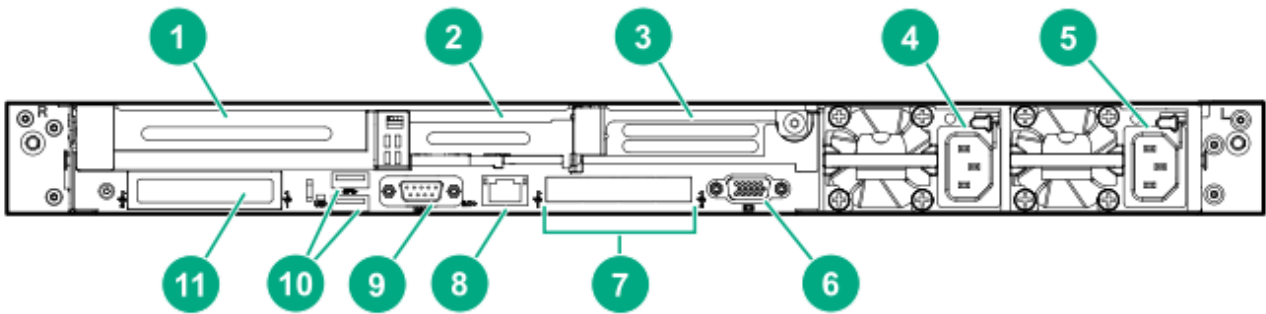
Systems Insight Display LED color	Health LED	System power LED	Status
			Power supply mismatch at POST or power supply mismatch through hot-plugging addition.
Power cap (off)	—	Amber	Standby.
Power cap (green)	—	Flashing green	Waiting for power.
Power cap (green)	—	Green	Power is available.
Power cap (flashing amber)	—	Amber	Power is not available.



IMPORTANT

If more than one DIMM slot LED is illuminated, further troubleshooting is required. Test each bank of DIMMs by removing all other DIMMs. Isolate the failed DIMM by replacing each DIMM in a bank with a known working DIMM.

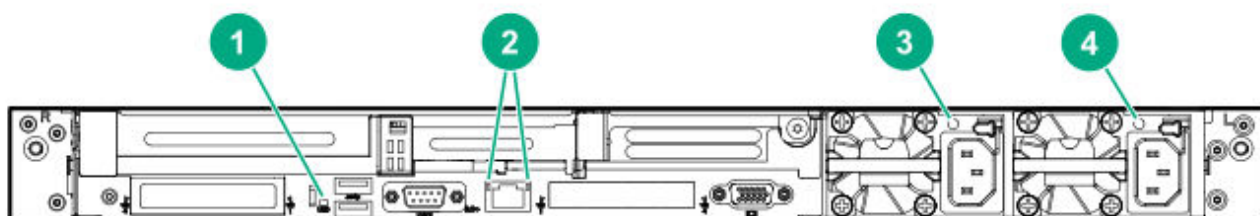
Rear panel components



Item	Description
1	Slot 1 PCIe3
2	Slot 2 PCIe3
3	Slot 3 PCIe3 (optional - requires second processor)
4	Power supply 2 (PS2)
5	Power supply 1 (PS1)

Item	Description
6	Video port
7	NIC ports (if equipped)
8	iLO Management Port
9	Serial port (optional)
10	USB 3.0 ports
11	FlexibleLOM (optional)

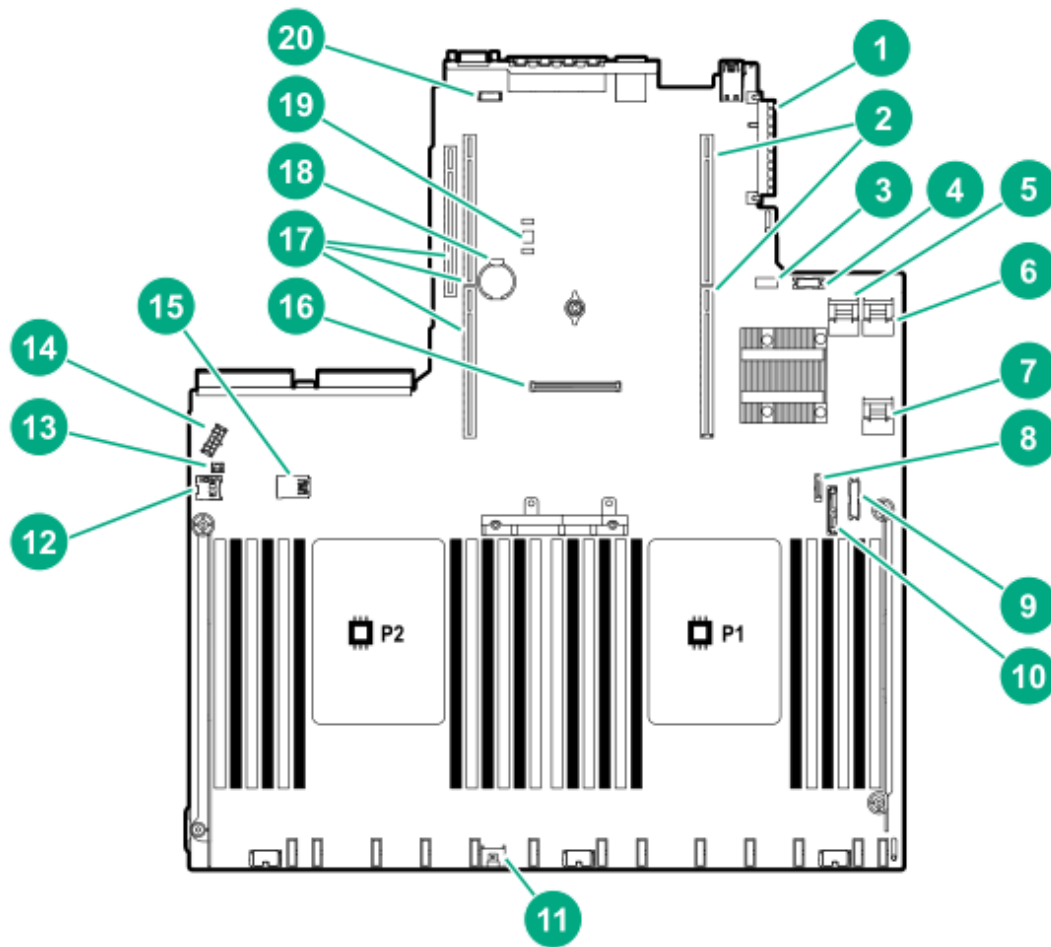
Rear panel LEDs



Item	Description	Status
1	UID LED	Solid blue = Identification is activated. Flashing blue = System is being managed remotely. Off = Identification is deactivated.
2R	iLO 5/standard NIC activity LED	Solid green = Activity exists. Flashing green = Activity exists. Off = No activity exists.
2L	iLO 5/standard NIC link LED	Solid green = Link exists. Off = No link exists.
3	Power supply 2 LED	Solid green = Normal Off = One or more of the following conditions exists:

Item	Description	Status
		• AC power unavailable • Power supply failed • Power supply in standby mode • Power supply exceeded current limit.
4	Power supply 1 LED	Solid green = Normal Off = One or more of the following conditions exists: • AC power unavailable • Power supply failed • Power supply in standby mode • Power supply exceeded current limit.

System board components



Item	Description
1	FlexibleLOM connector
2	Primary (processor 1) PCIe riser connector
3	System maintenance switch
4	Front display port/USB 2.0 connector
5	x4 SATA port 1
6	x4 SATA port 2
7	x2 SATA port 3
8	x1 SATA port 4
9	Front power/USB 3.0 connector
10	Optical/SATA port 5
11	Energy pack connector

Item	Description
12	microSD card slot
13	Chassis Intrusion Detection connector
14	Drive backplane power connector
15	Dual internal USB 3.0 connector
16	Type-a SmartArray connector
17	Secondary (processor 2) PCIe riser connector
18	System battery
19	TPM connector (optional)
20	Serial port connector (optional)

Subtopics

System maintenance switch descriptions

NMI functionality

DIMM slot locations

DIMM label identification

NVDIMM identification

NVDIMM LED identification

Intel Optane persistent memory 100 series for HPE label identification

System maintenance switch descriptions

Position	Default	Function
S1 <u>1</u>	Off	Off = iLO security is enabled. On = iLO security is disabled.
S2	Off	Reserved
S3	Off	Reserved
S4	Off	Reserved
S5 <u>1</u>	Off	Off = Power-on password is enabled. On = Power-on password is disabled.
S6 <u>1</u> , <u>2</u> , <u>3</u>	Off	Off = No function On = Restore default manufacturing settings

Position	Default	Function
S7	Off	Reserved
S8	—	Reserved
S9	—	Reserved
S10	—	Reserved
S11	—	Reserved
S12	—	Reserved

- ¹ To access the redundant ROM, set S1, S5, and S6 to On.
- ² When the system maintenance switch position 6 is set to the On position, the system is prepared to restore all configuration settings to their manufacturing defaults.
- ³ When the system maintenance switch position 6 is set to the On position and Secure Boot is enabled, some configurations cannot be restored. For more information, see [Secure Boot](#).

NMI functionality

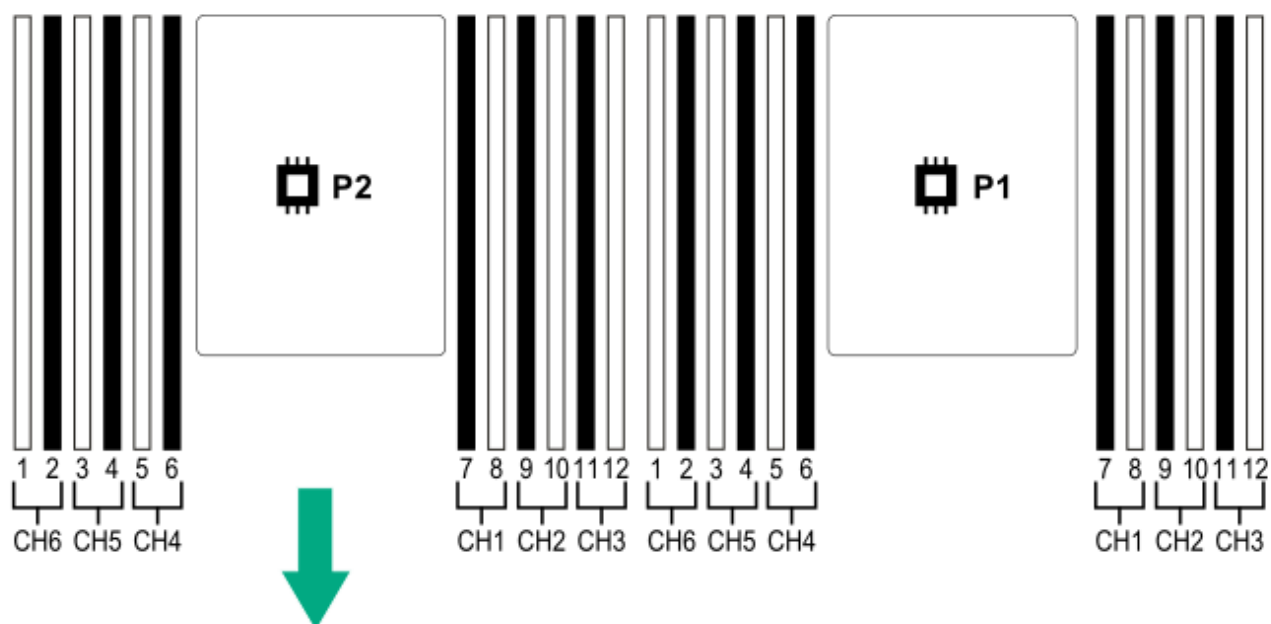
An NMI crash dump enables administrators to create crash dump files when a system is hung and not responding to traditional debugging methods.

An analysis of the crash dump log is an essential part of diagnosing reliability problems, such as hanging operating systems, device drivers, and applications. Many crashes freeze a system, and the only available action for administrators is to cycle the system power. Resetting the system erases any information that could support problem analysis, but the NMI feature preserves that information by performing a memory dump before a hard reset.

To force the OS to initiate the NMI handler and generate a crash dump log, the administrator can use the iLO Generate NMI feature.

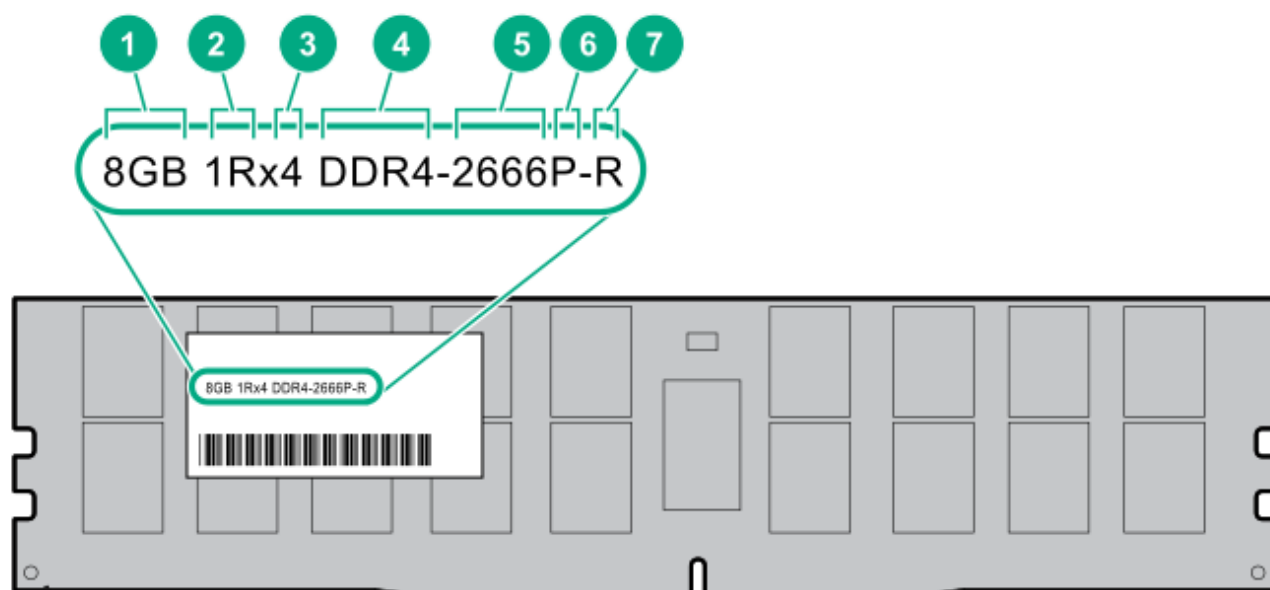
DIMM slot locations

DIMM slots are numbered sequentially (1 through 12) for each processor.



DIMM label identification

To determine DIMM characteristics, see the label attached to the DIMM. The information in this section helps you to use the label to locate specific information about the DIMM.



Item	Description	Example
1	Capacity	8 GB 16 GB 32 GB 64 GB 128 GB
2	Rank	1R = Single rank 2R = Dual rank 4R = Quad rank 8R = Octal rank
3	Data width on DRAM	x4 = 4-bit x8 = 8-bit x16 = 16-bit
4	Memory generation	PC4 = DDR4
5	Maximum memory speed	2133 MT/s 2400 MT/s 2666 MT/s 2933 MT/s
6	CAS latency	P = CAS 15-15-15 T = CAS 17-17-17 U = CAS 20-18-18 V = CAS 19-19-19 (for RDIMM, LRDIMM) V = CAS 22-19-19 (for 3DS TSV LRDIMM) Y = CAS 21-21-21 (for RDIMM, LRDIMM) Y = CAS 24-21-21 (for 3DS TSV LRDIMM)
7	DIMM type	R = RDIMM (registered) L = LRDIMM (load reduced) E = Unbuffered ECC (UDIMM)

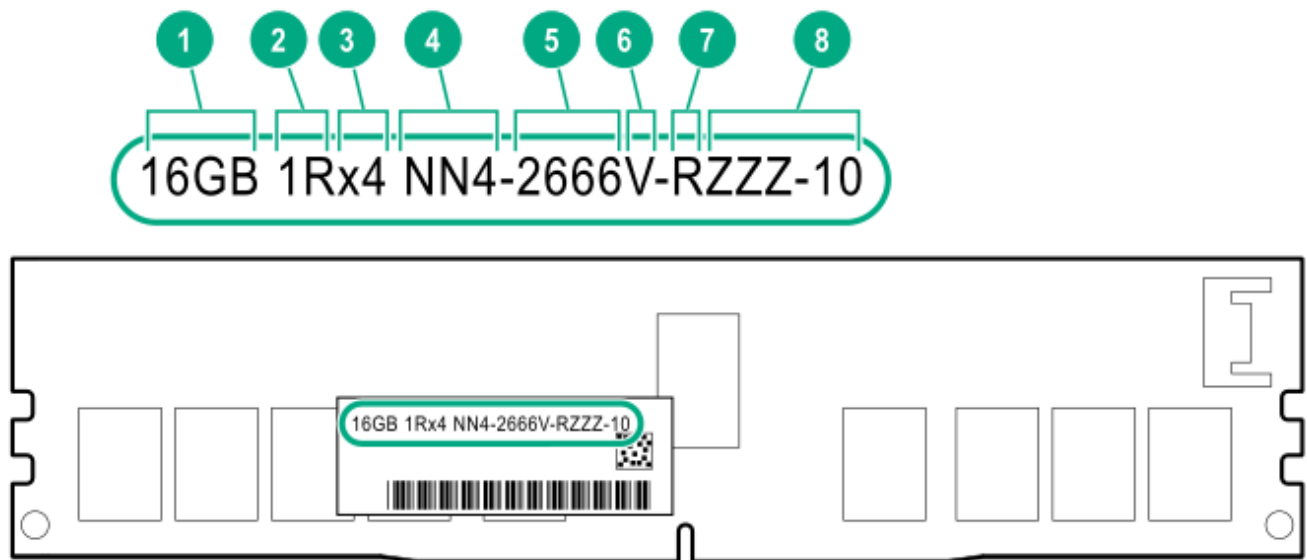
For more information about product features, specifications, options, configurations, and compatibility, see the HPE DDR4 SmartMemory QuickSpecs on the Hewlett Packard Enterprise website (<https://www.hpe.com/support/DDR4SmartMemoryQS>).

NVDIMM identification

NVDIMMs are supported only when first-generation Intel Xeon Scalable processors are installed on the server.

NVDIMM boards are blue instead of green. This change to the color makes it easier to distinguish NVDIMMs from DIMMs.

To determine NVDIMM characteristics, see the full product description as shown in the following example:



Item	Description	Definition
1	Capacity	16 GiB
2	Rank	1R (Single rank)
3	Data width per DRAM chip	x4 (4 bit)
4	Memory type	NN4=DDR4 NVDIMM-N
5	Maximum memory speed	2667 MT/s
6	Speed grade	V (latency 19-19-19)
7	DIMM type	RDIMM (registered)
8	Other	—

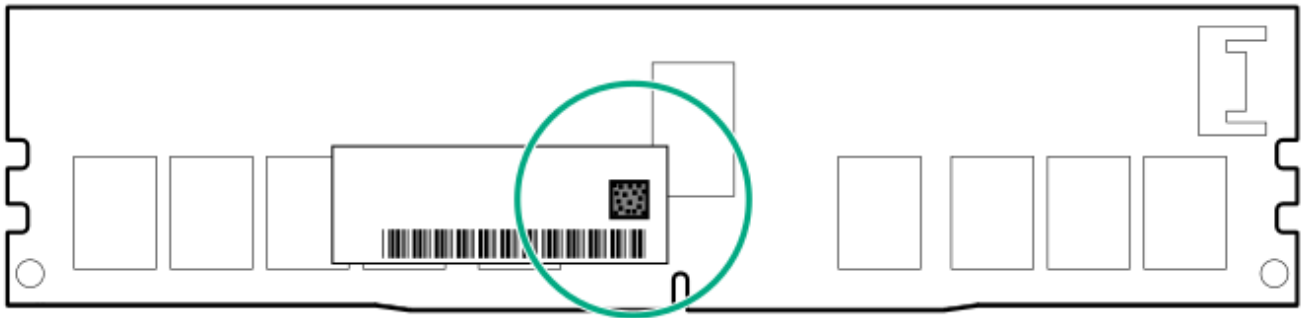
For more information about NVDIMMs, see the product QuickSpecs on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/qs>).

Subtopics

NVDIMM 2D Data Matrix barcode

NVDIMM 2D Data Matrix barcode

The 2D Data Matrix barcode is on the right side of the NVDIMM label and can be scanned by a cell phone or other device.

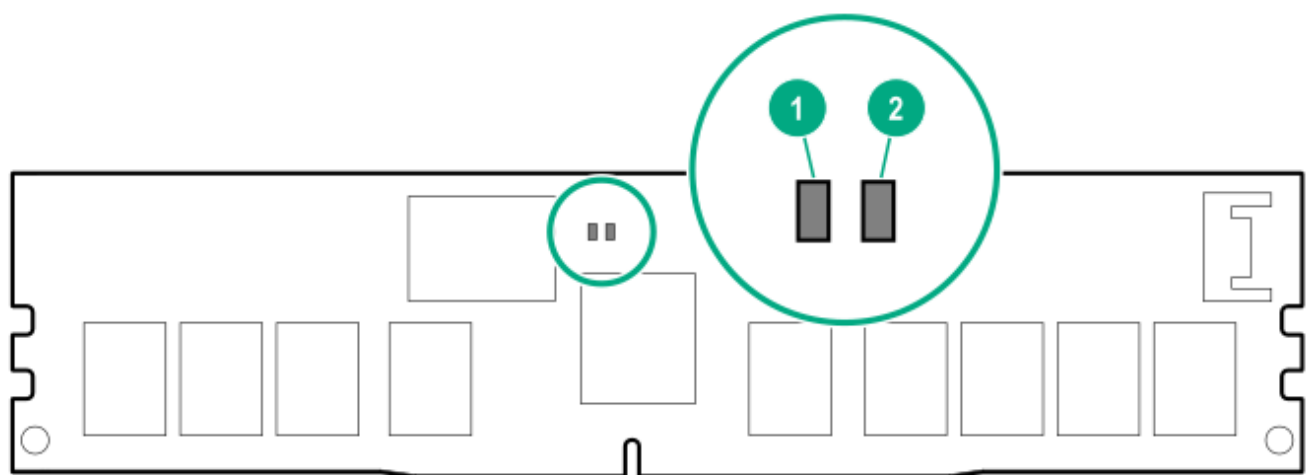


When scanned, the following information from the label can be copied to your cell phone or device:

- (P) is the module part number.
- (L) is the technical details shown on the label.
- (S) is the module serial number.

Example: (P)HMN82GR7AFR4N-VK (L)16GB 1Rx4 NN4-2666V-RZZZ-10(S)80AD-01-1742-11AED5C2

NVDIMM LED identification



Item	LED description	LED color
1	Power LED	Green
2	Function LED	Blue

Subtopics

- [NVDIMM-N LED combinations](#)
- [NVDIMM Function LED patterns](#)

NVDIMM-N LED combinations

State	Definition	NVDIMM-N Power LED (green)	NVDIMM-N Function LED (blue)
0	AC power is on (12V rail) but the NVM controller is not working or not ready.	On	Off
1	AC power is on (12V rail) and the NVM controller is ready.	On	On
2	AC power is off or the battery is off (12V rail off).	Off	Off

State	Definition	NVDIMM-N Power LED (green)	NVDIMM-N Function LED (blue)
3	AC power is on (12V rail) or the battery is on (12V rail) and the NVDIMM-N is active (backup and restore).	On	Flashing

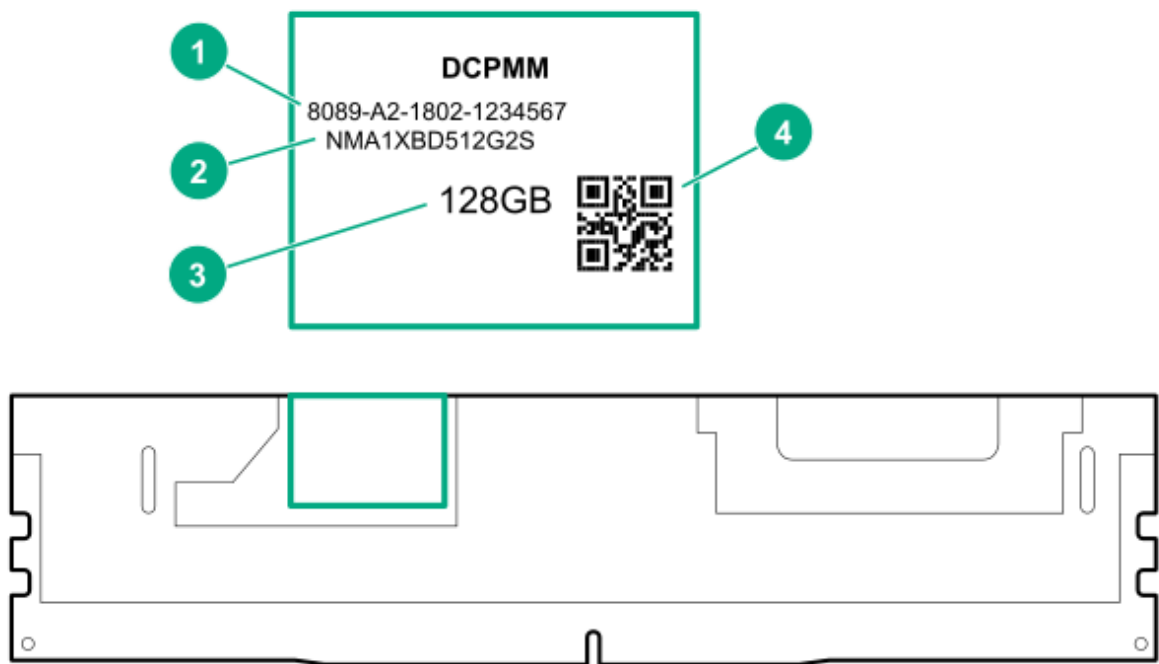
NVDIMM Function LED patterns

For the purpose of this table, the NVDIMM-N LED operates as follows:

- Solid indicates that the LED remains in the on state.
- Flashing indicates that the LED is on for 2 seconds and off for 1 second.
- Fast-flashing indicates that the LED is on for 300 ms and off for 300 ms.

State	Definition	NVDIMM-N Function LED
0	The restore operation is in progress.	Flashing
1	The restore operation is successful.	Solid or On
2	Erase is in progress.	Flashing
3	The erase operation is successful.	Solid or On
4	The NVDIMM-N is armed, and the NVDIMM-N is in normal operation.	Solid or On
5	The save operation is in progress.	Flashing
6	The NVDIMM-N finished saving and battery is still turned on (12 V still powered).	Solid or On
7	The NVDIMM-N has an internal error or a firmware update is in progress. For more information about an NVDIMM-N internal error, see the IML.	Fast-flashing

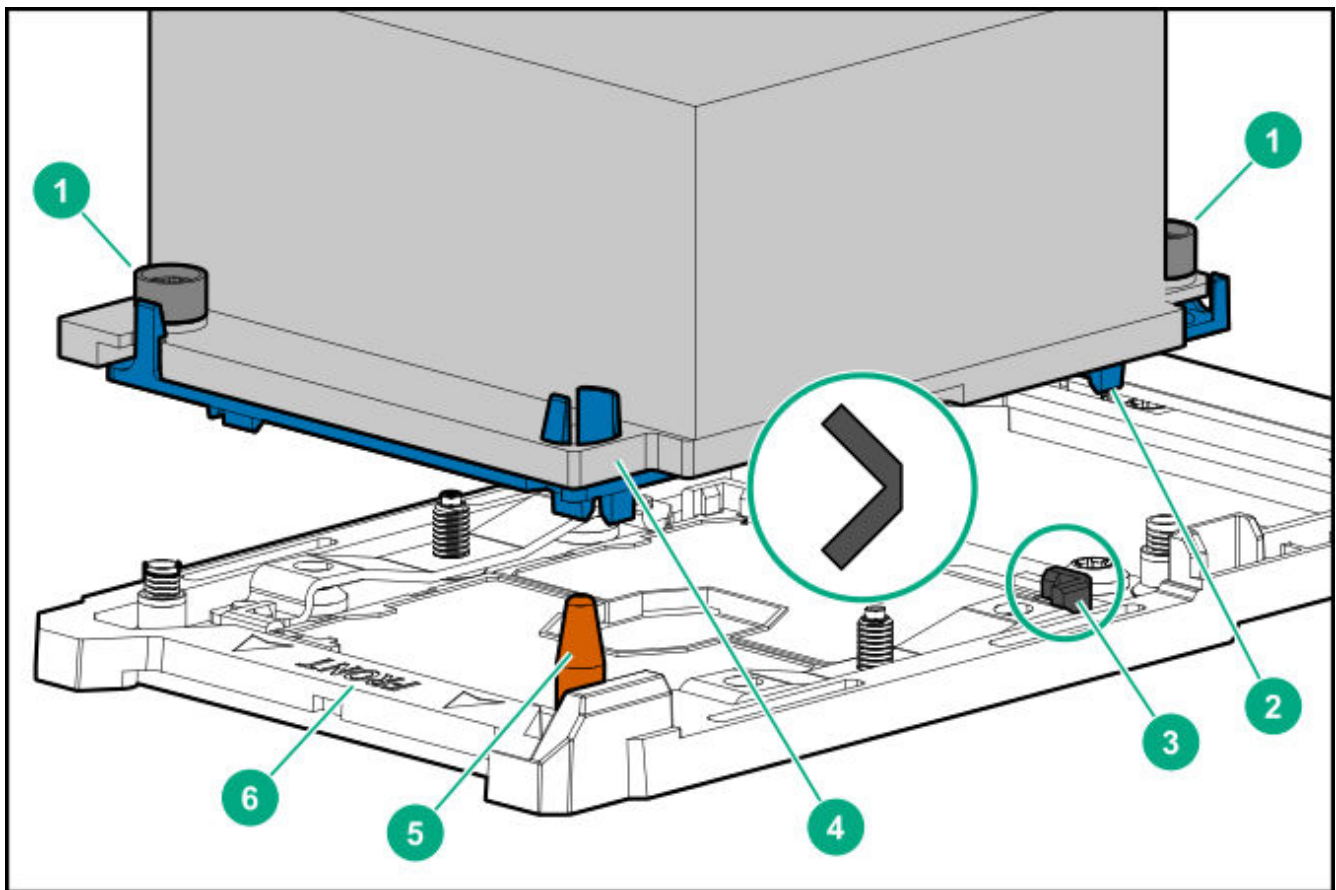
Intel Optane persistent memory 100 series for HPE label identification



Item	Description	Example
1	Unique ID number	8089-A2-1802-1234567
2	Model number	NMA1XBD512G2S
3	Capacity	128 GB 256 GB 512 GB
4	QR code	Includes part number and serial number

For more information about product features, specifications, options, configurations, and compatibility, see the product QuickSpecs on the Hewlett Packard Enterprise website (<https://www.hpe.com/support/persistentmemoryQS>).

Processor, heatsink, and socket components



Item	Description
1	Heatsink nuts
2	Processor carrier
3	Pin 1 indicator ¹ _—
4	Heatsink guide/keying feature
5	Alignment post
6	Heatsink keying frame

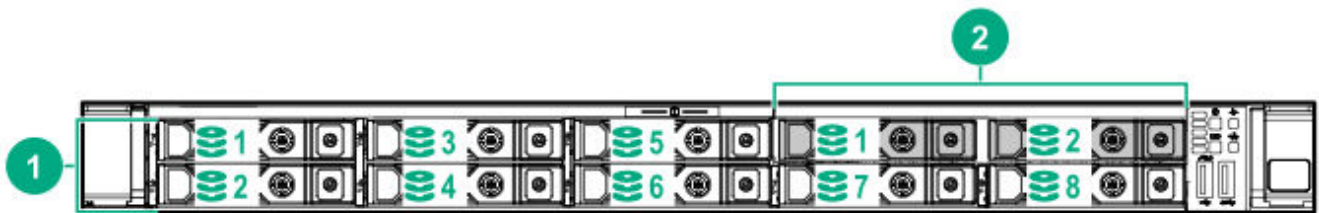
¹_— Symbol also on the processor and frame.

Device numbers

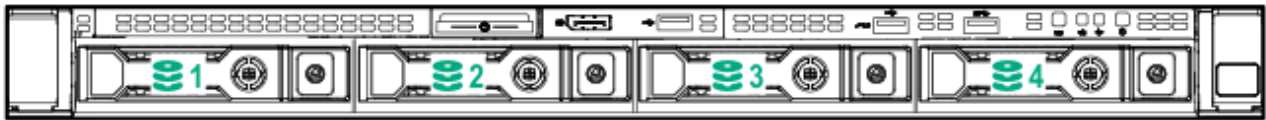
8 SFF device bay numbering



8 SFF + 2 SFF device bay numbering



Item	Description
1	Box 1, bays 1-8
2	Box 2, bays 1 and 2



10 SFF NVMe/SAS backplane option device bay numbering

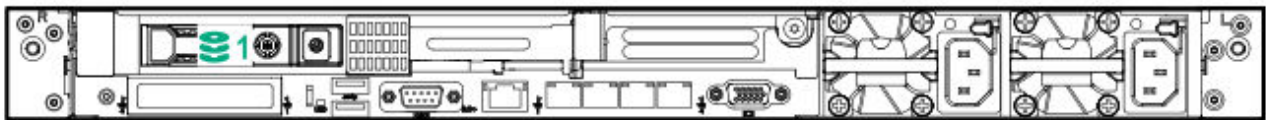
When the 10 SFF NVMe/SAS backplane option is installed, NVMe drives must be installed in bays 9 and 10. The other bays support a mix of NVMe and SAS drives.



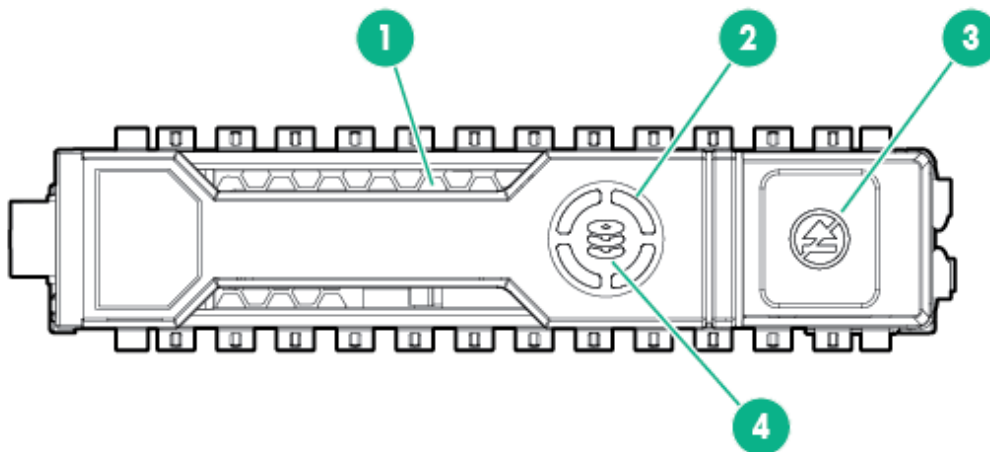
Optional rear device bay numbering

The optional rear device bay supports either 1 SFF drive in a SmartDrive carrier, or 2 uFF M.2 drives in an HPE Smart Carrier M.2 (SCM).

When the HPE SFF Flash Adapter is installed, the uFF drives are recognized as 1 and 101.



Hot-plug drive LED definitions



Item	LED	Status	Definition
1	Locate	Solid blue	The drive is being identified by a host application.
		Flashing blue	The drive carrier firmware is being updated or requires an update.
2	Activity ring	Rotating green	Drive activity.
		Off	No drive activity.
3	Do not remove	Solid white	Do not remove the drive. Removing the drive causes one or more of the logical drives to fail.
		Off	Removing the drive does not cause a logical drive to fail.
4	Drive status	Solid green	The drive is a member of one or more logical drives.
		Flashing green	The drive is doing one of the following: • Rebuilding • Performing a RAID migration • Performing a strip size migration • Performing a capacity expansion • Performing a logical drive extension • Erasing • Spare part activation

Item	LED	Status	Definition
		Flashing amber/green	The drive is a member of one or more logical drives and predicts the drive will fail.
		Flashing amber	The drive is not configured and predicts the drive will fail.
		Solid amber	The drive has failed.
		Off	The drive is not configured by a RAID controller or a spare drive.

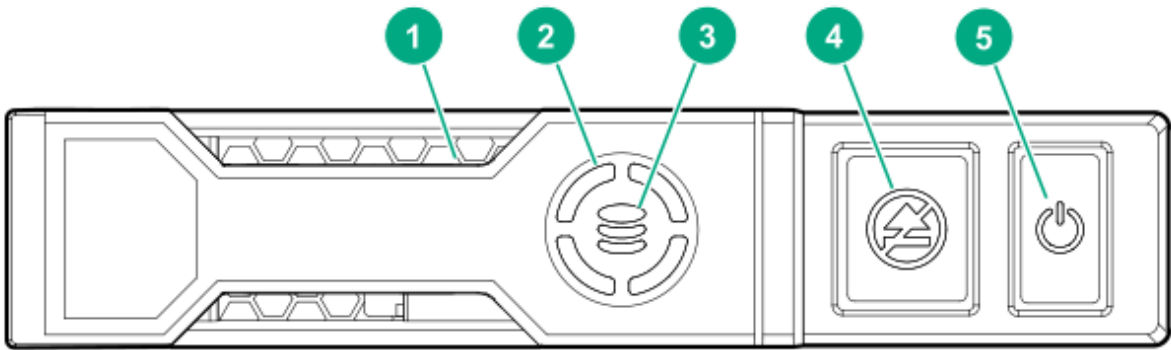
Smart Carrier NVMe (SCN) drive LED definitions

The NVMe drive is a PCIe bus device. A device attached to a PCIe bus cannot be removed without allowing the device and bus to complete and cease the signal/traffic flow.



CAUTION

Do not remove an NVMe drive from the drive bay while the Do not remove LED is flashing. The Do not remove LED flashes to indicate that the device is still in use. Removing the NVMe drive before the device has completed and ceased signal/traffic flow can cause loss of data.



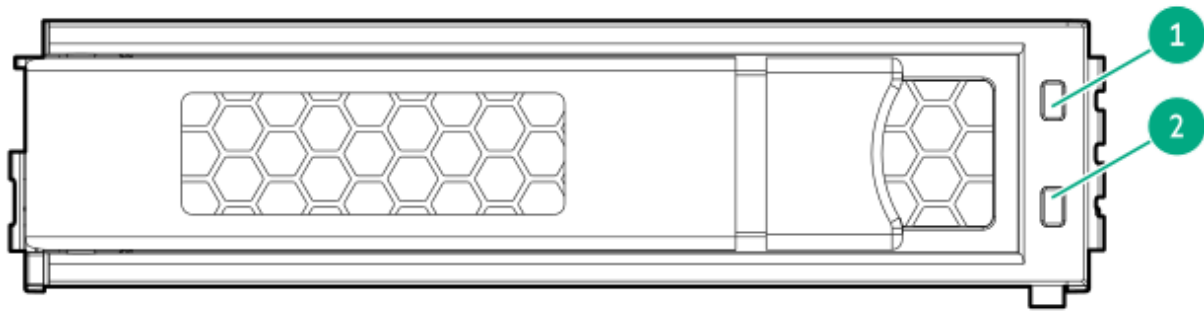
Item	LED	Status	Definition
1	Locate	Solid blue	The drive is being identified by a host application.
		Flashing blue	The drive carrier firmware is being updated or requires an update.

Item	LED	Status	Definition
2	Activity ring	Rotating green	Drive activity
		Off	No drive activity
3	Drive status	Solid green	The drive is a member of one or more logical drives.
		Flashing green	The drive is doing one of the following: • Rebuilding • Performing a RAID migration • Performing a stripe size migration • Performing a capacity expansion • Performing a logical drive extension • Erasing
		Flashing amber/green	The drive is a member of one or more logical drives and predicts the drive will fail.
		Flashing amber	The drive is not configured and predicts the drive will fail.
		Solid amber	The drive has failed.
4	Do not remove	Off	The drive is not configured by a RAID controller.
		Solid white	Do not remove the drive. The drive must be ejected from the PCIe bus prior to removal.
		Flashing white	The drive ejection request is pending.
		Off	The drive has been ejected.
5	Power	Solid green	Do not remove the drive. The drive must be ejected from the PCIe bus prior to removal.
		Flashing green	The drive ejection request is pending.
		Off	The drive has been ejected.

HPE Basic Drive LED definitions

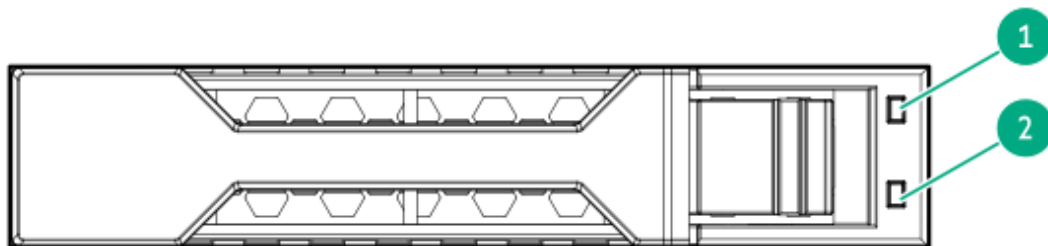
LFF low-profile drive carrier

The LFF low-profile drive carrier supports hot-plug SAS and SATA drives.



SFF basic drive carrier

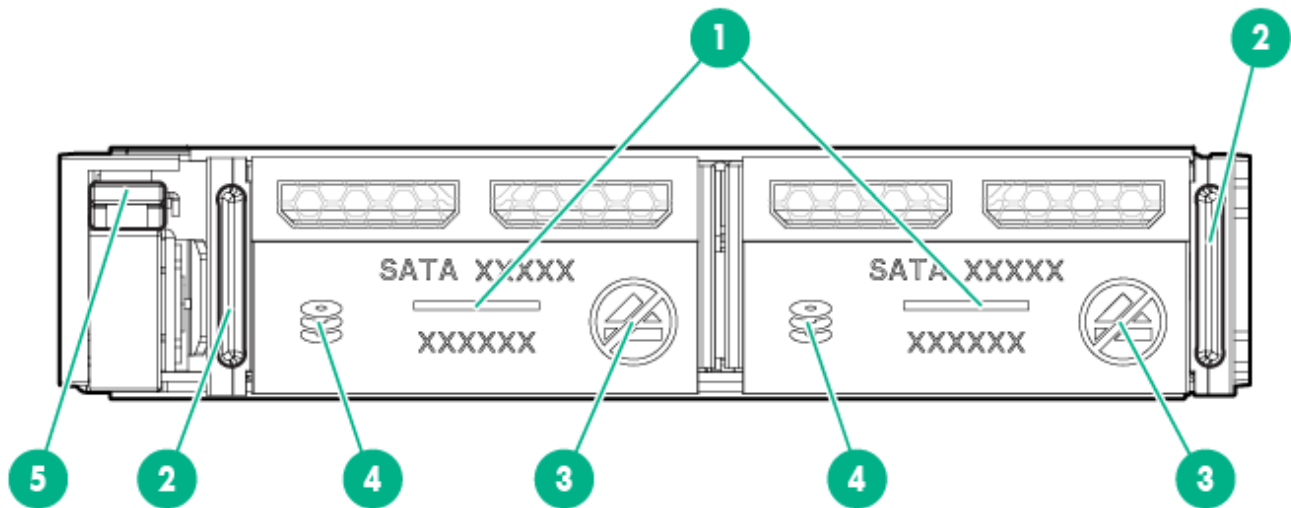
The SFF basic drive carrier supports hot-plug SAS, SATA, and NVMe drives.



Item	LED	Status	Definition
1	Fault/Locate	Solid amber	This drive has failed, is unsupported, or is invalid.
		Solid blue	The drive is operating normally and being identified by a management application.
		Flashing amber/blue (1 flash per second)	The drive has failed, or a predictive failure alert has been received for this drive. The drive has also been identified by a management application.
		Flashing amber (1 flash per second)	A predictive failure alert has been received for this drive. Replace the drive as soon as possible.
2	Online/Activity	Solid green	The drive is online and has no activity.

Item	LED	Status	Definition
		Flashing green (1 flash per second)	The drive is doing one of the following: - Rebuilding or performing a RAID - Performing a stripe size migration - Performing a capacity expansion - Performing a logical drive extension - Erasing - Spare part activation
		Flashing green (4 flashes per second)	The drive is operation normally and has activity.
		Off	The drive is not configured by a RAID controller or is a spare drive.

uFF drive components and LEDs



Item	Description	Status
1	Locate	Off—Normal

Item	Description	Status
		• Solid blue—The drive is being identified by a host application • Flashing blue—The drive firmware is being updated or requires an update
2	uFF drive ejection latch	Removes the uFF drive when released
3	Do not remove LED	• Off—OK to remove the drive. Removing the drive does not cause a logical drive to fail. • Solid white—Do not remove the drive. Removing the drive causes one or more of the logical drives to fail.
4	Drive status LED	• Off—The drive is not configured by a RAID controller • Solid green—The drive is a member of one or more logical drives • Flashing green (4 Hz)—The drive is operating normally and has activity • Flashing green (1 Hz)—The drive is rebuilding or performing a RAID migration, stripe size migration, capacity expansion, logical drive extension, or is erasing • Flashing amber/green (1 Hz)—The drive is a member of one or more logical drives that predicts the drive will fail • Solid amber—The drive has failed • Flashing amber (1 Hz)—The drive is not configured and predicts the drive will fail
5	Adapter ejection release latch and handle	Removes the SFF flash adapter when released

Hot-plug fans



CAUTION

To avoid damage to server components, fan blanks must be installed in fan bays 1 and 2 in a single-processor configuration.



CAUTION

To avoid damage to the equipment, do not operate the server for extended periods of time if the server does not have the optimal number of fans installed. Although the server might boot, Hewlett Packard Enterprise does not recommend operating the server without the required fans installed and operating.

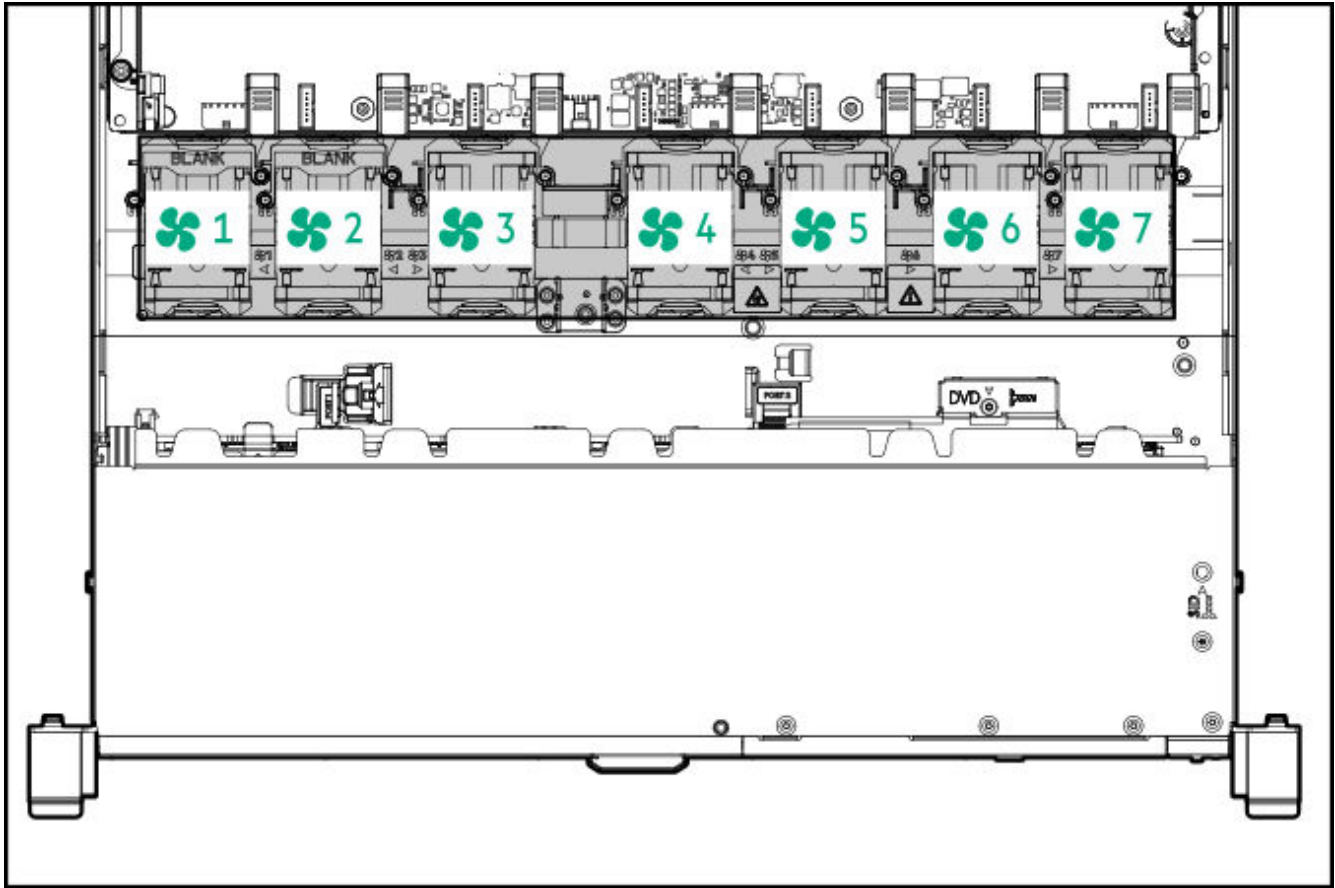
The valid fan configurations are listed in the following tables.

One-processor configuration

Fan bay 1	Fan bay 2	Fan bay 3	Fan bay 4	Fan bay 5	Fan bay 6	Fan bay 7
Fan blank	Fan blank	Fan	Fan	Fan	Fan	Fan

Two-processor configuration

Fan bay 1	Fan bay 2	Fan bay 3	Fan bay 4	Fan bay 5	Fan bay 6	Fan bay 7
Fan	Fan	Fan	Fan	Fan	Fan	Fan



The loss of a single fan rotor (one standard fan) causes loss of redundancy. The loss of two fan rotors (two standard fans or one high performance fan) causes the server to initiate a shutdown.

The high performance fans are required for some drive configurations. They are also required for ASHRAE-compliant configurations. For more information on ASHRAE-compliant configurations, see the Hewlett Packard Enterprise website <https://www.hpe.com/support/ASHRAEGen10>.

The server supports variable fan speeds. The fans operate at minimum speed until a temperature change requires a fan speed increase to cool the server. The server shuts down during the following temperature-related scenarios:

- At POST and in the OS, iLO performs an orderly shutdown if a cautionary temperature level is detected. If the server hardware detects a critical temperature level before an orderly shutdown occurs, the server performs an immediate shutdown.
- When the Thermal Shutdown feature is disabled in the BIOS/Platform Configuration (RBSU), iLO does not perform an orderly shutdown when a cautionary temperature level is detected. Disabling this

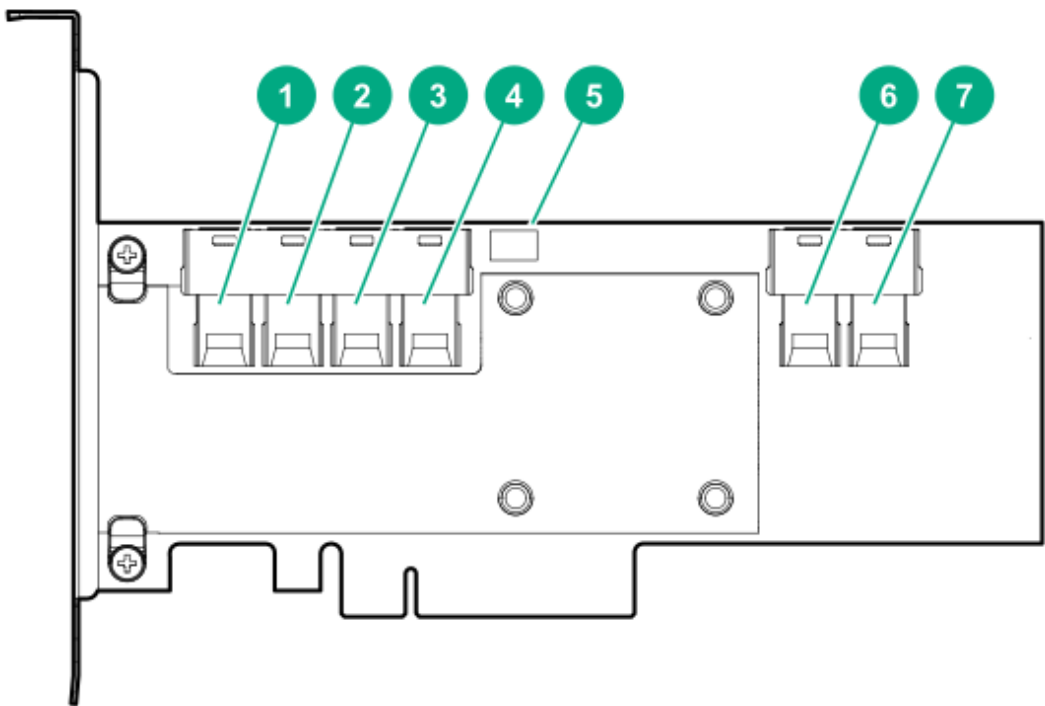
feature does not disable the server hardware from performing an immediate shutdown when a critical temperature level is detected.



CAUTION

A thermal event can damage server components when the Thermal Shutdown feature is disabled in the BIOS/Platform Configuration (RBSU).

HPE Smart Array P824i-p MR Gen10 Controller

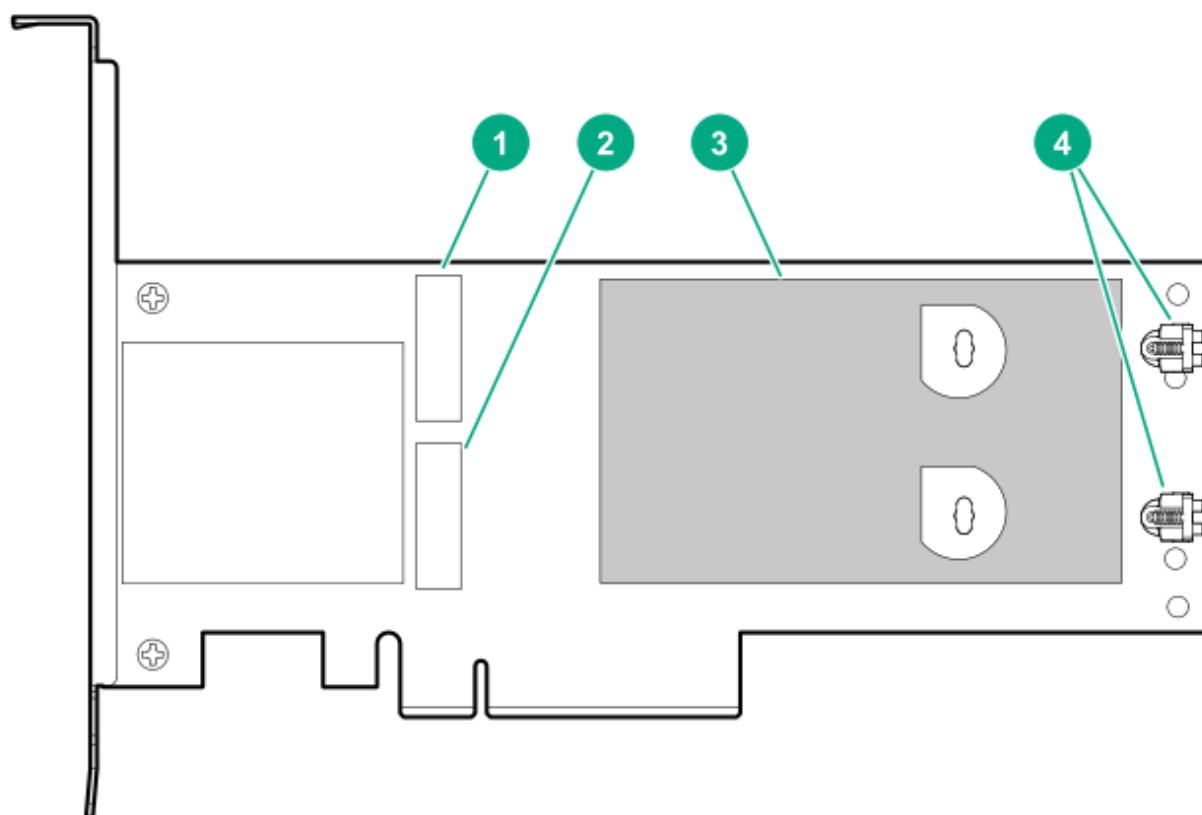


Components

Item	Description
1	Internal SAS port 1i
2	Internal SAS port 2i
3	Internal SAS port 3i
4	Internal SAS port 4i

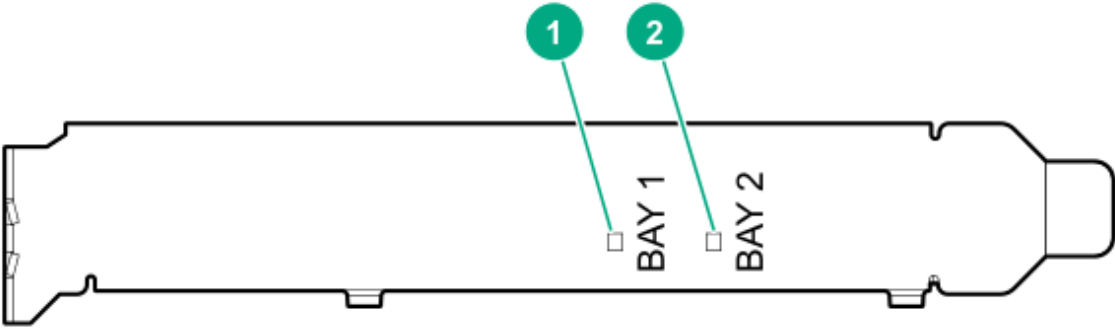
Item	Description
5	Controller backup power cable connector
6	Internal SAS port 5i
7	Internal SAS port 6i

HPE NS204i-p NVMe OS Boot Device components



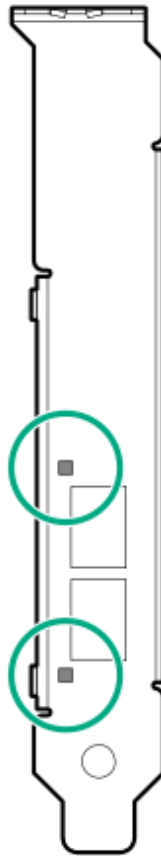
Item	Description
1	Drive bay 1
2	Drive bay 2
3	Thermal interface pad with removable liner
4	M.2 drive retaining latches

HPE NS204i-p NVMe OS Boot Device LED definitions



Item	Description	Fault LED status
1	Bay 1 LED	Off: Normal
2	Bay 2 LED	Flashing 1Hz: Drive predictive failure Amber: Drive failure

HPE InfiniBand HDR/Ethernet 940QSFP 56x16 adapter LEDs



Link LED status ¹	Description
Off	A link has not been established.
Solid amber	Active physical link exists
Blinking amber	4 Hz blinking amber indicates a problem with the physical link.
Solid green	A valid logical (data activity) link exists with no active traffic.
Blinking green	A valid logical link exists with active traffic.

¹ 2-port adapter LEDs are shown. The 1-port adapters have only a single LED.

DSC-25 2-port SFP28 card ports and LEDs

Ports

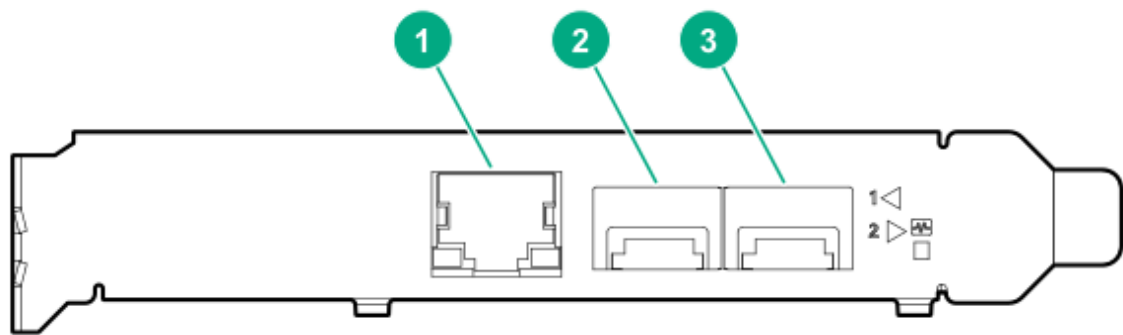


Table 1. Ports

Item	Port	Description
1	Management port	1GbE RJ45
2	Network interface port	10/25G SFP+ based
3	Network interface port	10/25G SFP+ based

LEDs

The HPE for Pensando DSP DSC-25 2p SFP28 card is a dual-port, single-slot, half-height, half-length (HHHL) SFP28 network adapter. It has LEDs for Link (L) and Activity (A) for each port. A half-height bracket is shown in the following illustration with SFP28 ports and LEDs.

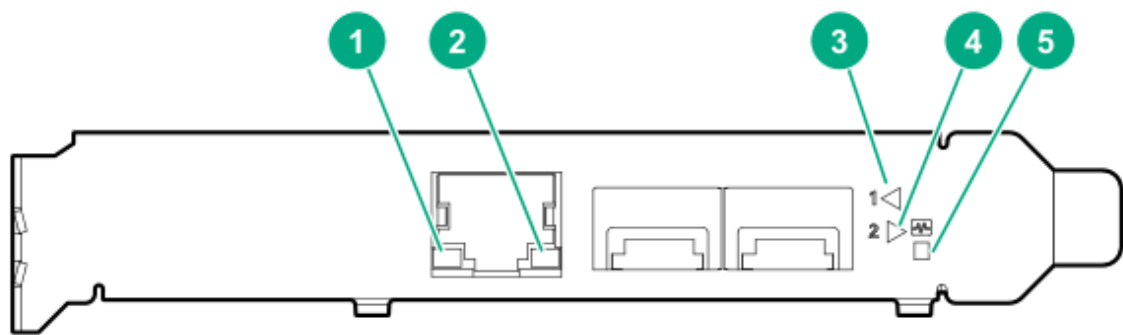


Table 2. LED indicators

Item	LED	Status	Description
1	Management Port Activity LED	Off	No activity
		Flashing	Passing traffic; flashing frequency indicates traffic intensity
2	Management Port Link LED	Off	A link has not been established
		Solid green	Valid Ethernet link
3	SFP Port 1 Link/Activity LED	Off	A link has not been established
		Solid green	Valid Ethernet link
		Flashing green	Passing traffic; flashing frequency indicates traffic intensity
		Solid amber	Link fault
4	SFP Port 2 Link/Activity LED	Off	A link has not been established
		Solid green	Valid Ethernet link
		Flashing green	Passing traffic; flashing frequency indicates traffic intensity
		Solid amber	Link fault
5	System status LED	Off	System is not powered
		Solid amber	Power is up, software has not booted yet
		Solid green	System is up and fully operational

Cabling

Subtopics

[Cabling overview](#)

[SFF cables](#)

[LFF cables](#)

Cabling overview

This section provides guidelines that help you make informed decisions about cabling the server and hardware options to optimize performance.



CAUTION

When routing cables, always be sure that the cables are not in a position where they can be pinched or crimped.

SFF cables

Option kit	Cable part number	Connects from	Connects to
SFF P824i-p Cable Kit	876864-001	8 SFF SAS/SATA Backplane	Port 1i/2i on the P824i-p Smart Array Controller
	876865-002	2 SFF SAS/SATA Backplane	Port 3i on the P824i-p Smart Array Controller
	876866-002	10 SFF SAS/SATA Backplane	Port 1i/2i on the P824i-p Smart Array Controller
SFF Internal Cables Kit	874618-001	2 SFF SAS/SATA Backplane	Embedded SATA
	869662-001	8 SFF SAS/SATA Backplane	Embedded SATA Type -a Smart Array Controller
	876867-001	2 SFF SAS/SATA Backplane	Type -p Smart Array Controller (Slot 1)
	874615-001	8 SFF SAS/SATA Backplane	Type -p Smart Array Controller (Slot 2)
2 SFF Backplane Option Kit	876867-001	2 SFF SAS/SATA Backplane	Embedded SATA Type -p Smart Array Controller
	869669-001	2 SFF SAS/SATA Backplane	P816i-a Smart Array Controller
2 SFF SCM Backplane Option Kit	869668-001	2 SFF SCM Backplane	Embedded SATA

Option kit	Cable part number	Connects from	Connects to
	869669-001	2 SFF SCM Backplane	P816i-a Smart Array Controller
		2 SFF SCM Backplane	Slot 1
1 SFF Rear Backplane Option Kit	869685-001	1 SFF Rear Backplane	Embedded SATA Type -p Smart Array Controller
	876868-001	1 SFF Rear Backplane	P816i-a Smart Array Controller
10 SFF SAS/SATA/NVMe Combo backplane	869675-001	10 SFF backplane, SAS port	P408i-a controller
	869676-001	10 SFF backplane, ports 1, 2, and 3	NVMe riser, ports 1, 2, and 3
	869681-001	10 SFF backplane, port 5	NVMe riser, port 5
	869680-001	10 SFF backplane, port 4	NVMe riser, port 4

Subtopics

SFF configuration cable routing

Additional SFF cabling

SFF configuration cable routing



NOTE

The following information describes the standard cable routing for this component. For more information on optional cable routing, see the HPE ProLiant DL380 Gen10 Servercabling matrix on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/CablingMatrixGen10>).

Subtopics

SFF backplane to P824i-p controller

8 SFF backplane to a P408i-a/P816i-a controller

2 SFF backplane to P816i-a controller

10 SFF NVMe backplane to NVMe riser

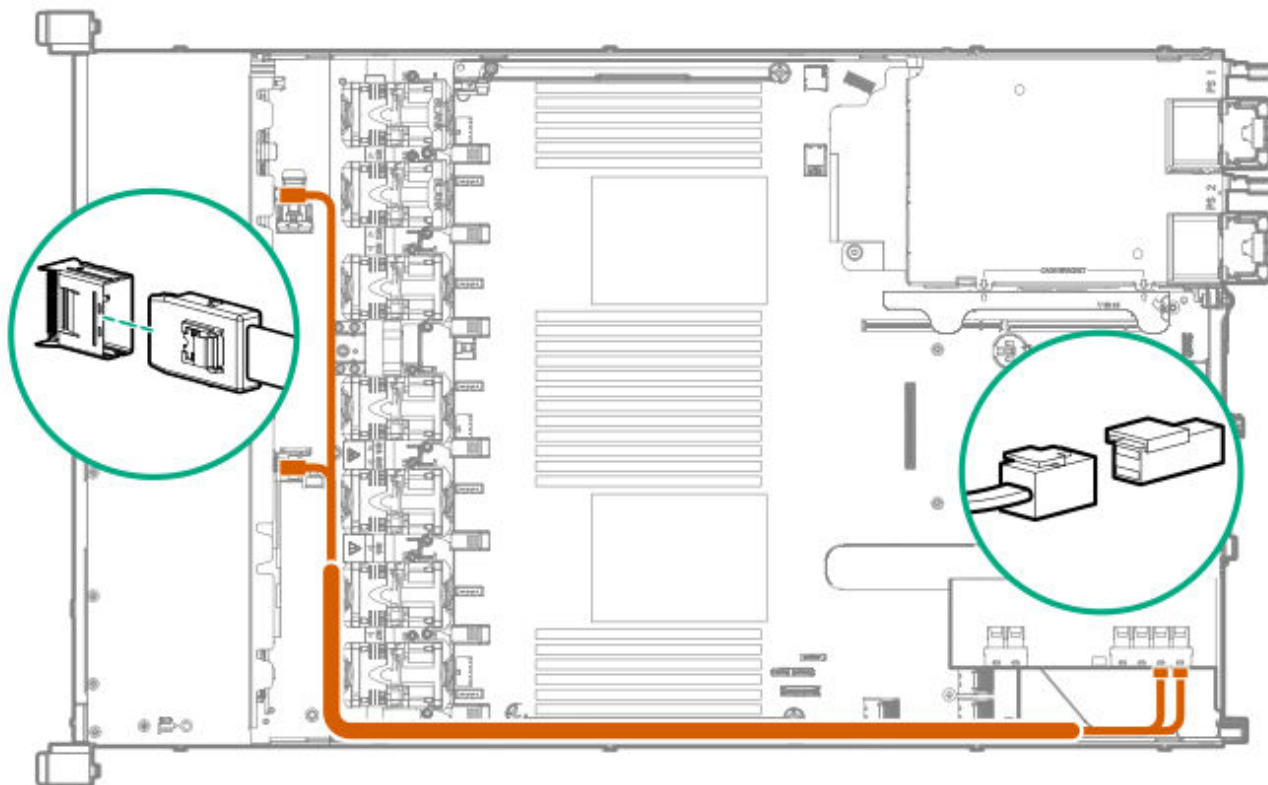
2 SFF NVMe backplane to primary riser

1 SFF rear backplane to system board SATA

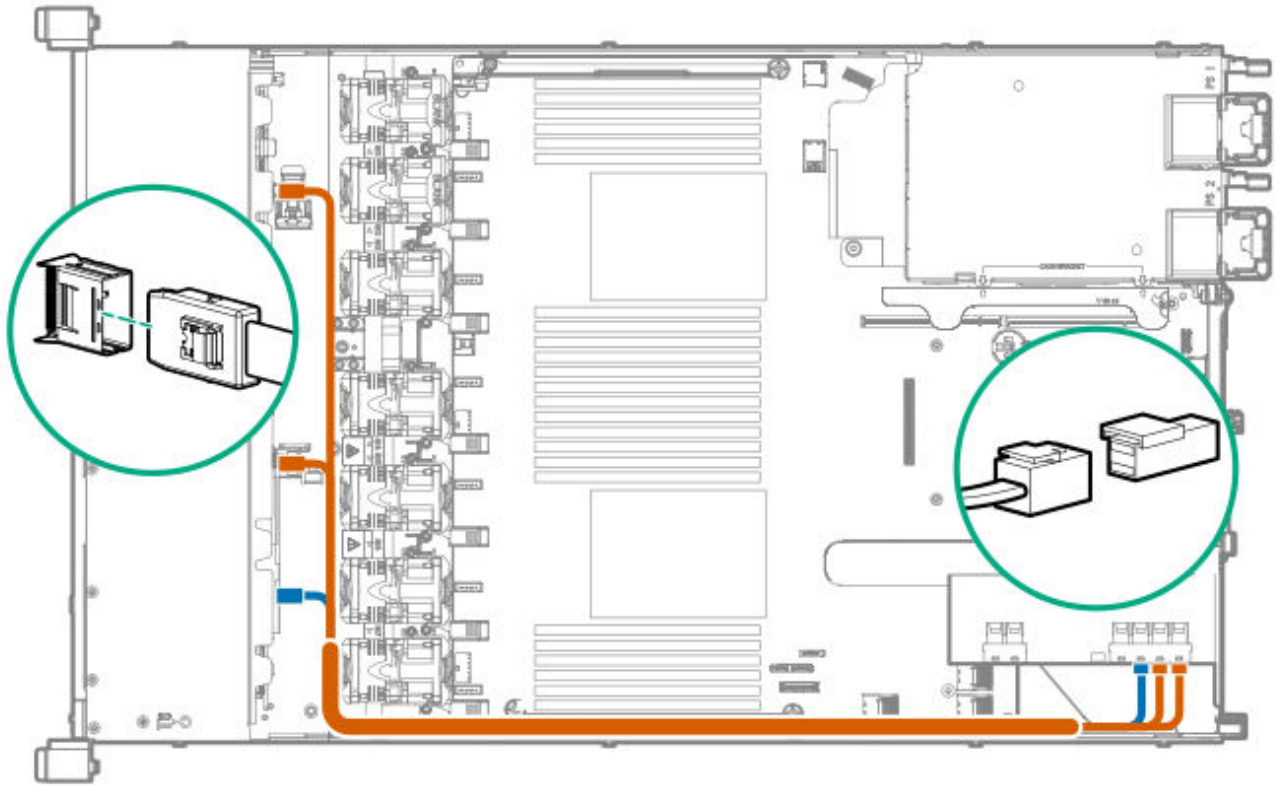
10 SFF backplane to P408i-a controller

SFF backplane to P824i-p controller

8 SFF to ports 1i/2i in the primary



8 SFF to ports 1i/2i + 2 SFF to port 3i in the primary



10 SFF premium backplane to ports 1i/2i in the primary



8 SFF backplane to a P408i-a/P816i-a controller

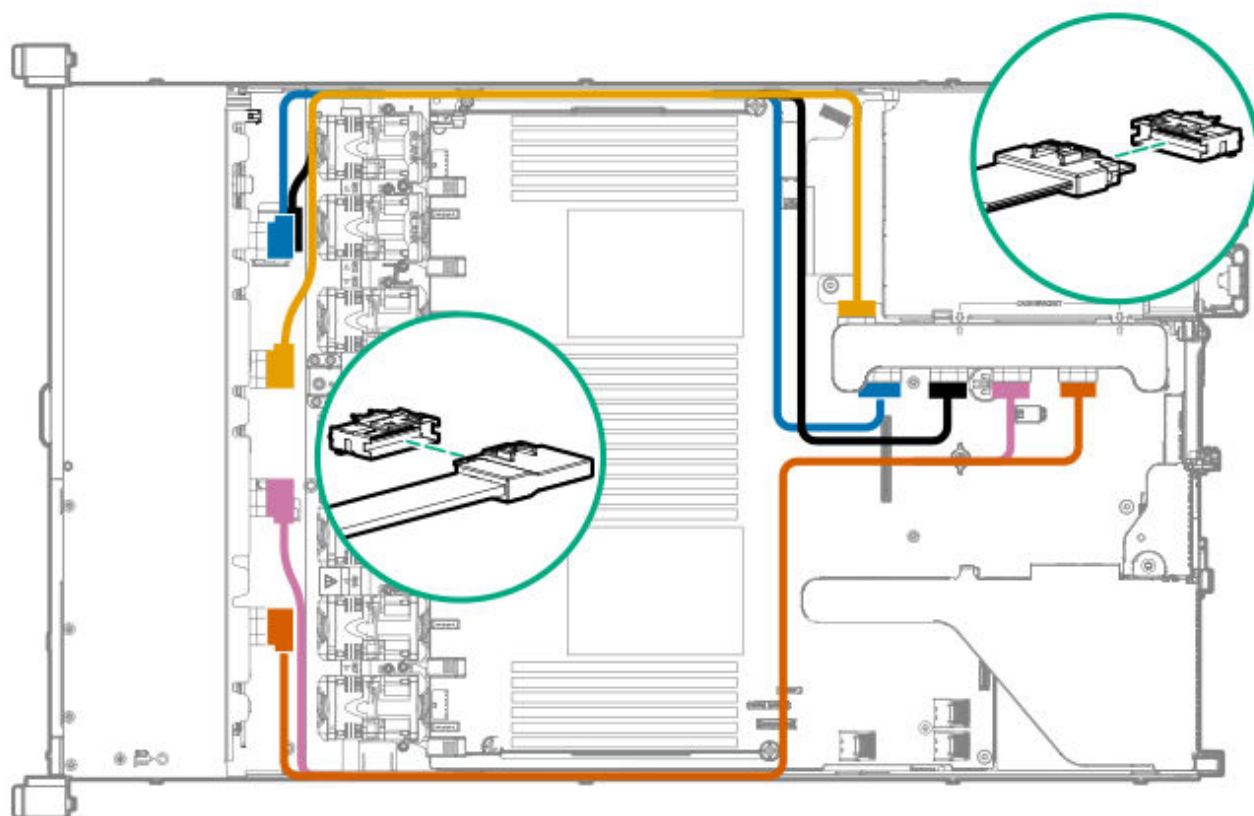
A P408i-a controller is shown, but the routing is the same for the P816i-a.



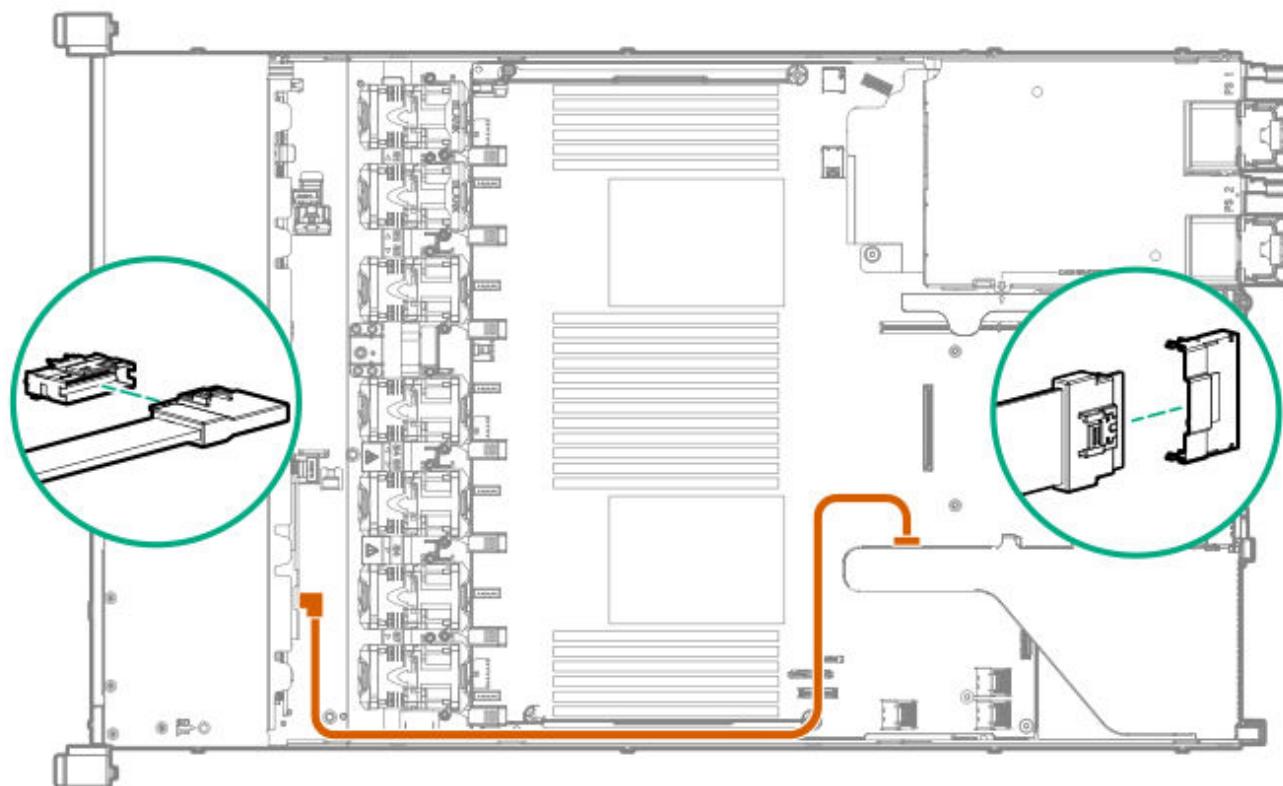
2 SFF backplane to P816i-a controller



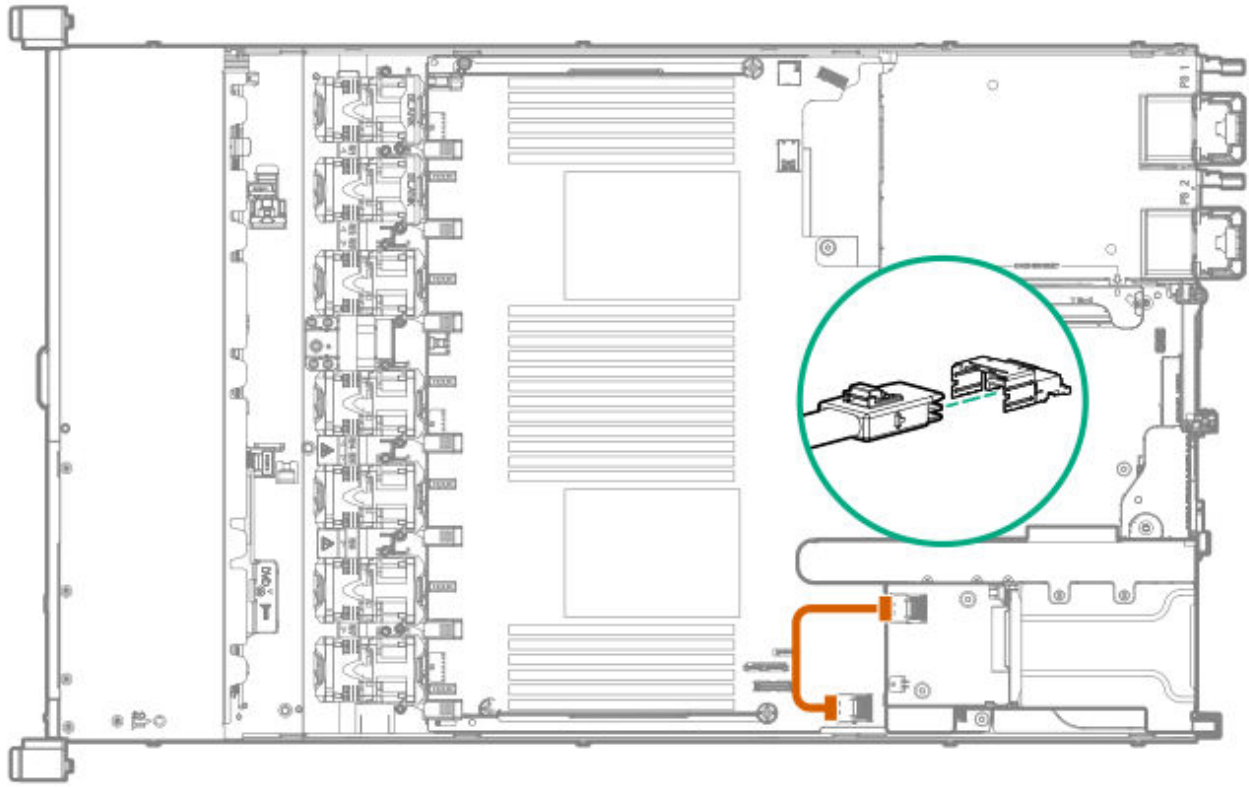
10 SFF NVMe backplane to NVMe riser



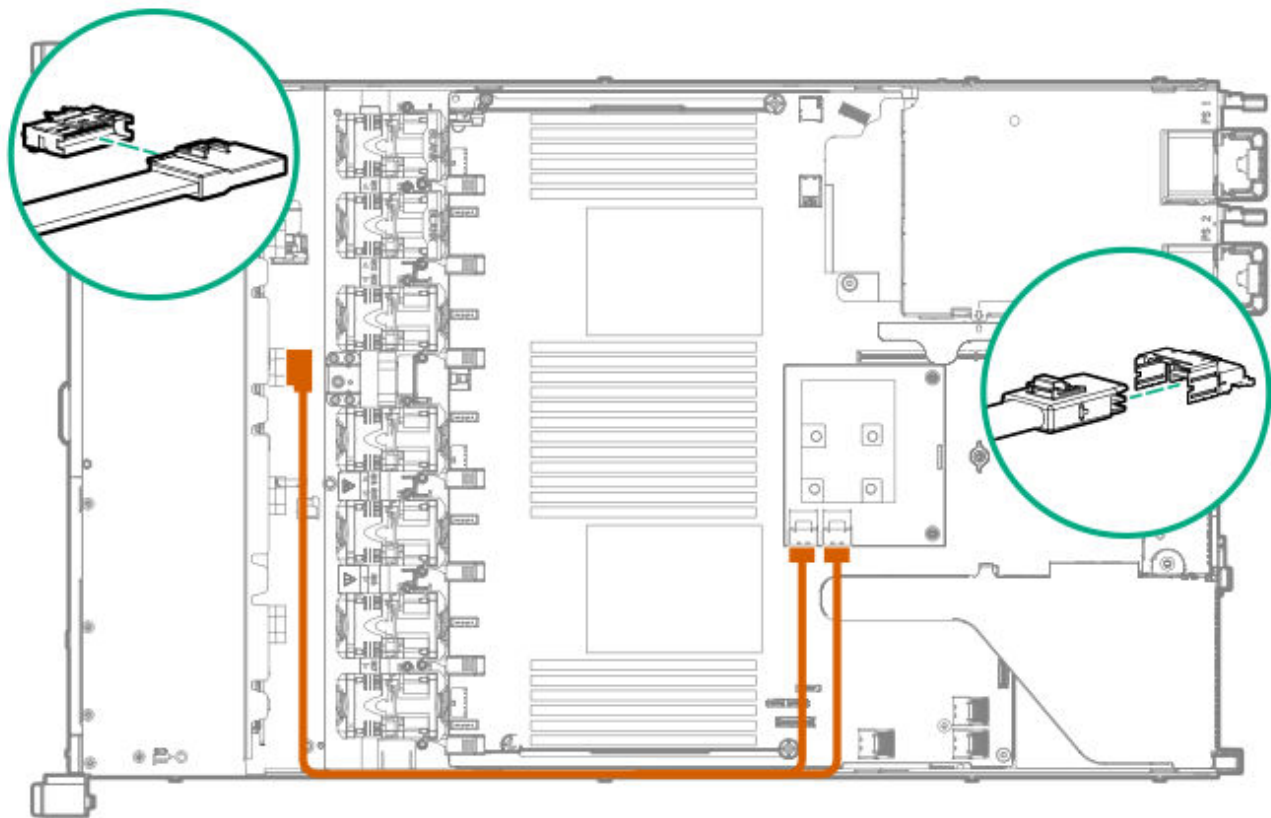
2 SFF NVMe backplane to primary riser



1 SFF rear backplane to system board SATA



10 SFF backplane to P408i-a controller



Additional SFF cabling

For more information on the following cables and cabling configurations, see the HPE DL360 Gen10 Server cabling matrix on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/CablingMatrixGen10>):

- 8 SFF backplane to P408i-p controller
- 8 SFF backplane to system board SATA
- 8 SFF backplane power cable
- 2 SFF backplane to P408i-p controller
- 2 SFF backplane to system board SATA
- 2 SFF to 8 SFF power cable
- 1 SFF rear backplane to P816i-a controller

- SFF system power/USB cable

LFF cables

Option kit	Cable part number	Connects from	Connects to
HPE Gen10 LFF Internal Cable Kit	869673-001	4 LFF backplane	Type -p Smart Array controller, slot 1 Type -a Smart Array controller Embedded SATA
	874616-001	4 LFF backplane	PCI controller, slot 2
HPE Gen9 LFF Optical Cable Kit	756914-001	LFF optical drive	System board SATA, port 5

Subtopics

[LFF configuration cable routing](#)

[Additional LFF cabling](#)

LFF configuration cable routing



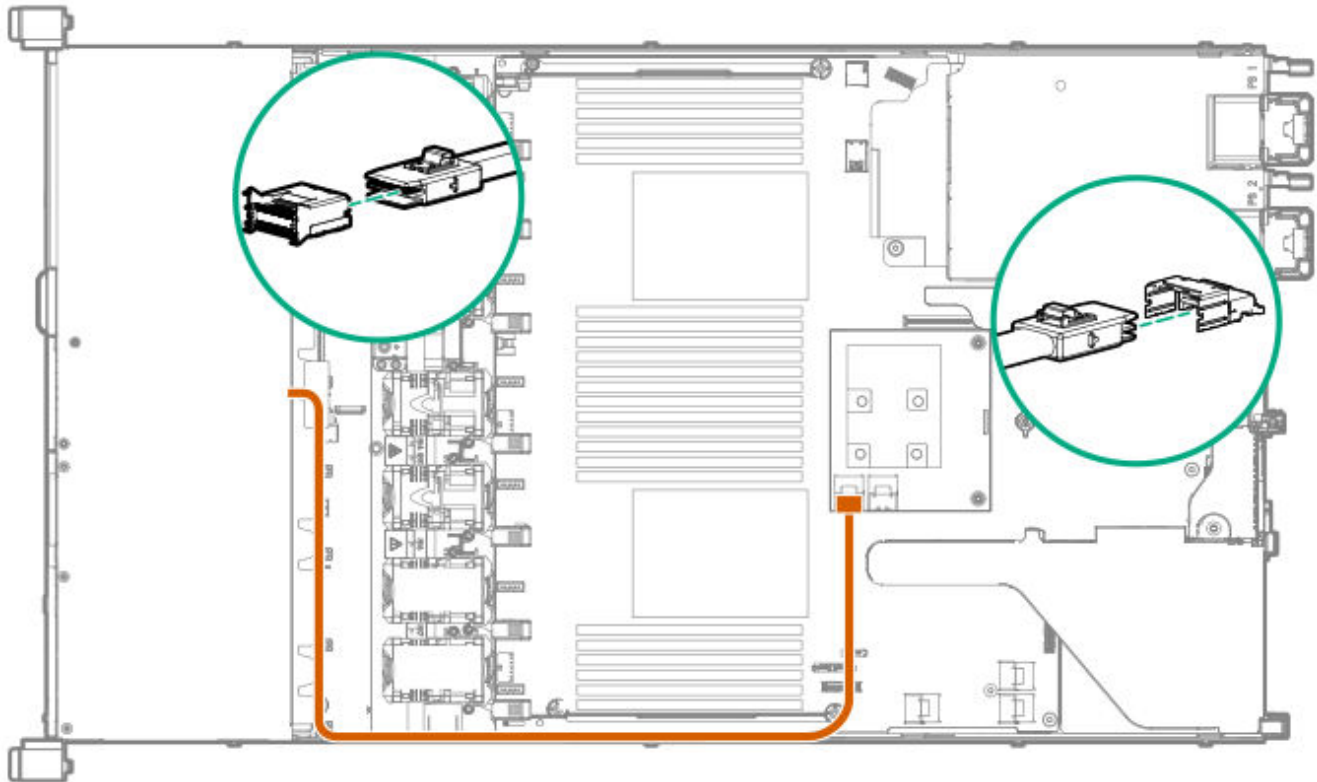
NOTE

The following information describes the standard cable routing for this component. For more information on optional cable routing, see the HPE ProLiant DL380 Gen10 Servercabling matrix on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/CablingMatrixGen10>).

Subtopics

[4 LFF backplane to P408i-a controller](#)

4 LFF backplane to P408i-a controller



Additional LFF cabling

For more information on the following cables and cabling configurations, see the HPE DL360 Gen10 Server cabling matrix on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/CablingMatrixGen10>):

- 4 LFF backplane to P408i-p controller
- 4 LFF backplane to system board SATA

Specifications

Subtopics

Environmental specifications

Specification	Value
Temperature range ¹ _—	—
Operating	10°C to 35°C (50°F to 95°F)
Nonoperating	-30°C to 60°C (-22°F to 140°F)
Relative humidity (noncondensing)	—
Operating	Minimum to be the higher (more moisture) of -12°C (10.4°F) dew point or 8% relative humidity Maximum to be 24°C (75.2°F) dew point or 90% relative humidity
Nonoperating	5% to 95% 38.7°C (101.7°F), maximum wet bulb temperature

¹_— All temperature ratings shown are for sea level. An altitude derating of 1.0°C per 304.8 m (1.8°F per 1000 ft) to 3048 m (10,000 ft) is applicable. No direct sunlight allowed. Maximum rate of change is 20°C per hour (36°F per hour). The upper limit and rate of change might be limited by the type and number of options installed.

For certain approved hardware configurations, the supported system inlet temperature range is extended:

- 5°C to 10°C (41°F to 50°F) and 35°C to 40°C (95°F to 104°F) at sea level with an altitude derating of 1.0°C per every 175 m (1.8°F per every 574 ft) above 900 m (2953 ft) to a maximum of 3048 m (10,000 ft).
- 40°C to 45°C (104°F to 113°F) at sea level with an altitude derating of 1.0°C per every 125 m (1.8°F per every 410 ft) above 900 m (2953 ft) to a maximum of 3048 m (10,000 ft).

The approved hardware configurations for this system are listed on the [Hewlett Packard Enterprise website](#).

Server specifications

Specification	Value
Height	4.29 cm (1.69 in)
Depth (chassis with SFF drive cage)	70.7 cm (27.83 in)
Depth (chassis with LFF drive cage)	74.98 cm (29.50 in)
Width	43.46 cm (17.11 in)
Weights	
SFF minimum (one drive, one processor, one power supply, two heatsinks, one Smart Array controller, five fans)	13.04 kg (28.74 lb)
SFF maximum (10 drives, two processors, two power supplies, two heatsinks, one Smart Array controller, seven fans)	16.27 kg (35.86 lb)
LFF minimum (one drive, one processor, one power supply, two heatsinks, one Smart Array controller, five fans)	13.77 kg (30.36 lb)
LFF maximum (Four drives, two processors, two power supplies, two heatsinks, one Smart Array controller, seven fans)	16.78 kg (37 lb)

Power supply specifications

Depending on the installed options and the regional location where the server was purchased, the server can be configured with one of the following power supplies:

- [HPE 500 W Flex Slot Platinum Hot-plug Low Halogen Power Supply](#)
- [HPE 800 W Flex Slot Platinum Hot-plug Low Halogen Power Supply](#)
- [HPE 800 W Flex Slot Titanium Hot-plug Low Halogen Power Supply](#)
- [HPE 800 W Flex Slot Universal Hot-plug Low Halogen Power Supply](#)
- [HPE 800 W Flex Slot -48 VDC Hot-plug Low Halogen Power Supply](#)
- [HPE 1600 W Flex Slot Platinum Hot-plug Low Halogen Power Supply](#)

For detailed power supply specifications, see the QuickSpecs on the Hewlett Packard Enterprise website (<https://www.hpe.com/info/proliant/powersupply>).

Subtopics

[HPE 500 W Flex Slot Platinum Hot-plug Low Halogen Power Supply](#)

[HPE 800 W Flex Slot Platinum Hot-plug Low Halogen Power Supply](#)

[HPE 800 W Flex Slot Titanium Hot-plug Low Halogen Power Supply](#)

[HPE 800 W Flex Slot Universal Hot-plug Low Halogen Power Supply](#)

[HPE 800 W Flex Slot -48 VDC Hot-plug Low Halogen Power Supply](#)

[HPE 1600 W Flex Slot Platinum Hot-plug Low Halogen Power Supply](#)

HPE 500 W Flex Slot Platinum Hot-plug Low Halogen Power Supply

Specification	Value
Input requirements	—
Rated input voltage	100 VAC to 240 VAC 240 VDC for China
Rated input frequency	50 Hz to 60 Hz Not applicable to 240 VDC
Rated input current	5.8 A at 100 VAC 2.8 A at 200 VAC 2.4 A at 240 VDC for China
Maximum rated input power	580 W at 100 VAC 560 W at 200 VAC 558 W at 240 VDC for China
BTUs per hour	1999 at 100 VAC 1912 at 200 VAC 1904 at 240 VDC for China
Power supply output	—
Rated steady-state power	500 W at 100 VAC to 127 VAC input 500 W at 100 VAC to 240 VAC input

Specification	Value
	500 W at 240 VDC input for China
Maximum peak power	500 W at 100 VAC to 127 VAC input 500 W at 100 VAC to 240 VAC input 500 W at 240 VDC input for China

HPE 800 W Flex Slot Platinum Hot-plug Low Halogen Power Supply

Specification	Value
Input requirements	—
Rated input voltage	100 VAC to 127 VAC 200 VAC to 240 VAC 240 VDC for China only
Rated input frequency	50 Hz to 60 Hz Not applicable to 240 VDC
Rated input current	9.4 A at 100 VAC 4.5 A at 200 VAC 3.8 A at 240 VDC for China only
Maximum rated input power	940 W at 100 VAC 900 W at 200 VAC 897 W at 240 VDC for China only
BTUs per hour	3067 at 100 VAC 2958 at 200 VAC 2949 at 240 VAC for China only
Power supply output	—
Rated steady-state power	800 W at 100 VAC to 127 VAC input 800 W at 100 VAC to 240 VAC input 800 W at 240 VDC input for China only

Specification	Value
Maximum peak power	800 W at 100 VAC to 127 VAC input 800 W at 100 VAC to 240 VAC input 800 W at 240 VDC input for China only

HPE 800 W Flex Slot Titanium Hot-plug Low Halogen Power Supply

Specification	Value
Input requirements	—
Rated input voltage	200 VAC to 240 VAC 240 VDC for China
Rated input frequency	50 Hz to 60 Hz Not applicable to 240 VDC
Rated input current	4.35 A at 200 VAC 3.62 A at 240 VAC 3.62 A at 240 VDC for China
Maximum rated input power	867 W at 200 VAC 848 W at 240 VAC 848 W at 240 VDC for China
BTUs per hour	2905 at 200 VAC 2893 at 240 VAC 2893 at 240 VDC for China
Power supply output	—
Rated steady-state power	800 W at 200 VAC to 240 VAC input 800 W at 240 VDC input for China
Maximum peak power	800 W at 200 VAC to 240 VAC input 800 W at 240 VDC input for China

HPE 800 W Flex Slot Universal Hot-plug Low Halogen Power Supply

Specification	Value
Input requirements	—
Rated input voltage	200 VAC to 277 VAC 380 VDC
Rated input frequency	50 Hz to 60 Hz
Rated input current	4.4 A at 200 VAC 3.8 A at 230 VAC 2.3 A at 380 VDC
Maximum rated input power	869 W at 200 VAC 865 W at 230 VAC 861 W at 277 VAC 863 W at 380 VDC
BTUs per hour	2964 at 200 VAC 2951 at 230 VAC 2936 at 277 VAC 2943 at 380 VDC
Power supply output	—
Rated steady-state power	800 W at 200 VAC to 277 VAC input
Maximum peak power	800 W at 200 VAC to 277 VAC input

HPE 800 W Flex Slot -48 VDC Hot-plug Low Halogen Power Supply

Specification	Value
Input requirements	—

Specification	Value
Rated input voltage	-40 VDC to -72 VDC -48 VDC nominal input
Rated input current	22.1 A at -40 VDC input 18.2 A at -48 VDC input, nominal input 12.0 A at -72 VDC input
Rated input power	874 W at -40 VDC input 865 W at -48 VDC input, nominal input 854 W at -72 VDC input
Rated input power (BTUs per hour)	2983 at -40 VDC input 2951 at -48 VDC input, nominal input 2912 at -72 VDC input
Power supply output	—
Rated steady-state power	800 W at -40 VDC to -72 VDC
Maximum peak power	800 W at -40 VDC to -72 VDC

Specification**Value**

Maximum peak power

800 W at -40 VDC to -72 VDC input

**WARNING**

To reduce the risk of electric shock or energy hazards:

- This equipment must be installed by trained service personnel.
- Connect the equipment to a reliably grounded secondary circuit source. A secondary circuit has no direct connection to a primary circuit and derives its power from a transformer, converter, or equivalent isolation device.
- The branch circuit overcurrent protection must be rated 27 A.

**CAUTION**

This equipment is designed to permit the connection of the earthed conductor of the DC supply circuit to the earthing conductor at the equipment.

If this connection is made, all of the following must be met:

- This equipment must be connected directly to the DC supply system earthing electrode conductor or to a bonding jumper from an earthing terminal bar or bus to which the DC supply system earthing electrode conductor is connected.
- This equipment must be located in the same immediate area (such as adjacent cabinets) as any other equipment that has a connection between the earthed conductor of the same DC supply circuit and the earthing conductor, and also the point of earthing of the DC system. The DC system must be earthed elsewhere.
- The DC supply source is to be located within the same premises as the equipment.
- Switching or disconnecting devices must not be in the earthed circuit conductor between the DC source and the point of connection of the earthing electrode conductor.

HPE 1600 W Flex Slot Platinum Hot-plug Low Halogen Power Supply

Specification	Value
Input requirements	—
Rated input voltage	200 VAC to 240 VAC 240 VDC for China only
Rated input frequency	50 Hz to 60 Hz
Rated input current	8.7 A at 200 VAC 7.5 A at 230 VAC 7.2 A at 240 VDC
Maximum rated input power	1734 W at 200 VAC 1720 W at 240 VAC
BTUs per hour	5918 at 200 VAC 5891 at 230 VAC
Power supply output	—
Rated steady-state power	1600 W at 200 VAC to 240 VAC input 1600 W at 240 VDC input
Maximum peak power	1600 W for 1 ms (turbo mode) at 200 VAC to 240 VAC input

Hot-plug power supply calculations

For hot-plug power supply specifications and calculators to determine electrical and heat loading for the server, see the Hewlett Packard Enterprise Power Advisor website (<https://www.hpe.com/info/poweradvisor/online>).

Websites

General websites

Hewlett Packard Enterprise Information Library

<https://www.hpe.com/info/EIL>

Single Point of Connectivity Knowledge (SPOCK) Storage compatibility matrix

<https://www.hpe.com/storage/spock>

Storage white papers and analyst reports

<https://www.hpe.com/storage/whitepapers>

For additional websites, see [Support and other resources](#).

Support and other resources

Subtopics

[Accessing Hewlett Packard Enterprise Support](#)

[Accessing updates](#)

[Remote support](#)

[Warranty information](#)

[Regulatory information](#)

[Documentation feedback](#)

Accessing Hewlett Packard Enterprise Support

- For live assistance, go to the Contact Hewlett Packard Enterprise Worldwide website:

<https://www.hpe.com/info/assistance>

- To access documentation and support services, go to the Hewlett Packard Enterprise Support Center website:

<https://www.hpe.com/support/hpesc>

Information to collect

- Technical support registration number (if applicable)

- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Accessing updates

- Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.
- To download product updates:

Hewlett Packard Enterprise Support Center

<https://www.hpe.com/support/hpesc>

Hewlett Packard Enterprise Support Center: Software downloads

<https://www.hpe.com/support/downloads>

My HPE Software Center

<https://www.hpe.com/software/hpesoftwarecenter>

- To subscribe to eNewsletters and alerts:
- <https://www.hpe.com/support/e-updates>
- To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://www.hpe.com/support/AccessToSupportMaterials>



IMPORTANT

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HPE Passport set up with relevant entitlements.

Remote support

Remote support is available with supported devices as part of your warranty or contractual support agreement. It provides intelligent event diagnosis, and automatic, secure submission of hardware event notifications to Hewlett Packard Enterprise, which will initiate a fast and accurate resolution based on your product's service level. Hewlett Packard Enterprise strongly recommends that you register your device for remote support.

If your product includes additional remote support details, use search to locate that information.

Remote support and Proactive Care information

HPE Get Connected

<https://www.hpe.com/services/getconnected>

HPE Proactive Care services

<https://www.hpe.com/services/proactivecare>

HPE Datacenter Care services

<https://www.hpe.com/services/datacentercare>

HPE Proactive Care service: Supported products list

<https://www.hpe.com/services/proactivecaresupportedproducts>

HPE Proactive Care advanced service: Supported products list

<https://www.hpe.com/services/proactivecareadvancedsupportedproducts>

Proactive Care customer information

Proactive Care central

<https://www.hpe.com/services/proactivecarecentral>

Proactive Care service activation

<https://www.hpe.com/services/proactivecarecentralgetstarted>

Warranty information

To view the warranty information for your product, see the links provided below:

HPE ProLiant and IA-32 Servers and Options

<https://www.hpe.com/support/ProLiantServers-Warranties>

HPE Enterprise and Cloudline Servers

<https://www.hpe.com/support/EnterpriseServers-Warranties>

HPE Storage Products

<https://www.hpe.com/support/Storage-Warranties>

HPE Networking Products

<https://www.hpe.com/support/Networking-Warranties>

Regulatory information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at the Hewlett Packard Enterprise Support Center:

<https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Hewlett Packard Enterprise is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements such as REACH (Regulation EC No 1907/2006 of the European Parliament and the Council). A chemical information report for this product can be found at:

<https://www.hpe.com/info/reach>

For Hewlett Packard Enterprise product environmental and safety information and compliance data, including RoHS and REACH, see:

<https://www.hpe.com/info/ecodata>

For Hewlett Packard Enterprise environmental information, including company programs, product recycling, and energy efficiency, see:

<https://www.hpe.com/info/environment>

Documentation feedback

Hewlett Packard Enterprise is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback@hpe.com). When submitting your feedback, include the document title, part number,

edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.